



2024 FRM[®]

Exam Prep

SchweserNotes[™]

Risk Management and Investment Management
& Current Issues in Financial Markets

PART II BOOK 5



KAPLAN SCHWESER

Book 5: Risk Management and Investment Management & Current Issues in Financial Markets

SchweserNotes™ 2024



FRM Part II



SCHWESERNOTES™ 2024 FRM® PART II BOOK 5: RISK MANAGEMENT AND INVESTMENT MANAGEMENT & CURRENT ISSUES IN FINANCIAL MARKETS

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STUDY SESSION 13

83. Factor Theory

Andrew Ang, *Asset Management: A Systematic Approach to Factor Investing* (New York, NY: Oxford University Press, 2014). Chapter 6.

After completing this reading, you should be able to:

- provide examples of factors that impact asset prices and explain the theory of factor risk premiums.
- discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM.
- explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM.
- describe multifactor models and compare and contrast multifactor models to the CAPM.
- explain how stochastic discount factors are created and apply them in the valuation of assets.
- describe efficient market theory and explain how markets can be inefficient.

84. Factors

Andrew Ang, *Asset Management: A Systematic Approach to Factor Investing* (New York, NY: Oxford University Press, 2014). Chapter 7.

After completing this reading, you should be able to:

- describe the process of value investing and explain why a value premium may exist.
- explain how different macroeconomic risk factors, including economic growth, inflation, and volatility, affect asset returns and risk premiums.
- assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk.
- explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-French model as an example.
- compare value and momentum investment strategies, including their return and risk profiles.

85. Alpha (and the Low-Risk Anomaly)

Andrew Ang, *Asset Management: A Systematic Approach to Factor Investing* (New York, NY: Oxford University Press, 2014). Chapter 10.

After completing this reading, you should be able to:

- describe and evaluate the low-risk anomaly of asset returns.
- define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio.
- explain the impact of benchmark choice on alpha and describe characteristics of an effective benchmark to measure alpha.
- describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the information ratio using this law.
- apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark.
- explain how to use style analysis to handle time-varying factor exposures.
- describe issues that arise when measuring alphas for nonlinear strategies.
- compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly.
- describe potential explanations for the risk anomaly.

86. Portfolio Construction

Richard Grinold and Ronald Kahn, *Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk, 2nd Edition* (New York, NY: McGraw-Hill, 2000). Chapter 14.

After completing this reading, you should be able to:

- distinguish among the inputs to the portfolio construction process.

- b. evaluate the motivation for and the methods used for refining alphas in the implementation process.
- c. describe neutralization and the different approaches used for refining alphas to be neutral.
- d. describe the implications of transaction costs on portfolio construction.
- e. describe practical issues in portfolio construction, including the determination of an appropriate risk aversion, aversions to specific risks, and proper alpha coverage.
- f. describe portfolio revisions and rebalancing, and analyze the tradeoffs between alpha, risk, transaction costs, and time horizon.
- g. determine the optimal no-trade region for rebalancing with transaction costs.
- h. evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming.
- i. describe dispersion, explain its causes, and describe methods for controlling forms of dispersion.

87. Portfolio Risk: Analytical Methods

Philippe Jorion, *Value-at-Risk: The New Benchmark for Managing Financial Risk, 3rd Edition* (New York, NY: McGraw-Hill, 2007). Chapter 7.

After completing this reading, you should be able to:

- a. define, calculate, and distinguish between the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR.
- b. explain the impact of correlation on portfolio risk.
- c. apply the concept of marginal VaR in making portfolio management decisions.
- d. explain the risk-minimizing position and the risk and return-optimizing position of a portfolio.
- e. explain the difference between risk management and portfolio management and describe how to use marginal VaR in portfolio management.

88. VaR and Risk Budgeting in Investment Management

Philippe Jorion, *Value-at-Risk: The New Benchmark for Managing Financial Risk, 3rd Edition* (New York, NY: McGraw-Hill, 2007). Chapter 17.

After completing this reading, you should be able to:

- a. define risk budgeting.
- b. describe the impact of horizon, turnover, and leverage on the risk management process in the investment management industry.
- c. describe the investment process of large investors such as pension funds.
- d. describe the risk management challenges associated with investments in hedge funds.
- e. distinguish among the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk.
- f. explain the use of VaR to check manager compliance and monitor risk.
- g. explain how VaR can be used in the development of investment guidelines and for improving the investment process.
- h. describe the risk budgeting process and calculate risk budgets across asset classes and active managers.

STUDY SESSION 14

89. Risk Monitoring and Performance Measurement

Robert Litterman and the Quantitative Resources Group, *Modern Investment Management: An Equilibrium Approach* (Hoboken, NJ: John Wiley & Sons, 2003). Chapter 17.

After completing this reading, you should be able to:

- a. describe the three fundamental dimensions behind risk management, and their relation to VaR and tracking error.
- b. describe risk planning, including its objectives, effects, and the participants in its development.
- c. describe risk budgeting and the role of quantitative methods in risk budgeting.
- d. describe risk monitoring and its role in an internal control environment.
- e. identify sources of risk consciousness within an organization.
- f. describe the objectives and actions of a risk management unit in an investment management firm.
- g. describe how risk monitoring can confirm that investment activities are consistent with expectations.

- h. describe the Liquidity Duration Statistic and how it can be used to measure liquidity.
- i. describe the objectives of performance measurement tools.
- j. describe the use of alpha, benchmarks, and peer groups as inputs in performance measurement tools.

90. Portfolio Performance Evaluation

Zvi Bodie, Alex Kane, and Alan J. Marcus, *Investments, 12th Edition* (New York, NY: McGraw-Hill, 2020). Chapter 24.

After completing this reading, you should be able to:

- a. differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses.
- b. describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio and identify the circumstances under which the use of each measure is most relevant.
- c. describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures.
- d. determine the statistical significance of a performance measure using standard error and the t-statistic.
- e. describe style analysis.
- f. explain the difficulties in measuring the performance of actively managed portfolios.
- g. describe performance manipulation and the problems associated with using conventional performance measures.
- h. describe techniques to measure the market timing ability of fund managers with a regression and with a call option model and compute return due to market timing.
- i. describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution.

91. Hedge Funds

George M. Constantinides, Milton Harris, and René M. Stulz, eds., *Handbook of the Economics of Finance, Volume 2B* (Oxford, UK: Elsevier, 2013). Chapter 17.

After completing this reading, you should be able to:

- a. explain biases that are commonly found in databases of hedge funds.
- b. explain the evolution of the hedge fund industry and describe landmark events that precipitated major changes in the development of the industry.
- c. explain the impact of institutional investors on the hedge fund industry and assess reasons for the growing concentration of assets under management (AUM) in the industry.
- d. explain the relationship between risk and alpha in hedge funds.
- e. compare and contrast the different hedge fund strategies, describe their return characteristics, and describe the inherent risks of each strategy.
- f. describe the historical portfolio construction and performance trends of hedge funds compared to those of equity indices.
- g. describe market events that resulted in a convergence of risk factors for different hedge fund strategies and explain the impact of such convergences on portfolio diversification strategies.
- h. describe the problem of risk sharing asymmetry between principals and agents in the hedge fund industry.

92. Performing Due Diligence on Specific Managers and Funds

Kevin R. Mirabile, *Hedge Fund Investing: A Practical Approach to Understanding Investor Motivation, Manager Profits, and Fund Performance, 2nd Edition* (Hoboken, NJ: Wiley Finance, 2016). Chapter 12.

After completing this reading, you should be able to:

- a. identify reasons for the failures of hedge funds in the past.
- b. explain elements of the due diligence process used to assess investment managers.
- c. identify themes and questions investors can consider when evaluating a hedge fund manager.
- d. describe criteria that can be evaluated in assessing a hedge fund's risk management process.
- e. explain how due diligence can be performed on a hedge fund's operational environment.
- f. explain how a hedge fund's business model risk and its fraud risk can be assessed.
- g. describe elements that can be included as part of a due diligence questionnaire.

93. Predicting Fraud by Investment Managers

Stephen G. Dimmock and William C. Gerken, (2012) “Predicting Fraud by Investment Managers,” *Journal of Financial Economics*, 105(1), 153–173.

After completing this reading, you should be able to:

- explain the use and efficacy of information disclosures made by investment advisors in predicting fraud.
- describe the barriers and the costs incurred in implementing fraud prediction methods.
- discuss ways to improve investors’ ability to use disclosed data to predict fraud.

STUDY SESSION 15

94. Review of the Federal Reserve’s Supervision and Regulation of Silicon Valley Bank

“Review of the Federal Reserve’s Supervision and Regulation of Silicon Valley Bank” (through page 66), Board of Governors of the Federal Reserve System.

After completing this reading, you should be able to:

- describe the events leading up to the failure of Silicon Valley Bank.
- describe shortfalls and deficiencies in the Federal Reserve’s supervisory oversight of Silicon Valley Bank during the period that the bank transitioned from the Fed’s Regional Banking Organization (RBO) portfolio to its Large and Foreign Banking Organization (LFBO) portfolio.
- identify Silicon Valley Bank’s specific risk issues which led to and accelerated its failure including deposit concentration, type of deposits, held-to-maturity securities, available-for-sale securities, the bank’s contingent funding plan and capacity, and its capital raising efforts.
- identify and describe the failures and shortfalls of Silicon Valley Bank in the areas of governance and risk management including those related to the CRO position and the bank’s internal audit function.
- identify the scope of Silicon Valley Bank’s liquidity risk management deficiencies and shortfalls, including its modeling and stress testing of its 30-day liquidity buffer, as well as the actions that management and regulators considered to address these specific liquidity issues.
- describe the deficiencies in Silicon Valley Bank’s interest rate risk management process, including its modelling process, and explain how proper use of metrics such as net interest income (NII) at risk and economic value of equity (EVE) could have improved its management of interest rate risk.

95. The Credit Suisse CoCo Wipeout: Facts, Misperceptions, and Lessons for Financial Regulation

“The Credit Suisse CoCo Wipeout: Facts, Misperceptions, and Lessons for Financial Regulation,” *Journal of Applied Corporate Finance*, June 2023.

After completing this reading, you should be able to:

- describe the features and mechanics of contingent convertible bonds (CoCos) and explain the rationale for banks to issue them.
- explain the rescue of Credit Suisse by Swiss regulators in 2023 and compare it to the rescue of Bear Stearns by U.S. regulators during the financial crisis in 2008.
- explain the rationale for the write-down of Credit Suisse CoCos that was engineered by regulators during the rescue of Credit Suisse and its takeover by UBS.
- describe the reactions by market participants to the write-down of the CoCos, and explain and evaluate different arguments and lessons learned related to the decision to write down the CoCos.

96. Artificial Intelligence and Bank Supervision

“Artificial Intelligence and Bank Supervision,” Federal Reserve Bank of Richmond, Second Quarter 2023.

After completing this reading, you should be able to:

- describe historical evolution and common types of AI-based applications used in the financial sector.
- explain the advantages of implementing AI-based applications to the banking services companies and their customers.
- discuss the disadvantages and difficulties for financial companies using AI.
- clarify the specific issues faced by banks and regulators arising from utilizing AI in modeling and valuation.

97. Financial Risk Management and Explainable, Trustworthy, Responsible AI

“Financial Risk Management and Explainable, Trustworthy, Responsible AI,” Fritz-Morgenthal S, Hein B and Papenbrock J (2022) Frontiers in Artificial Intelligence.

After completing this reading, you should be able to:

- a. describe the challenge posed by potential model bias and the ethical and responsible considerations surrounding the implementation of AI-driven solutions in financial risk management.
- b. analyze the potential benefits and challenges of utilizing AI while maintaining fairness and preventing biases in risk assessment and decision-making.
- c. explain the proposed considerations for the technical validation of decision-making algorithms to check for potential unfairness.
- d. describe the approaches and technologies that should be considered in the implementation and assessment of Trustworthy AI.
- e. examine the application of Explainable AI (XAI) in the field of credit risk management as presented in the use case of a European insurance group.

98. Artificial Intelligence Risk Management Framework

“Artificial Intelligence Risk Management Framework,” National Institute of Standards and Technology.

After completing this reading, you should be able to:

- a. describe how organizations can frame the risks related to AI and explain the challenges that should be considered in AI risk management.
- b. identify AI actors across the AI lifecycle dimensions and describe how these actors work together to manage risks and achieve the goals of trustworthy and responsible AI.
- c. describe the characteristics of trustworthy AI and analyze the proposed guidance to address them.
- d. explain the potential benefits of periodically evaluating AI risk management effectiveness.
- e. describe specific functions applied to help organizations address the risks of AI systems in practice.

99. Climate-Related Risk Drivers and Their Transmission Channels

“Climate-related risk drivers and their transmission channels,” Basel Committee on Banking Supervision Publication, April 2021.

After completing this reading, you should be able to:

- a. describe climate-related risk drivers and explain how those drivers give rise to different types of risks for banks.
- b. compare physical and transition risk drivers related to climate change.
- c. assess the potential impact of different microeconomic and macroeconomic drivers of climate risk.
- d. describe and assess factors that can amplify the impact of climate-related risks on banks as well as potential mitigants for these risks.

100. Climate-Related Financial Risks—Measurement Methodologies

“Climate-related financial risks—measurement methodologies,” Basel Committee on Banking Supervision Publication, April 2021.

After completing this reading, you should be able to:

- a. describe main issues in identifying and measuring climate-related financial risks.
- b. identify unique data needs inherent in the climate-related risks and describe candidate methodologies that could be used to analyze these types of data.
- c. describe current and developing methodologies for measuring climate-related financial risks employed by banks and supervisors.
- d. compare and contrast climate-measuring methodologies utilized by banks, regulators, and third-party providers.
- e. identify strengths and weaknesses of the main types of measurement approaches.
- f. assess gaps and challenges in designing a modeling framework to capture climate-related financial risk.

101. Principles for the Effective Management and Supervision of Climate-Related Financial Risks

“Principles for the effective management and supervision of climate-related financial risks,”

Basel Committee on Banking Supervision Publication, June 2022.

After completing this reading, you should be able to:

- a. describe the principles for managing climate-related financial risks related to corporate governance and internal control framework.
- b. describe the principles for managing climate-related financial risks related to capital and liquidity adequacy and the risk management process.
- c. describe the principles for the management of climate-related financial risks related to management, monitoring, and reporting, comprehensive management of credit risk and other risks, and scenario analysis.
- d. describe the principles for the supervision of climate-related financial risks related to prudential regulatory and supervisory requirements for banks and responsibilities, powers, and functions of supervisors.

102. The Crypto Ecosystem: Key Elements and Risks

“The Crypto Ecosystem: Key Elements and Risks,” Basel Committee on Banking Supervision Publication, July 2023.

After completing this reading, you should be able to:

- a. describe the key elements of the crypto ecosystem, including unbacked crypto, stablecoins, smart contracts, and DeFi services.
- b. describe the structural flaws inherent in various elements of the crypto ecosystem.
- c. describe the risks crypto poses to parties including crypto investors, governments, regulators, and traditional financial institutions; and identify potential policy actions that can be taken to mitigate these risks.

103. Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector

“Digital Resilience and Financial Stability. The Quest for Policy Tools in The Financial Sector” (April 13, 2023). Jose Ramon Martinez, Banco de Espana.

After completing this reading, you should be able to:

- a. describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions.
- b. assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk.
- c. describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one.

The following is a review of the Risk Management and Investment Management principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Ang, Chapter 6.

READING 83

FACTOR THEORY

Study Session 13

EXAM FOCUS

In this reading, we introduce factor theory and factor risk. A key point is that it is not the exposure to an asset that is rewarded, but the exposure to the underlying factors. The risk of these factors is being rewarded with risk premiums. Several factor theories are introduced, including the capital asset pricing model (CAPM) and multifactor models. For the exam, understand the key assumptions of the CAPM while recognizing the model's limitations in a real-world setting, and be able to contrast the CAPM with the assumptions of multifactor models. Through multifactor models, we introduce the concept of a stochastic discount factor, which is a random variable used in pricing an asset. Finally, be familiar with the efficient market hypothesis, since it identifies areas of market inefficiencies that can be exploited through active management.

MODULE 83.1: FACTORS THAT IMPACT ASSET PRICES AND THE CAPM

LO 83.a: Provide examples of factors that impact asset prices and explain the theory of factor risk premiums.

In the context of factor investing, it is easiest to think of assets as bundles of **factor risks**, where exposure to the different factor risks earns risk premiums. The underlying **factors** may include the market (which is a tradable investment factor), interest rates, or investing styles (including value/growth, low volatility, or momentum). Factors may also be classified as fundamental macroeconomic factors, such as inflation and economic growth.

Factor theory is based on an analysis of factor risks. Factor risks represent exposures to *bad times*, where these exposures are rewarded with risk premiums. Factor theory is based on three primary principles:

1. *Factors are important, not assets.* It is not exposure to the specific asset that matters, rather the exposure to the underlying risk factors. As a result, investors must look through assets and understand the underlying factor risks.

2. *Assets represent bundles of factors.* Assets typically represent bundles of risk factors, although some assets, like equities and government bonds, can be thought of as factors themselves. Other assets, including corporate bonds, private equity, and hedge funds, contain many factors, such as equity risk, interest rate risk, volatility risk, and default risk. Assets' risk premiums reflect these risk factors.
3. *Investors have differing optimal risk exposures.* Investors each have different optimal exposures to risk factors. One of the important factors is volatility. Higher volatility results in higher asset risks during bad times. One important recent example of bad times was the 2007–2009 financial crisis. In return for bearing factor risks, investors require compensation through a risk premium (e.g., a volatility premium for volatility risk) during normal times. Economic growth represents another factor to which investors want different exposures.

Bad times could represent economic bad times, including high inflation and low economic growth. They could also represent bad times for investing, including poorly performing investments or markets. Factors are all unique and each represents exposure to a different set of bad times.

LO 83.b: Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM.

The **capital asset pricing model (CAPM)** describes how an asset behaves not in isolation, but in relation to other assets and to the market. The CAPM views not the asset's own volatility as the relevant measure, but its covariance with the market portfolio, as measured by the asset's *beta*.

The CAPM assumes that the only relevant factor is the market portfolio, and risk premiums are determined solely by beta. As mentioned, risk premiums are important because they compensate investors for losses during bad times. Risk here is determined by the assets' movements relative to each other, and not by the assets in isolation.

LO 83.c: Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM.

Implications of Using the CAPM

The CAPM holds six important lessons.

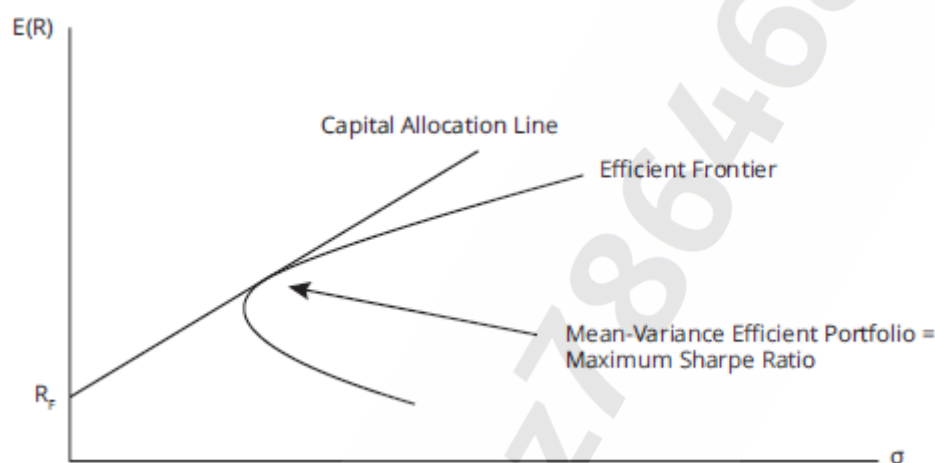
Lesson 1: Hold the factor, not the individual asset.

In a CAPM world, stocks are held in proportion to their market capitalization, where the sole factor is the market portfolio. The market portfolio can be constructed by holding many assets, which helps diversify away *idiosyncratic (firm-specific) risk*, leaving only *systematic (market) risk*. Individual stocks have risk premiums, which compensate investors for being exposed to the market factor. Market risk affects all investors exposed to the market portfolio.

According to the CAPM, investors do not wish to hold assets in isolation, because diversification improves the risk-return profile of a portfolio. The concept is simple: diversification helps ensure that bad returns from one asset will be offset by the returns of other assets that perform well. This also improves Sharpe ratios (i.e., risk premium divided by total risk). Investors continue to diversify until they are left with the market portfolio, which represents the optimal diversified portfolio.

Mean-variance efficient portfolio. Portfolio diversification and Sharpe ratios can be graphically represented by the mean-variance **efficient frontier**. When investors hold portfolios that combine the risky asset and the risk-free asset, the various risk-return combinations are represented by the **capital allocation line (CAL)**. The risky asset in this case is the *mean-variance efficient (MVE) market portfolio*, which is efficient because it represents the maximum Sharpe ratio given investors' preferences. The specific combination of the risk-free asset and MVE portfolio depends on investors' risk aversions.

Figure 83.1: Capital Allocation Line



Equilibrium. In equilibrium, demand for an asset equals supply, and since under the CAPM all investors hold the risky MVE market portfolio, the market is the factor. For equilibrium to happen, someone must hold the MVE portfolio as the risky asset. If no investor held the risky asset, the risky asset must be overpriced, and its expected return must be too low. This situation cannot represent an equilibrium state. Since under CAPM the expected payoff of an asset remains constant, the asset's expected return must increase as its price falls. In equilibrium, the risk factor is the market, and it has a risk premium. The market factor is a function of investor risk aversions and utilities, and risk premiums will not disappear since investors cannot use arbitrage to remove systematic risk.

Lesson 2: Investors have their own optimal factor risk exposures.

Every investor holds the same risky MVE market portfolio, but the proportion in which they hold it differs. Investors hold different combinations of the risk-free asset and the risky portfolio, representing various positions along the CAL.

Lesson 3: The average investor is fully invested in the market.

An investor with an average risk aversion would hold 100% of the risky MVE market portfolio, which represents the tangency point of the MVE frontier and the CAL. The average investor's risk aversion is, therefore, the risk aversion of the market.

Lesson 4: Exposure to factor risk must be rewarded.

When all investors invest in the same risky MVE portfolio, the CAL for an investor is called the **capital market line (CML)** in equilibrium. The risk premium of the CML depends on an investor's risk aversion and the volatility of the market portfolio:

$$E(R_M) - R_F = \bar{\gamma} \times \sigma_M^2$$

where $E(R_M) - R_F$ is the market risk premium, $\bar{\gamma}$ is the average investor's risk aversion, and σ_M^2 is the market portfolio's variance. During volatile market times (e.g., the 2007–2009 financial crisis), equity prices typically fall and expected returns increase. In the CAPM world, the risk premium is proportional to the market variance. Because market variance removes all idiosyncratic risk, the remaining systematic risk should be rewarded through the risk premium. When the average investor's risk aversion increases, the market risk premium should also increase.

Lesson 5: Risk is measured as beta exposure.

An individual asset's risk is measured as factor exposure to the asset, and higher factor exposures to the asset indicate higher expected returns (assuming the risk premium is positive). The risk premium of an individual asset is derived under the CAPM formula using beta pricing to construct the **security market line (SML)**. The formula states that:

$$E(R_i) - R_F = \frac{\text{cov}(R_i, R_M)}{\text{var}(R_M)} \times [E(R_M) - R_F] = \beta_i \times [E(R_M) - R_F]$$

where R_i is the individual stock's return, R_F is the risk-free rate, and **beta** is a function of the market variance and the asset's co-movement with the market: $[\beta_i = \text{cov}(R_i, R_M) / \text{var}(R_M)]$. Higher co-movements denote higher betas, which correspond to higher risk premiums. Whereas previously we looked at systematic risk and diversification, beta looks at idiosyncratic risk and the lack of diversification.

Higher betas imply lower diversification benefits. Investors tend to find high betas (high sensitivities to market returns) unattractive, and, therefore, want to be compensated with higher expected returns. On the other hand, low beta assets are valuable because they do comparatively well when markets perform poorly, offering significant diversification benefits. During the 2007–2009 financial crisis, certain assets (safe havens like gold and government bonds) became so attractive that they had negative expected returns. This meant investors actually paid to hold these assets!

Lesson 6: Valuable assets have low risk premiums.

The CAPM risk premium represents the reward investors receive for holding the asset in bad times. Since the market portfolio is the risk factor, bad times indicate low

market returns. Assets that have losses during periods of low market returns have high betas, which indicates they are risky and, therefore, should have high risk premiums. Low beta assets have positive payoffs when the market performs poorly, making them valuable to investors. As a result, investors do not require high risk premiums to hold these assets.

Shortcomings of the CAPM

The CAPM makes several simplifying assumptions that are necessary to make the model work; however, many of these assumptions are considered overly simplistic or not reflective of the real world. The assumptions of the CAPM break down especially in illiquid, inefficient markets where information may be costly and not available to all investors. We look at seven of these assumptions:

1. *Investors only have financial wealth.* Investors have unique income streams and liabilities. Liabilities are often denominated in real terms, and income streams are risky because incomes decline during periods of low economic growth. As a result, both inflation and income growth are important factors. In general, investors have many factors that contribute to wealth, including human capital (or labor income risk).
2. *Investors have mean-variance utility.* Mean-variance utility assumes a symmetric treatment of risk. In reality, investors have an asymmetric view of risk, disliking losses more than they like gains, which deviates from the CAPM assumptions. Therefore, in the real world, stocks exhibit different levels of downside risks. Those with higher downside risks should offer higher returns.
3. *Investors have a single period investment horizon.* While not a main assumption of the CAPM, a single period restriction does not hold in the real world. In the CAPM, all investors hold the market portfolio, which does not require rebalancing. However, the optimal strategy for long-term investors is to rebalance, which is a multi-period strategy.
4. *Investors have homogeneous (identical) expectations.* The assumption that all investors share the same expectations is not realistic in the real world, because investors have heterogeneous (differing) expectations. This can produce significant departures from the CAPM.
5. *Markets are frictionless (no taxes or transaction costs).* We all know that taxes and transaction costs affect investor returns; therefore, the CAPM assumption of frictionless markets does not hold in the real world. For illiquid securities, transaction costs can be very high, further heightening the deviations from the CAPM. In addition, investors have heterogeneous beliefs, but they may not be able to fully act on differing expectations if there are trading restrictions (e.g., a prohibition on short selling). When this happens, stock prices reflect only the expectations of those who believe stock prices will rise, causing asymmetries in the market. This is a deviation from the CAPM.
6. *All investors are price takers.* In the real world, investors are often price setters and not price takers. Large (institutional) investors frequently trade on special knowledge, and large trades will often move the market.

7. *Information is free and available to everyone.* In reality, information itself can be a factor. Information is often costly and unavailable to certain investors, which is a deviation from the CAPM.



MODULE QUIZ 83.1

1. Which of the following concepts would least likely meet the definition of a factor?
 - A. Market.
 - B. Volatility.
 - C. Hedge funds.
 - D. Momentum investing style.
2. According to the capital asset pricing model (CAPM), in equilibrium, all investors hold the mean-variance efficient portfolio. Which of the following investor types is an exception to this assumption?
 - A. Infinitely risk-averse investors.
 - B. Infinitely risk-tolerant investors.
 - C. Investors who hold some of the risk-free asset.
 - D. Investors who hold the market portfolio.
3. Assets that have losses during periods of low market returns have:
 - A. low betas and low risk premiums.
 - B. high betas and low risk premiums.
 - C. low betas and high risk premiums.
 - D. high betas and high risk premiums.
4. Which of the following statements best describes the relationship between asset payoffs and “bad times” events (high inflation, low economic growth, or both)?
 - A. The higher the expected payoff of an asset in bad times, the higher the asset’s expected return.
 - B. The higher the expected payoff of an asset in bad times, the lower the asset’s expected return.
 - C. The expected payoff of an asset in bad times is unrelated to the asset’s expected return, because it depends on investor preferences.
 - D. The expected payoff of an asset in bad times is unrelated to the asset’s expected return, because arbitrageurs eliminate any expected return potential.
5. Which of the following statements least likely represents a limitation of the capital asset pricing model (CAPM)?
 - A. All investors are price takers.
 - B. Information is costless to obtain.
 - C. All investors have the same expectations.

D. There are uniform taxes and transaction costs.

MODULE 83.2: MULTIFACTOR MODELS, PRICING KERNELS, AND EFFICIENT MARKET THEORY

Multifactor Models

LO 83.d: Describe multifactor models and compare and contrast multifactor models to the CAPM.

As mentioned, the CAPM is a single-factor model that looks at the market as the only factor and defines bad times as low returns to the market portfolio. By contrast, **multifactor models** incorporate other risk factors, including low economic growth, low GDP growth, or low consumption. One of the earliest multifactor models was **arbitrage pricing theory (APT)**, which describes expected returns as a linear function of exposures to common (i.e., macroeconomic) risk factors.

The lessons from multifactor models are similar to the lessons from the CAPM:

1. *Diversification is beneficial.* In the CAPM, the market removes (diversifies away) idiosyncratic risk. In multifactor models, it is the tradable version of a factor that removes this risk.
2. *Investors have optimal exposures.* Each investor has an optimal exposure to the market portfolio (in the CAPM) or to factor risks (in multifactor models).
3. *The average investor holds the market portfolio.* This is true under both the CAPM and multifactor models.
4. *Exposure to factor risk must be rewarded.* In the CAPM, the market factor is priced in equilibrium. In multifactor models, each factor has a risk premium, assuming no arbitrage or equilibrium.
5. *Risk is measured by a beta factor.* In the CAPM, an asset's risk is measured by its beta. In multifactor models, an asset's risk is measured by its factor exposures (i.e., factor betas).
6. *Valuable assets have low risk premiums.* Assets that have a positive payoff in bad times are attractive, and, therefore, have low risk premiums. In the CAPM, bad times are explicitly defined as low market returns.

Pricing Kernels

LO 83.e: Explain how stochastic discount factors are created and apply them in the valuation of assets.

Multifactor models define bad times over multiple factors. They use the concept of a **pricing kernel**, also known as the **stochastic discount factor (SDF)**, which represents a random variable used in pricing an asset. The SDF represents an index of bad times, where the bad times are indexed by a multitude of different factors and states. The SDF

is denoted as m in the multifactor model, where m is a single variable that captures all bad times for any given a and b constants:

$$m = a + b \times R_m$$

The CAPM is a special case of this model, where m moves linearly with the market return. However, modeling returns as linear is a shortcoming of the CAPM, which can be improved upon by using the pricing kernel which allows for the assumption of nonlinearity.

We can expand this model to include various factor exposures (f_1, f_2 , etc.) where SDF depends on a vector of these factors, where all the k factors represent different bad times:

$$m = a + b_1 f_1 + b_2 f_2 + \dots + b_k f_k$$

With multifactor pricing kernels, bad times can be defined as periods when an additional \$1 income becomes very valuable. Looking at bad times this way interprets SDF as a *marginal utility*. Periods of high marginal utility could arise from the loss of a job (resulting in low income, where the value of an extra dollar is high), low GDP growth, low consumption (resulting in current consumption below past consumption), or generally low economic growth.

Pricing Kernels vs. Discount Rate Models

In a traditional discount rate model, the price of an asset is determined by discounting its future cash flows at the appropriate discount rate:

$$P_i = E \left[\frac{\text{payoff}_i}{1 + E(R_i)} \right]$$

The discount rate is determined through the CAPM as:

$$E(R_i) = R_F + \beta_i \times [E(R_M) - R_F]$$

The SDF model can also be used to predict an asset's price, where we use the SDF as the relevant factor:

$$P_i = E[m \times \text{payoff}_i]$$

This equation helps explain the name “stochastic discount factor,” since the payoffs are discounted using m as the relevant factor. The SDF is called a “pricing kernel,” using the term kernel from statistics where we estimate m using the kernel estimator. Because the kernel is used to derive asset pricing, it is called a pricing kernel.

If we divide both sides of the equation by the asset's current price, P_i , the equation gives us a constant payoff formula, which we can then use to derive the risk-free asset:

$$\frac{P_i}{P_i} = E \left[m \times \frac{\text{payoff}_i}{P_i} \right]$$

$$1 = E[m \times (1 + R_i)]$$

$$\frac{1}{1 + R_F} = E[m \times 1], \text{ when payoffs are constant}$$

We can also model an asset's risk premium similar to the CAPM, where $[\beta_{i,m} = \text{cov}(R_i, m) / \text{var}(m)]$:

$$E(R_i) - R_F = \frac{\text{cov}(R_i, m)}{\text{var}(m)} \times \left(-\frac{\text{var}(m)}{E(m)} \right) = \beta_{i,m} \times \lambda_m$$

Beta is multiplied by the price of the “bad times” risk, determined as:

$$\lambda_m = -\frac{\text{var}(m)}{E(m)}$$

This equation represents the inverse of factor risk (denoted by the negative sign). In short, assets that have a positive payoff in bad times are valuable to hold, leading to high prices and low expected returns.

The equation for expected return can also be modeled as having exposure to the risk-free rate and multiple betas in the SDF model. Each beta represents a different macroeconomic factor, such as inflation, economic growth, the market portfolio, or investment strategy:

$$E(R_i) = R_F + \beta_{i,1} \times E(f_1) + \beta_{i,2} \times E(f_2) + \dots + \beta_{i,k} \times E(f_k)$$

Efficient Market Theory

LO 83.f: Describe efficient market theory and explain how markets can be inefficient.

The APT was one of the earliest forms of the **efficient market theory**. The APT is a multifactor model where market participants—including active managers and arbitrageurs—move an asset's expected return toward a value that represents an equilibrium risk-return tradeoff. The APT uses systematic factors that cannot be removed through arbitrage. As a result, investors demand to be compensated for this risk in the form of a risk premium.

Another efficient market theory was developed by Sanford Grossman and Joseph Stiglitz (1980).¹ In their theory, markets are near-efficient and information is costless. Market efficiency is in part caused by active managers searching for areas of inefficiency, making markets more efficient in the process. We can expect to find these areas of inefficiency in illiquid market segments where information does not move freely and where these inefficiencies make it difficult to earn large profits. Note, however, that the assumption of costless information creates a circular argument: if there is no cost to information and prices already reflect all information, there wouldn't be a need to collect information. However, if no one collects information, then it cannot be fully reflected in asset prices.

Market efficiency is also described in the **efficient market hypothesis (EMH)**. The EMH implies that speculative trading is costly, and active managers cannot generally beat the market. The average investor, who holds the market portfolio, can beat the market simply by saving on transaction costs. Even if markets cannot be perfectly

efficient, the EMH is still useful because it can help investors identify areas of market inefficiency that can be exploited through active management.

The EMH has been refined to improve upon the CAPM's shortcomings by allowing for imperfect information and various costs, including transaction, financing, and agency costs. Behavioral biases also represent inefficiencies, which have similar effects as frictions. Behavioral biases can be described either through a rational or behavioral explanation approach.

Under the *rational explanation* approach, losses during bad times are compensated by high returns. It is important to clearly define what bad times constitutes, and whether these bad times are actually bad for investors. For example, an investor who shorted the market would benefit, rather than incur losses, in a "bad times" scenario.

Under the *behavioral explanation* approach, it is agents' reactions (under/overreaction) to news that generates high returns. Perfectly rational investors are not prone to these biases, and they provide their own capital to take advantage of mispricing caused by biases. However, the markets may have barriers to the entry of capital that make it difficult to take advantage of mispricings, including structural barriers (e.g., certain investors are unable to take advantage of an opportunity) and regulatory barriers (e.g., minimum credit rating requirement of asset holdings). Structural barriers allow for behavioral biases to persist for a long time.

Ultimately, it is not the type of bias that matters, but whether the investor is different from the average investor who is subject to both rational and behavioral constraints, and whether return opportunities are expected to persist.



MODULE QUIZ 83.2

1. Market efficiency can be described with the efficient market hypothesis (EMH). Regarding the definition of EMH and the rational and behavioral explanations for this approach, the EMH suggests that:
 - A. speculative trading is costless.
 - B. active managers cannot generally beat the market.
 - C. under the behavioral explanation, losses during bad times are compensated for by high returns.
 - D. under the rational explanation, it is agents' under- or overreactions to news that generates high returns.

KEY CONCEPTS

LO 83.a

Exposure to different factor risks earns risk premiums. Underlying factors may include the market, interest rates, investing styles, inflation, and economic growth. Factor risks represent exposures to bad times, and this exposure must be compensated for with risk premiums. There are three important principles of factor risk:

1. It is not exposure to the specific asset that matters, rather the exposure to the underlying risk factors.

2. Assets represent bundles of factors, and assets' risk premiums reflect these risk factors.
3. Investors each have different optimal exposures to risk factors, including volatility.

LO 83.b

The capital asset pricing model (CAPM) is a single-factor model that describes how an asset behaves in relation to other assets and to the market. The CAPM incorporates an asset's covariance with the market portfolio, measured by the asset's beta. In the CAPM world, the only relevant factor is the market portfolio, and risk premiums are determined solely by beta.

LO 83.c

The CAPM has six important lessons:

1. Hold the factor, not the individual asset.
2. Investors have their own optimal factor risk exposures.
3. The average investor is fully invested in the market.
4. Exposure to factor risk must be rewarded.
5. Risk is measured as beta exposure.
6. Valuable assets have low risk premiums.

The CAPM has six main shortcomings (i.e., unrealistic simplifying assumptions):

1. Investors only have financial wealth.
2. Investors have mean-variance utility.
3. Investors have a single period investment horizon.
4. Investors have homogeneous (identical) expectations.
5. Markets are frictionless (no taxes or transaction costs).
6. All investors are price takers.

LO 83.d

There are six lessons from the multifactor models:

1. Diversification is beneficial.
2. Investors have optimal exposures, to factor risks in multifactor models.
3. The average investor holds the market portfolio.
4. Exposure to factor risks must be rewarded through risk premiums.
5. Risk is measured by factor betas.
6. Valuable assets have low risk premiums.

LO 83.e

Multifactor models define bad times over multiple factors using a pricing kernel, also known as the stochastic discount factor (SDF). The SDF represents an index of bad

times. The SDF is denoted as m in the multifactor model, representing a single variable that captures all bad times for any given a and b constants:

$$m = a + b \times R_m$$

The SDF model can also be set up using multiple factor exposures where factors represent different bad times.

The SDF model can be used to predict an asset's price, where SDF is the relevant factor m :

$$P_i = E[m \times \text{payoff}_i]$$

The asset's risk premium can be modeled using beta.

The risk premium equation can be set up using multiple factor exposures where factors represent different macroeconomic factors, including inflation, economic growth, the market portfolio, or investment strategy.

LO 83.f

Arbitrage pricing theory (APT) uses systematic factors that cannot be removed through arbitrage, and for which investors must be compensated for through risk premiums.

Another efficient market theory developed suggests that markets are near-efficient and information is costless. Active managers search for areas of inefficiency in illiquid market segments, making markets more efficient in the process.

The efficient market hypothesis (EMH) states that speculative trading is expensive, and active managers cannot beat the market on average. The EMH is useful because it helps investors identify areas of market inefficiency that active management can exploit. The EMH has been refined to allow for imperfect information, various costs (transaction, financing, and agency), and behavioral biases.

Under the rational explanation of behavioral biases, losses during bad times are compensated for by high returns. Under the behavioral explanation, it is agents' under- or overreactions to news that generates high returns. Market barriers may make it difficult to take advantage of mispricings.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 83.1

1. **C** Assets, including corporate bonds, private equity, and hedge funds, are not considered factors themselves, but contain many factors, such as equity risk, interest rate risk, volatility risk, and default risk.

Some assets, like equities and government bonds, can be thought of as factors themselves. Factors may also include the market (a tradable investment factor), interest rates, or investing styles (including value/growth, low volatility, or momentum). (LO 83.a)

2. **A** According to the CAPM, all investors hold a combination of the risky mean-variance efficient market portfolio and the risk-free asset. All investors hold the same market portfolio (therefore the mean-variance efficient portfolio is the market portfolio), and it is only the quantity of holdings that differs among investors. The only exception to this assumption is an *infinitely* risk-averse investor, who would only hold the risk-free asset. (LO 83.c)
3. **D** Assets that have losses during periods of low market returns have *high* betas (high sensitivity to market movements), which indicates they are risky and, therefore, should have *high* risk premiums. Low beta assets have positive payoffs when the market performs poorly, making them valuable to investors. As a result, investors do not require high risk premiums to hold these assets. (LO 83.c)
4. **B** The higher the expected payoff of an asset in bad times, the lower the asset's expected return. Assets that have a positive payoff in bad times are valuable to hold, leading to high prices and, therefore, low expected returns. (LO 83.c)
5. **D** The CAPM does not assume *uniform* taxes and transaction costs; it assumes there are *no* taxes or transaction costs (i.e., frictionless markets). The other limiting assumptions of the CAPM include:
1. Investors only have financial wealth.
 2. Investors have mean-variance utility.
 3. Investors have a single period investment horizon.
 4. Investors have homogeneous (identical) expectations.
 5. All investors are price takers.
- (LO 83.c)

Module Quiz 83.2

1. **B** The EMH implies that speculative trading is costly, and active managers cannot generally beat the market. Under the rational explanation of behavioral biases, losses during bad times are compensated for by high returns. Under the behavioral explanation, it is agents' under- or overreactions to news that generates high returns. Market barriers may make it difficult to take advantage of mispricings. (LO 83.f)

¹ Sanford J. Grossman and Joseph E. Stiglitz, "On the Impossibility of Efficient Markets," *American Economic Review* 70 (1980): 393–498.

The following is a review of the Risk Management and Investment Management principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Ang, Chapter 7.

READING 84

FACTORS

Study Session 13

EXAM FOCUS

Macroeconomic factors have been linked to asset returns. The most important macro factors that affect returns are economic growth, inflation, and volatility. Volatility risk can be mitigated by investing in low-volatility assets or buying volatility protection in the derivatives market (e.g., buying put options). The capital asset pricing model (CAPM) is a single-factor model that relates asset returns to market risk. The Fama-French model is a multifactor model that adds a size factor and a value factor to the original CAPM market factor to explain stock returns. A momentum factor can also help explain asset returns. The momentum strategy far outpaces the size and value-growth strategies in terms of returns. However, momentum strategies are prone to crashes. For the exam, understand the risk and return profiles of each factor. Also, be aware of rational and behavioral explanations for each factor.

MODULE 84.1: VALUE INVESTING AND MACROECONOMIC FACTORS

Value Investing

LO 84.a: Describe the process of value investing and explain why a value premium may exist.

Risk premiums are driven by **factors**. Economy-wide (i.e., fundamental-based) factors such as inflation, volatility, productivity, economic growth, and demographics drive risk premiums. Additionally, factors related to tradeable investment styles such as momentum investing, value investing, and investing based on firm size drive returns.

A company's **book value** (i.e., net worth) per share is equal to total assets minus total liabilities divided by shares outstanding. It indicates, on a per-share basis, what a company would be worth if it liquidated its assets and paid off its liabilities. Value stocks have high book-to-market ratios while growth stocks have low book-to-market

ratios, where “market” indicates the company’s stock price. An investment strategy that is long value stocks and short growth stocks is called a **value-growth strategy**.

Historically, value stocks have significantly outperformed growth stocks. One dollar invested in a value-growth strategy in 1965 would be worth more than \$6 around 2012, with a peak value of nearly \$8 in 2006 and 2007. During the more than 40-year period, value stock returns experienced a sharp downturn during the tech boom, during the late 1990s, during the financial crisis in 2007–2009, and again in 2011. Overall, however, value investing appears to work. Are returns higher than growth investing returns due to a systematic factor? Alternatively, is there a value risk premium? Risk factors offer premiums to investors to compensate them for bearing losses in bad times, like the late 1990s and 2007–2009. Rational and behavioral explanations for the value premium will be discussed in detail in LO 84.e.

Macroeconomic Factors

LO 84.b: Explain how different macroeconomic risk factors, including economic growth, inflation, and volatility, affect asset returns and risk premiums.

Macroeconomic factors, such as increasing inflation or slowing economic growth, affect all investors to varying degrees. Most, though not all, investors are hurt by rising inflation, slowing economic growth, or both. But it is not the level of the factor that matters, it is the shock (i.e., unanticipated changes) to a factor. For example, asset prices generally fall when inflation unexpectedly increases. Economic growth, inflation, and volatility are the three most important macro factors that affect asset prices.

Economic Growth

Risky assets like equities generally perform poorly during periods of low economic growth. Less-risky assets like bonds, and especially government bonds, tend to perform well during periods of slow growth. For the investor who can weather a downturn easily, she should invest in equities because returns will be greater over the long run. Periods of stronger growth generally last longer than downturns. For the investor who cannot bear large losses during a period of slow growth, she should invest in bonds. Her portfolio will likely perform better during the downturn but worse in the long run.

Figure 84.1 reports the returns of large and small stocks, as well as government, investment grade, and junk (high-yield) bonds during expansions and retractions as defined by the National Bureau of Economic Research (NBER). Returns are from Ibbotson Morningstar and cover the period 1952 through 2011. During periods of recession, government and investment grade bonds outperform equities and high-yield bonds, yielding 12.3% and 12.6%, respectively. During expansion periods, equities outperform bonds with large stocks yielding 12.4% and small stocks yielding 16.8%. High-yield bond returns appear indifferent to changes in economic growth, yielding 7.4% in recessions and 7.7% in expansions.

Figure 84.1 also reports returns based on quarter-on-quarter real GDP growth and quarter-on-quarter consumption growth (i.e., real personal consumption expenditures).

The patterns are similar to those exhibited by NBER expansion/recession data. Equities outperform in periods of high real GDP growth and high consumption growth, while bonds outperform in periods of low real GDP growth and low consumption growth. High-yield bonds perform slightly better in high-growth periods.

Figure 84.1: Investment Returns During Expansions and Recessions

	Large Stocks	Small Stocks	Government Bonds	Corporate Bonds	
				Investment Grade	High Yield
Returns					
Full sample	11.3%	15.3%	7.0%	7.0%	7.6%
Business Cycles					
Recessions	5.6%	7.8%	12.3%	12.6%	7.4%
Expansions	12.4%	16.8%	5.9%	6.0%	7.7%
Real GDP					
Low	8.8%	12.2%	10.0%	9.7%	7.0%
High	13.8%	18.4%	3.9%	4.4%	8.2%
Consumption					
Low	5.6%	5.6%	9.6%	9.1%	7.1%
High	17.1%	25.0%	4.4%	5.0%	8.2%
Inflation					
Low	14.7%	17.6%	8.6%	8.8%	9.2%
High	8.0%	13.0%	5.4%	5.3%	6.0%

In terms of volatility, both stocks and bonds are more volatile during downturns and periods of low growth. For example, large stock return volatility was 23.7% during recessions and 14.0% during expansions. Government bonds perform best during recessions but are also more volatile during these periods (15.5% volatility during recessions and 9.3% volatility during expansions).

Inflation

High inflation is generally bad for both stock and bond prices and returns. Figure 84.1 indicates that all categories perform better in low inflation versus high inflation periods. Volatilities are also higher in high inflation periods. Large and small stocks return 14.7% and 17.6%, respectively, during low inflation periods, and 8.0% and 13.0% during high inflation periods. Bond yields of 8.6%, 8.8%, and 9.2% (government, investment grade, and high-yield bonds, respectively) during low inflation periods exceeded returns during high inflation periods by approximately 3.0%. Bonds are fixed payment securities. As such, it is clear that bonds should perform poorly in high inflation times. Inflation lowers real bond returns. It is less clear that stocks perform poorly in high inflation times since they represent ownership of real, productive companies, not a claim to a stream of fixed cash flows.

Volatility

Volatility is an important risk factor for many asset classes. The CBOE Volatility Index (VIX) represents equity market volatility. The correlation between the VIX and stock returns has historically indicated a negative relationship (correlation coefficient of -0.39 between 1986 and 2011). This means that stock returns tend to drop when the VIX (equity volatility) increases.

The financial leverage of companies increases during periods of increased volatility because debt stays approximately the same while the market value of equity falls. The negative relationship between stock returns and volatility is called the **leverage effect**. As financial leverage increases, equities become riskier and volatility increases. Additionally, higher volatility increases the required rates of return on equities, pushing stock prices down. Thus, there are two paths to lower stock returns resulting from higher volatility:

1. When market volatility increases, the leverage effect suggests a negative relationship between stock returns and volatility.
2. When market volatility increases, discount rates increase and stock prices decline so that future stock returns can be higher (to compensate for the higher volatility). The capital asset pricing model (CAPM) supports this second path.

Other Macroeconomic Factors

Other macroeconomic factors, including productivity risk, demographic risk, and political risk, also affect asset returns. **Productivity shocks** affect firm output. In periods of falling productivity, stock prices fall (like in the 1960s and 1970s). In periods of improving productivity (like the 1980s and 1990s computer revolution) productivity shocks are positive and stock prices generally increase. The correlation between productivity shocks and stock returns is relatively high (approximately 50%).

New models, called *dynamic stochastic general equilibrium (DSGE)* macro models, indicate that economic variables change over time due to the actions of agents (i.e., consumers, firms, governments, and central banks), technologies (and their impact on how firms produce goods and services), and the way that agents interact (i.e., markets). A benchmark model created by Smets and Wouters (2007)¹ specifies seven shocks that impact the business cycle. They are: (1) productivity, (2) investment, (3) preferences, (4) inflation, (5) monetary policy, (6) government spending, and (7) labor supply.

Like productivity shocks, **demographic risk**, which can be interpreted as a shock to labor output, is a shock to firm production. Economic *overlapping generation (OLG)* models include demographic risk as a factor affecting investor returns. In these models, generations overlap. Young, middle-age, and retired workers exist in a system. Workers earn income and save during the young and middle-age stages. Retired workers disinvest. As a cohort progresses through life, they join others already in the cohort but born at an earlier time. According to several OLG models, events that shock the composition of the cohort, like World Wars I and II, infectious diseases, like the COVID-19 pandemic, and the baby boom, which followed World War II, impact returns. For example, one model predicts that stock prices will fall when baby boomers retire as

they liquidate assets to fund consumption. This would occur if there are relatively fewer young and middle-age investors to offset the asset liquidation of retirees. If there are a greater number of young and middle-age workers, relative to retirees, the impact will be lessened (or even overcome). Another study shows that risk aversion increases with age and that as the average age of the population increases, the equity risk premium should also increase. Note that it is important to use cross-country data in demographic studies.

Political (or sovereign) risk, once thought only important in emerging markets, increases risk premiums. The financial crisis of 2007–2009 made clear that political risk affects both developed and undeveloped countries.



MODULE QUIZ 84.1

1. A low book-to-market value ratio is indicative of a:
 - A. value stock.
 - B. growth stock.
 - C. small-cap stock.
 - D. large-cap stock.
2. Which of the following asset classes has approximately the same returns in high economic growth periods and low economic growth periods?
 - A. Small-cap stocks.
 - B. Large-cap stocks.
 - C. Government bonds.
 - D. High-yield bonds.

MODULE 84.2: MANAGING VOLATILITY RISK AND DYNAMIC RISK FACTORS

Managing Volatility Risk

LO 84.c: Assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk.

Volatility can be mitigated by investing in less volatile assets. As one would expect, bond returns are less impacted by volatility in equity markets (than equity returns). However, bonds are not necessarily a safe haven. Correlation between changes in the VIX and bond returns was 0.12 (between 1986 and 2011). This means bonds perform better than stocks (with a correlation coefficient of -0.39) when the VIX is rising, but the relationship is not highly positively correlated. For example, during the recent financial crisis, volatility was a factor that caused risky assets, bonds and stocks included, to fall simultaneously. The VIX can also capture uncertainty. Some research indicates that uncertainty risk is different from volatility risk, but the two risks are highly correlated.

Other investment approaches also perform poorly in periods of increased volatility. A number of strategies have a large exposure to volatility risk. For example, currency strategies perform poorly during periods of high volatility. For investors who want to

avoid volatility, they can buy put options (i.e., protection against volatility). Out-of-the-money puts, which pay off during periods of high volatility, provide hedges against volatility risk.

In sum, there are two basic approaches to mitigating volatility risk. They are:

- Invest in less volatile assets like bonds, understanding that they too can perform poorly during extreme circumstances such as the 2007–2009 financial crisis.
- Buy volatility protection in the derivatives market (e.g., buy out-of-the-money put options).

Volatility Premiums

Typically, an investor buys an asset, like a stock, and the long position produces a positive expected return. In other words, on average, assets have positive premiums. However, volatility has a negative premium. To collect the volatility premium, one must sell volatility protection (e.g., sell out-of-the money put options). Realized volatilities are lower on average (by approximately 2%–3%) than VIX implied volatilities. This means that, on average, options are expensive and investors can collect volatility premiums by shorting volatility strategies.

During normal economic periods, selling volatility provides high, stable payoffs. However, when there is a crash, like the 2007–2009 financial crisis, sellers of volatility suffer large, negative returns. A volatility (swap) index constructed by Merrill Lynch indicates steadily (with minor blips) increasing cumulative returns between January 1989 and December 2007, until the financial crisis. Between September and November 2008, losses were nearly 70%. Considering the data leading up to the crisis (through December 2007), selling volatility looked like easy money. Considering the whole sample period, including the crisis, the data indicates negative skewness of –8.26. Without the crisis (i.e., only considering the data up to December 2007), the negative skewness was a mere –0.37.



PROFESSOR'S NOTE

Selling volatility is like selling insurance. If you sell auto insurance, you collect stable premiums over time but occasionally face a large payout due to a car accident. The same is true for selling out-of-the-money put options. The seller collects option premiums for years, then a disaster happens, like the 2007–2009 financial crisis, and the seller faces massive losses. Option purchasers know in advance what they can lose, the option premium, but sellers do not. Thus, during a market crash, losses could be massive for volatility sellers. Only investors who can tolerate periods of high volatility, which often coincide with losses (sometimes very large losses), should sell volatility.

Academics have estimated a relationship between the expected market risk premium $E(R_M) - R_F$ and volatility. The equation is shown as follows:

$$E(R_M) - R_F = \bar{\gamma} \times \sigma_M^2$$

where σ_M^2 is equal to the variance of the market return and $\bar{\gamma}$ represents the average investor's risk aversion. While the coefficient $\bar{\gamma}$ is positive in theory, various studies have estimated it as either positive, negative, or zero. Again, though, whether positive or negative, only those investors who can withstand massive losses during periods of high volatility should sell volatility.

Dynamic Risk Factors

LO 84.d: Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-French model as an example.

The **capital asset pricing model (CAPM)** is a single-factor model. In the CAPM, the single risk factor is market risk. Stocks that have high exposure to the CAPM market factor perform well when the market performs well and poorly when the market performs poorly. Over the long run, stocks with high betas (i.e., a high market risk factor) should have higher returns than the market return. Returns are higher for high beta stocks to compensate investors for losses during bad periods.

The market portfolio can be readily traded via low-cost index funds, stock futures, and exchange-traded funds (ETFs). In general, macro factors, like political, inflation, and growth risks, are not directly traded (volatility risk is the exception). As a result, dynamic factors can be easily employed in portfolios. The best known example of a tradeable multifactor model is called the **Fama-French model**, introduced in 1993.²



PROFESSOR'S NOTE

In the academic finance literature “style factors,” “investment factors,” and “dynamic factors” are used interchangeably. Practitioners also refer to these factors as “smart beta” or “alternative beta.” Fama and French were the first to develop a multifactor model that captured these effects.

The Fama-French model (called the Fama-French three-factor model) explains asset returns based on three dynamic factors. The model includes:

- The traditional CAPM **market risk factor (MKT)**.
- A factor that captures the **size effect (SMB)**.
- A factor that captures the **value/growth effect (HML)**.

The Fama-French three-factor model is expressed as follows:

$$E(R_i) = R_F + \beta_{i,MKT} \times E(R_M - R_F) + \beta_{i,SMB} \times E(SMB) + \beta_{i,HML} \times E(HML)$$

Following the market factor, the second factor in the model is SMB. The SMB factor refers to the difference between the returns on small stocks (small market capitalization) versus big stocks (large market capitalization). In other words, the risk factor is small stock returns minus big stock returns, thus SMB. Historically, small-cap stocks have outperformed large-cap stocks. This factor captures the higher performance of small companies relative to large companies. Note, however, that the average stock only has market exposure. Every stock cannot be large, and every stock cannot be small.

The third factor in the model is HML. This factor captures the return differential of high book-to-market stocks versus low-book-to-market stocks. The ratios are calculated as book value divided by market capitalization. Recall that a value strategy consists of buying low-priced stocks (i.e., taking a long position in low-priced stocks) and selling high-priced stocks (i.e., shorting high-priced stocks), normalized by book value. Growth stocks have high stock prices relative to book values, and value stocks have low stock prices relative to book values. Historically, value stocks have outperformed growth stocks. Thus, the Fama-French factors are constructed to capture size (SMB) and value (HML) premiums (known as **factor-mimicking portfolios**).

A value investor, who buys stocks that are perceived as trading below their fundamental value, would have a positive HML beta. Relative to the CAPM expected return, the value investor's return adjusts upward by $\beta_{i,HML} \times E(HML)$. Thus, the overall risk premium increases above the single-factor CAPM risk premium. Likewise, the overall risk premium is adjusted down by $\beta_{i,HML} \times E(HML)$ for growth stocks. This is because growth stocks have negative HML betas, so expected returns are adjusted downward.

In the CAPM, both the average stock beta and the market beta equal one. In the Fama-French model, the HML and SMB betas are centered on zero. The average investor earns the market return as the average stock (or portfolio of stocks) does not have a value or size tilt. This means the investor must specifically choose a value play or a size play, to benefit from the HML and SMB factors. Also, the CAPM and Fama-French models assume betas are constant, but empirical research indicates they vary and increase during bad times.



MODULE QUIZ 84.2

1. Which of the following investment options provides a means of mitigating volatility risk?
 - A. Buying put options.
 - B. Selling put options.
 - C. Buying equities.
 - D. Buying call options.
2. Which of the following is not a factor in the Fama-French three-factor model?
 - A. The capital asset pricing model market risk factor.
 - B. The small capitalization minus big capitalization risk factor.
 - C. The winners minus losers risk factor.
 - D. The high book-to-market value minus low book-to-market value risk factor.

MODULE 84.3: VALUE AND MOMENTUM INVESTMENT STRATEGIES

LO 84.e: Compare value and momentum investment strategies, including their return and risk profiles.

The fact that small stocks tend to outperform big stocks, after adjusting for the firm's beta, was discovered by Banz (1981)³ and similarly by Reinganum (1981).⁴ Following

the publication of this finding, the effect disappeared. In other words, if you examine the returns to an SMB strategy from 1965 to 2011, returns to the strategy peak in the early 1980s, with no evidence of a small stock premium in subsequent years. The two possible explanations for the disappearing size effect are as follows:

- *Data mining.* Fischer Black (1993)⁵ suggested data mining following the publication of the Fama and French study. If a finding is discovered with *in-sample data* (i.e., in the data used in the original study) but is not substantiated in further studies using *out-of-sample data*, then data mining provides a possible explanation for the result.
- *Investor actions.* Upon the publication of the Banz and Reinganum studies, investors, acting rationally, bid up the prices of small-cap stocks until the SMB effect was removed. This is consistent with the efficient market hypothesis (EMH) in which investors exploit anomalies until they can no longer profit from them. If this is true, then size should be removed as a risk factor in the Fama-French model.

Note that small stocks do tend to have higher returns (i.e., weak size effect), partially because they are less liquid than large-cap stocks. Also, the value and momentum effects, discussed next, are stronger for small stocks. However, the ability to capture small-cap excess returns over the market (on a risk-adjusted basis) is no longer present.

Value Investing

Unlike the disappearing size premium, the value risk premium has provided investors with higher risk-adjusted returns for more than 50 years. Value strategies have suffered periods of loss, including the 1990s recession, the dot com bull market of the late 1990s, and the 2007–2009 financial crisis. The notion of value investing dates back to when Graham and Dodd (1934)⁶ published *Security Analysis* with a focus on finding stocks that had prices lower than their fundamental values.

There are generally two explanations for the value premium, one rational and the other behavioral.

Rational Theories of the Value Premium

Value stocks move with each other and co-vary with growth stocks in the rational story about the reason a value premium exists. They perform well together and poorly together. Value is risky and, as such, value stocks sometimes perform poorly. The value premium is compensation for these periods of poor performance, for losing money during bad times. Value did perform poorly during the bull market in the late 1990s. This means rational stories must define “bad times” and that value earns a premium on average, not all of the time. Also, not all value risk can be diversified away. The remaining value risk is captured in the value premium.

Labor income risk, investment growth, “luxury” consumption, long-run consumption risk, and housing risk are factors that have been used to explain the value premium. Value stock betas often increase during bad times defined by these risks, causing value stocks to be particularly risky. Macro-based and CAPM risk factors turn out to be the same factors that affect value firms.

Consider the difference between growth and value firms. Growth firms are more adaptable and can adjust when times change because the bulk of their capital is human capital. Value firms are more “old school” with capital in the form of fixed assets that cannot be redeployed when times change. Thus, value firms have *high and asymmetric adjustment costs*. This makes value stocks fundamentally more risky than growth stocks.

The average investor holds the market portfolio. Some investors choose a value tilt and others a growth tilt. The decision boils down to how well the investor can withstand bad times. Given the factors defined previously as bad for value (i.e., labor income risk, investment growth, etc.), the investor must ask himself, “Are these times bad for me (versus bad in general)?” If, for example, an investor can manage well during times of low investment growth, that is not a bad time for that investor relative to the average investor. The investor, who has a comparative advantage in holding value stocks, can bear value risk and, therefore, can earn the value premium.

Behavioral Theories of the Value Premium

Behavioral theories of the value premium revolve around two basic ideas: (1) overextrapolation and overreaction and (2) loss aversion and mental accounting.

Overextrapolation and overreaction. Investors have a tendency to assume that past growth rates will continue in the future. This is called **overextrapolation**. For example, a technology company may have a period of tremendous growth as it developed new products that are in high demand. Many investors may assume that this company will continue this growth into the future. Investors often bid up the prices of growth stocks beyond their intrinsic values due to unwarranted optimism. Prices fall when the high expected growth doesn’t materialize, leading to lower returns than those earned on value stocks.

Loss aversion and mental accounting. Investors dislike losses more than they like gains (i.e., **loss aversion**), and they tend to view investment gains and losses on a case-by-case basis rather than on a portfolio basis (known as *mental accounting*). Barberis and Huang (2001)⁷ use this notion to explain the value premium. They argue that the reason value stocks have high book-to-market values is that they have undergone a period of very poor performance. Loss-averse investors view the stock as riskier and, therefore, require a higher rate of return.



PROFESSOR’S NOTE

The extrapolation/overreaction behavioral explanation of the value premium is different from the rational one in that in the behavioral explanation, value stocks are not riskier; they are just cheap relative to growth stocks. Investors tend to underestimate the growth prospects of value stocks and overestimate the growth prospects of growth stocks. This bids up the prices of growth stocks and bids down the prices of value stocks, allowing value stocks to outperform on average. Investors must determine if they tend to overextrapolate or not. Investors who act like other average, non-overreacting or non-underreacting investors should hold the market

portfolio. Investors who overextrapolate will lean toward growth stocks, and those who underreact will lean toward value stocks.

Why are there not enough value investors in the market to push up prices and remove the value premium, as described in the section on the small-cap effect? Maybe investors find value investing difficult, although it is easy to sort stocks on a book-to-market basis using internet screening tools. Perhaps investment horizons must be too long to engage in value investing. The book-to-market value effect described here requires at least a three month to six month horizon. It is possible that not enough institutions have a long enough investment horizon to adopt a value investing approach.

Value investing exists in all asset classes. Strategies include the following:

- *Riding the yield curve* in fixed income (i.e., capturing the duration premium).
- *Roll return* in commodities (i.e., an upward- or downward-sloping futures curve determines the sign of the return).
- *Carry* in foreign exchange (e.g., long positions in currencies with high interest rates and short positions in currencies with low interest rates). In this case, high yields are akin to low prices in equity value strategies.

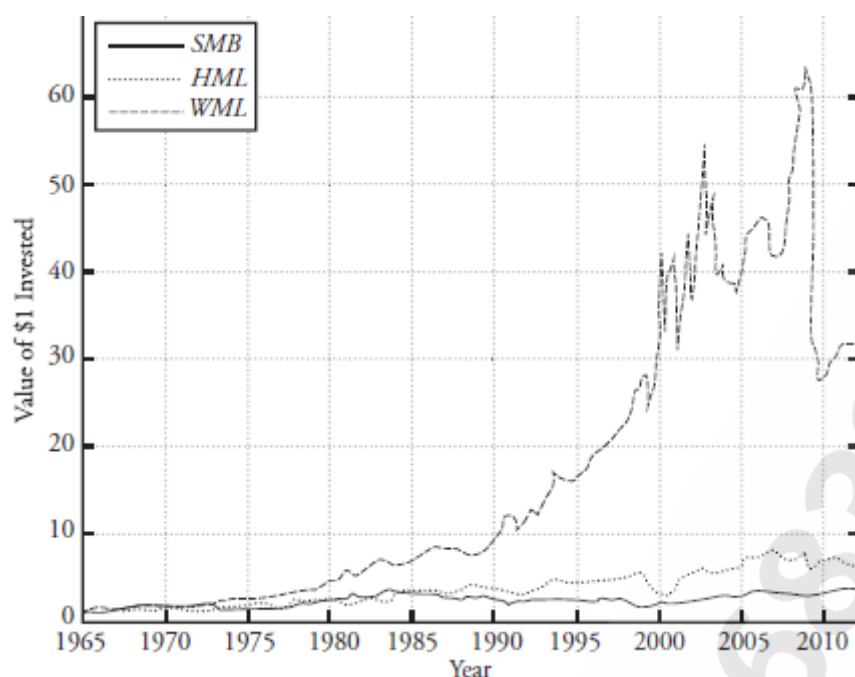
Retail investors can implement value strategies via low-cost index products. Large, institutional investors can, at least theoretically, cheaply implement value strategies across markets.

Momentum Investing

In 1993, the same year Fama and French captured the size and value/growth effects, Jegadeesh and Titman⁸ identified a **momentum effect**. Momentum strategies (also called **trend investing**) consist of buying stocks that have gone up over a period (e.g., six months or so) and short stocks that have fallen over the same period (i.e., buy past “winners” and sell past “losers”). The momentum factor, WML, stands for “winners minus losers.” It is also sometimes denoted UMD for “up minus down,” buying stocks that have gone up in price and selling stocks that have gone down in price. A momentum premium is observed in fixed income (government and corporate bonds), international equities, commodities, real estate, and specific industries and sectors.

The returns to momentum investing exceed size and value investing premiums by a wide margin. Figure 84.2 illustrates the differences in returns across the three strategies. One dollar invested in the WML premium in January 1965 reached a high of more than \$60 before following precipitously (below \$30) during the 2007–2009 financial crisis. Correlation between the value premium and the momentum premium was only approximately -0.16 during this period. This means that value returns are not opposite momentum returns.

Figure 84.2: Returns for SMB, HML, and WML Strategies



Value and momentum strategies are, however, opposite each other in the following sense. Value investing is inherently stabilizing. It is a *negative feedback strategy* where stocks that have fallen in value eventually are priced low enough to become value investments, pushing prices back up. Momentum is inherently destabilizing. It is a *positive feedback strategy* where stocks that have been increasing in value are attractive to investors, so investors buy them, and prices increase even more. Momentum investing can lead to crashes (e.g., the more than 50% drop during the 2007–2009 financial crisis). Notice that value and growth returns did not fall in quite so dramatic a fashion. An investor following a momentum strategy should still rebalance his portfolio.

Momentum is often added to the Fama-French model as follows:

$$E(R_i) = R_F + \beta_{i,MKT} \times E(R_M - R_F) + \beta_{i,SMB} \times E(SMB) + \beta_{i,HML} \times E(HML) + \beta_{i,WML} \times E(WML)$$

As mentioned, momentum can be riskier than value or size investing in that it is more prone to crashes. There have been 11 momentum crashes on record: seven during the 1930s Great Depression, three during the financial crisis starting in 2007, and one in 2001. During the 2007–2009 crisis, financial stocks were hit hard. Losers tend to keep losing, and they likely would have, but the government bailout put a floor on stock prices. Momentum investors were short these stocks. When the government bailed out financial firms and other firms that were hit hard, momentum investors experienced large losses as the market rebounded. During the Great Depression, policymakers also influenced asset prices, causing losses to momentum investors.

Momentum risk includes:

- Tendency toward crashes.

- Monetary policy and government risk (i.e., the government gets in the way of the natural progression of asset prices).
- Macro factors such as the business cycle, the state of the stock market, and liquidity risk.

Behavioral explanations suggest that investor biases explain momentum. Investors overreact (a delayed overreaction) to good news about firms. This causes prices to drift upward. Alternatively, investors may underreact to good news, causing prices to increase less than they should have given the good news. As investors acquire more information, prices go up in the next period. Thus, behavioral explanations for the momentum premium fall into two, difficult-to-distinguish camps:

1. *Overreaction to good news.* In some cases, overconfident, informed investors, like retail investors and hedge fund managers, observe positive signals in stock performance. They attribute the performance to their own skill. The overconfidence leads to overreaction, pushing prices up above their fundamental values.
2. *Underreaction to good news.* In this case, “news watchers” ignore information in the history of stock prices and other investors trade only on history (i.e., past price signals) and ignore fundamental information about the firm. In both cases, information is only partially incorporated into stock prices, causing an underreaction.

Whether there is momentum that results from overreaction or from underreaction, prices eventually revert to their fundamental values over the long run. An investor considering momentum investing must assess whether he leans toward overreaction or underreaction. Also, the investor must know that he can tolerate large losses during “crash” periods, historically concentrated around periods when policymakers (e.g., central banks) interrupt momentum, changing the course that asset prices would naturally take. In sum, assets are exposed to factor risks like value and momentum. Factor premiums compensate investors for losses during bad times.



MODULE QUIZ 84.3

1. Which of the following investment strategies stabilizes asset prices?
 - A. A value investment strategy.
 - B. A momentum investment strategy.
 - C. A size investment strategy.
 - D. Value, momentum, and size strategies all stabilize asset prices.

KEY CONCEPTS

LO 84.a

A value-growth investment strategy is long value stocks and short growth stocks. Value stocks have high book-to-market ratios, and growth stocks have low book-to-market ratios. Historically, value stocks have significantly outperformed growth stocks.

Risk premiums, including a value premium, exist to compensate investors for losses experienced during bad times. There are rational and behavioral explanations for why a

value premium may exist. Value stocks are risky, thus the value premium compensates investors for losses during bad times (rational explanation). Investors undervalue the growth prospects of value stocks and overextrapolate past growth into future prospects, overvaluing growth stocks. Value stocks are underpriced relative to their fundamental values, and growth stocks are overvalued, leading to a value premium (behavioral explanation).

LO 84.b

Macroeconomic factors, like inflation and economic growth, affect all investors to varying degrees. Economic growth, inflation, and volatility are the three most important macro factors that affect asset prices. It is unanticipated changes to a risk factor that affect asset prices, not the level of the factor. In other words, it is not the level of inflation, but an unanticipated increase or decrease in inflation that causes stock and bond prices to rise or fall.

- Risky assets generally perform poorly during periods of low economic growth.
- Stocks and bonds generally perform poorly in periods of high inflation.
- Stock returns drop when volatility (measured by the VIX) increases.

Other macroeconomic factors, like shocks to productivity, demographic risks, and sovereign risks, also affect asset returns.

LO 84.c

Volatility increases in periods of economic stress. There are two basic approaches to mitigating volatility risk:

- Invest in less-volatile assets like bonds. One challenge to managing volatility is that asset prices, including less volatile assets, tend to perform poorly during periods of economic stress (e.g., 2007–2009).
- Buy volatility protection in the derivatives market (e.g., buy out-of-the-money put options). Sellers of volatility protection (i.e., those selling put options) collect volatility premiums.

LO 84.d

The Fama-French model explains asset returns based on three dynamic factors. The factors are:

- The traditional CAPM market risk factor.
- A factor that captures the size effect (SMB or small cap minus big cap). Historically, small-cap stocks outperform large-cap stocks. The strategy is long small-cap stocks and short large-cap stocks.
- A factor that captures the value/growth effect (HML or high book-to-market value minus low book-to-market value). Value stocks tend to outperform growth stocks. The value-growth strategy is long value stocks and short growth stocks.

LO 84.e

A value strategy is long value stocks and short growth stocks. A momentum strategy is long “winners” (i.e., stocks that have gone up in value over the last six months or so)

and short “losers” (i.e., stocks that have gone down in value over the last six months or so). A momentum strategy has vastly outperformed both value-growth and size strategies since 1965. However, momentum strategies are subject to crashes. Rational and behavioral explanations can be used to describe both value and momentum risk premiums.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 84.1

1. **B** A company's book value per share is equal to total assets minus total liabilities all divided by shares outstanding. It indicates, on a per-share basis, what a company would be worth if it liquidated its assets and paid off its liabilities. Value stocks have high book-to-market ratios while growth stocks have low book-to-market ratios. (LO 84.a)
2. **D** During periods of recession, government and investment-grade bonds outperform equities and high-yield bonds. During expansion periods, equities outperform bonds. High-yield bond returns appear indifferent to changes in economic growth, yielding 7.4% in recessions and 7.7% in expansions. (LO 84.b)

Module Quiz 84.2

1. **A** There are two basic approaches to mitigating volatility risk. They are investing in less volatile assets like bonds (instead of stocks) or buying volatility protection in the derivatives market, such as buying out-of-the-money put options. (LO 84.c)
2. **C** The Fama-French model includes the following three risk factors:
 - The traditional capital asset pricing model market risk factor.
 - A factor that captures the size effect (SMB).
 - A factor that captures the value/growth effect (HML).The winners minus losers (WML) momentum factor was discovered by Jegadeesh and Titman. (LO 84.d)

Module Quiz 84.3

1. **A** Value and momentum are opposite each other in that value investing is inherently stabilizing. It is a *negative feedback strategy* where stocks that have fallen in value eventually are priced low enough to become value investments, pushing prices back up. Momentum is inherently destabilizing. It is a *positive feedback strategy* where stocks that have been increasing in value are attractive to investors, so investors buy them, and prices increase even more. (LO 84.e)

¹ Frank Smets and Rafael Wouters, “Shocks and Frictions in US Business Cycles: A Bayesian Dynamic Stochastic General Equilibrium Approach,” *American Economic Review* 97, no. 3 (2007): 586–606.

² Eugene F. Fama and Kenneth R. French, "Common Risk Factors in the Returns on Stocks and Bonds," *Journal of Financial Economics* 33 (1993): 3–56.

³ Rolf W. Banz, "The Relationship Between Return and Market Value of Common Stocks," *Journal of Financial Economics* 9 (1981): 3–18.

⁴ Marc R. Reinganum, "Misspecification of Capital Asset Pricing: Empirical Anomalies Based on Earnings' Yields and Market Values," *Journal of Financial Economics* 9, no. 1 (1981): 19–46.

⁵ Fischer Black, "Beta and Return," *Journal of Portfolio Management* 20, no. 1 (1993): 8–18.

⁶ Benjamin Graham and David Dodd, *Security Analysis* (New York: McGraw-Hill, 1934).

⁷ Nicholas Barberis and Ming Huang, "Mental Accounting, Loss Aversion, and Individual Stock Returns," *Journal of Finance* 56, no. 4 (2001): 1247–92.

⁸ Narasimhan Jegadeesh and Sheridan Titman, "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency," *Journal of Finance* 48, no. 1 (1993): 65–91.



The following is a review of the Risk Management and Investment Management principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Ang, Chapter 10.

READING 85

ALPHA (AND THE LOW-RISK ANOMALY)

Study Session 13

EXAM FOCUS

Investors are interested in generating alpha, which is the return earned in excess of a benchmark. It was traditionally thought that higher risk produced higher returns. However, in practice, strategies focused on lower volatility have actually been found to produce higher returns than higher-volatility investments. For the exam, be able to explain the impact of benchmark selection on alpha. Also, understand how to apply factor regression to construct a benchmark with multiple factors, and how to measure alpha against that benchmark. Finally, be familiar with the potential explanations for return anomalies with regard to low risk.

MODULE 85.1: LOW-RISK ANOMALY, ALPHA, AND THE FUNDAMENTAL LAW

Low-Risk Anomaly

LO 85.a: Describe and evaluate the low-risk anomaly of asset returns.

The capital asset pricing model (CAPM) from traditional finance states that there should be a positive relationship between risk and return. Higher risk, as measured by beta, should have a higher return. The **low-risk anomaly** appears to suggest the exact opposite. This anomaly finds that firms with lower betas and lower volatility have higher returns over time. For example, over a five-year period from 2011–2016, the cumulative return for a low volatility fund (iShares Edge MSCI Minimum Volatility USA ETF) was 68.75% relative to the cumulative return of 65.27% for the S&P 500 Index ETF.

Alpha, Tracking Error, the Information Ratio, and the Sharpe Ratio

LO 85.b: Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio.



PROFESSOR'S NOTE

We will demonstrate the calculations for these measures along with other performance metrics in Reading 90.

Alpha is often interpreted as a measure of investor skill, but it is really just a statement of average performance in excess of a benchmark. Excess return (R_t^{ex}) can be seen as the difference between the return of an asset (R_t) and the return of the asset's benchmark (R_t^B).

$$R_t^{ex} = R_t - R_t^B$$

Excess return is also sometimes called **active return**. This phrase assumes that the benchmark is passive and can be achieved without investment knowledge or human intervention. The S&P 500 Index and the Russell 1000 Index are commonly used large-cap benchmarks. If the benchmark is passive, then any additional return that the investor achieves is from doing something different from the benchmark, which by definition is active.

We compute alpha (α) by finding the average excess return for T observations.

$$\alpha = \frac{1}{T} \sum_{t=1}^T R_t^{ex}$$

To fully understand the concept of alpha, we also need to understand tracking error and the information ratio. **Tracking error** is the standard deviation of excess returns. It measures the dispersion of the investor's returns relative to their benchmark.

$$\text{tracking error} = \bar{\sigma} = \text{standard deviation}(R_t^{ex})$$

When a professional investment manager uses active strategies, there is often a constraint placed on the amount of tracking error permitted. Larger tracking errors indicate that the manager has more freedom in decision making.

One easy way to monitor alpha is to standardize it using tracking error. The ratio of alpha to tracking error is known as the **information ratio (IR)**, and it is a good way to monitor risk-adjusted returns for active asset managers. Active investment choices can be ranked based on their IR scores.

$$IR = \frac{\alpha}{\bar{\sigma}}$$

Sometimes the benchmark for an asset manager is the risk-free rate (R_F). In this case, alpha is measured as the return earned on an investment (R_t) in excess of the risk-free rate.

$$\alpha = R_t - R_F$$

When the risk-free rate is the appropriate benchmark, the best way to measure risk-adjusted returns is to use the **Sharpe ratio**. This measure has alpha in the numerator and the standard deviation of the asset in the denominator.

$$\text{Sharpe ratio} = \frac{\bar{R}_t - \bar{R}_F}{\sigma}$$

Benchmark Selection for Alpha

LO 85.c: Explain the impact of benchmark choice on alpha and describe characteristics of an effective benchmark to measure alpha.

The choice of benchmark has a significant impact on the calculated alpha for an investment. Strictly benchmarking to an identifiable index, like the S&P 500 Index, assumes that an asset has a beta of 1.0. What if the true beta is some value other than 1.0? Consider an investment that has a beta of 0.73 and tracking error of 6.16%. The alpha for this investment could be estimated by regressing the excess return of the fund ($R_t - R_F$) against the excess return of the benchmark ($R_t^{\text{SP500}} - R_F$). In the following regression equation, you see a calculated alpha of 3.44% and a placeholder for error term (ε_t) because we never know, in advance, how an individual observation will deviate from our model at any point in time.

$$R_t - R_F = 0.0344 + 0.73(R_t^{\text{SP500}} - R_F) + \varepsilon_t$$

We can rearrange this formula to isolate only the expected return on our investment. Doing so, we find that our customized benchmark should actually be invested 27% in the risk-free rate and 73% in the S&P 500 Index. Using a benchmark that recognizes the investment's beta as 0.73, we calculate an alpha of 3.44%, which translates into an IR of 0.5584 (= 0.0344 / 0.0616).

$$R_t = 0.0344 + 0.27(R_F) + 0.73(R_t^{\text{SP500}}) + \varepsilon_t$$

If this same investor were to wrongly regress their investment against only the S&P 500 Index, then they would calculate an alpha of 1.50%, which is incorrect because it assumes a beta of 1.0 when the actual beta is 0.73.

$$R_t = 0.015 + R_t^{\text{SP500}} + \varepsilon_t$$

Using the wrong benchmark would produce an IR of 0.2435 (= 0.0150 / 0.0616). This suggests that using an incorrect benchmark will understate both the expected alpha and the IR. Inaccurate information may cause an investor to pass on an investment that they otherwise should have accepted.

This illustration leads an investor to wonder: what is the best way to choose a benchmark? An appropriate benchmark can be selected by applying a few different complementary standards. First, the benchmark should be *well-defined*. It should be hosted by an independent index provider, which makes it both verifiable and free of ambiguity. The S&P 500 Index and the Russell 1000 Index are both examples of well-defined large-cap indices. Second, an index should be *tradeable*. If the benchmark is not a basket of tradeable securities that could be directly invested in as an alternative, then

the benchmark is not a very good comparison. Third, a benchmark must be *replicable*. This is closely related to the tradability standard. There are some benchmarks, like absolute return benchmarks, that are not feasible for an investor to replicate. If it cannot be replicated, then the tracking error will be very high. Fourth, the benchmark must be *adjusted for risk*. In the previous example, you can see that the alpha and the IR will be calculated too low if the risk level of the benchmark is too high for the investment in question.

Fundamental Law of Active Management

LO 85.d: Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the information ratio using this law.

Portfolio managers create value, and potentially create alpha, by making bets that deviate from their benchmark. Richard Grinold formalized this intuitive relationship in the **fundamental law of active management**.¹ This fundamental law does not provide a tool for searching for high IR plays, but it does present a good mechanism for systematically evaluating investment strategies. The law states that:

$$IR \approx IC \times \sqrt{BR}$$

The formula for Grinold's fundamental law shows that the information ratio (IR) is approximately equal to the product of the information coefficient (IC) and the square root of the breadth (BR) of an investor's strategy. The **information coefficient** is essentially the correlation between an investment's predicted and actual value. This is an explicit evaluation of an investor's forecasting skill. A higher IC score means that the predictions had a higher correlation (high-quality predictions). **Breadth** is simply the number of investments deployed.

Consider an example of an investor who requires an IR of 0.50. If this investor wants to time the market using an index and plans to only make four investments during the year, then he would need an IC of 0.25 as shown:

$$0.5 = 0.25 \times \sqrt{4}$$

What would happen if this same investor instead decided to deploy a stock selection strategy based on either value or momentum plays? These two strategies both involve taking a high number of bets every year. If they placed 200 bets in a given year, then they would only need an IC of 0.035 instead of 0.25. A lower IC means lower-quality predictions.

$$0.5 = 0.035 \times \sqrt{200}$$

Grinold's fundamental law teaches us about a central tradeoff in active management. Investors need to either play smart (a high IC shows high-quality predictions) or play often (a high BR shows a lot of trade activity). Essentially, investors can be very good at making forecasts and place a small number of bets, or they will need to simply place a lot of bets.

Grinold's framework ignores downside risk and makes a critical assumption that all forecasts are independent of one another. The Norwegian sovereign wealth fund has used Grinold's fundamental law in practice. Their philosophy is to take a high number of bets using a large list of entirely independent asset managers. This helps to keep forecasts independent and allows them to have reduced reliance on forecasting prowess while still endeavoring to achieve their benchmark IR goals.

In practice, it has also been noted that as assets under management go up, the IC tends to decline. This affects mutual funds, hedge funds, private equity firms, pension funds, and sovereign wealth funds alike. This is one reason why some mutual funds close to new investors and turn away new assets once they reach an internally set size.



MODULE QUIZ 85.1

1. Which of the following statements is correct concerning the relationship between the low-risk anomaly and the capital asset pricing model (CAPM)?
 - A. The low-risk anomaly provides support for the CAPM.
 - B. The notion that the low-risk anomaly violates the CAPM has not been proven empirically.
 - C. The low-risk anomaly violates the CAPM and suggests that low-beta stocks will outperform high-beta stocks.
 - D. Both CAPM and the low-risk anomaly point to a positive relationship between risk and reward.
2. Which of the following statements is not a characteristic of an appropriate benchmark? An appropriate benchmark should be:
 - A. tradeable.
 - B. replicable.
 - C. well-defined.
 - D. equally applied to all risky assets irrespective of their risk exposure.
3. Grinold's fundamental law of active management suggests that:
 - A. investors should focus on increasing only their predictive ability relative to stock price movements.
 - B. sector allocation is the most important factor in active management.
 - C. a small number of investment bets decreases the chances of making a mistake and, therefore, increases the expected investment performance.
 - D. to maximize the information ratio, active investors need to either have high-quality predictions or place a large number of investment bets in a given year.

MODULE 85.2: FACTOR REGRESSION AND PORTFOLIO SENSITIVITY

LO 85.e: Apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark.

Consider the CAPM formula, where $E(R_i)$ is the expected return for asset i for a given level of beta exposure, and $E(R_M)$ is the expected return on the market:

$$E(R_i) = R_F + \beta[E(R_M) - R_F]$$

If an investment has a beta of 1.3, then the following formulas demonstrate the algebraic evolution of this expression:

$$E(R_i) = R_F + 1.3[E(R_M) - R_F]$$

$$E(R_i) = R_F + 1.3E(R_M) - 1.3(R_F)$$

$$E(R_i) = -0.3R_F + 1.3E(R_M)$$

In this example, the expected return on a \$1 investment in asset i should be equal to a portfolio with a short position in the risk-free rate of \$0.30 and a long position in the market of \$1.30. Any return earned in excess of this unique blend will exceed our expectations and is, therefore, considered to be *alpha*. Using regression, the alpha is approximated as:

$$R_{i,t} - R_F = \alpha + \beta(R_M - R_F) + \varepsilon_{i,t}$$

This exact process was conducted on Berkshire Hathaway stock over the period of January 1990 to May 2012 relative to S&P 500 Index. The results are shown in Figure 85.1. The monthly alpha coefficient is statistically significant at a 95% confidence level due to the absolute value of the t -statistic being greater than 2.0. Most regressions do not produce a statistically significant alpha.

Figure 85.1: Regression of Excess Returns

	Coefficient	t -Statistic
Alpha	0.72%	2.02
Beta	0.51	6.51
Adjusted R^2	0.14	

This regression implies the following CAPM equation:

$$R_B = 0.49R_F + 0.51R_M$$

According to these regression results, a customized benchmark of 49% in the risk-free asset and 51% in the market would produce an expected alpha of 0.72% per month for Berkshire Hathaway. That is 8.6% ($= 0.72\% \times 12$) of annualized expected alpha! Since alpha is the excess return above the actual return, R_i , you can think of alpha using the following formula:

$$\alpha = R_i - [0.49R_F + 0.51E(R_M)]$$



PROFESSOR'S NOTE

For Berkshire, it is important to note that their market capitalization has grown from less than \$10 billion in the early 1990s to over \$220 billion in 2012. In his Annual Letter to Shareholders for 2010, Warren Buffet told shareholders that “the bountiful years, we want to emphasize, will never return. The huge sums of capital we currently manage eliminate any chance of exceptional performance.”² Thus, Berkshire Hathaway has acknowledged the law of declining marginal returns due to scale.

In 1993, Eugene Fama and Kenneth French extended the traditional CAPM-based regression to include additional factors. They controlled for the **size effect** (small

companies tend to outperform large companies) and for the **value/growth effect** (value stocks tend to perform better than growth stocks). They formally labeled the size premium as SMB, which stands for “small minus big” (the return on small stocks minus the return on big stocks), and they represented the value premium with HML, which stands for “high minus low” (high book-to-market stocks minus low book-to-market stocks). The factors for SMB and HML are long-short factors. The “small minus big” factor can be visualized as:

$$\text{SMB} = \$1 \text{ in small caps (long position)} - \$1 \text{ in large caps (short position)}$$

In a similar manner, we can visualize “high minus low” as:

$$\text{HML} = \$1 \text{ in value stocks (long position)} - \$1 \text{ in growth stocks (short position)}$$

The **Fama-French three-factor model** is constructed as follows:

$$R_i - R_F = \alpha + \beta_{i,\text{MKT}} \times (R_M - R_F) + \beta_{i,\text{SMB}} \times (\text{SMB}) + \beta_{i,\text{HML}} \times (\text{HML})$$

The SMB beta will be positive if there is co-movement with small stocks, and it will be negative if there is co-movement with large stocks. If a given asset does not co-move with either small or large companies (i.e., a medium company focus), then its beta coefficient will be zero. Likewise, the HML beta will be positive if the assets have a value focus, and it will be negative if the assets have a growth focus. Applying the Fama-French model to Berkshire Hathaway over the period of January 1990–May 2012 yields the results displayed in Figure 85.2.

Figure 85.2: Fama-French Three-Factor Model Results

	Coefficient	t-Statistic
Alpha (α)	0.65%	1.96
Market beta ($\beta_{i,\text{MKT}}$)	0.67	8.94
SMB beta ($\beta_{i,\text{SMB}}$)	−0.50	−4.92
HML beta ($\beta_{i,\text{HML}}$)	0.38	3.52
Adjusted R^2	0.27	

The results in Figure 85.2 show several interesting aspects. First, the alpha declined slightly but is still very high. Second, the market beta rose from 0.51 to 0.67. Third, the SMB beta is negative, which suggests a large company bias. Fourth, the HML beta is positive, which suggests a value focus for the fund. The adjusted R^2 also rose from 0.14 to 0.27, which suggests that SMB and HML do add value. Based on the results, the custom benchmark implied by the Fama-French three-factor model for Berkshire Hathaway is shown as follows:

$$R_B = 0.33(\text{T-bills}) + 0.67 \times (\text{market portfolio}) - 0.5(\text{small caps}) + 0.5(\text{large caps}) + 0.38(\text{value stocks}) - 0.38(\text{growth stocks})$$

All of the factor weights in this formula sum to 1.0, but adding the SMB and HML factors add explanatory ability to the regression equation. A test could also be added to account for the **momentum effect**, which is the theory that upward trending stocks will continue their upward movement while downward moving stocks will continue their downward trend. Thus, a fourth factor can be added to the Fama-French model. This fourth factor could be labeled as UMD, which stands for “up minus down” (upward

trending stocks minus downward trending stocks). A positive UMD beta would suggest a focus on upward trending stocks, while a negative UMD beta would suggest a focus on downward trending stocks. As with the SMB and HML betas, a beta of zero suggests no relationship. Figure 85.3 shows the UMD factor added to the previous results. Using this data, it can be discerned that Berkshire Hathaway does not have exposure to momentum investing.

Figure 85.3: Fama-French Three-Factor Model Results With UMD Factor

	Coefficient	<i>t</i> -Statistic
Alpha (α)	0.68%	2.05
Market beta ($\beta_{i,MKT}$)	0.66	8.26
SMB beta ($\beta_{i,SMB}$)	-0.50	-4.86
HML beta ($\beta_{i,HML}$)	0.36	3.33
UMD beta ($\beta_{i,UMD}$)	-0.04	-0.66
Adjusted R^2	0.27	

One core challenge with using the Fama-French model is replication of indices. Fama and French have created an SMB index and an HML index to increase explanatory power, but there is no way to directly trade an SMB or HML portfolio. These indices are conceptual and not directly tradeable. It is important to include only tradeable factors because the factors chosen will greatly influence the calculated alpha.



MODULE QUIZ 85.2

- Why would an investor include multiple factors in a regression study?
 - To attempt to improve the adjusted R^2 measure.
 - To reduce the *t*-stat value on the respective regression coefficients.
 - I only.
 - II only.
 - Both I and II.
 - Neither I nor II.

MODULE 85.3: TIME-VARYING FACTORS, VOLATILITY AND BETA ANOMALIES, AND ANOMALY EXPLANATIONS

Measurement of Time-Varying Factors

LO 85.f: Explain how to use style analysis to handle time-varying factor exposures.

Style analysis is a form of factor benchmarking where the factor exposures evolve over time. To illustrate time-varying factors, consider four investments: (1) LSV Value Equity (LSVEX), (2) Fidelity Magellan (FMAGX), (3) Goldman Sachs Capital Growth (GSCGX), and (4) Berkshire Hathaway (BRK). Figure 85.4 shows the regression data from monthly

returns on all four funds using the Fama-French three-factor model plus the UMD factor. The key difference between this information and data already presented is that the time period has been adjusted to January 2001 through December 2011.

Figure 85.4: Regression of Excess Returns for Multiple Funds

	LSVEX	FMAGX	GSCGX	BRK
Alpha (α)	0.00%	−0.27%	−0.14%	0.22%
<i>t-stat</i>	0.01	−2.23	−1.33	0.57
Market beta ($\beta_{i,MKT}$)	0.94	1.12	1.04	0.36
<i>t-stat</i>	36.9	38.6	42.2	3.77
SMB beta ($\beta_{i,SMB}$)	0.01	−0.07	−0.12	−0.15
<i>t-stat</i>	0.21	−1.44	−3.05	−0.97
HML beta ($\beta_{i,HML}$)	0.51	−0.05	−0.17	0.34
<i>t-stat</i>	14.6	−1.36	−4.95	2.57
UMD beta ($\beta_{i,UMD}$)	0.2	0.02	0.00	−0.06
<i>t-stat</i>	1.07	1.00	−0.17	−0.77

This data presents a different story about these funds than earlier. The only calculated alpha that is statistically significant is for Fidelity Magellan, but it is a −3.24% (= −0.27% × 12) in annualized terms. This was not good news for Fidelity investors, although it is time constrained to a period that ended in 2011. Berkshire’s alpha is nicely positive, but for this time period, it is not significant. According to the HML beta factors, LSV Value Equity is indeed a value-focused investment. The data also shows that FMAGX is a leveraged play on the market with a 1.12 market beta. The UMD beta confirms that none of these four funds are momentum plays.

Style analysis tries to solve some of the problems with standard multifactor regression. Unlike Fama and French’s untradeable SMB and HML indices, style analysis uses tradeable assets. For example, consider three funds: (1) SPDR S&P 500 ETF (SPY), (2) SPDR S&P 500 Value ETF (SPYV), and (3) SPDR S&P 500 Growth ETF (SPYG). These three exchange-traded funds (ETFs) are hosted by State Street Global Advisors and they all belong to the SPDR (pronounced “spider”) family of ETFs. Style analysis also adjusts for the fact that factor loadings (betas) change over time. A possible multifactor regression could be estimated for next period’s expected asset return (R_{t+1}) as follows:

$$R_{t+1} = \alpha_t + \beta_{SPY,t} SPY_{t+1} + \beta_{SPYV,t} SPYV_{t+1} + \beta_{SPYG,t} SPYG_{t+1} + \varepsilon_{t+1}$$

This formula has an imposed restriction that all factor loadings (i.e., factor weights) must sum to one:

$$1 = \beta_{SPY,t} + \beta_{SPYV,t} + \beta_{SPYG,t}$$

The time-varying portion of this equation comes into play with the respective factor loadings. This process uses estimates that incorporate information up to time t . Every new month ($t + 1$) requires a new regression to adjust the factor loadings. This means that the beta factors will change over time to reflect changes in the real world.

Issues With Alpha Measurement for Nonlinear Strategies

LO 85.g: Describe issues that arise when measuring alphas for nonlinear strategies.

Alpha is computed using regression, which operates in a linear framework. There are nonlinear strategies, such as uncovered long put options, that can make it appear that alpha exists when it actually does not. An uncovered long put option has a payoff profile that is L-shaped (nonlinear), but applying traditional regression tools will yield a positive alpha, which does not exist in reality. This situation is encountered when payoffs are quadratic terms, like R_t^2 or are option-like terms, such as $\max(R_t, 0)$. This can be a significant problem for hedge funds because merger arbitrage, pairs trading, and convertible bond arbitrage strategies all have nonlinear payoffs.

One reason that nonlinear strategies yield a false positive alpha is because the distribution of returns is not a normal distribution. Certain nonlinear strategies will also exhibit negative skewness in their distribution. This will increase loss potential in the left-hand tail and make the middle of the distribution appear thicker. Skewness is not factored into the calculation of alpha, which is an issue for nonlinear payoff strategies.

Volatility and Beta Anomalies

LO 85.h: Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly.

Using data from 1926–1971, Haugen and Heins (1975)³ found that “over the long run, stock portfolios with lesser variance in monthly returns have experienced greater average returns than ‘riskier’ counterparts.” Ang, Hodrick, Xing, and Zhang (2006)⁴ tested whether increased volatility, as measured by standard deviation, has a positive relationship with returns and Sharpe ratios. They organized their data, which comprised monthly return data from September 1963–December 2011, into quintiles and controlled for numerous variables including leverage, volume, bid-ask spreads, dispersion in analyst’s forecasts, and momentum. They observed a **volatility anomaly** which shows that as standard deviation increased, both the average returns and the Sharpe ratios decreased. For the lowest three quintiles, the average return was above 10%, but declined to 6.8% for quintile 4 and to 0.1% for the quintile with the highest volatility. Likewise, Sharpe ratios declined from 0.8 for the lowest volatility quintile to 0.0 for the highest volatility quintile. It was found that the most volatile stocks produce the lower returns, while the least volatile stocks performed the best.

When the capital asset pricing model (CAPM) was first tested in the 1970s, a positive relationship was found between beta and expected returns. Numerous academics have since retested this relationship with interesting results. Ang et al. (2006) found that stocks with high betas tend to have lower-risk-adjusted returns. Organizing monthly

return data from September 1963–December 2011 into quintiles, they found that the Sharpe ratio fell from 0.9 for stocks with the lowest betas to 0.4 for stocks with the highest betas. This **beta anomaly** does not suggest that stocks with higher betas have low return because they do not. It means they have lower Sharpe ratios (risk-adjusted performance) because higher betas are paired with higher volatility as measured by standard deviation, which is the denominator in the Sharpe ratio.

Interestingly, CAPM does *not* predict that lagged betas (measured over previous periods) should produce higher returns. It does predict that investors should find a contemporaneous relationship between beta and expected returns. This means that stocks with higher betas should also have higher returns during the same time period when the beta was measured. This is a confirming, not a predictive, metric. Following this logic, if investors could reliably predict future betas, then they could more accurately predict future expected returns. The trouble is that historical betas are not good predictors of future betas. Buss and Vilkov (2012)⁵ estimated future betas using implied volatility measures in option pricing models and found some improvement over using historical betas. The beta anomaly is less a mystery as it is a challenge to find a reliable way of predicting future betas to improve the risk perspective of beta.

LO 85.i: Describe potential explanations for the risk anomaly.

A comprehensive explanation for the risk anomaly is elusive. It has been speculated that the true explanation is some combination of data mining, investor leverage constraints, institutional manager constraints, and preference theory.

Some academics have wondered if the risk anomaly is the result of *data mining*. Ang et al. (2006) found that the risk anomaly appears during both recessions and expansions. Frazzini and Pedersen (2014)⁶ found that low beta portfolios have high Sharpe ratios in U.S. stocks, international stocks, Treasury bonds, and corporate bonds. Cao and Han (2013)⁷ also found evidence of the risk anomaly in option and commodity markets. The argument of data mining is not well supported.

Another possible explanation is the prevalence of *leverage constrained investors*. This is sometimes an occurrence with institutional investors, but it is very much a constraint of retail investors. Since certain investors are leverage constrained, meaning that they cannot borrow funds for investing, they choose to invest in stocks with built-in leverage in the form of high betas. The additional demand for high-beta stocks will bid up their respective prices until the assets are overvalued and they deliver a decreased risk-adjusted return with regard to lower beta stocks. This same theory works to lower the prices of low beta stocks and, therefore, results in higher risk-adjusted returns due to lower entry prices.

Institutional managers also have *constraints* that could help to explain the risk anomaly. Consider a scenario with two competing portfolios. Portfolio A has positive alpha because the portfolio is undervalued, while Portfolio B has a negative alpha because it is overvalued. In a perfect world, an investor would buy (go long) Portfolio A and short sell Portfolio B to capture the perceived alphas. Many institutional investors will have constraints against short selling. Most also have tracking error constraints

that only permit a specified deviation from their benchmark. Under either of these constraints, an institutional investor would not be able to capture the alpha that they think exists. One solution for the tracking error constraint is to change the benchmark or the tracking error tolerance bands, but this can be a difficult process requiring formal approval from the investment committee of the fund.

Sometimes investors simply have a *preference* for high-volatility and high-beta stocks. This could occur because their capital market expectations are very bullish, so they want to amplify their returns. The end result is that investors buy the higher-beta investments and bid up their prices to the point where future returns will be much lower. There will always be a group of investors that desire to shun “safe” and “boring” lower-volatility stocks. The good news is that this creates less emotionally driven entry points for long-term investors who desire lower volatility.

Investors holding heterogeneous preferences (disagreeing on investment potential) and having investment constraints could explain a portion of the risk anomaly. Hong and Sraer (2012)⁸ found that when disagreement is low and investors are long-only constrained, then the CAPM holds the best. When disagreement is high, some investments become overpriced and future returns are decreased. Significant disagreement can lead to an inverse relationship between beta and returns.



MODULE QUIZ 85.3

1. Which of the following characteristics is a potential explanation for the risk anomaly?
 - A. Investor preferences.
 - B. The presence of highly leveraged retail investors.
 - C. Lack of short selling constraints for institutional investors.
 - D. Lack of tracking error constraints for institutional investors.

KEY CONCEPTS

LO 85.a

The capital asset pricing model (CAPM) states that there should be a positive relationship between risk and return. Higher risk, as measured by beta, should have a higher return. The low-risk anomaly appears to suggest the exact opposite. This anomaly finds that firms with lower betas and lower volatility have higher returns over time.

LO 85.b

Alpha is the average performance of an investor in excess of their benchmark. Excess return is often called active return, and the standard deviation of the active return is known as tracking error.

The ratio of active return to tracking error is called the information ratio, which is one way to easily rank competing investment alternatives.

$$IR = \frac{\alpha}{\sigma}$$

If an investor is using the risk-free rate as their benchmark, then their alpha is any return earned in excess of the risk-free rate, and the best risk-adjusted return measurement is the Sharpe ratio.

$$\text{Sharpe ratio} = \frac{\bar{R}_t - \bar{R}_F}{\sigma}$$

LO 85.c

A benchmark is very important for investment comparisons. If the benchmark is riskier than the investment in question, then both the alpha and the information ratio will be too low. The best combination for a benchmark is for it to be well-defined, tradeable, replicable, and adjusted for the risk of the underlying pool of investments.

LO 85.d

Grinold's fundamental law of active management suggests a tradeoff between the number of investment bets placed (breadth) and the required degree of forecasting accuracy (information coefficient).

$$IR \approx IC \times \sqrt{BR}$$

An investor either needs to place a large number of bets and not be very concerned with forecasting accuracy, or he needs to be very good at forecasting if he places only a small number of bets.

LO 85.e

The traditional capital asset pricing model only accounts for co-movement with a market index. Multifactor models, like the Fama and French three-factor model, add other explanatory factors in an attempt to better predict the alpha for an asset. Multifactor models have been shown to enhance the informational value of regression output. The Fama-French three-factor model is expressed as:

$$R_i - R_F = \alpha + \beta_{i,MKT} \times (R_M - R_F) + \beta_{i,SMB} \times (SMB) + \beta_{i,HML} \times (HML)$$

This model adds a size premium (SMB) and a value premium (HML) to the CAPM single-factor model. A momentum effect (UMD) could also be added to help explain excess returns. This factor suggests that upward trending stocks will continue their upward movement while downward moving stocks will continue their downward trend.

LO 85.f

Style analysis is a form of factor benchmarking where the factor exposures evolve over time. The traditional Fama-French three-factor model can be improved by using indices that are tradeable, such as the SPDR S&P Value ETF (SPYV), and incorporating time-varying factors that change over time.

LO 85.g

Alpha is computed using regression, which operates in a linear framework. There are nonlinear strategies that can make it appear that alpha exists when it actually does not. This situation is encountered when payoffs are quadratic terms or option-like terms.

This may be a significant problem for hedge funds because merger arbitrage, pairs trading, and convertible bond arbitrage strategies all have nonlinear payoffs.

LO 85.h

The volatility anomaly and the beta anomaly both agree that stocks with higher risk, as measured by either high standard deviation or high beta, produce lower risk-adjusted returns than stocks with lower risk.

LO 85.i

A comprehensive explanation for the risk anomaly is elusive. It has been speculated that the true explanation is some combination of data mining, investor leverage constraints, institutional manager constraints, and preference theory.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 85.1

1. **C** The low-risk anomaly violates the CAPM and suggests that low beta stocks will outperform high-beta stocks. This has been empirically proven with several studies. The CAPM points to a positive relationship between risk and reward, but the low-risk anomaly suggests an inverse relationship. (LO 85.a)
2. **D** An appropriate benchmark should be well-defined, replicable, tradeable, and risk-adjusted. If the benchmark is not on the same risk scale as the assets under review, then there is an unfair comparison. (LO 85.c)
3. **D** Grinold's fundamental law of active management focuses on the tradeoff of high quality predictions relative to placing a large number of investment bets. Investors can focus on either action to maximize their information ratio, which is a measure of risk-adjusted performance. While sector allocation is a very important component of the asset allocation decision, Grinold focused only on the quality of predictions and the number of investment bets made. (LO 85.d)

Module Quiz 85.2

1. **A** An investor should consider adding multiple factors to the regression analysis to potentially *improve* the adjusted R^2 measurement, potentially *increase* the tests of statistical significance, and to search for a benchmark that is more representative of a portfolio's investment style. (LO 85.e)

Module Quiz 85.3

1. **A** Potential explanations for the risk anomaly include: the preferences of investors, leverage constraints on retail investors that drive them to buy pre-leveraged investments in the form of high-beta stocks, and institutional investor constraints like prohibitions against short selling and tracking error tolerance bands. (LO 85.i)

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- ¹ Richard C. Grinold, "The Fundamental Law of Active Management," *Journal of Portfolio Management* 15, no. 3 (1989): 30–37.
- ² Berkshire Hathaway Annual Letter to Shareholders, 2010.
- ³ Robert A. Haugen and A. James Heins, "Risk and the Rate of Return on Financial Assets: Some Old Wine in New Bottles," *Journal of Financial and Quantitative Analysis* 10, no. 5 (1975): 775–84.
- ⁴ Andrew Ang, Robert J. Hodrick, Yuhang Xing, and Xiaoyan Zhang, "High Idiosyncratic Volatility and Low Returns: International and Further U.S. Evidence," *Journal of Financial Economics* 91 (2009): 1–23.
- ⁵ Adrian Buss and Grigory Vilkov, "Measuring Equity Risk With Option-Implied Correlations," *The Review of Financial Studies* 25, no. 10 (2012): 3113–40.
- ⁶ Andrea Frazzini and Lasse Heje Pedersen, "Betting Against Beta," *The Journal of Financial Economics* 111, no. 1 (2014): 1–25.
- ⁷ Jie Cao and Bing Han, "Cross Section of Option Returns and Idiosyncratic Stock Volatility," *The Journal of Financial Economics* 108, no. 1 (2013): 231–49.
- ⁸ Harrison Hong and David Sraer, "Speculative Betas," NBER Working Paper 18548, November 2012.

The following is a review of the Risk Management and Investment Management principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Grinold and Kahn, Chapter 14.

READING 86

PORTFOLIO CONSTRUCTION

Study Session 13

EXAM FOCUS

This reading addresses techniques for optimal portfolio construction. We will discuss important inputs into the portfolio construction process as well as ways to refine the alpha inputs as an alternative to imposing constraints directly into the portfolio optimization calculations. The role of transaction costs in determining optimal rebalancing is also explained. For the exam, pay attention to the discussions of refining alphas and the implications of transaction costs for both rebalancing and dispersion of returns across separately managed portfolios. Also, be prepared to compare and contrast the various methods of portfolio construction: screening, stratification, linear programming, and quadratic programming.

MODULE 86.1: THE PORTFOLIO CONSTRUCTION PROCESS AND TRANSACTION COSTS

The Portfolio Construction Process

LO 86.a: Distinguish among the inputs to the portfolio construction process.

The process of constructing an optimal investment portfolio requires several inputs:

- *Current portfolio*: The assets and their weights in the current portfolio. Relative to the other inputs, the current portfolio input can be measured with the most certainty.
- *Alphas*: The expected excess returns of portfolio stocks (relative to their expected returns). This input is subject to forecast error and bias.
- *Covariances*: Estimates of covariances are subject to estimation error.
- *Transaction costs*: Transaction costs are estimated and increase as more frequent portfolio changes are made.
- *Active risk aversion*: Refers to the strength of the preference for lower volatility of the difference between actively managed portfolio returns and benchmark portfolio returns.

LO 86.b: Evaluate the motivation for and the methods used for refining alphas in the implementation process.

A portfolio can be optimized, based on the inputs, using mean-variance analysis. In most cases there are significant constraints imposed on the asset weights, either by client or manager requirements. A client (or regulations) may prohibit short sales. A manager may impose an upper limit on active risk or on maximum deviations from benchmark weights. As more constraints are introduced, simple mean-variance analysis, maximizing active return minus a penalty for active risk, can become quite complex.

An alternative approach is to adjust the manager's estimated alphas (an input into a mean-variance optimization analysis) in ways that effectively impose the various constraints. Consider an account for which short selling is prohibited. Rather than performing an optimization that constrains asset weights to be nonnegative, we can use the optimization equations (in reverse) to solve for the set of alphas that would produce nonnegative weights in an unconstrained mean-variance optimization. The optimal weights are moved toward benchmark weights. This method allows us to focus on the effects of a specific constraint on alphas, the key input for active portfolio construction.

Before we examine refining alphas to satisfy other constraints, such as a constraint on the beta of the active portfolio, we consider two techniques that are often employed after refining alphas for various client or manager imposed constraints: **scaling** and **trimming**.

An often used equation for **alpha** is:

$$\text{alpha} = (\text{volatility}) \times (\text{information coefficient}) \times (\text{score})$$

Where *volatility* refers to residual risk, the *information coefficient* (IC) measures the linear relationship between the manager's forecasted asset alphas and actual asset returns, and *score* is expected to be approximately normally distributed with a mean of 0 and a standard deviation of 1. Considering that volatility (residual risk) and information coefficient (IC) are relatively constant, we can see that the standard deviation (scale) of portfolio alphas is proportional to the standard deviation of the score variable. Alphas will have a mean of zero and a scale approximately equal to volatility $\times$ information coefficient when score follows a standard normal distribution. With an information coefficient of 0.10 and residual risk of 30%, the scale of the alphas will be $0.10 \times 30\% = 3\%$; the alphas will have a mean of zero and a standard deviation of 3%.

If we compare the scale (standard deviation) of the refined alphas from our earlier discussion of a prohibition on short sales to the scale of the original unconstrained alphas, we can calculate the decrease in the information coefficient that results from the decrease in the scale of the alphas due to the imposition of the constraint. If the adjusted alphas do not have the appropriate scale, they can be rescaled.

Another refinement to manager alphas is to reduce large positive or negative alpha values, a process called trimming. The threshold for “large” values might be three times the scale of the alphas. For large alpha values, the reasons supporting these values are re-examined. Any alphas found to be the result of questionable data are set to zero. Additionally, the remaining large alphas may be reduced to some maximum value, typically some multiple of the scale of the alphas.

LO 86.c: Describe neutralization and the different approaches used for refining alphas to be neutral.

Neutralization is the process of removing biases and undesirable bets from alpha. There are several types of neutralization: benchmark, cash, and risk-factor. In all cases, the type of neutralization and the strategy for the process should be specified before the process begins.

Benchmark neutralization eliminates any difference between the benchmark beta and the beta of the active portfolio. In this case, we say the portfolio alpha of the active portfolio is zero. Consider an active portfolio that has a beta of 1.1 when the benchmark portfolio has a beta of 1. This represents an active bet on market (and benchmark portfolio) returns. When market returns are high, the active portfolio will outperform the benchmark portfolio; when returns are low (less than the risk-free rate) the active portfolio will underperform the benchmark portfolio. The alphas can be adjusted so that the active portfolio beta is the same as the benchmark portfolio beta, unless the manager intends to make an active bet by increasing or decreasing the active portfolio beta relative to that of the benchmark. Matching the beta of the active portfolio to the beta of the benchmark portfolio is referred to as **benchmark neutralization**. Note that this neutralization is equivalent to adding a constraint on portfolio beta in a mean-variance optimization.

Computing modified benchmark-neutral alpha involves subtracting (benchmark alpha $\times$ active position beta) from the alpha of the active position. For example, assume benchmark alpha is equal to 0.013%. If an active position has an alpha of 0.5% and a beta of 1.2, the modified benchmark-neutral alpha will equal: $0.5\% - (0.013\% \times 1.2) = 0.48\%$.

In the explanation, we used a single risk factor, market risk. There may be other risk factors, such as those from a multi-factor returns generating model, that lead to unintended risk exposures relative to the benchmark. For example, consider the risk factor small-cap returns minus large-cap returns. The alpha inputs may produce an active portfolio with a greater sensitivity to this risk factor if the portfolio's weight on small-cap firms is larger than that of the benchmark portfolio. Again, if this is unintended, alphas can be adjusted so that the beta of the active portfolio with respect to this risk factor matches that of the benchmark portfolio.

The active portfolio may also be neutralized with respect to industry risk factors, by matching the portfolio weights of each industry to those of the benchmark portfolio. In this case, subtracting the average alpha for an industry from the alphas of each firm within that industry will result in an active portfolio that is neutral relative to the

benchmark with respect to industry risk factors. In each of our examples, neutralization reduces active risk by matching the factor risks of the active portfolio to those of the benchmark portfolio.

An active portfolio can also be made **cash neutral**, by adjusting the alphas so that the portfolio has no active cash position. It's possible to make the alpha values both cash- and benchmark-neutral.

Transaction Costs

LO 86.d: Describe the implications of transaction costs on portfolio construction.

Transaction costs are the costs changing portfolio allocations, primarily trading commissions and spreads. Transaction costs reduce active portfolio returns relative to those of the benchmark portfolio and are uncertain, although typically less so than alphas. Because of this, transaction costs are an important input into the portfolio optimization process. Including transaction costs in portfolio optimization increases the importance of both precision in estimating alphas and of the choice of scale.

Transaction costs occur at points in time, while the benefits (i.e., additional return) are realized over time. Consider two stocks, one of which will return 2% over 6 months, at which time it can be replaced by another stock that returns 2% over 6 months, and another stock which will return 4% over 1 year. Also, assume transaction costs are 1%. The annual returns on the first stock will be approximately $(2\% - 1\%) \times 2 = 2\%$ and the annual returns on the second stock will be approximately $4\% - 1\% = 3\%$. With uncertain holding periods across portfolio holdings, the question arises over what period transaction costs should be amortized. Precision in scale is important in addressing the tradeoff between alphas and transaction costs. Annual transaction costs will be the cost of a round-trip trade divided by the holding period in years.



MODULE QUIZ 86.1

1. The most measurable of the inputs into the portfolio construction process is(are) the:
 - A. position alphas.
 - B. transaction costs.
 - C. current portfolio.
 - D. active risk aversion.
2. Which of the following is correct with respect to adjusting the optimal portfolio for portfolio constraints?
 - A. No reliable method exists.
 - B. By refining the alphas and then optimizing, it is possible to include constraints of both the investor and the manager.
 - C. By refining the alphas and then optimizing, it is possible to include constraints of the investor, but not the manager.
 - D. By optimizing and then refining the alphas, it is possible to include constraints of both the investor and the manager.
3. Which of the following statements most correctly describes a consideration that complicates the incorporation of transaction costs into the portfolio construction process?

- A. The transaction costs and the benefits always occur in two distinct time periods.
- B. The transaction costs are uncertain while the benefits are relatively certain.
- C. There are no complicating factors from the introduction of transaction costs.
- D. The transaction costs must be amortized over the horizon of the benefit from the trade.

MODULE 86.2: PRACTICAL ISSUES IN PORTFOLIO CONSTRUCTION

LO 86.e: Describe practical issues in portfolio construction, including the determination of an appropriate risk aversion, aversions to specific risks, and proper alpha coverage.

We need a measure of **active risk aversion** as an input to determine the optimal portfolio. As a practical matter, a portfolio manager does not likely have an intuitive idea of optimal active risk aversion in mind, but will have good intuition about his information ratio (the ratio of alpha to standard deviation) and the amount of active risk (as opposed to active risk aversion) he is willing to accept in pursuit of active returns. An equation that translates those values into a measure of active risk aversion is:

$$\text{risk aversion} = \frac{\text{information ratio}}{2 \times \text{active risk}}$$

For example, if the information ratio is 0.8 and the desired level of active risk is 10%, then the implied level of risk aversion is:

$$\frac{0.80}{2 \times 10} = 0.04$$

The utility function for the optimization is: utility = active return – (0.04 × variance). Of course, the accuracy of the estimate of active risk aversion is dependent on the accuracy of the inputs, the information ratio, and the preferred level of active risk.



PROFESSOR'S NOTE

Remember that active risk is just another name for tracking error. Also note that in the risk aversion equation, the desired level of active risk is measured in percentage points rather than in decimal form.

In addition to active risk aversion, **aversion to specific factor risk** is important for two reasons. First, it can help the manager address the risks associated with having a position with the potential for large losses. For example, the risk from a portfolio with sector risks that do not match those of the benchmark portfolio. Second, appropriately high risk aversion values for specific factor risks will reduce dispersion (of holdings and performance) across portfolios when the manager manages more than one portfolio. Setting high risk aversion values for factor specific risks will increase the similarity of client portfolios so that they will tend to hold the same assets. Considering these two effects of specific factor risk aversion values will help a manager determine appropriate values to include in portfolio optimization.

Proper **alpha coverage** refers to addressing situations where the manager has forecasts of stocks that are not in the benchmark or where the manager does not have alpha forecasts for stocks in the benchmark. When the manager has information on stocks not in the benchmark, a benchmark weight of zero should be assigned for benchmarking, but active weights can be assigned to these stocks to generate active alpha.

When there is not an alpha forecast for stocks in the benchmark, adjusted alphas can be inferred from the alphas of stocks for which there are forecasts. One approach is to first compute the following two measures:

$$\text{value-weighted fraction of stocks with forecasts} = \frac{\text{sum of active holdings with forecasts}}{\text{with forecasts}}$$

$$\begin{aligned} &\text{average alpha for the stocks with forecasts} \\ &= \frac{(\text{weighted average of the alphas with forecasts})}{(\text{value-weighted fraction of stocks with forecasts})} \end{aligned}$$

The second step is to subtract this measure from each alpha for which there is a forecast and set alpha to zero for assets that do not have forecasts. This provides a set of benchmark-neutral forecasts where assets without forecasts have alphas of zero.

Portfolio Revisions and Rebalancing

LO 86.f: Describe portfolio revisions and rebalancing, and analyze the tradeoffs between alpha, risk, transaction costs, and time horizon.

LO 86.g: Determine the optimal no-trade region for rebalancing with transaction costs.

If transaction costs are zero, a manager should revise a portfolio every time new information arrives. However, as a practical matter, a manager should make trading decisions based on expected active return, active risk, and transaction costs. The manager may wish to be conservative because these measures are uncertain. Underestimating transaction costs, for example, will lead to trading too frequently. In addition, the frequent trading and short time horizons would cause alpha estimates to exhibit a great deal of uncertainty. Therefore, the manager must choose an optimal time horizon where the certainty of the alpha is sufficient to justify a trade given the transaction costs.

The rebalancing decision depends on the tradeoff between transaction costs and the value added from changing the position. Portfolio managers must be aware of the existence of a *no-trade region* where the benefits of rebalancing are less than the costs. The benefit of adjusting the number of shares of a given portfolio asset is given by the following expression:

$$\text{marginal contribution to value added} = (\text{alpha of asset}) - [2 \times (\text{risk aversion}) \times (\text{active risk}) \times (\text{marginal contribution to active risk of asset})]$$

If this value is between the negative cost of selling and the cost of purchase, the manager would not trade that particular asset. In other words, the no-trade region is as follows:

$$-(\text{cost of selling}) < (\text{marginal contribution to value added}) < (\text{cost of purchase})$$

Rearranging this relationship with respect to alpha gives a no-trade region for alpha:

$$[2 \times (\text{risk aversion}) \times (\text{active risk}) \times (\text{marginal contribution to active risk})] - (\text{cost of selling}) < \alpha \text{ of asset} < [2 \times (\text{risk aversion}) \times (\text{active risk}) \times (\text{marginal contribution to active risk})] + (\text{cost of purchase})$$

The size of the no-trade region is determined by transaction costs, risk aversion, alpha, and the riskiness of the assets.

Portfolio Construction Techniques

LO 86.h: Evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming.

The following four procedures comprise most of the institutional portfolio construction techniques: screens, stratification, linear programming, and quadratic programming. In each case the goal is the same: high alpha, low active risk, and low transaction costs.

An active manager's value depends on her ability to increase returns relative to the benchmark portfolio that are greater than the penalty for active risk and the additional transaction costs of active management.

$$(\text{portfolio alpha}) - (\text{risk aversion}) \times (\text{active risk})^2 - (\text{transaction costs})$$

Screens

Screens are just what you would expect; they allow some stocks "through" but not the rest. A screen can be designed in many ways, but two examples will illustrate how a screen might be used with alpha values to select portfolio stocks (given a universe of 200 stocks). Consider a screen that selects the 60 benchmark stocks with the greatest alphas. We could then construct a portfolio of these high-alpha stocks, either equal- or value-weighted.

Another screening method is based on assigning buy, hold, or sell ratings to all the stocks in the manager's universe of investable stocks. For example, we could assign a buy rating to the 60 stocks with the greatest alphas, a hold rating to the 40 remaining stocks with the next highest alphas, and a sell rating to the remaining stocks under consideration. One way to rebalance the current portfolio would be to purchase any stocks on the buy list not currently in the portfolio and to sell any portfolio stocks on the sell list. Portfolio turnover can be adjusted by adjusting the sizes of the categories.

Stratification

Stratification refers to dividing stocks into multiple mutually exclusive categories prior to screening the stocks for inclusion in the portfolio. For example, we could divide the portfolio into large-cap, medium-cap, and small-cap stocks and further divide these categories into six different industry categories; giving us 18 different size-sector categories. By using percentage weights of these size-sector categories in the benchmark portfolio we can match the benchmark portfolio's size and sector coverage.

Stratification is a method of risk control. If the size and sector categories are chosen in such a way that they capture the risk dimensions of the benchmark well, portfolio risk control will be significant. If they are not, risk control will not be achieved.

Stratification will reduce the effects of bias in estimated alphas across the categories of firm size and sector. However, it takes away the possibility of adding value by deviating from benchmark size-sector weights. Using stratification, any value from information about actual alphas (beyond their category) and about possible sector alphas is lost.

Linear Programming

Linear programming is an improvement on stratification, in that it uses several risk characteristics, for example, firm size, returns volatility, sector, and beta. Unlike stratification, it does not require mutually exclusive categories of portfolio stocks. The linear programming methodology will choose assets for the optimal portfolio so that category weights in the active portfolio closely resemble those of the benchmark portfolio. This technique can also include the effects of transaction costs (which reduces turnover) and limits on position sizes.

Linear programming's strength is creating a portfolio that closely resembles the benchmark. However, the result can be a portfolio that is very different from the benchmark with respect to the number of assets included and any unincluded dimensions of risk.

Quadratic Programming

Quadratic programming can be designed to include alphas, risks, and transaction costs. Additionally, any number of constraints can be imposed. Theoretically, this is the best method of optimization, as it efficiently uses the information in alphas to produce the optimal (constrained) portfolio. However, estimation error is an important consideration. Consider that for a universe of 400 stocks, quadratic programming will require estimates of 400 stock volatilities and 79,800 covariances. The quadratic program will use the information in the estimates to reduce active risk.

Small estimation error in covariances will not necessarily reduce value added significantly. But even moderate levels of estimation error for the covariances can significantly reduce value added; with 5% estimation error, value added may actually be negative. The importance of good estimates of the relevant inputs, especially covariances, cannot be over emphasized.

Portfolio Return Dispersion

LO 86.i: Describe dispersion, explain its causes, and describe methods for controlling forms of dispersion.

For portfolio managers, **dispersion** refers to the variability of returns across client portfolios. One dispersion measure is the difference between the maximum return and minimum return over a period for separately managed client accounts.

Managers can reduce dispersion by reducing differences in asset holdings between portfolios and differences in portfolio betas through better supervision and control. Other causes of dispersion are outside the manager's control. Different portfolio constraints for different accounts will unavoidably increase dispersion (e.g., not being able to invest in derivatives or other asset classes).

Of course, if all client accounts were identical there would be no dispersion. All accounts will not be identical in the presence of transaction costs, however. The existence of transaction costs implies that there is some optimal level of dispersion. To understand the tradeoff between transaction costs and dispersion, consider a managed portfolio that is currently 60% stocks and 40% bonds. The manager knows the optimal portfolio is 62% stocks and 38% bonds, but transaction costs from rebalancing would reduce returns more than the change to optimal weights would increase them.

If the manager acquires a second client, he can set portfolio weights to 62% and 38% for that client's account. Because one client has a 60/40 portfolio and the other has a 62/38 portfolio, there will be dispersion. Clearly, higher transaction costs lead to greater dispersion. If the manager eliminates dispersion by matching the new client portfolio to the existing client portfolio, returns from the new information will be sacrificed. If the manager eliminates dispersion by rebalancing the existing client portfolio, the transaction costs of this rebalancing will reduce overall portfolio return. Given transaction costs, there is an optimal level of dispersion that balances transaction costs and gains from rebalancing.

A greater number of portfolios and higher active risk will both increase optimal dispersion, and for a given number of portfolios, dispersion is proportional to active risk. As long as alphas and risk are not constant (an unlikely occurrence) dispersion will decrease over time and eventually convergence (of account returns) will occur. However, there is no certainty as to the rate at which it will occur.



MODULE QUIZ 86.2

1. An increase in which of the following factors will increase the no-trade region for the alpha of an asset?
 - I. Risk aversion.
 - II. Marginal contribution to active risk.
 - A. I only.
 - B. II only.
 - C. Both I and II.

- D. Neither I nor II.
2. A manager has forecasts of Stocks A, B, and C, but not of Stocks D and E. Stocks A, B, and D are in the benchmark portfolio. Stocks C and E are not in the benchmark portfolio. Which of the following is correct concerning specific weights the manager should assign in tracking the benchmark portfolio?
- A. $w_C = 0$.
 - B. $w_D = 0$.
 - C. $w_C = (w_A + w_B) / 2$.
 - D. $w_C = w_D = w_E$.

KEY CONCEPTS

LO 86.a

The inputs into the portfolio construction process are the current portfolio, the alphas, covariance estimates, transaction costs, and active risk aversion. Except for the current portfolio, these inputs are all subject to estimation error and possible bias.

LO 86.b

Refining alphas is an alternative to including constraints (e.g., no short selling or maximum deviations from benchmark weights) in the portfolio optimization process. Using refined alphas and then performing optimization can achieve the same goal as a constrained optimization approach, but has the advantage of focusing on the alpha inputs and the effects of individual constraints on portfolio returns.

LO 86.c

Neutralization can remove undesirable portfolio risks. Benchmark neutralization can reduce active risk by matching active portfolio beta to that of the benchmark portfolio. Cash neutralization eliminates any active cash position in the portfolio. Risk-factor neutralization matches specific factor risks in the active portfolio to those of the benchmark.

LO 86.d

Transaction costs have several implications. First, they may make it optimal not to rebalance even with the arrival of new information. Second, transaction costs increase the importance of robust alpha estimates. The fact that transaction costs occur at a point in time while the benefits of the portfolio adjustments occur over the investment horizon complicates analysis and makes rebalancing decisions dependent on the estimated holding period of portfolio assets.

LO 86.e

Practical issues in portfolio construction include determining the level of risk aversion, the optimal risk, and the alpha coverage. The inputs in computing the level of risk aversion must be accurate. Including the aversion to specific risk factors can help a manager address the risks of a position with a large potential loss and the dispersion across separately managed portfolios. Proper alpha coverage addresses situations in which the manager has alpha estimates for stocks that have zero weight in (are not

included in) the benchmark or does not have alpha estimates for some stocks in the benchmark portfolio.

LO 86.f

In the process of portfolio revisions and rebalancing, there are tradeoffs between alpha, risk, transaction costs, and the investment horizon. The manager may choose to be conservative, given the uncertainty regarding these inputs. Also, the shorter the horizon, the more uncertain the alpha, which means the manager should choose an optimal time horizon where the certainty of the alpha is sufficient to justify a trade given the transaction costs.

LO 86.g

Because of transaction costs, there will be an optimal no-trade region when new information arrives concerning the alpha of an asset, the costs of rebalancing the portfolio outweigh the expected incremental returns. That region is determined by the level of risk aversion, a portfolio's active risk, the marginal contribution of rebalancing to active risk, and transaction costs.

LO 86.h

A screen may be as simple as "screening" for assets with the highest estimated alphas or as a method of assigning relative ranks based on estimated alphas.

Stratification applies screening separately to categories of stocks and weights the active portfolio across these categories with their weights in the benchmark portfolio.

Linear programming is an improvement on stratification in that the optimal portfolio is structured to closely resemble the benchmark with respect to such characteristics (risk factors) as industry groups, firm size, volatility, and beta.

Quadratic programming improves on the linear programming methodology by explicitly considering alpha, risk, and transaction costs. It is theoretically the best optimization method, incorporating the most information; however, the value added in the active portfolio is quite sensitive to the level of estimation error in the covariance inputs.

LO 86.i

For a manager with separately managed accounts, dispersion of client returns will result when the portfolios are not identical. The basic causes of dispersion are the different histories and cash flows of each of the clients. A manager can control this source of dispersion by trying to increase the proportion of assets that are common to all portfolios.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 86.1

1. C The current portfolio is the only input that is directly observable. (LO 86.a)

2. **B** The approach of first refining alphas and then optimizing can replace even the most sophisticated portfolio construction process. With this technique, both the investor and manager constraints are considered. (LO 86.b)
3. **D** A challenge is to correctly assign the transaction costs to projected future benefits. The transaction costs must be amortized over the horizon of the benefit from the trade. The benefits (e.g., the increase in alpha) occur over time while the transaction costs generally occur at a specific time when the portfolio is adjusted. (LO 86.d)

Module Quiz 86.2

1. **C** This is evident from the definition of the no-trade region for the alpha of the asset.

$$[2 \times (\text{risk aversion}) \times (\text{active risk}) \times (\text{marginal contribution to active risk})] - (\text{cost of selling}) < \text{alpha of asset} < [2 \times (\text{risk aversion}) \times (\text{active risk}) \times (\text{marginal contribution to active risk})] + (\text{cost of purchase})$$
 (LO 86.g)
2. **A** The manager should assign a tracking portfolio weight equal to zero for stocks for which there is a forecast but that are not in the benchmark. A weight should be assigned to Stock D, and it should be a function of the alphas of the other assets. (LO 86.e)

The following is a review of the Risk Management and Investment Management principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Jorion, Chapter 7.

READING 87

PORTFOLIO RISK: ANALYTICAL METHODS

Study Session 13

EXAM FOCUS

Due to diversification, the value at risk (VaR) of a portfolio will be less than or equal to the sum of the VaRs of the positions in the portfolio. If all positions are perfectly correlated, then the portfolio VaR equals the sum of the individual VaRs. A manager can make optimal adjustments to the risk of a portfolio with such measures as marginal VaR, incremental VaR, and component VaR. This reading is highly quantitative. Be able to find the optimal portfolio using the excess-return-to-marginal VaR ratios. For the exam, understand how correlations impact the measure of portfolio VaR. Also, it is important that you know how to compute incremental VaR and component VaR using the marginal VaR measure. We have included several examples to help with application of these concepts.

MODULE 87.1: VaR MEASURES

LO 87.a: Define, calculate, and distinguish between the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR.

LO 87.b: Explain the impact of correlation on portfolio risk.

Portfolio theory depends a lot on statistical assumptions. In finance, researchers and analysts often assume returns are normally distributed. Such an assumption allows us to express relationships in concise expressions such as beta. Actually, beta and other convenient concepts can apply if returns follow an elliptical distribution, which is a broader class of distributions that includes the normal distribution. In what follows, we will assume returns follow an elliptical distribution unless otherwise stated.

Diversified Portfolio VaR

Diversified VaR is simply the VaR of the portfolio where the calculation takes into account the diversification effects. The basic formula is:

$$\text{VaR}_p = Z_c \times \sigma_p \times P$$

where:

Z_c = z-score associated with the level of confidence c

σ_p = standard deviation of the portfolio return

P = nominal value invested in the portfolio

Examining the formula for the variance of the portfolio returns is important because it reveals how the correlations of the returns of the assets in the portfolio affect volatility.

The variance formula is:

$$\sigma_p^2 = \sum_{i=1}^N w_i^2 \sigma_i^2 + 2 \sum_{i=1}^N \sum_{j < i}^N w_i w_j \rho_{ij} \sigma_i \sigma_j$$

where:

σ_p^2 = variance of the portfolio returns

w_i = portfolio weight invested in position i

σ_i = standard deviation of the return in position i

ρ_{ij} = correlation between the returns of asset i and asset j

The standard deviation, denoted σ_p , is:

$$\sigma_p = \sqrt{\sigma_p^2} = \sqrt{\sum_{i=1}^N w_i^2 \sigma_i^2 + 2 \sum_{i=1}^N \sum_{j < i}^N w_i w_j \rho_{ij} \sigma_i \sigma_j}$$

Clearly, the variance and standard deviation are lower when the correlations are lower.

In order to calculate delta-normal VaR with more than one risk factor, we need a covariance matrix that incorporates correlations between each risk factor in the portfolio and volatilities of each risk factor. If we know the volatilities and correlations, we can derive the standard deviation of the portfolio and the corresponding VaR measure. We will discuss how to calculate VaR using matrix multiplication later in this reading.

Individual VaR is the VaR of an individual position in isolation. If the proportion or weight in the position is w_i , then we can define the individual VaR as:

$$\text{VaR}_i = Z_c \times \sigma_i \times |P_i| = Z_c \times \sigma_i \times |w_i| \times P$$

where:

P = portfolio value

P_i = nominal amount invested in position i

We use the absolute value of the weight because both long and short positions pose risk.

The Role of Correlation

In a two-asset portfolio, the equation for the standard deviation is:

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2}$$

and the VaR is:

$$\text{VaR}_p = Z_c P \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2}$$

We can square Z_c and P and put them under the square-root sign. This allows us to express VaR_p as a function of the VaRs of the individual positions, which we express as VaR_i for each position i . For a two-asset portfolio we will have VaR_1 and VaR_2 . If the correlation is zero, $\rho_{1,2} = 0$, then the third term under the radical is zero and:

$$\text{VaR for uncorrelated positions: } \text{VaR}_p = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$$

The other extreme is when the correlation is equal to unity, $\rho_{1,2} = +1$. With perfect correlation, there is no benefit from diversification. For the two-asset portfolio, we find:

$$\begin{aligned} \text{Undiversified VaR} = \text{VaR}_p &= \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\text{VaR}_1\text{VaR}_2} \\ &= \text{VaR}_1 + \text{VaR}_2 \end{aligned}$$

In general, undiversified VaR is the sum of all the VaRs of the individual positions in the portfolio when none of those positions are short positions.

Notice how evaluating VaR using both uncorrelated positions and perfectly correlated positions will place a lower and upper bound on the total (or portfolio) VaR. Total VaR will be less if the positions are uncorrelated and greater if the positions are correlated. The following examples illustrate this point.

EXAMPLE: Computing portfolio VaR (Part 1)

An analyst computes the VaR for the two positions in her portfolio. The VaRs: $\text{VaR}_1 = \$2.4$ million and $\text{VaR}_2 = \$1.6$ million. **Compute** VaR_p if the returns of the two assets are uncorrelated.

Answer:

For uncorrelated assets:

$$\begin{aligned} \text{VaR}_p &= \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2} \\ &= \sqrt{(2.4^2 + 1.6^2)(\$millions)^2} = \sqrt{8.32(\$millions)^2} \\ \text{VaR}_p &= \$2.8844 \text{ million} \end{aligned}$$

EXAMPLE: Computing portfolio VaR (Part 2)

An analyst computes the VaR for the two positions in her portfolio. The VaRs: $\text{VaR}_1 = \$2.4$ million and $\text{VaR}_2 = \$1.6$ million. **Compute** VaR_p if the returns of the two assets are perfectly correlated.

Answer:

For perfectly correlated assets:

$$\text{VaR}_p = \text{VaR}_1 + \text{VaR}_2 = \$2.4 \text{ million} + \$1.6 \text{ million} = \$4 \text{ million}$$

Under certain assumptions, the portfolio standard deviation of returns for a portfolio with more than two assets has a very concise formula. The assumptions are:

- The portfolio is equally weighted.
- All the individual positions have the same standard deviation of returns.
- The correlations between each pair of returns are the same.

The formula is then:

$$\sigma_P = \sigma \sqrt{\frac{1}{N} + \left(1 - \frac{1}{N}\right)\rho}$$

where:

N = number of positions

σ = standard deviation that is equal for all N positions

ρ = correlation between the returns of each pair of positions



PROFESSOR'S NOTE

This formula greatly simplifies the process of having to calculate portfolio standard deviation with a covariance matrix.

To demonstrate the benefits of diversification, we can simply set up a 2×2 table where there is a small and large correlation (ρ) column and a small and large sample size (N) row. Assuming that the standard deviation of returns is 20% for both assets, we see how the portfolio variance is affected by the different inputs.

Figure 87.1: Portfolio Standard Deviation

Sample Size/ Correlation	$\rho = 0.1$	$\rho = 0.5$
$N = 4$	$\sigma_P = 11.40\%$	$\sigma_P = 15.81\%$
$N = 10$	$\sigma_P = 8.72\%$	$\sigma_P = 14.83\%$

EXAMPLE: Computing portfolio VaR (Part 3)

A portfolio has five positions of \$2 million each. The standard deviation of the returns is 30% for each position. The correlations between each pair of returns is 0.2. **Calculate** the VaR using a z-score of 2.33.

Answer:

The standard deviation of the portfolio returns is:

$$\sigma_P = 30\% \sqrt{\frac{1}{5} + \left(1 - \frac{1}{5}\right)0.2}$$

$$\sigma_P = 30\% \sqrt{0.36}$$

$$\sigma_P = 18\%$$

The VaR in nominal terms is:

$$\text{VaR}_P = Z_c \times \sigma_P \times V = (2.33)(18\%)(\$10 \text{ million})$$

$$\text{VaR}_P = \$4,194,000$$

Marginal VaR

Marginal VaR applies to a particular position in a portfolio, and it is the *per unit change in a portfolio VaR that occurs from an additional investment in that position*.

Mathematically speaking, it is the partial derivative of the portfolio VaR with respect to the position:

$$\begin{aligned}\text{Marginal VaR} = \text{MVar}_i &= \frac{\partial \text{VaR}_p}{\partial (\text{monetary investment in } i)} \\ &= Z_c \frac{\partial \sigma_p}{\partial w_i} = Z_c \frac{\text{cov}(R_i, R_p)}{\sigma_p}\end{aligned}$$

Using CAPM methodology, we know a regression of the returns of a single asset i in a portfolio on the returns of the entire portfolio gives a beta, denoted β_i , which is a concise measure that includes the covariance of the position's returns with the total portfolio:

$$\beta_i = \frac{\text{cov}(R_i, R_p)}{\sigma_p^2}$$

Using the concept of beta gives another expression for marginal VaR:

$$\text{Marginal VaR} = \text{MVar}_i = \frac{\text{VaR}_p}{\text{portfolio value}} \times \beta_i$$

EXAMPLE: Computing marginal VaR

Assume Portfolio X has a VaR of €400,000. The portfolio is made up of four assets: Asset A, Asset B, Asset C, and Asset D. These assets are equally weighted within the portfolio and are each valued at €1,000,000. Asset A has a beta of 1.2. **Calculate** the marginal VaR of Asset A.

Answer:

$$\text{Marginal VaR}_A = (\text{VaR}_p / \text{portfolio value}) \times \beta_A$$

$$\text{Marginal VaR}_A = (400,000 / 4,000,000) \times 1.2 = 0.12$$

Thus, portfolio VaR will change by 0.12 for each euro change in Asset A.

Incremental VaR

Incremental VaR is the change in VaR from the addition of a new position in a portfolio. Since it applies to an entire position, it is generally larger than marginal VaR and may include nonlinear relationships, which marginal VaR generally assumes away. The problem with measuring incremental VaR is that, in order to be accurate, a full revaluation of the portfolio after the addition of the new position would be necessary. The incremental VaR is the difference between the new VaR from the revaluation minus the VaR before the addition. The revaluation requires not only measuring the risk of the position itself, but it also requires measuring the change in the risk of the other positions that are already in the portfolio. For a portfolio with hundreds or thousands

of positions, this would be time consuming. Clearly, VaR measurement becomes more difficult as portfolio size increases given the expansion of the covariance matrix. Using a shortcut approach for computing incremental VaR would be beneficial.

For small additions to a portfolio, we can approximate the incremental VaR with the following steps:

Step 1: Estimate the risk factors of the new position and include them in a vector $[\eta]$.

Step 2: For the portfolio, estimate the vector of marginal VaRs for the risk factors $[MVar_j]$.

Step 3: Take the cross product.

This probably requires less work and is faster to implement because it is likely the managers already have estimates of the vector of $MVar_j$ values in Step 2.

Before we take a look at how to calculate incremental VaR, let's review the calculation of delta-normal VaR using matrix notation (i.e., using a covariance matrix).

EXAMPLE: Computing VaR using matrix notation

A portfolio consists of Assets A and B. These assets are the risk factors in the portfolio. The volatilities are 6% and 14%, respectively. There are \$4 million and \$2 million invested in them, respectively. If we assume they are uncorrelated with each other, **compute** the VaR of the portfolio using a confidence parameter of 1.65.

Answer:

We can use matrix notation to derive the dollar variance of the portfolio:

$$\sigma_P^2 V^2 = \begin{bmatrix} \$4 & \$2 \end{bmatrix} \begin{bmatrix} 0.06^2 & 0 \\ 0 & 0.14^2 \end{bmatrix} \begin{bmatrix} \$4 \\ \$2 \end{bmatrix} = 0.0576 + 0.0784 = 0.136$$

This value is in (\$ millions)². VaR is then the square root of the portfolio variance times 1.65:

$$VaR = (1.65)(\$368,782) = \$608,490$$



PROFESSOR'S NOTE

Matrix multiplication consists of multiplying each row by each column. For example: $(4 \times 0.06^2) + (2 \times 0) = 0.0144$; $0.0144 \times 4 = 0.0576$. Had the positions been positively correlated, some positive value would replace the zeros in the covariance matrix.

EXAMPLE: Computing incremental VaR

A portfolio consists of Assets A and B. The volatilities are 6% and 14%, respectively. There are \$4 million and \$2 million invested in them respectively. If we assume they are uncorrelated with each other, **compute** the incremental VaR for an increase of \$10,000 in Asset A. Assume a z-score of 1.65.

Answer:

To find incremental VaR, we compute the per dollar covariances of each risk factor:

$$\begin{bmatrix} \text{cov}(R_A, R_P) \\ \text{cov}(R_B, R_P) \end{bmatrix} = \begin{bmatrix} 0.06^2 & 0 \\ 0 & 0.14^2 \end{bmatrix} \begin{bmatrix} \$4 \\ \$2 \end{bmatrix} = \begin{bmatrix} 0.0144 \\ 0.0392 \end{bmatrix}$$

These per dollar covariances represent the covariance of a given risk factor with the portfolio. Thus, we can substitute these values into the marginal VaR equations for the risk factors as follows.

The marginal VaRs of the two risk factors are:

$$\text{MVaR}_A = Z_c \times \frac{\text{cov}(R_A, R_P)}{\sigma_P} = 1.65 \times \frac{0.0144}{\sqrt{0.136}} = 0.064428$$

$$\text{MVaR}_B = Z_c \times \frac{\text{cov}(R_B, R_P)}{\sigma_P} = 1.65 \times \frac{0.0392}{\sqrt{0.136}} = 0.175388$$

Since the two assets are uncorrelated, the incremental VaR of an additional \$10,000 investment in Position A would simply be \$10,000 times 0.064428, or \$644.28.

Component VaR

Component VaR for position i , denoted CVaR_i , is the amount of risk a particular fund contributes to a portfolio of funds. It will generally be less than the VaR of the fund by itself (i.e., stand-alone VaR) because of diversification benefits at the portfolio level. In a large portfolio with many positions, the approximation is simply the marginal VaR multiplied by the dollar weight in position i :

$$\begin{aligned} \text{CVaR}_i &= (\text{MVaR}_i) \times (w_i \times P) = \text{VaR} \times \beta_i \times w_i \\ &= (\alpha \times \sigma_P \times P) \times \beta_i \times w_i = (\alpha \times \sigma_i \times w_i \times P) \times \rho_i = \text{VaR}_i \times \rho_i \end{aligned}$$

The last two components consider the fact that $\beta_i = (\rho_i \times \sigma_i) / \sigma_P$.

Using CVaR_i , we can express the total VaR of the portfolio as:

$$\text{VaR} = \sum_{i=1}^N \text{CVaR}_i = \text{VaR} \left(\sum_{i=1}^N w_i \times \beta_i \right)$$

Given the way the betas were computed we know: $\left(\sum_{i=1}^N w_i \times \beta_i \right) = 1$

EXAMPLE: Computing component VaR (Example 1)

Assume Portfolio X has a VaR of €400,000. The portfolio is made up of four assets: Asset A, Asset B, Asset C, and Asset D. These assets are equally weighted within the portfolio and are each valued at €1,000,000. Asset A has a beta of 1.2. **Calculate** the component VaR of Asset A.

Answer:

$$\text{Component VaR}_A = \text{VaR}_P \times \beta_A \times \text{asset weight}$$

$$\begin{aligned}\text{Component VaR}_A &= 400,000 \times 1.2 \times (1,000,000 / 4,000,000) \\ &= €120,000\end{aligned}$$

Thus, portfolio VaR will decrease by €120,000 if Asset A is removed.

EXAMPLE: Computing component VaR (Example 2, Part 1)

Recall our previous incremental VaR example of a portfolio invested \$4 million in A and \$2 million in B. Using their respective marginal VaRs, 0.064428 and 0.175388, **compute** the component VaRs.

Answer:

$$\text{CVaR}_A = (\text{MVar}_A) \times (w_A \times P) = (0.064428) \times (\$4 \text{ million}) = \$257,713$$

$$\text{CVaR}_B = (\text{MVar}_B) \times (w_B \times P) = (0.175388) \times (\$2 \text{ million}) = \$350,777$$

**PROFESSOR'S NOTE**

The values have been adjusted for rounding.

EXAMPLE: Computing component VaR (Example 2, Part 2)

Using the results from the previous example, **compute** the percentage of contribution to VaR of each component.

Answer:

The answer is the sum of the component VaRs divided into each individual component VaR:

$$\% \text{ contribution to VaR from A} = \frac{\$257,713}{(\$257,713 + \$350,777)} = 42.35\%$$

$$\% \text{ contribution to VaR from B} = \frac{\$350,777}{(\$257,713 + \$350,777)} = 57.65\%$$

Normal distributions are a subset of the class of distributions called elliptical distributions. As a class, elliptical distributions have fewer assumptions than normal distributions. Risk management often assumes elliptical distributions, and the procedures to estimate component VaRs up to this point have applied to elliptical distributions.

If the returns do not follow an elliptical distribution, we can employ other procedures to compute component VaR. If the distribution is homogeneous of degree one, for example, then we can use Euler's theorem to estimate the component VaRs. The return of a portfolio of assets is homogeneous of degree one because, for some constant, k , we can write:

$$k \times R_p = \sum_{i=1}^N k \times w_i \times R_i$$

The following steps can help us find component VaRs for a non-elliptical distribution using historical returns:

Step 1: Sort the historical returns of the portfolio.

Step 2: Find the return of the portfolio, which we will designate $R_{P(VaR)}$, that corresponds to a return that would be associated with the chosen VaR.

Step 3: Find the returns of the individual positions that occurred when $R_{P(VaR)}$ occurred.

Step 4: Use each of the position returns associated with $R_{P(VaR)}$ for component VaR for that position.

To improve the estimates of the component VaRs, an analyst should probably obtain returns for each individual position for returns of the portfolio slightly above and below $R_{P(VaR)}$. For each set of returns for each position, the analyst would compute an average to better approximate the component VaR of the position.



MODULE QUIZ 87.1

- Which of the following is the best synonym for diversified VaR?
 - Vector VaR.
 - Position VaR.
 - Portfolio VaR.
 - Incidental VaR.
- When computing individual VaR, it is proper to:
 - use the absolute value of the portfolio weight.
 - use only positive weights.
 - use only negative weights.
 - compute VaR for each asset within the portfolio.
- A portfolio consists of two positions. The VaR of the two positions are \$10 million and \$20 million. If the returns of the two positions are not correlated, the VaR of the portfolio would be closest to:
 - \$5.48 million.
 - \$15.00 million
 - \$22.36 million.
 - \$25.00 million.
- Which of the following is true with respect to computing incremental VaR? Compared to using marginal VaRs, computing with full revaluation is:
 - more costly, but less accurate.
 - less costly, but more accurate.
 - less costly, but also less accurate.
 - more costly, but also more accurate.

MODULE 87.2: MANAGING PORTFOLIOS WITH VaR

LO 87.c: Apply the concept of marginal VaR in making portfolio management

decisions.

LO 87.d: Explain the risk-minimizing position and the risk and return-optimizing position of a portfolio.

A manager can lower a portfolio VaR by lowering allocations to the positions with the highest marginal VaR. If the manager keeps the total invested capital constant, this would mean increasing allocations to positions with lower marginal VaR. Portfolio risk will be at a global minimum where all the marginal VaRs are equal for all i and j :

$$MVaR_i = MVaR_j$$

We can use our earlier example to see how we can use marginal VaRs to make decisions to lower the risk of the entire portfolio. In the earlier example, Position A has the smaller MVaR; therefore, we will compute the marginal VaRs and total VaR for a portfolio which has \$5 million invested in A and \$1 million in B. The portfolio variance is:

$$\sigma_P^2 V^2 = \begin{bmatrix} \$5 & \$1 \end{bmatrix} \begin{bmatrix} 0.06^2 & 0 \\ 0 & 0.14^2 \end{bmatrix} \begin{bmatrix} \$5 \\ \$1 \end{bmatrix} = 0.0900 + 0.0196 = 0.1096$$

This value is in (\$ millions)². VaR is then the square root of the portfolio variance times 1.65 (95% confidence level):

$$VaR = (1.65)(\$331,059) = \$546,247$$

The VaR of \$546,247 is less than the VaR of \$608,490, which was produced when Portfolio A had a lower weight. We can see that the marginal VaRs are now much closer in value:

$$\begin{bmatrix} \text{cov}(R_A, R_P) \\ \text{cov}(R_B, R_P) \end{bmatrix} = \begin{bmatrix} 0.06^2 & 0 \\ 0 & 0.14^2 \end{bmatrix} \begin{bmatrix} \$5 \\ \$1 \end{bmatrix} = \begin{bmatrix} 0.0180 \\ 0.0196 \end{bmatrix}$$

The marginal VaRs of the two positions are:

$$MVaR_A = Z_c \times \frac{\text{cov}(R_A, R_P)}{\sigma_P} = 1.65 \times \frac{0.0180}{\sqrt{0.1096}} = 0.08971$$

$$MVaR_B = Z_c \times \frac{\text{cov}(R_B, R_P)}{\sigma_P} = 1.65 \times \frac{0.0196}{\sqrt{0.1096}} = 0.09769$$

LO 87.e: Explain the difference between risk management and portfolio management and describe how to use marginal VaR in portfolio management.

As the name implies, risk management focuses on risk and ways to reduce risk; however, minimizing risk may not produce the optimal portfolio. Portfolio management requires assessing both risk measures and return measures to choose the optimal portfolio. Traditional efficient frontier analysis tells us that the minimum variance portfolio is not optimal. We should note that the **efficient frontier** is the plot of portfolios that have the lowest standard deviation for each expected return (or highest return for each standard deviation) when plotted on a plane with the vertical axis measuring return and the horizontal axis measuring the standard deviation. The

optimal portfolio is represented by the point where a ray from the risk-free rate is just tangent to the efficient frontier. That optimal portfolio has the highest Sharpe ratio:

$$\text{Sharpe ratio} = \frac{(\text{portfolio return} - \text{risk-free rate})}{(\text{standard deviation of portfolio return})}$$

We can modify this formula by replacing the standard deviation with VaR so that the focus then becomes the excess return of the portfolio over VaR:

$$\frac{(\text{portfolio return} - \text{risk-free rate})}{(\text{VaR of portfolio})}$$

This ratio is maximized when the excess return in each position divided by its respective marginal VaR equals a constant. In other words, at the optimum:

$$\frac{(\text{position } i \text{ return} - \text{risk-free rate})}{(\text{MVaR}_i)} = \frac{(\text{position } j \text{ return} - \text{risk-free rate})}{(\text{MVaR}_j)}$$

for all positions i and j .



PROFESSOR'S NOTE

Equating the excess return/MVaR ratios will obtain the optimal portfolio. This differs from equating just the MVaRs, which obtains the portfolio with the lowest portfolio VaR.

Assuming that the returns follow elliptical distributions, we can represent the condition in a more concise fashion by employing betas, β_i , which are obtained from regressing each position's return on the portfolio return:

$$\frac{(\text{position } i \text{ return} - \text{risk-free rate})}{\beta_i} = \frac{(\text{position } j \text{ return} - \text{risk-free rate})}{\beta_j}$$

for all positions i and j .

The portfolio weights that make these ratios equal will be the optimal portfolio. We now turn our attention to determining the optimal portfolio for our example portfolio of A and B. We will assume the expected excess return of A is 6% and that of B is 11%. Even without this information, we should know that the optimal portfolio will have an allocation in A less than \$5 million and in B greater than \$1 million. This is because the marginal VaRs were almost equal with those allocations. *Thus, the resulting portfolio would be close to the minimum variance*, which will not be optimal. We might want to find out how to adjust the allocation with respect to the original values of \$4 million in A and \$2 million in B. By comparing the ratios of the two assets we find:

$$\frac{\text{excess return of A}}{\text{MVaR}_A} = \frac{0.06}{0.064428} = 0.9313$$

$$\frac{\text{excess return of B}}{\text{MVaR}_B} = \frac{0.11}{0.175388} = 0.6272$$

We see that there is too much allocated in B. Before we adjust the portfolio, we compute the excess-return-to-VaR ratio for the entire portfolio. The return is:

$$\% \text{ excess return on portfolio} = 7.67\% = \frac{\$4 \text{ million}}{\$6 \text{ million}} (6\%) + \frac{\$2 \text{ million}}{\$6 \text{ million}} (11\%)$$

The return to VaR (scaled by the size of the portfolio) is:

$$0.7559 = \frac{0.0767}{\$608,490} \times \$6 \text{ million}$$

Now, because the return to MVaR ratio was greater for A, we will increase the allocation in A to \$4.5 million and decrease that in B to \$1.5 million. With those changes, the portfolio variance is:

$$\sigma_P^2 V^2 = \begin{bmatrix} \$4.5 & \$1.5 \end{bmatrix} \begin{bmatrix} 0.06^2 & 0 \\ 0 & 0.14^2 \end{bmatrix} \begin{bmatrix} \$4.5 \\ \$1.5 \end{bmatrix} = 0.0729 + 0.0441 = 0.1170$$

This value is in (\$ millions)². VaR is then the square root of the portfolio variance times 1.65 (95% confidence level):

$$\text{VaR} = (1.65)(\$342,053) = \$564,387$$

In this case, the marginal VaRs are found by:

$$\begin{bmatrix} \text{cov}(R_A, R_P) \\ \text{cov}(R_B, R_P) \end{bmatrix} = \begin{bmatrix} 0.06^2 & 0 \\ 0 & 0.14^2 \end{bmatrix} \begin{bmatrix} \$4.5 \\ \$1.5 \end{bmatrix} = \begin{bmatrix} 0.0162 \\ 0.0294 \end{bmatrix}$$

The marginal VaRs of the two positions are then:

$$\text{MVaR}_A = Z_c \times \frac{\text{cov}(R_A, R_P)}{\sigma_P} = 1.65 \times \frac{0.0162}{\sqrt{0.1170}} = 0.0781$$

$$\text{MVaR}_B = Z_c \times \frac{\text{cov}(R_B, R_P)}{\sigma_P} = 1.65 \times \frac{0.0294}{\sqrt{0.1170}} = 0.1418$$

We see the expected excess-return-to-marginal VaR ratios are much closer:

$$\frac{0.06}{0.0781} = 0.7678$$

$$\frac{0.11}{0.1418} = 0.7756$$

The portfolio return is now:

$$\% \text{ excess return on portfolio} = 7.25\% = \frac{\$4.5 \text{ million}}{\$6 \text{ million}} (6\%) + \frac{\$1.5 \text{ million}}{\$6 \text{ million}} (11\%)$$

The portfolio return divided by the portfolio VaR has risen. The return to VaR (scaled by the size of the portfolio) is:

$$0.7707 = \frac{0.0725}{\$564,387} \times \$6 \text{ million}$$

This is greater than the 0.7559 value associated with the original \$4 million and \$2 million allocations. The result is a more optimal portfolio allocation.



MODULE QUIZ 87.2

1. A portfolio has an equal amount invested in two positions, X and Y. The expected excess return of X is 9% and that of Y is 12%. Their marginal VaRs are 0.06 and 0.075, respectively. To move toward the optimal portfolio, the manager will probably:
 - A. increase the allocation in Y and/or lower that in X.
 - B. increase the allocation in X and/or lower that in Y.
 - C. do nothing because the information is insufficient.

D. not change the portfolio because it is already optimal.

KEY CONCEPTS

LO 87.a

Diversified VaR is simply the VaR of the portfolio where the calculation takes into account the diversification effects. The basic formula is:

$$\text{VaR}_p = Z_c \times \sigma_p \times P$$

where:

Z_c = z-score associated with the level of confidence c

σ_p = standard deviation of the portfolio return

P = nominal value invested in the portfolio

Individual VaR is the VaR of an individual position in isolation. If the proportion or weight in the position is w_i , then we can define the individual VaR as:

$$\text{VaR}_i = Z_c \times \sigma_i \times |P_i| = Z_c \times \sigma_i \times |w_i| \times P$$

where:

P = portfolio value

P_i = nominal amount invested in position i

Marginal VaR is the change in a portfolio VaR that occurs from an additional one unit investment in a given position. Useful representations are:

$$\text{Marginal VaR} = \text{MVar}_i = Z_c \frac{\text{cov}(R_i, R_p)}{\sigma_p}$$

$$\text{Marginal VaR} = \text{MVar}_i = \frac{\text{VaR}}{P} \times \beta_i$$

Incremental VaR is the change in VaR from the addition of a new position in a portfolio. It can be calculated precisely from a total revaluation of the portfolio, but this can be costly. A less costly approximation is found by (1) breaking down the new position into risk factors, (2) multiplying each new risk factor times the corresponding partial derivative of the portfolio with respect to the risk factor, and then (3) adding up all the values.

Component VaR for position i , denoted CVaR_i , is the amount a portfolio VaR would change from deleting that position in a portfolio. In a large portfolio with many positions, the approximation is simply the marginal VaR multiplied by the dollar weight in position i :

$$\text{CVaR}_i = (\text{MVar}_i) \times (w_i \times P) = \text{VaR} \times \beta_i \times w_i$$

There is a method for computing component VaRs for distributions that are not elliptical. The procedure is to sort the historical returns of the portfolio and designate a portfolio return that corresponds to the loss associated with the VaR and then find the returns of each of the components associated with that portfolio loss. Those position returns can be used to compute component VaRs.

LO 87.b

For a two-asset portfolio, two special cases are:

1. VaR for uncorrelated positions:

$$\text{VaR}_p = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$$

2. VaR for perfectly correlated positions:

$$\begin{aligned}\text{Undiversified VaR} = \text{VaR}_p &= \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\text{VaR}_1 \text{VaR}_2} \\ &= \text{VaR}_1 + \text{VaR}_2\end{aligned}$$

LO 87.c

Portfolio risk will be at a global minimum where all the marginal VaRs are equal for all i and j :

$$\text{MVaR}_i = \text{MVaR}_j$$

LO 87.d

Equating the MVaRs will obtain the portfolio with the lowest portfolio VaR. Equating the excess return/MVaR ratios will obtain the optimal portfolio.

LO 87.e

The optimal portfolio is the one for which all excess-return-to-marginal VaR ratios are equal:

$$\frac{(\text{position } i \text{ return} - \text{risk-free rate})}{(\text{MVaR}_i)} = \frac{(\text{position } j \text{ return} - \text{risk-free rate})}{(\text{MVaR}_j)}$$

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 87.1

1. **C** Portfolio VaR should include the effects of diversification. None of the other answers are types of VaRs. (LO 87.a)
2. **A** The expression for individual VaR is $\text{VaR}_i = Z_c \times \sigma \times |P_i| = Z \times \sigma_i \times |w_i| \times P$. The absolute value signs indicate that we need to measure the risk of both positive and negative positions, and risk cannot be negative. (LO 87.a)
3. **C** For uncorrelated positions, the answer is the square root of the sum of the squared VaRs:

$$\text{VaR}_p = \sqrt{(10^2 + 20^2)} \times (\$ \text{ million}) = \$22.36 \text{ million}$$

(LO 87.b)

4. **D** Full revaluation means recalculating the VaR of the entire portfolio. The marginal VaRs are probably already known, so using them is probably less costly, but will not be as accurate. (LO 87.a)

Module Quiz 87.2

1. **A** The expected excess-return-to-MVaR ratio for X is $0.09/0.06 = 1.5$ and for Y is $0.12/0.075 = 1.6$. Therefore, the portfolio weight in Y should increase to move the portfolio toward the optimal portfolio. (LO 87.e)



The following is a review of the Risk Management and Investment Management principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Jorion, Chapter 17.

READING 88

VAR AND RISK BUDGETING IN INVESTMENT MANAGEMENT

Study Session 13

EXAM FOCUS

Banks on the “sell side” of the investment industry have long used risk budgeting and value at risk (VaR). There is a trend for the “buy side” investment firms to increasingly use VaR. One reason for increased demand for risk budgeting is the increased complexity, dynamics, and globalization of the investment industry. Use of VaR can help set better guidelines than more traditional limits. By measuring marginal and incremental VaRs, a manager can make better decisions concerning portfolio weights. For the exam, be comfortable with the concept of surplus at risk (SaR). Also, understand how to budget risk across asset classes and active managers.

MODULE 88.1: BUDGETING AND MANAGING RISK WITH VAR

Risk Budgeting

LO 88.a: Define risk budgeting.

Risk budgeting is a top-down process that involves choosing and managing exposures to risk. The main idea is that the risk manager establishes a risk budget for the entire portfolio and then allocates risk to individual positions based on a predetermined fund risk level. The risk budgeting process differs from market value allocation since it involves the allocation of risk.

Managing Risk With VaR

LO 88.b: Describe the impact of horizon, turnover, and leverage on the risk management process in the investment management industry.

The “sell side” of the investment industry largely consists of banks that have developed VaR techniques and have used them for many years. Investors make up the “buy side” of the investment industry. Investors are now using VaR techniques, but they have to adapt them to the different nature of that side of the business. To understand why the needs are different, we should compare the characteristics of the two “sides.” Figure 88.1 makes direct comparisons.

Figure 88.1: Sell Side and Buy Side Characteristics

Characteristic	Sell Side	Buy Side
Horizon	Short-term (days)	Long-term (month or more)
Turnover	Fast	Slow
Leverage	High	Low
Risk measures	VaR	Asset allocation
	Stress tests	Tracking error
Risk controls	Position limits	Diversification
	VaR limits	Benchmarking
	Stop-loss rules	Investment guidelines

Banks trade rapidly, which is why they cannot rely on traditional measures of risk that are based on historical data. For banks, yesterday’s risk may not have anything to do with today’s positions. Investors usually try to hold positions for longer periods of time (e.g., years).

Having a more dynamic method for measuring risk such as VaR is also important for banks because of their high leverage. Institutional investors often have much stronger constraints with respect to leverage; therefore, they have a much lower need to control downside risk.

The Investment Process

LO 88.c: Describe the investment process of large investors such as pension funds.

The *first step* in the investment process is to determine the long-term, strategic asset allocations. Usually, the goal of the first step is to balance returns and risks using methods like mean-variance portfolio optimization. This step determines the allocations to asset classes such as domestic and foreign stocks, domestic and foreign bonds, and alternative investments such as real estate, venture capital, and hedge funds. Making this allocation relies on passive indices and other benchmarks to help measure the properties of the investment, and the availability of passive indices helps make the allocations feasible.

The *second step* in the investment process is to choose the managers who may either passively manage the fund (i.e., simply track the benchmarks) or actively manage the fund in an effort to outperform the benchmarks. The investors should review the managers' activities and performance periodically. Their activities should conform to a list of guidelines, which includes the types of investments and risk exposure restrictions such as beta and duration. Managers' performance can be evaluated by analyzing their tracking error.

VaR risk management systems are beginning to become more important because of the globalization of available investments and the increased complexity of investments. Also, investment companies are becoming more dynamic, which makes it more difficult to assess risk. With many managers, for example, each of the managers may make changes within his constraints, but the collective changes could be difficult to gauge with historical measures. In sum, because of increased globalization, complexity, and the dynamic nature of the investment industry, simply measuring risk using historical measures is no longer adequate, which has increased the need for VaR.

Hedge Fund Issues

LO 88.d: Describe the risk management challenges associated with investments in hedge funds.

Hedge funds are a very heterogeneous class of assets that include a variety of trading strategies. Since they often use leverage and trade a great deal, their risk characteristics may be more similar to the "sell side" of the industry. Hedge funds have some other risks like liquidity and low transparency. Liquidity risk has many facets. First, there is the obvious potential loss from having to liquidate too quickly. Second, there is the difficulty of measuring the exact value of the fund to be able to ascertain its risk. Furthermore, the low liquidity tends to lower the volatility of historical prices as well as the correlations of the positions. These properties will lead to an underestimation of traditional measures of risk. In addition to these risks, there is the low level of transparency. This makes the risk measurement difficult with respect to both the size and type. Not knowing the type of risk increases the difficulty of risk management for the entire portfolio in which an investor might include hedge funds.

Absolute Risk vs. Relative Risk and Policy Mix vs. Active Risk

LO 88.e: Distinguish among the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk.

Absolute or asset risk refers to the total possible losses over a horizon. It is simply measured by the return over the horizon. **Relative risk** is measured by excess return, which is the dollar loss relative to a benchmark. The shortfall is measured as the difference between the fund return and that of a benchmark in dollar terms. VaR

techniques can apply to tracking error (i.e., standard deviation of excess return) if the excess return is normally distributed.



PROFESSOR'S NOTE

The author's definition of tracking error differs from the definition of tracking error in other assigned readings. Jorion defines tracking error as active return minus the benchmark return. In other readings, this value is simply the excess return and tracking error is the volatility (i.e., standard deviation) of the excess return. Throughout this reading, we have expressed excess return as portfolio return minus benchmark return and tracking error as the volatility of the excess return. This methodology follows the definition of tracking error on previous FRM exams.

Distinguishing **policy mix** from **active risk** is important when an investment firm allocates funds to different managers in various asset classes. This breaks down the risk of the total portfolio into that associated with the target policy (i.e., the weights assigned to the various funds in the policy) and the risk from the fact that managers may make decisions which lead to deviations from the designated weights. VaR analysis is especially useful here because it can show the risk exposure associated with the two types of risk and how they affect the overall risk of the entire portfolio. Often, active management risk is not much of a problem for several reasons:

- For well-managed funds, it is usually fairly small for each of the individual funds.
- There will be diversification effects across the deviations.
- There can be diversification effects with the policy mix VaR to actually lower the total portfolio VaR.

Funding Risk

Funding risk refers to being able to meet the obligations of an investment company (e.g., a pension's payout to retirees). Put another way, funding risk is the risk that the value of assets will not be sufficient to cover the liabilities of the fund. The level of funding risk varies dramatically across different types of investment companies. Some have zero, while defined benefit pension plans have the highest.

The focus of this analysis is the surplus, which is the difference between the value of the assets and the liabilities, and the change in the surplus, which is the difference between the change in the assets and liabilities:

$$\text{surplus} = \text{assets} - \text{liabilities}$$

$$\Delta \text{surplus} = \Delta \text{assets} - \Delta \text{liabilities}$$

Typically, in managing funding risk, an analyst will transform the nominal return on the surplus into a return on the assets, and break down the return as indicated:

$$\begin{aligned} R_{\text{surplus}} &= \frac{\Delta \text{surplus}}{\text{assets}} = \frac{\Delta \text{assets}}{\text{assets}} - \left(\frac{\Delta \text{liabilities}}{\text{liabilities}} \right) \left(\frac{\text{liabilities}}{\text{assets}} \right) \\ &= R_{\text{asset}} - R_{\text{liabilities}} \left(\frac{\text{liabilities}}{\text{assets}} \right) \end{aligned}$$

Evaluating this expression requires assumptions about the liabilities, which are in the future and uncertain. For pension funds, liabilities represent “accumulated benefit obligations,” which are the present value of pension benefits owed to the employees and other beneficiaries. Determining the present value requires a discount rate, which is usually tied to some current level of interest rates in the market. An ironic aspect of funding risk is that assets for meeting the obligations like equities and bonds usually increase in value when interest rates decline, but the present value of future obligations may increase even more. When assets and liabilities change by different amounts, this affects the surplus, and the resulting volatility of the surplus is a source of risk. If the surplus turns negative, additional contributions will be required. This is called **surplus at risk (SaR)**.

One answer to this problem is to immunize the portfolio by making the duration of the assets equal that of the liabilities. This may not be possible since the necessary investments may not be available, and it may not be desirable because it may mean choosing assets with a lower return.

EXAMPLE: Determining a fund's risk profile

The XYZ Retirement Fund has \$200 million in assets and \$180 million in liabilities. Assume that the expected return on the surplus, scaled by assets, is 4%. This means the surplus is expected to grow by \$8 million over the first year. The volatility of the surplus is 10%. Using a z-score of 1.65, **compute** VaR and the associated deficit that would occur with the loss associated with the VaR.

Answer:

First, we calculate the expected value of the surplus. The current surplus is \$20 million (= \$200 million – \$180 million). It is expected to grow another \$8 million to a value of \$28 million. As for the VaR:

$$\text{VaR} = (1.65)(10\%)(\$200 \text{ million}) = \$33 \text{ million}$$

If this decline in value occurs, the deficit would be the difference between the VaR and the expected surplus value: \$33 million – \$28 million = \$5 million.



PROFESSOR'S NOTE

According to the assigned reading, the surplus at risk (SaR) is the VaR amount calculated previously. Note that SaR on previous exams has been approached differently, as illustrated in the following example. Be prepared for either approach on the actual exam. In the example to follow, we will illustrate how to calculate the volatility of surplus growth. On previous FRM exams, this value has not been provided.

EXAMPLE: Surplus at risk (via computing volatility of surplus)

The XYZ Retirement Fund has \$200 million in assets and \$180 million in liabilities. Assume that the expected annual return on the assets is 4% and the expected annual growth of the liabilities is 3%. Also assume that the volatility of the asset return is 10% and the volatility of the liability growth is 7%. **Compute** 95%

surplus at risk assuming the correlation between asset return and liability growth is 0.4.

Answer:

First, compute the expected surplus growth:

$$200 \times (0.04) - 180 \times (0.03) = \$2.6 \text{ million}$$

Next, compute the volatility of the surplus growth. To compute the volatility you need to recall one of the properties of covariance discussed in the FRM Part I curriculum. The variance of assets minus liabilities [i.e., $\text{Var}(A - L)$] = $\text{Var}(A) + \text{Var}(L) - 2 \times \text{Cov}(A, L)$. Where covariance is equal to the standard deviation of assets times the standard deviation of liabilities times the correlation between the two. The asset and liability amounts will also need to be applied to this formula.

$$\text{Variance}(A - L) = 200^2 \times 0.10^2 + 180^2 \times 0.07^2 - 2 \times 200 \times 180 \times 0.10 \times 0.07 \times 0.4 = 400 + 158.76 - 201.6 = \$357.16 \text{ million}$$

$$\text{Standard deviation} = \sqrt{357.16} = \$18.89$$

Thus, SaR can be calculated by incorporating the expected surplus growth and standard deviation of the growth.

$$95\% \text{ SaR} = 2.6 - 1.65 \times 18.89 = \$28.57 \text{ million}$$



PROFESSOR'S NOTE

Like VaR, SaR is a negative value since it is the surplus amount that is at risk. As a result, the negative sign is usually not presented since a negative amount is implied.

Plan Sponsor Risk

The plan sponsor risk is an extension of surplus risk and how it relates to those who ultimately bear responsibility for the pension fund. We can distinguish between the following risk measures:

- **Economic risk** is the variation in the total economic earnings of the plan sponsor. This takes into account how the risks of the various components relate to each other (e.g., the correlation between the surplus and operating profits).
- **Cash flow risk** is the variation of contributions to the fund. Being able to absorb fluctuations in cash flow allows for a more volatile risk profile.

Ultimately, from the viewpoint of the sponsor, the focus should be on the variation of the economic value of the firm. The management should integrate the various risks associated with the movement of the assets and surplus with the overall financial goals of the sponsor. This is aligned with the current emphasis on enterprise-wide risk management.



MODULE QUIZ 88.1

1. With respect to the buy side and sell side of the investment industry, the:
I. buy side uses more leverage.

- II. sell side has relied more on VaR measures.
- A. I only.
 - B. II only.
 - C. Both I and II.
 - D. Neither I nor II.
2. Compared to policy risk, which of the following is not a reason that management risk is not much of a problem?
- A. There will be diversification effects across the deviations.
 - B. Managers tend to make the same style shifts at the same time.
 - C. For well-managed funds, it is usually fairly small for each of the individual funds.
 - D. There can be diversification with the policy mix VaR to actually lower the total portfolio VaR.

MODULE 88.2: MONITORING RISK WITH VAR

LO 88.f: Explain the use of VaR to check manager compliance and monitor risk.

There are many types of risks that can increase dramatically in a large firm. For example, the “rogue trader” phenomenon is more likely in a large firm. This occurs when a manager of one of the accounts or funds within the larger portfolio deviates from her guidelines in terms of portfolio weights or even trades in unauthorized investments. Such deviations from compliance can be very short-term, and regular reporting measures may not catch the violations.

Risk management is necessary for all types of portfolios—even passively managed portfolios. Some analysts erroneously believe that passive investing, or benchmarking, does not require risk monitoring. This is not true because the risk profiles of the benchmarks change over time. In the late 1990s, a portfolio benchmarked to the S&P 500 would clearly have seen a change in risk exposures (e.g., an increase in the exposure to risks associated with the high-tech industry). A forward-looking risk measurement system would pick up on such trends.

Monitoring the risk of actively managed portfolios should help identify the reasons for changes in risk. Three explanations for dramatic changes in risk are (1) a manager taking on more risk, (2) different managers taking similar bets, and (3) more volatile markets. Thus, when there is an increase in the overall risk of a portfolio, top management would want to investigate the increase by asking the following questions.

- *Has the manager exceeded her risk budget?* VaR procedures and risk management can allocate a risk budget to each manager. The procedures should give an indication if and why the manager exceeds the risk budget. Is it a temporary change from changes in the market? Has the manager unintentionally let the weights of the portfolio drift so as to increase risk? Or, more seriously, has the manager engaged in unauthorized trades?
- *Are managers taking too many of the same style bets?* If the managers are acting independently, it is possible that they all start pursuing strategies with the same risk exposures. This could happen, for example, if all managers forecast lower interest rates. Bond managers would probably begin moving into long-term bonds, and equity

managers would probably begin moving into stocks that pay a high and stable dividend like utility companies and REITs. This would drastically increase the interest rate risk of the overall portfolio.

- *Have markets become more volatile?* If the risk characteristics of the entire market have changed, top management will have to decide if it is worth accepting the volatility or make decisions to reduce it by changing the target portfolio weights.

VaR can also be reverse engineered by utilizing the VaR tools outlined in the previous reading, such as component VaR and marginal VaR. These tools provide insight on how the overall portfolio will be affected by individual position changes. This method can be used provided that all relevant risks have been identified within the risk management system.

In the risk management process, there is a problem with measuring the risk of some unique asset classes like real estate, hedge funds, and venture capital. Also, there may be limited information on investments in a certain class (e.g., emerging markets and initial public offerings).

There is a trend in the investment industry toward management choosing a **global custodian** for the firm. Such a choice means an investor aggregates the portfolios with a single custodian, which more easily allows a consolidated picture of the total exposures of the fund. The custodian can combine reports on changes in positions with market data to produce forward-looking risk measures. Thus, the global custodian is an easy choice in pursuing centralized risk management. Along with the trend toward global custodians, there has been a trend in the “custodian industry” toward fewer custodians that can provide more services. Large custodian banks such as Citibank, Deutsche Bank, and State Street are providing risk management products.

Those that choose not to use a global custodian have done so because they feel that they have a tighter control over risk measures and can better incorporate VaR systems into operations. There are often economies of scale for larger firms in that they can spread the cost of risk management systems over a large asset base. Also, they can require tighter control when their assets are partly managed internally.

Increasingly, clients are asking money managers about their risk management systems. The clients are no longer satisfied with quarterly performance reports. Many investment managers have already incorporated VaR systems into their investment management process. Widely used risk standards for institutional investors recommend measuring the risk of the overall portfolio and measuring the risk of each instrument. It may be the case that those who do not have comprehensive risk management systems will soon be at a significant disadvantage to those who do have such systems. There also seems to be some attempt by managers to differentiate themselves with respect to risk management.

VaR Applications

LO 88.g: Explain how VaR can be used in the development of investment guidelines and for improving the investment process.

Investment Guidelines

VaR can help move away from the ad hoc nature and overemphasis on notionals and sensitivities that characterize the guidelines many managers now use. Clearly, ad hoc procedures will generally be inferior to formal guidelines using established principles. Also, limits on notionals and sensitivities have proven insufficient when leverage and positions in derivatives exist. The limits do not account for variations in risk nor correlations. VaR limits include all of these factors.

The problem with controlling positions and not risk is that there are many rules and restrictions, which in the end may not achieve the main goal. There is no measure of the possible losses that can occur in a given time period—a good quantity to identify in order to know how much capital to have on hand to meet liquidity needs. Furthermore, simple restrictions on certain positions can be easily evaded with the many instruments that are now available. As a wider range of products develop, obviously, the traditional and cumbersome position-by-position guidelines will become even less effective.

Investment Process

VaR can help in the first step of the investment process, which is the strategic asset-allocation decision. Since this step usually uses mean-variance analysis, as does the most basic VaR measures, VaR can help in the portfolio allocation process. Furthermore, VaR can measure specific changes in risk that can result as managers subjectively adjust the weights from those recommended by pure quantitative analysis.

VaR is also useful at the trading level. A trader usually focuses on the return and stand-alone risk of a proposed position. The trader may have some idea of how the risk of the position will affect the overall portfolio, but an adequate risk management system that uses VaR can give a specific estimate of the change in risk. In fact, the risk management system should stand ready to automatically calculate the marginal VaR of each existing position and proposed position. When the trader has the choice between adding one of two positions with similar return characteristics, the trader would choose the one with the lower marginal VaR. VaR methodology can help make choices between different assets too. The optimal portfolio will be the one that has the excess-return-to-marginal VaR ratios equal for all asset types, as seen in the previous reading. Thus, when a trader is searching for the next best investment, the trader will look at securities in the asset classes that currently have the higher returns-to-marginal-VaR ratios.

Budgeting Risk

LO 88.h: Describe the risk budgeting process and calculate risk budgets across asset classes and active managers.

Risk budgeting should be a top down process. The first step is to determine the total amount of risk, as measured by VaR, that the firm is willing to accept. The next step is to choose the optimal allocation of assets for that risk exposure. As an example, a firm's management might set a return volatility target equal to 20%. If the firm has \$100 million in assets under management and assuming the returns are normally distributed, at a 95% confidence level, this translates to:

$$\text{VaR} = (1.65) \times (20\%) \times (\$100 \text{ million}) = \$33 \text{ million}$$

The goal will be to choose assets for the fund that keep VaR less than this value. Unless the asset classes are perfectly correlated, the sum of the VaRs of the individual assets will be greater than the actual VaR of the portfolio. Thus, the budgeting of risk across asset classes should take into account the diversification effects. Such effects can be carried down to the next level when selecting the individual assets for the different classes.

EXAMPLE: Budgeting risk across asset classes (Part 1)

A manager has a portfolio with only one position: a \$500 million investment in W. The manager is considering adding a \$500 million position X or Y to the portfolio. The current volatility of W is 10%. The manager wants to limit portfolio VaR to \$200 million at the 99% confidence level. Position X has a return volatility of 9% and a correlation with W equal to 0.7. Position Y has a return volatility of 12% and a correlation with W equal to zero. **Determine** which of the two proposed additions, X or Y, will keep the manager within his risk budget.

Answer:

Currently, the VaR of the portfolio with only W is:

$$\text{VaR}_W = (2.33)(10\%)(\$500 \text{ million}) = \$116.5 \text{ million}$$

When adding X, the return volatility of the portfolio will be:

$$8.76\% = \sqrt{(0.5^2)(10\%)^2 + (0.5^2)(9\%)^2 + (2)(0.5)(0.5)(0.7)(10\%)(9\%)}$$

$$\text{VaR}_{W+X} = 2.33(8.76\%)(\$1,000 \text{ million}) = \$204 \text{ million}$$

When adding Y, the return volatility of the portfolio will be:

$$7.81\% = \sqrt{(0.5^2)(10\%)^2 + (0.5^2)(12\%)^2}$$

$$\text{VaR}_{W+Y} = (2.33)(7.81\%)(\$1,000 \text{ million}) = \$182 \text{ million}$$

Thus, Y keeps the total portfolio within the risk budget.

EXAMPLE: Budgeting risk across asset classes (Part 2)

Using the information provided in the previous example, **demonstrate** why focusing on the stand-alone VaR of X and Y would have led to the wrong choice.

Answer:

Obviously, the VaR of X is less than that of Y.

$$\text{VaR}_X = (2.33)(9\%)(\$500 \text{ million}) = \$104.9 \text{ million}$$

$$\text{VaR}_Y = (2.33)(12\%)(\$500 \text{ million}) = \$139.8 \text{ million}$$

The individual VaRs would have led the manager to select X over Y; however, the high correlation of X with W gives X a higher incremental VaR, which puts the portfolio of W and X over the limit. The zero correlation of W and Y makes the incremental VaR of Y much lower, and the portfolio of W with Y keeps the risk within the limit.

The traditional method for evaluating active managers is by measuring their excess return and tracking error and using it to derive a measure known as the information ratio. Excess return is the active return minus the benchmark return. The **information ratio** of manager i is:

$$\text{IR}_i = \frac{(\text{expected excess return of the manager})}{(\text{the manager's tracking error})}$$

For a portfolio of funds, each managed by a separate manager, the top management of the entire portfolio would be interested in the portfolio information ratio:

$$\text{IR}_P = \frac{(\text{expected excess return of the portfolio})}{(\text{the portfolio's tracking error})}$$

If the excess returns of the managers are independent of each other, it can be shown that the optimal allocation across managers is found by allocating weights to managers according to the following formula:

$$\text{weight of portfolio managed by manager } i = \frac{\text{IR}_i \times (\text{portfolio's tracking error})}{\text{IR}_P \times (\text{manager's tracking error})}$$

One way to use this measure is to “budget” portfolio tracking error. Given the IR_P , the IR_i , and the manager’s tracking error, top management can calculate the respective weights to assign to each manager. The weights of the allocations to the managers do not necessarily have to sum to one. Any difference can be allocated to the benchmark itself because, by definition, $\text{IR}_{\text{benchmark}} = 0$.

Determining the precise weights will be an iterative process in that each selection of weights will give a different portfolio expected excess return and tracking error. Figure 88.2 illustrates a set of weights derived from the given inputs that satisfy the condition.

Figure 88.2: Budgeting Risk Across Active Managers

	Tracking Error	Information Ratio	Weights
Manager A	5.0%	0.70	51%
Manager B	5.0%	0.50	37%
Benchmark	0.0%	0.00	12%
Portfolio	3.0%	0.82	100%

Although we have skipped the derivation, we can see that the conditions for optimal allocation hold true:

$$\text{For A: } 51\% = \frac{(3\%)(0.70)}{(5\%)(0.82)}$$

$$\text{For B: } 37\% = \frac{(3\%)(0.50)}{(5\%)(0.82)}$$

The difference between 100% and the sum of the weights 51% and 37% is the 12% invested in the benchmark.



MODULE QUIZ 88.2

- Using VaR to monitor risk is important for a large firm with many types of managers because:
 - it can help catch rogue traders and it can detect changes in risk from changes in benchmark characteristics.
 - although it cannot help catch rogue traders, it can detect changes in risk from changes in benchmark characteristics.
 - although it cannot detect changes in risk from changes in benchmark characteristics, it can help detect rogue traders.
 - of no reason. VaR is not useful for monitoring risk in large firms.
- VaR can be used to compose better guidelines for investment companies by:
 - relying less on notionals.
 - focusing more on overall risk.
 - I only.
 - II only.
 - Both I and II.
 - Neither I nor II.
- In making allocations across active managers, which of the following represents the formula that gives the optimal weight to allocate to a manager denoted i , where IR_i and IR_P are the information ratios of the manager and the total portfolio, respectively?

- A. $\frac{IR_p \times (\text{portfolio's tracking error})}{IR_i \times (\text{manager's tracking error})}$
- B. $\frac{IR_i \times (\text{manager's tracking error})}{IR_p \times (\text{portfolio's tracking error})}$
- C. $\frac{IR_i \times (\text{portfolio's tracking error})}{IR_p \times (\text{manager's tracking error})}$
- D. $\frac{IR_p \times (\text{manager's tracking error})}{IR_i \times (\text{portfolio's tracking error})}$

KEY CONCEPTS

LO 88.a

Risk budgeting is a top-down process that involves choosing and managing exposures to risk.

LO 88.b

Compared to banks on the “sell side,” investors on the “buy side” have a longer horizon, slower turnover, and lower leverage. They have tended to use historical risk measures and focus on tracking error, benchmarking, and investment guidelines. Banks use forward-looking VaR risk measures and VaR limits. Investors seem to be using VaR more and more, but they have to adapt it to their needs.

LO 88.c

Investors are relying more on VaR because of increased globalization, complexity, and dynamics of the investment industry. They have found simply measuring risk from historical measures is no longer adequate.

LO 88.d

Hedge funds have risk characteristics that make them more similar to the “sell side” of the industry like the use of leverage and high turnover. In addition to that, they have other risks such as low liquidity and low transparency. Low liquidity leads to problems in measuring risk because it tends to put a downward bias on volatility and correlation measures.

LO 88.e

Absolute or asset risk refers to the total possible losses over a horizon. Relative risk is measured by excess return, which is the dollar loss relative to a benchmark. VaR measures can apply to both.

The risk from the policy mix is from the chosen portfolio weights, and active risk is from individual managers deviating from the chosen portfolio weights.

Funding risk is the risk that the value of assets will not be sufficient to cover the liabilities of the fund. It is important for pension funds. In applying VaR, a manager will add the expected increase in the surplus to the surplus and subtract the VaR of the

assets from it. The difference between the expected surplus and the portfolio VaR is the shortfall associated with the VaR.

Two components of sponsor risk are cash flow risk, which addresses variations of contributions to the fund, and economic risk, which is the variation of the earnings.

LO 88.f

Risk monitoring is important in large firms to catch “rogue traders” whose activities may go undetected with simple periodic statements. It is also needed for passive portfolios because the risk characteristics of the benchmarks can change. Risk monitoring can also determine why changes in risk have occurred (e.g., individual managers exceeding their budget, different managers taking on the same exposures, or the risk characteristics of the whole market changing).

There is a trend toward using a global custodian in the risk management of investment firms. It is an easy means to the goal of centralized risk management. The custodians can combine reports on changes in positions with market data to produce forward-looking risk measures. Those that choose not to use a global custodian have done so because they feel they have tighter control over risk measures and can better incorporate VaR systems into operations.

LO 88.g

There is a trend of investment managers incorporating VaR systems into their investment management process. There is evidence that money managers are differentiating themselves with respect to their risk management systems, and those that do not use such systems are at a competitive disadvantage.

VaR techniques can help move away from the ad hoc nature and overemphasis on notionals and sensitivities that characterize the guidelines many managers now use. Such guidelines are cumbersome and ineffective in that they focus on individual positions and can be easily circumvented.

VaR is useful for the investment process. When a trader has a choice between two new positions for a portfolio, the trader can compare the marginal VaRs to make the selection. When deciding whether to increase one existing position over another, the trader can compare the excess-return-to-MVaR ratios and increase the position in the one with the higher ratio.

LO 88.h

Budgeting risk across asset classes means selecting assets whose combined VaRs are less than the total allowed. The budgeting process would examine the contribution each position makes to the portfolio VaR.

For allocating across active managers, it can be shown that the optimal allocation is achieved with the following formula:

$$\text{weight of portfolio managed by manager } i = \frac{IR_i \times (\text{portfolio's tracking error})}{IR_p \times (\text{manager's tracking error})}$$

For a given group of active managers, the weights may not sum to one. The remainder of the weight can be allocated to the benchmark, which has no tracking error.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 88.1

1. **B** Compared to banks on the “sell side,” investors on the “buy side” have a longer horizon, slower turnover, and lower leverage. Banks use forward-looking VaR risk measures and VaR limits. (LO 88.b)
2. **B** If managers make the same style shifts, then that would actually increase management risk. All the other reasons are valid. (LO 88.e)

Module Quiz 88.2

1. **A** Both of these are reasons large firms find VaR and risk monitoring useful. (LO 88.f)
2. **C** Investment companies have been focusing on limits on notionals, which is cumbersome and has proved to be ineffective. (LO 88.g)
3. **C** weight of portfolio managed by

$$\text{manager } i = \frac{IR_i \times (\text{portfolio's tracking error})}{IR_P \times (\text{manager's tracking error})}$$

(LO 88.h)

The following is a review of the Risk Management and Investment Management principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Litterman, Chapter 17.

READING 89

RISK MONITORING AND PERFORMANCE MEASUREMENT

Study Session 14

EXAM FOCUS

Most of this reading is qualitative in nature, however, it does contain several testable concepts. Many of the concepts covered here are also covered in other assigned readings, so this reading should serve as reinforcement of those concepts. For the exam, focus on the three dimensions of effective risk management: planning, budgeting, and monitoring. Understand the concept of a risk management unit (RMU) and be able to discuss its appropriate role within a company. Always keep in mind while reviewing this reading that it is the amount of risk taken that ultimately drives the level of returns—risk is the “cost” of returns.

MODULE 89.1: RISK PLANNING, BUDGETING, AND MONITORING

LO 89.a: Describe the three fundamental dimensions behind risk management, and their relation to VaR and tracking error.

Value at risk (VaR) is defined to be the *largest* loss possible for a *certain* level of confidence over a *specific* period of time. For example, a firm could express its VaR as being 95% certain that they will lose a maximum of \$5 million in the next ten days. Delta-normal VaR assumes a normal distribution, and its calculation reflects losses in the lower tail of the returns distribution.

Tracking error is defined as the standard deviation of excess returns. Excess return is defined as the portfolio return less the benchmark return (i.e., alpha). Assuming a normal distribution of excess returns, 95% of the outcomes will fall within the mean benchmark return plus or minus roughly two standard deviations.

VaR and tracking error are both measures of risk. An organization’s objective is to maximize profits for a given level of risk taken. Too much risk taken suggests a VaR

level that is too high and a willingness to accept large losses to produce unnecessarily high returns. Too little risk taken suggests that there is not enough active management.

VaR may be used to identify the maximum dollar value of losses for a specific level of confidence over a specific time. From a portfolio management perspective, VaR could be determined for each asset class, and capital allocation decisions could be made amongst the asset classes depending on risk and return preferences. This will help to achieve targeted levels of dollar VaR. In contrast, tracking error may be used to determine the relative amount of discretion that can be taken by the portfolio manager (away from benchmark returns) in his or her attempts at active management.

An effective risk management process has three related dimensions: risk planning, risk budgeting, and risk monitoring. A risk plan creates expectations for return and risk (e.g., VaR and tracking error). A risk budget allocates capital to meet those expectations, and risk monitoring identifies any variations from budget.

Risk Planning

LO 89.b: Describe risk planning, including its objectives, effects, and the participants in its development.

There are five risk planning objectives for any entity to consider:

1. *Setting expected return and expected volatility goals*

Examples of an entity's goals could include specifying the acceptable amounts of VaR and tracking error for a given period of time. Scenario analysis could be employed to determine potential sources of failure in the plan as well as ways to respond should those sources occur.

2. *Defining quantitative measures of success or failure*

Specific guidelines should be stated. For example, one could state an acceptable level of return on equity (ROE) or return on risk capital (RORC). This would help regulatory agencies assess the entity's success or failure from a risk management perspective.

3. *Generalizing how risk capital will be utilized to meet the entity's objectives*

Objectives relating to return per unit of risk capital need to be defined. For example, the minimum acceptable RORC should be defined for each activity where risk is allocated from the budget. The correlations between the RORCs should also be considered within an entity-wide risk diversification context.

4. *Defining the difference between events that cause ordinary damage versus serious damage*

Specific steps need to be formulated to counter any event that threatens the overall long-term existence of the entity, even if the likelihood of occurrence is remote. The choice between seeking external insurance (i.e., put options) versus self-insurance for downside portfolio risk has to be considered from a cost-benefit perspective, taking into account the potential severity of the losses.

5. *Identifying mission critical resources inside and outside the entity and discussing what should be done in case those resources are jeopardized*

Examples of such resources would include key employees and financing sources. Scenario analysis should be employed to assess the impact on those resources in both good and bad times. Specifically, adverse events often occur together with other adverse (and material) events.

In general, the risk planning process frequently requires the input and approval of the entity's owners and its management team. An effective plan requires very active input from the entity's highest level of management so as to ensure risk and return issues are addressed, understood, and communicated within the entity, to key stakeholders, and to regulatory agencies.

Risk Budgeting

LO 89.c: Describe risk budgeting and the role of quantitative methods in risk budgeting.

The risk budget quantifies the risk plan. There needs to be a structured budgeting process to allocate risk capital to meet the entity's objectives and minimize deviations from the plan. Each specific allocation from the risk budget comes with a reasonable return expectation. The return expectation comes with an estimate of variability around that expectation.

With risk budgets, an amount of VaR could be calculated for each item on the income statement. This allows RORC to be calculated individually and in aggregate.

Quantitative methods (i.e., mathematical modeling) may be used in risk budgeting as follows:

1. Set the minimum acceptable levels of RORC and ROE over various time periods. This is to determine if there is sufficient compensation for the risks taken (i.e., risk-adjusted profitability).
2. Apply mean-variance optimization (or other quantitative methods) to determine the weights for each asset class.
3. Simulate the portfolio performance based on the weights and for several time periods. Apply sensitivity analysis to the performance by considering changes in estimates of returns and covariances.

Risk Monitoring

LO 89.d: Describe risk monitoring and its role in an internal control environment.

Within an entity's internal control environment, risk monitoring attempts to seek and investigate any significant variances from budget. This is to ensure, for example, that

there are no threats to meeting its ROE and RORC targets. Risk monitoring is useful in that it should detect and address any significant variances in a timely manner.

LO 89.e: Identify sources of risk consciousness within an organization.

The increasing sense of risk consciousness within and among organizations is mainly derived from the following three sources:

1. *Banks* who lend funds to investors are concerned with where those funds are invested.
2. *Boards of investment clients, senior management, and plan sponsors* have generally become more versed in risk management issues and more aware of the need for effective oversight over asset management activities.
3. *Investors* have become more knowledgeable about their investment choices. For example, beneficiaries of a defined contribution plan are responsible for selecting their individual pension investments.



MODULE QUIZ 89.1

1. Which of the following statements about tracking error and value at risk (VaR) is least accurate?
 - A. Tracking error and VaR are complementary measures of risk.
 - B. Both tracking error and VaR may assume a normal distribution of returns.
 - C. Tracking error is the standard deviation of the excess of portfolio returns over the return of the peer group.
 - D. VaR can be defined as the maximum loss over a given period at a given level of confidence.
2. Which of the following statements about the use of quantitative methods in risk budgeting is least accurate? They may be used:
 - A. to simulate the performance of portfolios.
 - B. to set levels of return on equity (ROE) and return on risk capital (RORC).
 - C. in a scenario analysis context to determine the weights for each asset class.
 - D. in a sensitivity analysis context to consider changes in estimates of returns and covariances.

MODULE 89.2: RISK MANAGEMENT UNITS, LIQUIDITY CONSIDERATIONS, AND PERFORMANCE MEASUREMENT

Risk Management Units

LO 89.f: Describe the objectives and actions of a risk management unit in an investment management firm.

A **risk management unit (RMU)** monitors an investment management entity's portfolio risk exposure and ascertains that the exposures are authorized and consistent with the risk budgets previously set. To ensure proper segregation of duties, it is crucial

that the risk management function has an independent reporting line to senior management.

The objectives of an RMU include:

- Gathering, monitoring, analyzing, and distributing risk data to managers, clients, and senior management. Accurate and relevant information must be provided to the appropriate person(s) at the appropriate time(s).
- Assisting the entity in formulating a systematic and rigorous method as to how risks are identified and dealt with. Promotion of the entity's risk culture and best risk practices is crucial here.
- Going beyond merely providing information by taking the initiative to research relevant risk topics that will affect the firm.
- Monitoring trends in risk on a continual basis and promptly reporting unusual events to management before they become significant problems.
- Promoting discussion throughout the entity and developing a process as to how risk data and issues are discussed and implemented within the entity.
- Promoting a greater sense of risk awareness (culture) within the entity.
- Ensuring that transactions that are authorized are consistent with guidance provided to management and with client expectations.
- Identifying and developing risk measurement and performance attribution analytical tools.
- Gathering risk data to be analyzed in making portfolio manager assessments and market environment assessments.
- Providing the management team with information to better comprehend risk in individual portfolios as well as the source of performance.
- Measuring risk within an entity. In other words, measuring how consistent portfolio managers are with respect to product objectives, management expectations, and client objectives. Significant deviations are brought to the attention of appropriate management to provide a basis for correction.



PROFESSOR'S NOTE

You may see references elsewhere to an Independent Risk Oversight Unit. This is the same concept as RMU. Both measure and manage risk exposure and operate as an independent business unit.

LO 89.g: Describe how risk monitoring can confirm that investment activities are consistent with expectations.

Is the manager generating a forecasted level of tracking error that is consistent with the target?

The forecasted tracking error is an approximation of the potential risk of a portfolio using statistical methods. For each portfolio, the forecast should be compared to budget using predetermined guidelines as to how much variance is acceptable, how much

variance requires further investigation, and how much variance requires immediate action. Presumably, the budget was formulated taking into account client expectations.

Tracking error forecast reports should be produced for all accounts that are managed similarly in order to gauge the consistency in risk levels taken by the portfolio manager.

Is risk capital allocated to the expected areas?

Overall tracking risk is not sufficient as a measure on its own; it is important to break down the tracking risk into “subsections.” If the analysis of the risk taken per subsection does not suggest that risk is being incurred in accordance with expectations, then there may be **style drift**. Style drift may manifest itself in a value portfolio manager who attains the overall tracking error target but allocates most of the risk (and invests) in growth investments.

Therefore, by engaging in risk decomposition, the RMU may ensure that a portfolio manager’s investment activities are consistent with the predetermined expectations (i.e., stated policies and manager philosophy). Also, by running the report at various levels, unreasonably large concentrations of risk (that may jeopardize the portfolio) may be detected.

Liquidity Considerations

LO 89.h: Describe the Liquidity Duration Statistic and how it can be used to measure liquidity.

Liquidity considerations are important because a portfolio’s liquidity profile could change significantly in the midst of a volatile market environment or an economic downturn, for instance. Therefore, measuring portfolio liquidity is a priority in stress testing.

One potential measure is **liquidity duration**. It is an approximation of the number of days necessary to dispose of a portfolio’s holdings without a significant market impact. For a given security, the liquidity duration could be calculated as follows:

$$LD = \frac{Q}{(0.10 \times V)}$$

where:

LD = liquidity duration for the security on the assumption that the desired maximum daily volume of any security is 10%

Q = number of shares of the security

V = daily volume of the security

Performance Measurement

LO 89.i: Describe the objectives of performance measurement tools.

Performance measurement looks at a portfolio manager’s actual results and compares them to relevant comparables such as benchmarks and peer groups. Therefore,

performance measurement seeks to determine whether a manager can consistently outperform (through excess returns) the benchmark on a risk-adjusted basis. Similarly, it seeks to determine whether a manager consistently outperforms its peer group on a risk-adjusted basis.

Furthermore, performance measurement may help to determine whether the returns achieved are commensurate with the risk taken. Finally, performance measurement provides a basis for identifying managers who are able to generate consistent excess risk-adjusted returns. Such superior processes and performance could be replicated on an on-going basis, thereby maximizing the entity's long-run returns and profitability.

Comparison of Performance With Expectations

From a risk perspective (e.g., tracking error), portfolio managers should be assessed on the basis of being able to produce a portfolio with risk characteristics that are expected to approximate the target. In addition, they should also be assessed on their ability to actually achieve risk levels that are close to target.

From a returns perspective (e.g., performance), portfolio managers could be assessed on their ability to earn excess returns.

Goldman Sachs Asset Management utilizes a so-called "green zone" to identify instances of actual tracking error or performance that are outside of normal expectations. An acceptable amount of deviation (from a statistical perspective) is determined, and any deviations up to that amount are considered a green zone event. Unusual events that are expected to occur with some regularity are considered "yellow zone" events. Truly unusual events that require immediate investigation are considered "red zone" events. In using this simple color-coded system, the various zones are predefined and provide clear expectations for the portfolio managers. The movements of portfolios into yellow or red zones are triggering events that require further investigation and discussion.

Return Attribution

The source of returns can be attributed to specific factors or securities. For example, it is important to ensure that returns result from decisions where the manager intended to take risk and not simply from sheer luck.

Variance analysis is used to illustrate the contribution to overall portfolio performance by each security. The securities can be regrouped in various ways to conduct analysis by industry, sector, and country, for example.

In performing return attribution, factor risk analysis and factor attribution could be used. Alternatively, risk forecasting and attribution at the security level could also be used.

Sharpe Ratio and Information Ratio

The **Sharpe ratio** is calculated by taking the portfolio's actual return and subtracting the risk-free rate in the numerator. The denominator is the portfolio's standard deviation. The **information ratio** is calculated by taking the portfolio's excess returns and subtracting the benchmark's excess returns (if applicable) in the numerator. The

denominator is the portfolio's tracking error. These two measures are both considered risk-adjusted return measures.

Strengths of these metrics include the following: (1) easy to use as a measure of relative performance compared to a benchmark or peer group; (2) easy to determine if the manager has generated sufficient excess returns in relation to the amount of risk taken; and (3) easy to apply to industrial sectors and countries.

Weaknesses of these metrics include the following: (1) insufficient data available to perform calculations; and (2) the use of realized risk (instead of potential risk) may result in overstated performance calculations.

Comparisons With Benchmark Portfolios and Peer Groups

LO 89.j: Describe the use of alpha, benchmarks, and peer groups as inputs in performance measurement tools.

One could use linear regression analysis to regress the excess returns of the investment against the excess returns of the **benchmark**. One of the outputs from this regression is **alpha**, and it could be tested for statistical significance to determine whether the excess returns are attributable to manager skill or just pure luck. The other output is **beta**, and it relates to the amount of leverage used or underweighting/overweighting in the market compared to the benchmark.

The regression also allows a comparison of the absolute amount of excess returns compared to the benchmark. Furthermore, there is the ability to separate excess returns due to leverage and excess returns due to skill. One limitation to consider is that there may not be enough data available to make a reasonable conclusion as to the manager's skill.

One could also regress the excess returns of the manager against the excess returns of the manager's **peer group**. The features of this regression are generally similar to that for the benchmark, except that the returns of the peer group suffer from **survivorship bias**, and there is usually a wide range of funds under management amongst the peers (that reduces the comparability).



MODULE QUIZ 89.2

1. A risk management unit (RMU) is most likely to be active in which of the following contexts?
 - A. Risk monitoring.
 - B. Risk measurement.
 - C. Risk budgeting.
 - D. Risk planning.
2. Which of the following statements does not help explain the purpose of risk decomposition?
 - A. To ensure that there is no style drift.
 - B. To detect large concentrations of risk.
 - C. To detect excessive amounts of tracking risk.

- D. To ensure that investment activities are consistent with expectations.
3. Which of the following statements regarding alphas and betas is incorrect?
- A. Alpha is the excess return attributable to pure luck.
 - B. Alpha is the excess return attributable to managerial skill.
 - C. Beta suggests the relative amount of leverage used.
 - D. Beta suggests whether some of the returns are attributable to over or under weighting the market.

KEY CONCEPTS

LO 89.a

VaR and tracking error are both measures of risk. VaR is defined to be the largest loss possible for a certain level of confidence over a specific period of time. Tracking error is defined as the standard deviation of excess returns.

LO 89.b

There are five risk planning objectives to consider:

- Setting expected return and expected volatility goals
- Defining quantitative measures of success or failure
- Generalizing how risk capital will be utilized to meet the entity's objectives
- Defining the difference between events that cause ordinary damage versus serious damage
- Identifying mission critical resources inside and outside the entity and discussing what should be done in case those resources are jeopardized

The risk planning process frequently requires the input and approval of the entity's owners and its management team.

LO 89.c

The risk budget quantifies the risk plan. There needs to be a structured budgeting process to allocate risk capital to meet the corporate objectives and minimize deviations from plan.

Quantitative methods may be used in risk budgeting. Activities include: setting the minimum acceptable levels of RORC and ROE, applying mean-variance optimization, simulating portfolio performance, and applying sensitivity analysis.

LO 89.d

Within an entity's internal control environment, risk monitoring attempts to seek and investigate any significant variances from budget.

LO 89.e

Sources of risk consciousness include: (1) banks; (2) boards of investment clients, senior management, and plan sponsors; and (3) investors.

LO 89.f

A risk management unit (RMU) monitors an investment management entity's portfolio risk exposure and ascertains that the exposures are authorized and consistent with the risk budgets previously set. To ensure proper segregation of duties, it is crucial that the risk management function has an independent reporting line to senior management.

LO 89.g

The risk monitoring process attempts to confirm that investment activities are consistent with expectations. Specifically, is the manager generating a forecasted level of tracking error that is consistent with the target? And is risk capital allocated to the expected areas?

LO 89.h

Liquidity considerations are important because a portfolio's liquidity profile could change significantly in the midst of a volatile market environment or an economic downturn, for instance.

LO 89.i

Performance measurement looks at a portfolio manager's actual results and compares them to relevant comparables such as benchmarks and peer groups.

A performance measurement framework includes: (1) comparison of performance with expectations, (2) return attribution, (3) calculation of metrics such as the Sharpe ratio and the information ratio, and (4) comparisons with benchmark portfolios and peer groups.

LO 89.j

The excess returns of an investment can be regressed against the excess returns of its benchmark (e.g., S&P 500 Index). An output from this regression is alpha, which determines whether the investment's excess returns are due to skill or luck.

The excess returns of a manager can be regressed against the excess returns of the manager's peer group. This is similar to the linear regression with a benchmark portfolio, but differs since it suffers from survivorship bias.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 89.1

1. **C** All of the statements are accurate with the exception of the one relating to the peer group. Tracking error is the standard deviation of the excess of portfolio returns over the return of an appropriate benchmark, not peer group. (LO 89.a)
2. **C** All of the statements are accurate with the exception of the one relating to scenario analysis. One should apply mean-variance optimization (and not scenario analysis) to determine the weights for each asset class. (LO 89.c)

Module Quiz 89.2

1. **A** An RMU monitors an investment management firm's portfolio risk exposure and ascertains that the exposures are authorized and consistent with the risk budgets previously set. (LO 89.f)
2. **C** Risk decomposition is not designed to detect excessive amounts of tracking risk. In fact, it is the forecasted tracking error amount that should be compared to budget to ensure that there is not excessive tracking risk. All the other reasons are consistent with the purpose of risk decomposition. (LO 89.g)
3. **A** Alpha is a measure of the excess return of a manager over the peer group/benchmark that relates to skill as opposed to pure luck. Beta is a measure of the amount of leverage used compared to the peer group or a measure of the underweighting or overweighting of the market compared to the benchmark. (LO 89.j)



The following is a review of the Risk Management and Investment Management principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Bodie, Kane, and Marcus, Chapter 24.

READING 90

PORTFOLIO PERFORMANCE EVALUATION

Study Session 14

EXAM FOCUS

Professional money managers are routinely evaluated using a wide array of metrics. In this reading, alternative methods of computing portfolio returns will be presented, and contrasts will be made between time-weighted and dollar-weighted returns for portfolios experiencing cash redemptions and contributions. For the exam, be sure to understand differences in the risk-adjusted performance measures, including the Sharpe ratio, Treynor ratio, Jensen's alpha, information ratio, and M^2 , and how the trading practices of hedge funds complicates the evaluation process. Be able to apply Sharpe's regression-based style analysis to conduct performance attributions.

MODULE 90.1: TIME-WEIGHTED AND DOLLAR-WEIGHTED RETURNS

LO 90.a: Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses.

The **dollar-weighted rate of return** is defined as the internal rate of return (IRR) on a portfolio, taking into account all cash inflows and outflows. The beginning value of the account is an inflow as are all deposits into the account. All withdrawals from the account are outflows, as is the ending value.

EXAMPLE: Dollar-weighted rate of return

Assume an investor buys a share of stock for \$100 at $t = 0$, and at the end of the next year ($t = 1$), she buys an additional share for \$120. At the end of Year 2, the investor sells both shares for \$130 each. At the end of each year in the holding period, the stock paid a \$2.00 per share dividend. What is the investor's dollar-weighted rate of return?

Answer:

Step 1: Determine the timing of each cash flow and whether the cash flow is an inflow (+) or an outflow (-).

t = 0: purchase of first share	=	-\$100
t = 1: dividend from first share	=	+\$2
purchase of second share	=	-\$120
subtotal, t = 1	=	-\$118
t = 2: dividend from two shares	=	+\$4
proceeds from selling shares	=	+\$260
subtotal, t = 2	=	+\$264

Step 2: Net the cash flows for each time period, and set the PV of cash inflows equal to the present value of cash outflows.

$$PV_{\text{inflows}} = PV_{\text{outflows}}$$
$$\$100 + \frac{\$120}{(1+r)} = \frac{\$2}{(1+r)} + \frac{\$264}{(1+r)^2}$$

Step 3: Solve for r to find the dollar-weighted rate of return. This can be done using trial and error or by using the IRR function on a financial calculator or spreadsheet.

The intuition here is that we deposited \$100 into the account at t = 0, then added \$118 to the account at t = 1 (which, with the \$2 dividend, funded the purchase of one more share at \$120), and ended with a total value of \$264.

To compute this value with a financial calculator, use these net cash flows and follow the procedure described in Figure 90.1 to calculate the IRR.

$$\text{Net cash flows: } CF_0 = -100; CF_1 = -120 + 2 = -118;$$
$$CF_2 = 260 + 4 = 264$$

Figure 90.1: Calculating Dollar-Weighted Return With the TI BA II Plus™ Calculator

Key Strokes	Explanation	Display
[CF] [2nd] [CLR WORK]	Clear cash flow registers	CF0 = 0.00000
100 [+/-] [ENTER]	Initial cash outlay	CF0 = -100.00000
[↓] 118 [+/-] [ENTER]	Period 1 cash flow	C01 = -118.00000
[↓] [↓] 264 [ENTER]	Period 2 cash flow	C02 = 264.00000
[IRR] [CPT]	Calculate IRR	IRR = 13.86122

The dollar-weighted rate of return for this problem is 13.86%.

Time-weighted rate of return measures compound growth. It is the rate at which \$1.00 compounds over a specified time horizon. Time-weighting is the process of averaging a set of values over time. The *annual* time-weighted return for an investment may be computed by performing the following steps:

Step 1: Value the portfolio immediately preceding significant addition or withdrawals. Form subperiods over the evaluation period that correspond to the dates of

deposits and withdrawals.

Step 2: Compute the holding period return (HPR) of the portfolio for each subperiod.

Step 3: Compute the product of $(1 + \text{HPR}_t)$ for each subperiod t to obtain a total return for the entire measurement period [i.e., $(1 + \text{HPR}_1) \times (1 + \text{HPR}_2) \dots (1 + \text{HPR}_n)$].

If the total investment period is greater than one year, you must take the geometric mean of the measurement period return to find the annual time-weighted rate of return.

EXAMPLE: Time-weighted rate of return

A share of stock is purchased at $t = 0$ for \$100. At the end of the next year, $t = 1$, another share is purchased for \$120. At the end of Year 2, both shares are sold for \$130 each. At the end of Years 1 and 2, the stock paid a \$2.00 per share dividend. What is the time-weighted rate of return for this investment? (This is the same data as presented in the dollar-weighted rate of return example.)

Answer:

Step 1: Break the evaluation period into two subperiods based on timing of cash flows.

Holding period 1: beginning price = \$100.00
dividends paid = \$2.00
ending price = \$120.00

Holding period 2: beginning price = \$240.00 (2 shares)
dividends paid = \$4.00 (\$2 per share)
ending price = \$260.00 (2 shares)

Step 2: Calculate the HPR for each holding period.

$$\text{HPR}_1 = [(\$120 + 2) / \$100] - 1 = 22\%$$

$$\text{HPR}_2 = [(\$260 + 4) / \$240] - 1 = 10\%$$

Step 3: Take the geometric mean of the annual returns to find the annualized time-weighted rate of return over the measurement period.

$$(1 + \text{time-weighted rate of return})^2 = (1.22)(1.10)$$

$$\text{time-weighted rate of return} = [\sqrt{(1.22)(1.10)}] - 1 = 15.84\%$$

In the investment management industry, the time-weighted rate of return is the preferred method of performance measurement for a portfolio manager because it is not affected by the timing of cash inflows and outflows, which may be beyond the manager's control.

In the preceding examples, the time-weighted rate of return for the portfolio was 15.84%, while the dollar-weighted rate of return for the same portfolio was 13.86%. The difference in the results is attributable to the fact that the procedure for determining the dollar-weighted rate of return gave a larger weight to the Year 2 HPR, which was 10% versus the 22% HPR for Year 1.

If funds are contributed to an investment portfolio just before a period of relatively poor portfolio performance, the dollar-weighted rate of return will tend to be depressed. Conversely, if funds are contributed to a portfolio at a favorable time, the dollar-weighted rate of return will increase. The use of the time-weighted return removes these distortions, providing a better measure of a manager's ability to select investments over the period. If a private investor has complete control over money flows into and out of an account, the dollar-weighted rate of return may be the more appropriate performance measure.

Therefore, the dollar-weighted return will exceed the time-weighted return for a manager who has superior market timing ability.



MODULE QUIZ 90.1

Use the following information to answer Questions 1 and 2.

Assume you purchase a share of stock for \$50 at time $t = 0$ and another share at \$65 at time $t = 1$, and at the end of Year 1 and Year 2, the stock paid a \$2.00 dividend. Also at the end of Year 2, you sold both shares for \$70 each.

1. The dollar-weighted rate of return on the investment is:
 - A. 10.77%.
 - B. 15.45%.
 - C. 15.79%.
 - D. 18.02%.
2. The time-weighted rate of return on the investment is:
 - A. 18.04%.
 - B. 18.27%.
 - C. 20.13%.
 - D. 21.83%.

MODULE 90.2: RISK-ADJUSTED PERFORMANCE MEASURES

LO 90.b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio and identify the circumstances under which the use of each measure is most relevant.

LO 90.c: Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures.

Universe Comparisons

Portfolio rankings based merely on returns ignore differences in risk across portfolios. A popular alternative is to use a comparison universe. This approach classifies portfolios according to investment style (e.g., small-cap growth, small-cap value, large-cap growth, large-cap value) and, then, ranks portfolios based on rate of return within the appropriate style universe. The rankings are now more meaningful because they

have been standardized on the investment style of the funds. This method will fail, however, if risk differences remain across the funds within a given style.

The Sharpe Ratio

The **Sharpe ratio** uses standard deviation (total risk) as the relevant measure of risk. It shows the amount of excess return (over the risk-free rate) earned per unit of total risk. Hence, the Sharpe ratio evaluates the performance of the portfolio in terms of both overall return and diversification.

The Sharpe ratio is defined as:

$$S_A = \frac{\bar{R}_A - \bar{R}_F}{\sigma_A}$$

where:

$\bar{R}_A$ = average account return

$\bar{R}_F$ = average risk-free return

σ_A = standard deviation of account returns

The Treynor Measure

The **Treynor measure** is very similar to the Sharpe ratio except that it uses beta (systematic risk) as the measure of risk. It shows the excess return (over the risk-free rate) earned per unit of systematic risk.

The Treynor measure is defined as:

$$T_A = \frac{\bar{R}_A - \bar{R}_F}{\beta_A}$$

where:

$\bar{R}_A$ = average account return

$\bar{R}_F$ = average risk-free return

β_A = average beta



PROFESSOR'S NOTE

Ideally, the Treynor measure should be calculated using the actual beta for the portfolio over the measurement period. Since beta is subject to change due to varying covariance with the market, using the premeasurement period beta may not yield reliable results. The beta for the measurement period is estimated by regressing the portfolio's returns against the market returns.

For a well-diversified portfolio, the difference in risk measurement between the Sharpe ratio and the Treynor measure becomes irrelevant as the total risk and systematic risk will be very close. For a less than well-diversified portfolio, however, the difference in rankings based on the two measures is likely due to the amount of diversification in the portfolio. Used along with the Treynor measure, the Sharpe ratio provides additional information about the degree of diversification in a portfolio.

If a portfolio was not well-diversified over the measurement period, it may be ranked relatively higher using Treynor than using Sharpe because Treynor considers only the beta (i.e., systematic risk) of the portfolio over the period. When the Sharpe ratio is calculated for the portfolio, the excess total risk (standard deviation) due to diversifiable risk will cause rankings to be lower. Although we do not get an absolute measure of the lack of diversification, the change in the rankings shows the presence of unsystematic risk, and the greater the difference in rankings, the less diversified the portfolio.

Jensen's Alpha

Jensen's alpha, also known as Jensen's measure, is the difference between the actual return and the return required to compensate for systematic risk. To calculate the measure, we subtract the return calculated by the capital asset pricing model (CAPM) from the account return. Jensen's alpha is a direct measure of performance (i.e., it yields the performance measure without being compared to other portfolios).

$$\alpha_A = R_A - E(R_A)$$

where:

$$\alpha_A = \text{alpha}$$

$$R_A = \text{return on the account}$$

$$E(R_A) = R_F + \beta_A[E(R_M) - R_F]$$

A superior manager would have a statistically significant and positive alpha. Jensen's alpha uses the portfolio return, market return, and risk-free rate for each time period separately. The Sharpe and Treynor measures use only the average of portfolio return and risk-free rate. Furthermore, like the Treynor measure, Jensen's alpha only takes into account the systematic risk of the portfolio and, hence, gives no indication of the diversification in the portfolio.

Information Ratio

The Sharpe ratio can be changed to incorporate an appropriate benchmark instead of the risk-free rate. This form is known as the **information ratio** or **appraisal ratio**:

$$IR_A = \frac{\bar{R}_A - \bar{R}_B}{\sigma_{A-B}}$$

where:

$$\bar{R}_A = \text{average account return}$$

$$\bar{R}_B = \text{average benchmark return}$$

$$\sigma_{A-B} = \text{standard deviation of excess returns measured as the difference between account and benchmark returns}$$

The information ratio is the ratio of the surplus return (in a particular period) to its standard deviation. It indicates the amount of risk undertaken (denominator) to achieve a certain level of return above the benchmark (numerator). An active manager makes specific cognitive bets to achieve a positive surplus return. The variability in the surplus return is a measure of the risk taken to achieve the surplus. The ratio computes

the surplus return relative to the risk taken. A higher information ratio indicates better performance.



PROFESSOR'S NOTE

The version of the information ratio presented here is the most common. However, you should be aware that an alternative calculation of this ratio exists that uses alpha over the expected level of unsystematic risk over the time period, $\frac{\alpha_A}{\sigma(\varepsilon_A)}$.

M-Squared (M^2) Measure

A relatively new measure of portfolio performance developed by Leah Modigliani and her grandfather, 1985 Nobel Prize recipient Franco Modigliani, has become quite popular. The M^2 measure compares the return earned on the managed portfolio against the market return, after adjusting for differences in standard deviations between the two portfolios.

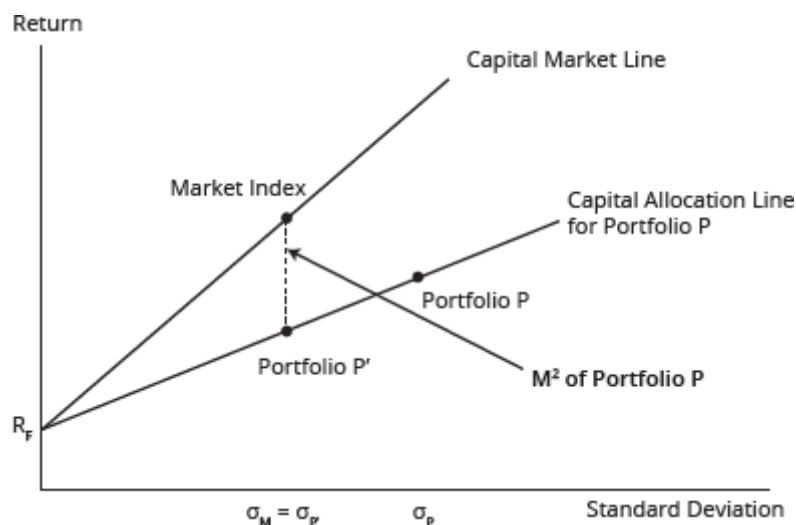


PROFESSOR'S NOTE

There are no squared terms in the M-squared calculation. The term “M-squared” merely refers to the last names of its originators (Leah and Franco Modigliani).

The M^2 measure can be illustrated with a graph comparing the capital market line for the market index and the capital allocation line for managed Portfolio P. In Figure 90.2, notice that Portfolio P has a higher standard deviation than the market index. But, we can easily create a Portfolio P' that has standard deviation equal to the market standard deviation by investing appropriate percentages in both the risk-free asset and Portfolio P. The difference in return between Portfolio P' and the market portfolio, equals the M^2 measure for Portfolio P.

Figure 90.2: The M^2 Measure of Portfolio Performance



EXAMPLE: Calculating the M^2 performance measure

Calculate the M^2 measure for Portfolio P:

■ Portfolio P mean return	10%
■ Portfolio P standard deviation	40%
■ Market portfolio mean return	12%
■ Market portfolio standard deviation	20%
■ Risk-free rate	4%

Answer:

To answer the question, first note that a portfolio, P' , can be created that allocates 50/50 to the risk-free asset and to Portfolio P such that the standard deviation of Portfolio P' equals the standard deviation of the market portfolio:

$$\sigma_{P'} = w_P \sigma_P = 0.50(0.40) = 0.20$$

Therefore, a 50/50 allocation between Portfolio P and the risk-free asset provides risk identical to the market portfolio. What is the difference in return between Portfolio P' and the market portfolio? To answer this question, first we must derive the mean return on Portfolio P' :

$$R_{P'} = w_F R_F + w_P R_P = 0.50(0.04) + 0.50(0.10) = 0.07$$

Alternatively, the mean return for Portfolio P' can be derived by using the equation of the capital allocation line for Portfolio P:

$$\begin{aligned} R_{P'} &= R_F + \left(\frac{R_P - R_F}{\sigma_P} \right) \sigma_{P'} = R_F + \left(\frac{R_P - R_F}{\sigma_P} \right) \sigma_M \\ &= 0.04 + \left(\frac{0.10 - 0.04}{0.40} \right) 0.20 = 0.04 + (0.15)0.20 = 0.07 \end{aligned}$$

Therefore, we now have created a portfolio, P' , that matches the risk of the market portfolio (standard deviation equals 20%). All that remains is to calculate the difference in returns between Portfolio P' and the market portfolio:

$$M^2 = R_{P'} - R_M = 0.07 - 0.12 = -0.05$$

Clearly, Portfolio P is a poorly performing portfolio. After controlling for risk, Portfolio P provides a return that is 5 percentage points below the market portfolio.



PROFESSOR'S NOTE

Unfortunately, a consistent definition of M^2 does not exist. Sometimes M^2 is defined as equal to the return on the risk-adjusted Portfolio P' rather than equal to the difference in returns between P' and M . However, portfolio rankings based on the return on P' or on the difference in returns between P' and M will be identical. Therefore, both definitions provide identical portfolio performance rankings.

M^2 will produce the same conclusions as the Sharpe ratio. As stated earlier, Jensen's alpha will produce the same conclusions as the Treynor measure. However, M^2 and Sharpe may not give the same conclusion as Jensen's alpha and Treynor. A discrepancy could occur if the manager takes on a large proportion of unsystematic risk relative to systematic risk. This would lower the Sharpe ratio but leave the Treynor measure unaffected.

EXAMPLE: Risk-adjusted performance appraisal measures

The data in Figure 90.3 has been collected to appraise the performance of four asset management firms:

Figure 90.3: Performance Appraisal Data

	Fund 1	Fund 2	Fund 3	Fund 4	Market Index
Return	6.45%	8.96%	9.44%	5.82%	6%
Beta	0.88	1.02	1.36	0.80	1.00
Standard deviation	2.74%	4.54%	3.72%	2.64%	2.80%
Standard deviation of excess returns	5.6%	6.1%	12.5%	5.3%	N/A

The market index return and risk-free rate of return for the relevant period were 6% and 3%, respectively. **Calculate** and rank the funds using Jensen's alpha, the Treynor measure, the Sharpe ratio, the information ratio, and M^2 .

Answer:

Evaluation Tool	Fund 1	Fund 2	Fund 3	Fund 4
Jensen's Alpha	$6.45 - 5.64 = 0.81\%$	$8.96 - 6.06 = 2.90\%$	$9.44 - 7.08 = 2.36\%$	$5.82 - 5.40 = 0.42\%$
Rank	3	1	2	4
Treynor	$\frac{6.45 - 3}{0.88} = 3.92$	$\frac{8.96 - 3}{1.02} = 5.84$	$\frac{9.44 - 3}{1.36} = 4.74$	$\frac{5.82 - 3}{0.80} = 3.53$
Rank	3	1	2	4
Sharpe	$\frac{6.45 - 3}{2.74} = 1.26$	$\frac{8.96 - 3}{4.54} = 1.31$	$\frac{9.44 - 3}{3.72} = 1.73$	$\frac{5.82 - 3}{2.64} = 1.07$
Rank	3	2	1	4
Information Ratio	$\frac{6.45 - 6}{5.6} = 0.08$	$\frac{8.96 - 6}{6.1} = 0.49$	$\frac{9.44 - 6}{12.5} = 0.28$	$\frac{5.82 - 6}{5.3} = -0.03$
Rank	3	1	2	4
M ²	$3 + (1.26) \times (2.8) = 6.53\% - 6\% = 0.53\%$	$3 + (1.31) \times (2.8) = 6.67\% - 6\% = 0.67\%$	$3 + (1.73) \times (2.8) = 7.84\% - 6\% = 1.84\%$	$3 + (1.07) \times (2.8) = 6\% - 6\% = 0$
Rank	3	2	1	4

Note that Jensen's alpha and the Treynor measures give the same rankings, and the Sharpe and M² measures give the same rankings. However, when comparing the alpha/Treynor rankings to the Sharpe/M² measures, Funds 2 and 3 trade places.

Fund 2 has a much higher total risk (standard deviation) than Fund 3 but has a much lower beta. Relatively speaking, a smaller proportion of Fund 2's total risk relates to systematic risk, which is reflected in the low beta. Compared to Fund 3, it must have a bigger proportion of risk relating to nonsystematic risk factors.

Hence, Fund 2 does better in the alpha/Treynor measures, as those measures only look at systematic risk (beta). It fares less well when it comes to the Sharpe/M² measures that look at total risk.

**MODULE QUIZ 90.2**

1. The following information is available for Funds ABC, RST, JKL, and XYZ:

Fund	Annual Rate of Return	Beta	Volatility
ABC	15%	1.25	20%
RST	18%	1.00	25%
JKL	25%	1.20	15%
XYZ	11%	1.36	9%

The average risk-free rate was 5%. Rank the funds from best to worst according to their Treynor measure.

- A. JKL, RST, ABC, XYZ.
 - B. JKL, RST, XYZ, ABC.
 - C. RST, JKL, ABC, XYZ.
 - D. XYZ, ABC, RST, JKL.
2. The following data has been collected to appraise Funds A, B, C, and D:

	Fund A	Fund B	Fund C	Fund D	Market Index
Return	8.25%	7.21%	9.44%	10.12%	8.60%
Beta	0.91	0.84	1.02	1.34	1.00
Standard deviation	3.24%	3.88%	3.66%	3.28%	3.55%

The risk-free rate of return for the relevant period was 4%. Calculate and rank the funds from best to worst using Jensen's alpha.

- A. B, D, A, C.
- B. A, C, D, B.
- C. C, A, D, B.
- D. C, D, A, B.

MODULE 90.3: ALPHA, HEDGE FUNDS, DYNAMIC RISK, MARKET TIMING, AND STYLE

Statistical Significance of Alpha Returns

LO 90.d: Determine the statistical significance of a performance measure using standard error and the t-statistic.

Alpha (α) plays a critical role in determining portfolio performance. A positive alpha produces an indication of superior performance; a negative alpha produces an indication of inferior performance; and zero alpha produces an indication of normal performance matching the benchmark. The performance indicated by alpha, however, could be a result of luck and not skill. In order to assess a manager's ability to generate alpha, we conduct a *t*-test under the following hypotheses.

Null (H_0): True alpha is zero.

Alternative (H_A): True alpha is not zero.

$$t = \frac{\alpha - 0}{\sigma / \sqrt{N}}$$

where:

α = alpha estimate

σ = alpha estimate volatility

N = sample number of observations

standard error of alpha estimate = $\sigma / \sqrt{N}$

In order to compute the ***t*-statistic**, we will need to know the alpha estimate, the sample number of observations, and the alpha estimate of volatility. From the volatility and sample size estimates, we can compute the **standard error** of the alpha estimate, which is shown in the denominator of the *t*-statistic calculation.

At a 95% confidence level (5% significance level) we reject the null hypothesis if we estimate a *t*-value of 2 or larger. That is, the probability of observing such a large estimated alpha by chance is only 5%, assuming returns are normally distributed.



PROFESSOR'S NOTE

Using a *t*-value of 2 is a general test of statistical significance. From the FRM Part I curriculum, we know that the actual *t*-value with a 95% confidence level given a large sample size is 1.96.

If we assume an excess (alpha) return of 0.09% and a standard error of the alpha of 0.093%, the *t*-statistic would be equal to 0.97 ($t = 0.09\% / 0.093\%$); therefore, we fail to reject H_0 and conclude that there is no evidence of superior (or inferior) performance.



PROFESSOR'S NOTE

Using statistical inference when evaluating performance is extremely challenging in practice. By the time you are reasonably confident that a manager's returns are in fact due to skill, the manager may have moved elsewhere.

Measuring Hedge Fund Performance

LO 90.f: Explain the difficulties in measuring the performance of actively managed portfolios.

Long-short hedge funds are often used to complement an investor's well-diversified portfolio. For example, the investor might allocate funds to a passively managed index fund and an actively managed long-short hedge fund. The hedge fund is designed to provide positive alpha with zero beta to the investor's overall composite portfolio. The hedge fund creates **portable alpha** in the sense that the alpha does not depend on the performance of the broad market and can be ported to any existing portfolio. Because the long-short fund is market-neutral, the alpha may be generated outside the investor's desired asset class mix.

Unfortunately, hedge fund performance evaluation is complicated because:

- Hedge fund risk is not constant over time (nonlinear risk).
- Hedge fund holdings are often illiquid (data smoothing).

- Hedge fund sensitivity with traditional markets increases in times of a market crisis and decreases in times of market strength.

The latter problem necessitates the use of estimated prices for hedge fund holdings. The values of the hedge funds, therefore, are not transactions-based. The estimation process unduly smooths the hedge fund “values,” inducing serial correlation into any statistical examination of the data.

Performance Evaluation With Dynamic Risk Levels

LO 90.g: Describe performance manipulation and the problems associated with using conventional performance measures.

The Sharpe ratio is useful when evaluating the portfolio performance of a passive investment strategy, where risk and return characteristics are relatively constant over time. However, the application of the Sharpe ratio is challenged when assessing the performance of active investment strategies, where risk and return characteristics are more dynamic. Changes in volatility will likely bias the Sharpe ratio, and produce incorrect conclusions when comparing portfolio performance to a benchmark or index.

Take for example a low-risk portfolio with an alpha return of 1% and a standard deviation of 3%. The manager implements this strategy for one-year, producing quarterly returns of -2%, 4%, -2%, and 4%. The Sharpe ratio for this portfolio is calculated as: $1\% / 3\% = 0.3333$. If the market index has a Sharpe ratio of 0.3, we would conclude that this portfolio has superior risk-adjusted performance. In the following year, the portfolio manager decides to switch to a high-risk strategy. The alpha return and risk correspondingly increase to 5% and 15%, respectively. For the second year, quarterly returns were -10%, 20%, -10%, and 20%. The Sharpe ratio in this case is still 0.3333 ($= 5\% / 15\%$), which still indicates superior performance compared to the market index. However, if the Sharpe ratio is evaluated over the two-year time frame, considering both the low-risk and high-risk strategies, the measure will drop to 0.2727 since average excess return over both years was 3% with volatility of 11%. The lower Sharpe ratio now suggests underperformance relative to the market index.

In this example, the Sharpe ratio was biased downward due to the perceived increase in risk in portfolio returns. In isolation, both the low-risk and high-risk strategies produced higher Sharpe ratios than the market index. However, when analyzed together, the Sharpe ratio suggests that the portfolio excess returns are inferior to the market. Therefore, it is important to consider changes in portfolio composition when using performance measures, as dynamic risk levels can lead to incorrect ranking conclusions.

Measuring Market Timing Ability

LO 90.h: Describe techniques to measure the market timing ability of fund managers with a regression and with a call option model and compute return

Measuring Market Timing With Regression

Extending basic return regression models offers a tool to assess superior market timing skills of a portfolio manager. A market timer will include high (low) beta stocks in her portfolio if she expects an up (down) market. If her forecasts are accurate, her portfolio will outperform the benchmark portfolio. Using a market timing regression model, we can empirically test whether there is evidence of superior market timing skills exhibited by the portfolio manager. The regression equation used for this test is as follows:

$$R_P - R_F = \alpha + \beta_P(R_M - R_F) + M_P(R_M - R_F)D + \varepsilon_P$$

In this equation, D is a dummy variable that is assigned a value of 0 for down markets (i.e., when $R_M < R_F$) and 1 for up markets (i.e., when $R_M > R_F$). M_P is the difference between the up market and down market betas and will be positive for a successful market timer. In a bear market, beta is simply equal to β_P . In a bull market, beta is equal to $\beta_P + M_P$. Empirical evidence of mutual fund return data suggests that M_P is actual negative for most funds. Thus, researchers have concluded that fund managers exhibit little, if any, ability to correctly time the market.

Measuring Market Timing With a Call Option Model

Consider an investor who has 100% perfect market forecasting ability and holds a portfolio allocated either 100% to Treasury bills or 100% to the S&P 500 equity market index, depending on the forecast performance of the S&P 500 versus the Treasury bill return. The investor's portfolio will be:

$$\begin{aligned} &100\% \text{ invested in the S\&P 500 if } E(R_M) > R_F \\ &100\% \text{ invested in Treasury bills if } E(R_M) < R_F \end{aligned}$$

If the investor has perfect forecasting ability, then his return performance will be as follows:

$$\begin{aligned} &R_M \text{ if } R_M > R_F \\ &R_F \text{ if } R_M < R_F \end{aligned}$$

Now consider an investor who invests S_0 (the current value of the S&P 500) in Treasury bills and also owns a call option on the S&P 500 with exercise price equal to the current value of the index times $(1 + R_F)$, or $S_0(1 + R_F)$. Note that the exercise price equals the value of the S&P 500 if it grows at a rate equal to the risk-free rate.

What are the return possibilities for this investor? To answer this question, note that if the S&P 500 holding period return exceeds the risk-free rate, then the ending value of the call option will be:

$$S_T - X = S_0(1 + R_M) - S_0(1 + R_F)$$

The investor also owns Treasury bills with face value equal to $S_0(1 + R_F)$. Therefore, the face value (FV) of the Treasury bills will perfectly offset the exercise price of the call

option. In the up-market scenario, the ending value of the calls plus bills portfolio equals:

$$S_T - X + FV = S_0(1 + R_M) - S_0(1 + R_F) + S_0(1 + R_F) = S_0(1 + R_M)$$

Therefore, the return performance on the calls plus bills portfolio will equal:

$$R_M \text{ if } R_M > R_F$$

If the market rises by less than the risk-free rate, the call option has no value, but the risk-free asset will still return R_F . Therefore, the down-market scenario return for the calls plus bills portfolio is:

$$R_F \text{ if } R_M < R_F$$

In summary, the returns to the calls plus bills portfolio are identical to the 100% perfect foresight returns. Therefore, the value or appropriate fee for perfect foresight should equal the price of the call option on the market index.

Style Analysis

LO 90.e: Describe style analysis.

LO 90.i: Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution.

William Sharpe introduced the concept of style analysis. From January 1985 to December 1989 he analyzed the returns on Fidelity's Magellan Fund for style and selection bets. His study concluded that 97.3% of the fund's returns were explained by style bets (asset allocation), and 2.7% were due to selection bets (individual security selection and market timing). The importance of long-run asset allocation has been well established empirically. These results suggest that the returns to market timing and security selection are minimal at best and at worst insufficient to cover the associated operating expenses and trading costs.

The steps for Sharpe's style analysis are as follows:

1. Run a regression of portfolio returns against an exhaustive and mutually exclusive set of asset class indices:

$$R_P = b_{P1}R_{B1} + b_{P2}R_{B2} + \dots + b_{Pn}R_{Bn} + e_P$$

where:

R_P = return on the managed portfolio

R_{Bj} = return on passive benchmark asset class j

b_{Pj} = sensitivity or exposure of Portfolio P return to passive asset class j return

e_P = random error term

In Sharpe's style analysis, the slopes/weights are constrained to be nonnegative and to sum to 100%. In that manner, the slopes can be interpreted to be "effective" allocations of the portfolio across the asset classes.

2. Conduct a performance attribution (return attributable to asset allocation and to selection):

- The percentage of the performance attributable to asset allocation = R^2 (the coefficient of determination).
- The percentage of the performance attributable to selection = $1 - R^2$.

The **asset allocation attribution** equals the difference in returns attributable to active asset allocation decisions of the portfolio manager:

$$\sum (\text{portfolio weight} - \text{benchmark weight}) \times R_B$$

Notice if the weights (estimated allocations) for the managed portfolio equal those within the benchmark (passive asset allocation), then the asset allocation attribution will be zero.

The **selection attribution** equals the difference in returns attributable to superior individual security selection (correct selection of mispriced securities) and sector allocation (correct over and underweighting of sectors within asset classes):

$$\sum \text{portfolio weight} \times (R_P - R_B)$$

Notice if the manager has no superior selection ability, then portfolio returns earned within each asset class will equal the benchmark asset class returns and the selection attribution will equal zero. Also, note that the combination of the two attribution components (asset allocation plus selection) equals the aggregate contribution.

$$\sum (\text{portfolio weight} \times R_P) - \sum (\text{benchmark weight} \times R_B)$$

3. Uncover the investment style of the portfolio manager: the regression slopes are used to infer the investment style of the manager. For example, assume the following results are derived:

$$R_P = 0.75R_{LCG} + 0.15R_{LCV} + 0.05R_{SCG} + 0.05R_{SCV}$$

where:

R_{LCG} = return on the large-cap growth index

R_{LCV} = return on the large-cap value index

R_{SCG} = return on the small-cap growth index

R_{SCV} = return on the small-cap value index

The regression results indicate that the manager is pursuing primarily a large-cap growth investment style.



MODULE QUIZ 90.3

1. Sharpe's style analysis, used to evaluate an active portfolio manager's performance, measures performance relative to:
 - A. a passive benchmark of the same style.
 - B. broad-based market indices.
 - C. the performance of an equity index fund.
 - D. an average of similar actively managed investment funds.

KEY CONCEPTS

LO 90.a

The dollar-weighted rate of return is defined as the internal rate of return (IRR) on a portfolio, taking into account all cash inflows and outflows. The beginning value of the account is an inflow as are all deposits into the account. All withdrawals from the account are outflows, as is the ending value.

Time-weighted rate of return measures compound growth. It is the rate at which \$1 compounds over a specified time horizon. Time-weighting is the process of averaging a set of values over time.

LO 90.b

The Sharpe ratio uses standard deviation (total risk) as the relevant measure of risk. It shows the amount of excess return (over the risk-free rate) earned per unit of total risk.

The Treynor measure is very similar to the Sharpe ratio except that it uses beta (systematic risk) as the measure of risk. It shows the excess return (over the risk-free rate) earned per unit of systematic risk.

Jensen's alpha is the difference between the actual return and the return required to compensate for systematic risk. To calculate the measure, we subtract the return calculated by the capital asset pricing model (CAPM) from the account return.

The information ratio is the ratio of the surplus return (in a particular period) to its standard deviation. It indicates the amount of risk undertaken to achieve a certain level of return above the benchmark.

LO 90.c

The M^2 measure compares the return earned on the managed portfolio against the market return, after adjusting for differences in standard deviations between the two portfolios.

LO 90.d

A positive alpha produces an indication of superior performance; a negative alpha produces an indication of inferior performance; and zero alpha produces an indication of normal performance matching the benchmark.

LO 90.e

William Sharpe introduced the concept of style analysis. From January 1985 to December 1989 he analyzed the returns on Fidelity's Magellan Fund for style and selection bets. His study concluded that 97.3% of the fund's returns were explained by style bets (asset allocation), and 2.7% were due to selection bets (individual security selection and market timing).

LO 90.f

Hedge fund performance evaluation is complicated because:

- Hedge fund risk is not constant over time (nonlinear risk).
- Hedge fund holdings are often illiquid (data smoothing).

- Hedge fund sensitivity with traditional markets increases in times of a market crisis and decreases in times of market strength.

LO 90.g

Changes in volatility will likely bias the Sharpe ratio, and produce incorrect conclusions when comparing portfolio performance to a benchmark or index.

LO 90.h

Extending basic return regression models offers a tool to assess superior market timing skills of a portfolio manager. A market timer will include high (low) beta stocks in her portfolio if she expects an up (down) market. If her forecasts are accurate, her portfolio will outperform the benchmark portfolio.

LO 90.i

The importance of long-run asset allocation has been well established empirically. Historical results suggest that the returns to market timing and security selection are minimal at best and at worst insufficient to cover the associated operating expenses and trading costs.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 90.1

1. **D** One way to do this problem is to set up the cash flows so that the PV of inflows = PV of outflows and then to plug in each of the multiple choices.

$$50 + 65 / (1 + \text{IRR}) = 2 / (1 + \text{IRR}) + 144 / (1 + \text{IRR})^2 \rightarrow \text{IRR} = 18.02\%$$

Alternatively, on your financial calculator, solve for IRR:

$$-50 - \frac{65 - 2}{1 + \text{IRR}} + \frac{2(70 + 2)}{(1 + \text{IRR})^2} = 0$$

Calculating Dollar-Weighted Return		
Key Strokes	Explanation	Display
[CF] [2nd] [CLR WORK]	Clear CF Memory Registers	CF0 = 0.00000
50 [+/-] [ENTER]	Initial cash inflow	CF0 = -50.00000
[↓] 63 [+/-][ENTER]	Period 1 cash inflow	C01 = -63.00000
[↓] [↓] 144 [ENTER]	Period 2 cash outflow	C02 = 144.00000
[IRR] [CPT]	Calculate IRR	IRR = 18.02210

(LO 90.a)

2. **D** $\text{HPR}_1 = (65 + 2) / 50 - 1 = 34\%$, $\text{HPR}_2 = (140 + 4) / 130 - 1 = 10.77\%$

$$\text{time-weighted return} = [(1.34)(1.1077)]^{0.5} - 1 = 21.83\%$$

(LO 90.a)

Module Quiz 90.2

1. A Treynor measures:

$$T_{ABC} = \frac{0.15 - 0.05}{1.25} = 0.08 = 8$$
$$T_{RST} = \frac{0.18 - 0.05}{1.00} = 0.13 = 13$$
$$T_{JKL} = \frac{0.25 - 0.05}{1.20} = 0.1667 = 16.7$$
$$T_{XYZ} = \frac{0.11 - 0.05}{1.36} = 0.0441 = 4.4$$

The following table summarizes the results:

Fund	Treynor Measure	Rank
ABC	8.00%	3
RST	13.00%	2
JKL	16.67%	1
XYZ	4.41%	4

(LO 90.b)

2. C CAPM returns:

$$R_A = 4 + 0.91(8.6 - 4) = 8.19\%$$
$$R_B = 4 + 0.84(8.6 - 4) = 7.86\%$$
$$R_C = 4 + 1.02(8.6 - 4) = 8.69\%$$
$$R_D = 4 + 1.34(8.6 - 4) = 10.16\%$$

	Fund A	Fund B	Fund C	Fund D
Alpha	8.25% – 8.19% = +0.06	7.21% – 7.86% = –0.65%	9.44% – 8.69% = +0.75%	10.12% – 10.16% = –0.04%
Ranking	2	4	1	3

(LO 90.b)

Module Quiz 90.3

1. A Sharpe's style analysis measures performance relative to a passive benchmark of the same style. (LO 90.h)

The following is a review of the Risk Management and Investment Management principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Constantinides, Harris, and Stulz, Chapter 17.

READING 91

HEDGE FUNDS

Study Session 14

EXAM FOCUS

The reading examines two decades of hedge fund performance. Significant events that shaped the hedge fund industry are discussed, including the growth of institutional investments. Different hedge fund strategies are explained, along with the continuing growth of assets under management. Performance is analyzed to see if the rewards justify the risks, and performance is compared with the broad equity markets. The performance of top fund managers is also compared to the performance across the hedge fund industry.

MODULE 91.1: HEDGE FUND INDUSTRY, ALPHA-BETA SEPARATION, AND HEDGE FUND STRATEGIES

Characteristics of Hedge Funds

There are important distinctions between hedge funds and mutual funds. Hedge funds are private, much less regulated investment vehicles, not available to the general public. On the other hand, mutual funds are more structured and regulated. Hedge funds are highly leveraged, and managers obtain profits from both long and short positions. Hedge fund managers tend to take large bets based on perceived relative price discrepancies of assets.

Privacy is a hallmark of hedge funds. There is little transparency in the hedge fund industry because managers do not want their methods copied. A hedge fund charges a fixed management fee plus a healthy share of new profits from the fund, generally around 10–20%.

Evolution of the Hedge Fund Industry

LO 91.a: Explain biases that are commonly found in databases of hedge funds.

LO 91.b: Explain the evolution of the hedge fund industry and describe landmark events that precipitated major changes in the development of the industry.

Historical data on hedge fund performance was difficult to obtain prior to the early 1990s. In early 1994, dramatic losses triggered by a Federal Reserve change in interest rate policy had a large impact on hedge fund performance reporting. This prompted the development of hedge fund databases so that participants could better obtain and analyze hedge fund performance.

Assets under management have increased 10 times from 1997 to 2010 as the number of funds has quadrupled. There are some hedge funds that do not participate in commercial databases, which impacts aggregate hedge fund performance. Thus, there is **selection bias**, also known as **self-reporting bias**, contained in hedge fund databases.

There is evidence that suggests that selection bias in large hedge fund databases is actually small. The average return of funds-of-hedge funds (FoFs), comprised of managers who theoretically invest across all hedge funds, not just funds reported to commercial databases, is highly correlated to the average return of hedge funds in commercial databases.

However, there are still concerns about possible measurement errors and various biases in reported hedge fund returns. The consensus is that hedge fund index returns became increasingly reliable beginning in 1996. Prior to 1996, looking at the period from 1987 to 1996, 27 large hedge funds substantially outperformed the S&P 500 index. The outperformance is high, which is more than enough to account for any measurement biases.

The collapse of Long-Term Capital Management (LTCM) in 1998 was a watershed event in the hedge fund industry. It was a reminder that higher returns are accompanied by higher risk. The LTCM collapse had a much greater effect on hedge fund performance compared to equity performance.

The time period of 2000 to 2001 brought the dot-com bubble collapse. During this period, the hedge fund industry experienced a 20% net asset inflow and there was a major shift in the hedge fund industry structure. Hedge funds outperformed the S&P 500 with half of the S&P 500 standard deviation. As a result, institutional investors poured money into hedge funds.

Impact of Institutional Investors

LO 91.c: Explain the impact of institutional investors on the hedge fund industry and assess reasons for the growing concentration of assets under management

(AUM) in the industry.

Beginning in 1999, institutional investor funds flowed into hedge funds. Assets under management in the hedge fund industry grew from \$197 billion at 1999 year-end to \$1.39 trillion by 2007 year-end. Thus, investors in hedge funds shifted from exclusively private wealth to institutions, including foundations, endowments, pension funds, and insurance companies.

Institutional investors were rewarded for allocating capital to a much higher fee environment. Three hedge fund performance databases, DJCSI, HFRI, and HFRFOFI, respectively, reported cumulative performance of 72.64%, 69.82%, and 38.18% from the 2002 to 2010 time period, compared to the S&P 500 index return of 13.5%. The S&P 500 index had a 16% standard deviation during that period, versus annualized standard deviations of return of 5.84%, 6.47%, and 5.51%, for the respective hedge fund indices.

With the increase of institutional investment came greater demands on hedge fund management for operational integrity and governance. Some institutional investors were seeking absolute performance, while others were seeking alternative sources of return beyond equities. There is some concern that there is no identifiable alpha associated with hedge fund investing, so it is increasingly important that hedge fund managers differentiate themselves from their peers.

Alpha-Beta Separation

LO 91.d: Explain the relationship between risk and alpha in hedge funds.

Alpha is a risk-adjusted measure of return often used to assess the performance of active managers. It is the return in excess of the compensation for risk. It is important to identify how much of a strategy's return results from risk (i.e., beta) and how much results from active management (i.e., alpha). This is known as **distinguishing alpha and beta**. A manager who uses statistical techniques, quantitative tools, and benchmarking to discern whether high returns are the result of the superior performance of an active manager or a function of bearing high levels of systematic risk is attempting to distinguish alpha from beta.

A hedge fund may attempt to independently manage alpha and beta. The firm may manage beta exposure while separately managing the portfolio's alpha. This is known as **separating alpha and beta**. Managers can use investment tools to pursue alpha while sustaining a target beta for the portfolio. Managers typically seek to limit beta while trying to optimize alpha. Derivatives are often used to minimize or eliminate undesired systematic risk.

For example, assume a manager's benchmark is the S&P 500. He would like to pursue opportunities that increase alpha, but the result is beta exposure different from the benchmark. He can use futures contracts to hedge all systematic risks other than exposure to the S&P 500 such that the portfolio's beta relative to the S&P 500 is 1.0. He does this while simultaneously pursuing an alpha optimizing strategy. In this way, he is independently managing, or separating, alpha from beta.

Hedge Fund Strategies

LO 91.e: Compare and contrast the different hedge fund strategies, describe their return characteristics, and describe the inherent risks of each strategy.

Managed Futures and Global Macro

Managed futures funds focus on investments in bond, equity, commodity futures, and currency markets around the world. Systematic trading programs are used which rely on historical pricing data and market trends. A high degree of leverage is employed because futures contracts are used. With managed futures, there is no net long or net short bias.

Many managed futures funds are market timing funds, which switch between stocks and Treasuries. When both short and long positions are considered, the payoff function of this strategy is similar to a lookback straddle, which is a combination of a lookback call option and a lookback put option. The lookback call option gives the owner the right to purchase the underlying instrument at the lower price during the call option's life, while the lookback put option gives the owner the right to sell the underlying instrument at the highest price during the put option's life.

Global macro fund managers make large bets on directional movements in interest rates, exchange rates, commodities, and stock indices. They are dynamic asset allocators, betting on various risk factors over time.

Both managed futures and global macro funds have *trend following* behavior (i.e., directional styles). Global macro funds do better during extreme moves in the currency markets. Both of these strategies are essentially *asset allocation* strategies, since the managers take opportunistic bets in different markets. They also both have a low return correlation to equities.

Merger/Risk Arbitrage and Distressed Securities

Merger (or risk) arbitrage strategies try to capture spreads in merger/acquisition transactions involving public companies, following public announcement of a transaction. The primary risk is **deal risk**, or the risk that the deal will fail to close.

Examining merger arbitrage returns, the largest negative monthly returns in this strategy are after the S&P 500 index has had a large negative return. This equates to being long deal risk. The logic is that when the market has a large decline, mergers have a greater tendency to be called off.

Distressed hedge funds is another event-driven hedge fund style. This strategy invests across the capital structure of firms that are under financial or operational distress, or are in the middle of bankruptcy. The strategy tends to have a long bias. With this strategy, hedge fund managers try to profit from an issuer's ability to improve its operation, or come out of a bankruptcy successfully.

A key feature of the strategy is long exposure to credit risk of corporations with low credit ratings. A good proxy for these types of returns is publicly traded high-yield

bonds since the correlation between the DJCS Distressed index and high-yield bonds is 0.55.

In sum, both of these event-driven strategies exhibit nonlinear return characteristics, since tail risk appears under extreme market conditions. With merger arbitrage, the tail risk is a large drop in equity investments. With distressed hedge funds, the tail risk is a big move in short-term rates. Unlike trend following strategies, event-driven funds are hurt by extreme market movements.

Fixed-Income Arbitrage

Fixed-income arbitrage funds attempt to obtain profits by exploiting inefficiencies and price anomalies between related fixed-income securities. The fund managers try to limit volatility by hedging exposure to interest rate risk. An example of this strategy is leveraging long/short positions in fixed-income securities that are related—mathematically or economically.

The sectors traded under fixed-income arbitrage include:

- Credit yield curve relative value trading of swaps, government securities, and futures.
- Volatility trading using options.
- Mortgage-backed securities arbitrage.

A **swap spread trade** is a bet that the fixed side of the spread will stay higher than the floating side of the spread, and stay in a reasonable range according to historical trends. With **yield-curve spread trades**, the hope is that bond prices will deviate from the overall yield curve only in the short term, and will revert to normal spreads over time. **Mortgage spread trades** are bets on prepayment rates, while **fixed-income volatility trades** are bets that the implied volatility of interest rate caps have a tendency to be higher than the realized volatility of, for example, a Eurodollar futures contract. **Capital structure** or **credit arbitrage trades** try to capitalize on mispricing among different types of securities (e.g., equity and debt).

Convertible Arbitrage

Convertible arbitrage funds attempt to profit from the purchase of convertible securities and the shorting of corresponding stock, taking advantage of a perceived pricing error made in the security's conversion factor. The number of shares shorted is based on a delta neutral or market neutral ratio. The plan is for the combined position to be insensitive to underlying stock price fluctuations under normal market conditions.

The return to convertible arbitrage hedge funds comes from the liquidity premium paid by issuers of convertible bonds to hedge fund managers, for holding convertible bonds and managing the inherent risk by hedging the equity part of the bonds.

Long/Short Equity

Long/short equity funds take both long and short positions in the equity markets, diversifying or hedging across sectors, regions, or market capitalizations. Examples are

shifts from value to growth, small- to mid-cap stocks, and net long to net short. Trades in equity futures and options can also take place.

The percentage of hedge funds that are long/short is 30%–40%. Long/short managers are stock pickers with varying opinions and abilities, so performance tends to be very idiosyncratic. Underpriced or under-researched stocks are favored, as are small stocks, on the long side. On the short side, low liquidity makes small stocks and foreign stocks less attractive. Long/short equity funds have directional exposure to the overall market and also have exposure to long small-cap/short large-cap positions.

Dedicated Short Bias

Funds with a dedicated short bias tend to take net short positions in equities. Sometimes the short position strategy is implemented by selling forward. To manage risk, managers take offsetting long positions and stop-loss positions. The returns are negatively correlated with equities.

Emerging Markets

Emerging market funds invest in currencies, debt, equities, and other instruments in countries with emerging or developing markets. These markets are usually identified in terms of gross national product (GNP) per capita. China, India, Latin America, Southeast Asia, parts of Eastern Europe, and parts of Africa are examples of emerging markets. These funds have a long bias because it is more difficult to short securities in emerging markets.

Equity Market Neutral

When reviewing equity market neutral hedge fund strategies, research shows that there is not one common component (or risk factor) in their returns. Different funds utilize different trading strategies, but they all have a similar goal of trying to achieve zero beta(s) against a broad set of equity indices.



MODULE QUIZ 91.1

1. What critical shift occurred in the hedge fund industry following the collapse of Long-Term Capital Management (LTCM) in 1998 and the dot-com bubble burst in 2001?
 - A. There was a significant drop in assets under management in the hedge fund industry.
 - B. There was a large influx of institutional investors investing in hedge funds.
 - C. Reporting within the hedge fund industry became more regulated than mutual funds.
 - D. There was a significant increase in hedge fund failures.
2. Which of the following hedge fund strategies would be characterized as an “asset allocation” strategy that performs best during extreme moves in the currency markets?
 - A. Global macro.
 - B. Risk arbitrage.
 - C. Dedicated short bias.

- D. Long/short equity.
3. Jamie Chen, FRM, is considering investing a client into distressed hedge funds. Which of the following investments would serve as the best proxy for the types of returns to expect?
- A. Convertible bonds.
 - B. Small-cap equities.
 - C. Managed futures.
 - D. High-yield bonds.

MODULE 91.2: HEDGE FUND PERFORMANCE, RISK FACTORS, AND RISK SHARING

Hedge Fund Performance

LO 91.f: Describe the historical portfolio construction and performance trends of hedge funds compared to those of equity indices.

Twenty-seven large hedge funds were identified in 2000, and research has been done to determine if these hedge funds are truly a separate asset class, not correlated to equity or bond indices. Hedge fund returns were regressed against an 8-factor model used to analyze hedge fund performance. Findings were that hedge fund portfolios had no significant exposure to stocks and bonds. As an equally weighted portfolio, this portfolio of 27 top performing hedge funds had a large alpha of 1.48% per month. There was a persistent exposure to emerging markets, but other factor betas showed a lot of variability. Also, alpha declined over time, and there was not a persistent directional exposure to the U.S. equity market. Measurement bias may have affected these results somewhat.

Alternatively, a strategy of investing in a portfolio of the top 50 large hedge funds was tested using data from 2002 to 2010. Two test portfolios were constructed:

- The first test portfolio attempted to mimic performance of a strategy of investing in the top funds in equal dollar amounts, and rebalancing at the end of each calendar year. The funds were selected based on the assets under management at year-end 2001.
- A similar portfolio was constructed using top funds based on year-end 2010, rather than 2001.

For the first portfolio, the intent was to give a lower and upper bound of performance which investors could achieve, by just following a strategy of investing equally in the top 50 large hedge funds, and rebalancing yearly. The second portfolio was “foresight assisted.”

In evaluating performance characteristics, the first portfolio did not have a significant alpha, while the foresight-assisted portfolio had a monthly alpha of 0.53%, and was statistically significant at the 1% level. Compared to hedge fund returns prior to 2002, the decline in alpha is consistent with the thinking that there is more competition in

the hedge fund industry. It should, however, be noted that there is no significant negative alpha.

Looking at the top 50 hedge funds versus all hedge funds, the top 50 portfolios (both versions) demonstrated statistically significant alpha relative to the DJCSI and HFRI hedge fund indices. The strategy of buying large hedge funds appears to deliver superior performance compared to just investing in hedge fund indices.

During the 2002 to 2010 time period, the top 50 hedge fund portfolios (with the exception of the foresight-assisted portfolio), and the two broad hedge fund indices, DJCSI and HFRI, all outperformed the equity market, as measured by the S&P 500 index. In sum, analysis of large hedge funds shows that managers are still delivering alpha return relative to peers, and also have low exposure to the U.S. equity market. These factors continue to attract institutional investors.

Convergences of Risk Factors

LO 91.g: Describe market events that resulted in a convergence of risk factors for different hedge fund strategies and explain the impact of such convergences on portfolio diversification strategies.

Theoretically, diversification among hedge fund strategies should protect investors, but there are certain events that affect all, or mostly all, strategies, as they all undergo stress at the same time. Portfolio diversification implodes, and seemingly diverse hedge fund portfolios *converge* in terms of risk factors during times of stress.

The first recorded major market event for hedge funds was in March and April of 1994 when unexpected changes in interest rate policy were set by the Federal Reserve. This caused two months of losses by seven of the ten style-specific sub-indices in the DJCS family. Exceptions were short sellers and managed futures funds. Merger arbitrage funds earned a positive return in March, while equity market neutral funds had a positive return in April.

Another major event was in August 1998 right before the collapse of LTCM. Eight of the ten niche DJCS style sub-indices had large losses. Short sellers and managed futures funds avoided losses. The losses occurred primarily due to market-wide liquidation of risky assets and the high amount of leverage on LTCM's balance sheet.

With hedge fund investing, leverage has a magnifying effect on gains and losses, and risk is on both sides of the balance sheet. There were events prior to the 2007–2009 financial crisis that illustrated how much a market-wide funding crisis can significantly impair leveraged positions. In August 2007, for the first time, all nine specialist style sub-indices lost money. The only positive return was from short sellers. During the peak of the financial crisis from July to October 2008, July to September brought losses for all hedge fund styles (excluding short sellers). When leveraged positions are forced to liquidate, losses can be high.

The point is that when there is a market-wide funding crisis, it is difficult to mitigate risk by simply spreading capital among different hedge fund strategies. There is

significant credit-driven tail risk in a hedge fund portfolio. The use of managed futures may be a partial solution—it has been a strategy with a convex performance profile relative to other hedge fund strategies. Hedge fund investors need to consider portfolio risks associated with dramatic market events.

Risk Sharing Asymmetry

LO 91.h: Describe the problem of risk sharing asymmetry between principals and agents in the hedge fund industry.

In the hedge fund industry, risk sharing asymmetry between the principal (investor) and the agent (fund manager) is a concern due to variable compensation schemes.

The problem occurs when the incentive fee that a hedge fund manager is entitled to, typically 15–20% of new profits [profits above a high water mark (HWM)], encourages a fund manager to take outsized risks. This tends to increase the future loss-carried-forward if and when these bets fail. If the fund fails, the same fund manager can start up a new hedge fund.

However, there is an opportunity cost involved in cases where a hedge fund manager closes a fund. It is costly in terms of harming the track record of the manager and affects reputation risk of both the manager and the fund company. All things considered, this cost does not totally mitigate the basic principal/agent conflict.

Investors may be best served to invest in funds for which the fund managers invest a good portion of their own wealth. As much as this issue has been discussed, the basic structure of how fund managers are compensated has not changed.



MODULE QUIZ 91.2

1. Comparing hedge fund performance during the time period 2002–2010 to earlier time periods, how would monthly alpha compare, if looking at large hedge funds?
 - A. Alpha was higher in the 2002–2010 time period.
 - B. Alpha remained constant over both time periods.
 - C. A “foresight-assisted” portfolio did not have a statistically significant alpha during the 2002–2010 time period.
 - D. There was a decline in alpha in the 2002–2010 time period.
2. What would be an ideal approach for a hedge fund investor who is concerned about the problem of risk sharing asymmetry between principals and agents within the hedge fund industry?
 - A. Focus on investing in funds for which the fund managers have a good portion of their own wealth invested.
 - B. Focus on diversifying among the various niche hedge fund strategies.
 - C. Focus on funds with improved operational efficiency and transparent corporate governance.
 - D. Focus on large funds from the “foresight-assisted” group.

KEY CONCEPTS

LO 91.a

Hedge funds are private investments and have very little financial regulation. They tend to be highly leveraged, and managers make large bets. On the other hand, mutual funds are regulated and more structured.

There are some hedge funds that do not participate in commercial databases, which impacts aggregate hedge fund performance. Thus, there is selection bias contained in hedge fund databases.

LO 91.b

There have been major events affecting the hedge fund industry, including large losses following a change in Fed policy in 1994, the LTCM collapse in 1998, and the dot-com collapse in 2001.

LO 91.c

Institutional investors flocked to hedge funds beginning in 1999. With the increase of institutional investment came greater demands on hedge fund management for operational integrity and governance.

LO 91.d

Alpha is the return in excess of the compensation for risk. Beta is a measure of the systematic risk of the security or portfolio relative to the market as a whole. Firms may independently manage alpha and beta. This is known as separating alpha and beta. Managers can use investment tools to pursue alpha while sustaining a target beta for the portfolio.

LO 91.e

Managed futures funds focus on investments in bond, equity, commodity futures, and currency markets around the world. The payoff function of this strategy is similar to a lookback straddle.

Global macro managers make large bets on directional movements in interest rates, exchange rates, commodities, and stock indices, and do better during extreme moves in the currency markets.

Merger arbitrage funds bet on spreads related to proposed merger and acquisition transactions, and perform poorly during major market declines.

Distressed hedge funds invest across the capital structure of firms that are under financial or operational distress, or are in the middle of bankruptcy. The strategy tends to have a long-bias. These hedge fund managers try to profit from an issuer's ability to improve its operation, or come out of a bankruptcy successfully.

Fixed-income arbitrage funds try to obtain profits by exploiting inefficiencies and price anomalies between related fixed-income securities. Their performance is correlated to changes in the convertible bond default spread.

Convertible arbitrage funds attempt to profit from the purchase of convertible securities and the shorting of corresponding stock.

Long/short equity funds take both long and short positions in the equity markets, diversifying or hedging across sectors, regions, or market capitalizations, and have directional exposure to the overall market and also have exposure to long small-cap/short large-cap positions.

Dedicated short bias funds tend to take net short positions in equities, and their returns are negatively correlated with equities.

Emerging market funds invest in currencies, debt, equities, and other instruments in countries with emerging or developing markets.

Equity market neutral funds attempt to achieve zero beta(s) against a broad set of equity indices.

LO 91.f

The top 50 hedge funds demonstrated statistically significant alpha relative to the DJCSI and HFRI hedge fund indices. The strategy of buying large hedge funds appears to deliver superior performance compared to just investing in hedge fund indices. Hedge fund managers are still delivering alpha relative to peers, and also have low exposure to the U.S. equity market.

LO 91.g

Diversification among hedge fund strategies may not always be effective due to the convergence of risk during times of extreme market stress. There is significant credit-driven tail risk in a hedge fund portfolio. The use of managed futures may be a partial solution—it has been a strategy with a convex performance profile relative to other hedge fund strategies. Hedge fund investors need to consider portfolio risks associated with dramatic market events.

LO 91.h

In the hedge fund industry, risk sharing asymmetry between the principal (investor) and the agent (fund manager) is a concern due to variable compensation schemes.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 91.1

1. **B** During the time period following the dot-com collapse, hedge funds outperformed the S&P 500 with a lower standard deviation, which attracted institutional investment. (LO 91.b)
2. **A** A global macro fund does better if there are extreme moves in the currency markets. Along with managed futures, global macro is an asset allocation strategy. Managers take opportunistic bets in different markets. The strategy has a low correlation to equities. (LO 91.e)
3. **D** Distressed hedge funds have long exposure to credit risk of corporations with low credit ratings. Publicly traded high-yield bonds are a good proxy for the returns to expect. (LO 91.e)

Module Quiz 91.2

1. **D** Comparing the two different time periods, there was a decline in alpha due to more competition in the hedge fund industry. (LO 91.f)
2. **A** The incentive fee structure within the hedge fund industry has not really changed over the years, and there is incentive for managers to take undue risks in order to earn fees. Thus, there should be a focus on investing in funds for which the fund managers have a good portion of their own wealth invested. (LO 91.h)

The following is a review of the Risk Management and Investment Management principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Mirabile, Chapter 12.

READING 92

PERFORMING DUE DILIGENCE ON SPECIFIC MANAGERS AND FUNDS

Study Session 14

EXAM FOCUS

This reading emphasizes the reasons investors should perform due diligence on potential investments. It provides a thorough list of items to consider in the due diligence process. For the exam, understand in detail the steps involved in evaluating a manager, a fund's risk management process, and a fund's operational environment.

MODULE 92.1: PAST FUND FAILURES, DUE DILIGENCE, AND EVALUATION

Past Fund Failures

LO 92.a: Identify reasons for the failures of hedge funds in the past.

Investors should be familiar with the reasons past funds have failed to ensure they can avoid investing in a failing fund. Following is a concise list of reasons past funds have failed.

1. **Poor investment decisions.** Could be a series of decisions (“domino effect”) or a very calculated risk on a specific investment that backfired.
2. **Fraud.** Fraud could occur in several forms including accounting (e.g., misstating asset book values or misstating income), valuation (e.g., misstating asset market values), and theft of funds.
3. **Extreme events.** Events occurring that would otherwise occur with very low probability or were unexpected (e.g., market crashes).
4. **Excess leverage.** Related to making poor investment decisions. Leverage goes both ways. That is, it magnifies gains but also magnifies losses.
5. **Lack of liquidity.** Too many capital withdrawals and redemptions to honor at once, thereby creating a squeeze on cash flow and an inability to meet all capital

withdrawals and redemptions.

6. **Poor controls.** Closely related to fraud. Lack of supervision could result in excessive risks being taken that lead to losses large enough to bankrupt the fund.
7. **Insufficient questioning.** Often in a committee-style decision-making process, there may be a dominant member who sways the decision and/or members who are afraid to voice any valid concerns over information they have discovered that would question the merits of the investment manager and/or investment. Ideally, all due diligence team members should be encouraged to play the role of “devil’s advocate” when appropriate and raise reasonable concerns as early as possible, especially before they reach the committee stage.
8. **Insufficient attention to returns.** Investment funds attempting to reduce operational risk sometimes overcompensate by implementing excessive controls and may end up bearing too many expenses and not generating enough returns. Ideally, there is a healthy balance between generating strong returns while taking on a reasonable level of risk.

Due Diligence Elements

LO 92.b: Explain elements of the due diligence process used to assess investment managers.

Prior to investing, an investor performs due diligence on a potential investment manager, which involves assessing the manager, the fund, and the investment strategy. Information such as the investment background, manager’s reputation (e.g., education, employers), and past performance have always been key considerations but are insufficient on their own.

An additional element of due diligence involves assessing the investment process and risk controls. The starting point is a review of the fund’s prospectus or offering memorandum. Additionally, an attribution analysis could be performed to determine how the returns were generated. Were they generated through the skill and control of the manager, luck, and/or factors beyond the manager’s control? In addition, was the amount of return in line with the amount of risk taken?

Another related element is assessing the fund’s operations and business model. In general, are there internal controls and policies in place to preserve the investors’ funds? Specifically, are the controls in place sufficiently robust to detect and prevent fraudulent activities or are limits imposed on managers to seek higher level approval for transactions exceeding a certain dollar amount or frequency? Is there appropriate segregation of duties between the front office and the back office? What is the process and frequency of asset valuations? What is the fee structure and are there any additional fees after a specific threshold? Are there any limitations or blackout periods on redemptions?

In the end, investors should assess potential managers and their investment strategies with an objective and unbiased mind. They should not get caught up with a manager’s past successes.

Manager Evaluation

LO 92.c: Identify themes and questions investors can consider when evaluating a hedge fund manager.

Manager evaluation is not a task that should be taken lightly by potential investors. This process can be broken down into four areas including strategy, ownership, track record, and investment management.

Strategy

General questions regarding a manager's strategy may include:

- Does the manager follow a particular investment style (e.g., growth, value)?
- Are there any current "trends" in the fund or specializations in specific securities, industries, or sectors?
- How has the fund changed its investment style or rebalanced its holdings over the past year? What changes are contemplated in light of anticipated market conditions?
- What is the extent of turnover and liquidity in the fund? What market signals are used to determine whether to exit or enter a position?
- What mechanisms are in place to limit any potential losses in the fund?
- To what extent is quantitative analysis and modeling utilized in the investment process? Have any models been developed or tested to date?
- Are short sales used to generate excess profits or to hedge? How successful or detrimental have they been so far?
- Are derivatives used in the portfolio? If so, are they used for hedging or speculative purposes?
- How does the trade execution process work? Does a central trading desk exist for maximum efficiency?
- What is the extent of any investment in private company securities and their role in the overall investment strategy?
- What is the tradeoff between maximizing current returns versus long-term fund growth?
- Has the fund ever been closed or provided investors with a return of capital?

Ownership

Ownership interests often help align the interests of the investment team and the investors. They can be useful in attracting and maintaining quality staff, thereby enhancing and/or continuing to generate strong investment returns for investors.

Therefore, potential investors should inquire as to whether any members of the investment team (e.g., traders, portfolio managers, research analysts) have ownership interests in the firm.

Track Record

Specific questions about the manager's and fund's track records may include:

- How does the past performance of the manager and/or fund compare to its peers and/or funds that follow the same or similar investment philosophy?
- Has past performance been audited or verified by a third party?
- Is there sufficient performance history to perform trend and/or attribution analysis? How did the manager or fund perform during market downturns?
- What were the investment returns relative to the size of the investment assets?
- Are most or all of the staff on the investment team that generated those past results still employed by the firm?

Investment Management

Inquiries during manager interviews may include:

- What is/was the manager's investment strategy for generating excess returns?
- How did the manager cope with tough market periods?

Reference checks on managers could include the following individuals:

- Former employers: Was the manager a leader or follower? Proactive or reactive? A team player or individualist?
- Current and former colleagues, clients, and other independent parties: Ensure consistency but if there are mixed reviews, follow up for explanations and/or obtain clarification from the manager.
- Current and former investors: What good and bad investment experiences did they have with the manager?

Background checks on managers may include the following questions/activities:

- Obtaining comprehensive background check reports on the manager.
- Review the Form ADV filed by the manager with the SEC and state securities authorities. It contains general information about the business as well as more detailed information such as fees, services provided, conflicts of interest, and background of key personnel.
- Has the manager consistently demonstrated herself to be a person of integrity? This could be verified by examining public databases and the SEC website to look for any past or current instances of litigation or criminal behavior.
- Has the manager demonstrated strong personal financial responsibility? This could be verified by examining personal credit reports and bankruptcy reports.
- Are the manager's stated representations accurate? This could be verified by inquiring with auditors and brokers who are currently working with the manager or have worked with the manager in the past.
- What is the extent of the manager's involvement in any related party transactions?

Risk Management Evaluation

LO 92.d: Describe criteria that can be evaluated in assessing a hedge fund's risk management process.

A proper risk management process should contain an assessment of the following areas: risk, security valuation, portfolio leverage and liquidity, tail risk exposure, risk reports, and consistency of the fund terms with the investment strategy.

Risk

- Assess the applicable systematic risk factors (i.e., regular market risks common to most or all funds) and unsystematic risk factors (i.e., risks specific to the manager, fund, or strategy).
- Determine whether written policies and procedures exist regarding measuring and monitoring risk.
- Determine whether a risk committee exists that would receive such measurements. If so, how often are they reported?
- Evaluate the extent of the risk management culture among the various types of employees. For example, how actively involved are employees with managing and mitigating the firm's risks on a day- to-day basis?
- Assess the information technology resources used to quantify the risks. For example, are they reliable and do they measure items consistently between traders and portfolio managers?
- Identify the existence and structure of any risk models. What are their inputs and assumptions? Have the models been tested and are they robust?

Security Valuation

- Identify the proportion of fund assets that are objectively valued through reliable market prices versus those that are more subjectively valued by the broker or through simulation.
- Examine the independence of valuations. Is valuation performed by the fund administrator (generally more independent) or by the fund manager (generally less independent)?
- Determine if prices may be overridden for valuation purposes. If so, by whom? Is there documentation or an approval process?

Portfolio Leverage and Liquidity

- Assess the sources of leverage as well as the current and historical levels of leverage.
- Calculate the current level of liquidity and observe how it has changed over time. The current level is especially relevant because of the impact on portfolio investment capacity and whether it can take on more investment capital.
- Within a stated investment strategy, excessive leverage and/or illiquidity could generate actual returns that are significantly different than expected (i.e., no longer comparing apples to apples), thereby requiring an adjustment in expected returns.

Exposure to Tail Risk

- Analyze information about the fund to conclude whether the fund's return distribution possesses skewness or kurtosis.
- Discuss the possibility of tail risk with the manager and determine whether the manager has sufficiently mitigated the risk or whether further action is required by the investor.

Risk Reports

- Review risk reports prior to investing in the fund. Investors should receive these risk reports on a regular basis (e.g., monthly, quarterly, annually) whether they are prepared in-house or by a third party.
- Analyze key risk metrics and compare them to other similar funds for benchmarking purposes and for determining if any unusual risks exist in the fund.

Consistency of the Fund Terms With the Investment Strategy

- Examine the general fee structure of the fund and determine whether it is consistent with similar funds.
- Identify the existence of any additional fees after a specific threshold (e.g., high-water mark, hurdle rate).
- Evaluate whether high fees are being paid to managers in search of market alpha (fair) as opposed to beta (unfair).
- Identify the existence of any limitations or blackout periods on redemptions.



MODULE QUIZ 92.1

1. Based on historical evidence, which of the following factors is least likely to result in the eventual failure of a hedge fund?
 - A. Excessive controls in place.
 - B. Taking on more systematic risk.
 - C. Making decisions in a committee setting.
 - D. Materially misstated financial statements.
2. In performing due diligence on a potential investment manager, which of the following factors is the least important for the investor to consider?
 - A. Risk controls.
 - B. Business model.
 - C. Past performance.
 - D. Investment process.
3. Which of the following statements regarding the assessment of a fund's risk management process is correct?
 - A. The periodic valuation of a fund's securities is best performed by the fund manager.
 - B. The existence of written policies and procedures for internal controls is useful in measuring and monitoring risk.
 - C. The risk reports received by investors are preferably prepared by a third-party risk provider instead of by the fund itself.

- D. The key requirement for information technology resources used to quantify the risks is that they measure items consistently.
4. Lisa Tahara, FRM, is considering an institutional investment in a hedge fund that has experienced volatile and generally positive returns in the past. Which of the following considerations about the fund's track record is least relevant for consideration in her investment decision?
- A. Size of investment assets.
 - B. Absolute level of past returns.
 - C. Verification of returns by a third party.
 - D. Employment continuity of the investment team.

MODULE 92.2: OPERATIONAL DUE DILIGENCE

LO 92.e: Explain how due diligence can be performed on a hedge fund's operational environment.

Investors should focus on several key areas when performing operational due diligence on a fund. The focus areas are internal control assessment, documents and disclosure, and service provider evaluation.

Internal Control Assessment

A starting point in due diligence is examining the qualifications and attitudes of the personnel. For instance, does the CEO believe in controls and compliance with the rules? An analyst must also assess whether the internal control staff have sufficient technical and work experience to perform their compliance duties properly. Have they been properly trained and do they continue to expand their skills in compliance? Some assurance may be required regarding whether the back and middle office managers are sufficiently experienced in performing supervisory duties. Finally, background checks on critical internal control staff members might be required.

Examining the fund's policies and procedures may also be useful. Related documents may cover areas such as trading, derivatives usage, and transaction processing. One drawback is that these documents tend to be general and only demonstrate the intention to have a strong control environment. In other words, merely reading the documents provides little assurance that the policies and procedures are actually being followed or are effective. It is usually a good sign if a fund has been proactive and obtained an audit report and opinion on the effectiveness of its controls. If this report is available, it should be reviewed.

The due diligence process should include an examination of the in-house or outsourced compliance system that is in place. Examples of specific items to consider include the code of ethics (if one exists) and any restrictions on employee trading and related-party transactions.

There should be an investigation into how the funds deal with counterparty risk arising from OTC derivatives and other counterparties. Is such risk mitigated by dealing with more than one counterparty? Are the counterparties monitored for risk on a daily basis?

Finally, there should be an assessment as to the effectiveness of corporate governance. Is it pervasive throughout the organization? Are examples of internal control “breaches” followed up with appropriate actions to remedy and prevent future recurrence?

Documents and Disclosure

As part of the due diligence process, investors must confirm with the fund’s legal counsel its involvement in preparing the original version of the fund documents as well as any subsequent revisions. The investor should also confirm if the law firm remains as the fund’s legal counsel. A physical check of the documents should be made to look for any changes made after the date indicated on the documents.

The investor should corroborate the terms of the offering memorandum by examining other documents such as the Form ADV, subscription agreement, and investment management agreement. Consistency is important here. Terms relating to fees, redemption rights, liquidity, and lockups should be examined closely and clarified with the manager if required.

Conflicts of interest that are disclosed in the offering memorandum should be scrutinized carefully. Lack of clarity in the disclosure may be a red flag and warrant further discussion with the manager and/or require independent information.

Similarly, lack of clarity or sufficiency in the disclosure of risks may warrant further investigation. The discussion of very general or irrelevant risk factors may be cause for concern.

The focus of any due diligence should be on the manager. As a starting point, the potential investor should determine the extent of the manager’s authority. Are the provisions very broad (potentially more risky) or quite specific? Is the manager subject to limitations on the amount of leverage employed or on the percentage of the fund invested in specific securities, sectors, or industries? Can the manager be indemnified for his actions outside of fraud, gross negligence, or malicious intent? Additionally, there should be a consideration of the manager’s reporting duties to investors (e.g., audited financial statements, disclosure of the tax treatment of the fund’s income and transactions).

In analyzing the financial statements, the investor should begin by ensuring the audit opinion is unqualified (i.e., the auditor believes the financial statements contain no material misstatements). The balance sheet and income statement should be examined for consistency with the fund’s investment strategy (e.g., a high leverage fund should have high interest expense on the income statement and high liabilities on the balance sheet). Any inconsistencies should be discussed with the manager on a timely basis. In addition, the footnotes (which are also audited) should be examined carefully since they provide more detailed information on key items (e.g., contingent liabilities, related-party transactions) than the corresponding financial statements.

Fees paid to the manager by the fund should be scrutinized and recalculated. They should be corroborated with the offering memorandum. Specifically, there should be a check of any incentive fees paid in loss years.

Finally, there should be a check for the level of net contributions to the fund by the general partner. Any fund withdrawals should be questioned.

Service Provider Evaluation

Third-party service providers may be hired by a fund for trade execution, information technology, valuation, verification, and asset safeguarding purposes.

A starting point for assessing the actual service providers is to examine the internal control letters issued by its auditors and its audited financial statements. Further due diligence could be performed through in-person discussions regarding the service provider's role.

LO 92.f: Explain how a hedge fund's business model risk and its fraud risk can be assessed.

In addition to the previous due diligence, potential investors need to closely examine the fund to ensure that the risks associated with its business model and potential fraud are not excessive.

Business Model Risk

Evaluating business model risk requires assessing whether managers know how to operate the business as well as generate high returns. Typical risks, potentially leading to failure and closure of the fund, include a lack of cash and working capital, a lack of a succession plan, and excessive redemptions in a short period of time.

A fund's business model risk can be assessed by performing the following tasks:

- Examining the nature of the revenues and expenses. For example, are revenue items stable, recurring, or one-time? Can costs be reduced or are they increasing uncontrollably?
- Calculating the percentage of revenues derived from variable incentive or performance fees (that may not materialize in market downturns).
- Assessing the significance of the gap between management fees (revenue) and operating expenses.
- Considering the sufficiency of the amount of working capital (especially cash) in place to cover revenue shortfalls and/or expense overages for a reasonable period of time.
- Determining how frequently budgets are created and for what period of time.
- Determining the fund's breakeven points in terms of assets under management and required performance level. Comparing those amounts to current (actual) and future (projected) amounts.
- Ascertaining if there is sufficient personnel or capacity to increase the fund's investment asset base.
- Ascertaining the existence of key person insurance on relevant individuals and the existence of a succession plan.

Fraud Risk

Fraud risk can always exist even though extensive due diligence has been performed on the manager and fund prior to investing. A fund's fraud risk can be assessed by determining the existence of the following factors:

- Frequent related-party transactions, including trading through a broker or using a valuator who is a related party.
- Frequent instances of illiquidity, including significant concentrations of illiquid investments (especially those that are valued by the manager only).
- Frequent litigation as a defendant, especially regarding claims of fraud.
- Unreasonably high (stated) investment returns.
- Frequent personal trading by the manager of the same or similar securities as those held by the fund.
- Frequent shorting transactions.

Fraud risk may be mitigated by performing the following actions:

- Check the SEC website for any prior regulatory infractions.
- Check court records for any prior litigation and bankruptcy records for examples of financial irresponsibility.
- Inquire with service providers for assurance over their competence and independence from the manager.
- Perform extensive background checks on the manager.

Due Diligence Questionnaire

LO 92.g: Describe elements that can be included as part of a due diligence questionnaire.

Properly designed due diligence questionnaires that are thoroughly and honestly answered by respondents can yield valuable information to a potential investor and may provide a list of concerns that need further assessment. The questionnaire should make the following inquiries:

1. Inquiry into general information on the manager provides a starting point in the due diligence process. Examples of such information include:
 - Confirmation of proper registration with regulatory authorities.
 - Determination of ownership form (e.g., corporation) and structure.
 - Identification of key shareholders.
 - Reference checks.
 - Information on past performance.
 - Business contact information.
2. Inquiry into general information on the fund also is critical. Examples of general information that should be collected include:

- Fees.
 - Lockup periods.
 - Redemption policies.
 - Primary broker.
 - Fund director.
 - Administrator.
 - Compliance: auditor and legal advisor.
 - Financial: assets under administration, investment capacity, and historical performance (also see financial statements).
 - Historical drawdown levels.
3. Inquiry into execution and trading as well as service providers may provide some insight on the speed and accuracy of transaction processing and the existence of related-party service providers, the latter of which may raise red flags with potential investors as discussed earlier.
 4. Inquiry regarding the firm's third-party research policy may be useful to determine a fund's sources of research information, thereby allowing the assessment of the extent and quality of the due diligence performed by the fund in its investment process.
 5. Inquiry regarding compliance processes, the existence and degree of involvement of in-house legal counsel, and the existence of anti-money laundering policy and procedures may help provide comfort that the fund and its managers have a desire to operate in an ethical manner and/or within the boundaries of the law.
 6. Inquiry into the existence of information regarding disaster recovery and business continuity plans as well as insurance coverage and key person provisions may provide some assurance regarding the stability of the firm and, therefore, the safety of any invested funds.
 7. Inquiry into the investment process and portfolio construction provides the potential investor with information required to make an informed decision whether the overall risk and return profile of the fund is consistent with the investor's investment objectives.
 8. Inquiry into risk controls such as leverage, liquidity, asset concentrations, portfolio diversification, and market risk factors give the investor a more complete picture of the investment risks and how the managers attempt to manage and mitigate them.

The existence of financial statements, especially if audited with an unqualified opinion, provide objective and historical financial information on the fund that can be used to assess performance. Information on the composition of the invested assets may also be helpful to the potential investor. Finally, interim statements (not necessarily audited) may provide more timely information to make a more current assessment of the fund by the potential investor.



MODULE QUIZ 92.2

1. Which of the following items is least likely to be included as requested information on a due diligence questionnaire?
 - A. Insurance coverage.
 - B. Returns attribution analysis.
 - C. Disaster recovery procedures.
 - D. Anti-money laundering policy.

KEY CONCEPTS

LO 92.a

Past fund failures can be attributed to poor investment decisions, fraud, extreme events, excess leverage, lack of liquidity, poor controls, insufficient questioning, and insufficient attention to returns.

LO 92.b

The due diligence process for assessing investment managers should include information on the investment background and reputation of the managers and past performance. In addition, there should be an assessment of the fund's investment process, risk controls, operations, and business model.

LO 92.c

In evaluating a manager, investors should consider four broad themes including strategy (e.g., evolution, risk management, quantification, types of investments), ownership, track record (e.g., comparison with peers, independent verification of results), and investment management (e.g., manager interviews, reference checks, background checks).

LO 92.d

Criteria that could be used in assessing a fund's risk management process includes risk (e.g., types, culture, quantification/models), security valuation, portfolio leverage and liquidity, tail risk exposure, risk reports, and consistency of the fund terms with the investment strategy.

LO 92.e

Performing due diligence on a fund's operating environment focuses on:

- Internal control assessment (i.e., qualifications and attitude of personnel, written policies and procedures, compliance system, counterparty risk, effectiveness of governance).
- Documents and disclosure (i.e., confirmations with the fund's legal counsel regarding fund documents, corroborating terms of the offering memorandum, conflicts of interest, disclosure of risks, manager's authority, manager's reporting duties to investors, financial statements, and fees paid to the manager, net contributions/withdrawals by the general partner).
- Service provider evaluation.

LO 92.f

Business model risk can be assessed by considering revenues and expenses (detailed examination), sufficiency of working capital, existence of budgets, computation of breakeven points, ability to increase investment asset base, existence of key person insurance, and existence of a succession plan.

Fraud risk can be assessed by considering the existence of related-party transactions, illiquidity, litigation, unreasonably high (stated) investment returns, personal trading by the manager of the same or similar securities as those held by the fund, and shorting transactions.

LO 92.g

Items to include as part of the due diligence questionnaire include general information on the manager and the fund, execution and trading, service providers, third-party research policy, compliance processes, existence and degree of involvement of in-house legal counsel, existence of anti-money laundering policy and procedures, existence of information regarding disaster recovery and business continuity plans, insurance coverage, key person provisions, details of the investment process and portfolio construction, risk controls, and information contained in the fund's financial statements.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 92.1

1. **B** If a fund takes on more systematic risk (i.e., regular market risk), it is less likely to result in a failure unless there is a significant market downturn. Taking on more unsystematic risk, however, is more likely to result in a failure. Excessive controls to reduce operational risk may be a good idea but may also result in excessive expenses and insufficient returns, thereby leading to a possible failure of the fund.

In a committee-style decision-making process, there may be a dominant member who sways the decision and/or members who are afraid to voice any valid concerns. Materially misstated financial statements are a form of accounting fraud, which significantly increases the risk of the eventual failure of a fund. (LO 92.a)
2. **C** Investors should assess potential managers and their investment strategies with an objective and unbiased mind. They should not be unduly concerned with a manager's past successes given that past performance is not always indicative of future performance. Risk controls, the business model, and the investment process are all fundamental parts of the due diligence process. (LO 92.b)
3. **D** It is very important for the information technology resources used to quantify risks to measure items consistently. Securities valuation is an important and potentially subjective task, therefore, independence and objectivity is critical. Policies and procedures tend to be general and only demonstrate the intention to have a strong control environment. Their existence alone provides little

assurance that they are properly measuring and monitoring risk. In general, the reporting of risk measures is a more objective task and as a result, there is little or no preference for the reporting to be done internally or externally. (LO 92.d)

4. **B** The absolute level of past returns is least relevant here given the volatile returns in the past. Also, past returns are not an assurance of similar returns in the future. The relative level of returns is more important than the absolute level. Verification of returns by a third party provides assurance that the return calculations were computed fairly and accurately by the fund. It is relevant to ascertain whether most or all of the staff on the investment team that generated the past results are still currently employed by the fund. It provides some (but not absolute) assurance that similar returns may be generated in the future. (LO 92.c)

Module Quiz 92.2

1. **B** A returns attribution analysis could be performed to determine how a fund's returns were generated. Return attributions are not generally part of a due diligence questionnaire but such an analysis could subsequently be performed based on some of the information received from the questionnaire. The other items (insurance coverage, disaster recovery procedures, and anti-money laundering policy) are all standard items that would be found in most, if not all, due diligence questionnaires. (LO 92.g)

The following is a review of the Risk Management and Investment Management principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Dimmock and Gerken.

READING 93

PREDICTING FRAUD BY INVESTMENT MANAGERS

Study Session 14

EXAM FOCUS

Investors should utilize investment adviser disclosures to avoid fraud. For the exam, know that investment firms that have conflicts of interests, have an association with their broker-dealer, are registered under the Investment Company Act (ICA) of 1940, have smaller investors, have clients who are agents, and have past violations are more likely to experience fraud in the future. Also, understand the problems that prevent fraud prediction methods from being implemented sooner.

MODULE 93.1: PREDICTING FRAUD BY INVESTMENT MANAGERS

Using Disclosures to Predict Fraud

LO 93.a: Explain the use and efficacy of information disclosures made by investment advisors in predicting fraud.

It is important that investors be able to predict fraud for two reasons. First, in many cases, fraud results in a total loss for the investor. When fraud is disclosed, an investment firm has a much greater probability of losing investor capital and going bankrupt.

Second, in the U.S., there are no requirements for investment advisor experience or accreditation. There are also no laws against investment advisors' conflicts of interests, but they simply must be disclosed. Therefore, it is up to investors to examine the required disclosures that investment advisors must make. The main federal law protecting investors is the Investment Advisers Act of 1940, which requires advisors to file **Form ADV**. This form requires disclosure of past regulatory and legal violations and possible conflicts of interest. Investors should use the information disclosed to predict and avoid fraud.

Predictive Variables From Form ADV

In the U.S., investment advisors who manage more than \$25 million are required to file (with the Securities and Exchange Commission [SEC]) Form ADV annually or whenever there is a material change.

In addition to past fraudulent activity, there are other items reported on Form ADV of which potential investors should be aware.

- *Conflicts of interest the firm may have with its investors*: including whether the firm recommends securities to investors in which the investment manager has an ownership interest, transacts with clients, or receives referral fees. The existence of conflicts of interest suggests a higher likelihood of fraud.
- *Soft dollar arrangements*: whether the investment firm receives benefits from the broker who executes client trades. The concern here is that the firm may do business with a more costly broker to benefit the firm at the expense of the investor. However, despite the potential conflict of interest, disclosed soft dollar arrangements do not predict fraud.
- *The broker-dealer*: whether the firm is associated with the broker-dealer. The concern here is that the firm may front run its investors or overcharge them. Without the use of an unrelated broker, the investors lose an external monitor of the firm. If the firm is associated with the broker-dealer, fraud is much more likely to occur.
- *The custodian*: whether the firm serves as the custodian for client cash or securities. If the firm serves as the custodian, third-party oversight is absent and there is greater risk of the firm expropriating investor wealth, even with an audit requirement (e.g., at least one unannounced annual asset verification visit). However, a firm serving as a custodian does not support greater likelihood of fraud.
- *Registration under the Investment Company Act (ICA) of 1940*: despite the fact that registered firms have greater regulatory requirements, they may also have vulnerable clients who could easily be defrauded. Therefore, although counterintuitive, registration under the ICA is a predictor for fraud from theft (*but it does suggest a reduction in the risk of fraudulent misrepresentation—see following discussion*).
- *Chief compliance officer (CCO)*: whether the firm has a dedicated CCO who does not hold other positions. Having a CCO does not significantly reduce fraud from theft (*but it does suggest a reduction in the risk of fraudulent misrepresentation—see following discussion*).
- *Majority employee owned*: it may be argued that when the majority of the firm is employee owned, there is more risk of fraud because externally-owned firms would likely be monitoring employees and that is lost when the firm is internally-owned. However, there is not significant evidence to suggest greater likelihood of fraud with majority employee-owned firms.
- *Investor size*: the smaller (larger) the firm's investors, the more (less) likely fraud will occur.
- *Clients who are agents*: an example would be a pension fund manager who is the client but not a direct beneficiary of the funds. Firms that have more clients who are agents are more likely to engage in fraud.

- *Firms that manage hedge funds*: the opaque nature of hedge funds would suggest a greater likelihood of fraud. However, there is no significant evidence to support that conclusion, although the opaque nature may reduce fraud detection and understate the actual fraud.

In summary, investors could use much of the information currently reported on Form ADV to reasonably predict fraud in the future. An informal and rough estimate of Form ADV compliance costs of \$500 million is small compared with the fraud losses of over \$4 billion experienced by investors in the 5% of firms identified in advance as having the greatest likelihood of experiencing fraud.

Predicting Different Types of Fraud

Two general categories of fraud include theft and fraudulent misrepresentation.

Using Form ADV disclosures, firms with any past regulatory violations, firms that pay referral fees, firms with mainly smaller clients, and firms with mainly clients who are agents would have a high probability of committing future fraud in the form of **theft** (e.g., Ponzi schemes, self-dealing, misappropriation, overstating asset values).

Using Form ADV disclosures, past regulatory violations are a strong predictor of future **fraudulent misrepresentation**. The fact that violations occur would suggest underlying issues with unethical behavior and internal control weaknesses. The additional attention given to firms with past violations may result in the discovery of misrepresentations. At the same time, firms that are registered under the ICA of 1940 are less likely to suffer from fraudulent misrepresentation, perhaps due to the audit requirements. The same goes for firms with an appointed CCO, perhaps due to greater internal monitoring.

Firm-wide fraud perpetrated by senior management is more challenging to predict than fraud perpetrated by a rogue employee. In that regard, firms with noted weaknesses in internal controls are more likely to experience rogue employee fraud.

Costs of Fraud Prediction

LO 93.b: Describe the barriers and the costs incurred in implementing fraud prediction methods.

Implementing fraud prediction methods is a cost-benefit decision—although zero fraud is often desired, it is likely the case that the costs involved with 100% eradication of fraud would exceed the benefits. In the past, cost considerations likely kept the emphasis on disclosure, but in the post-Bernie Madoff environment, more attention has been given to securities law enforcement.

It is also possible that some investors are willing to assume more fraud risk in exchange for lower fees or higher returns. In that case, having strong fraud prediction methods may lower the amount of fraud but also hurt those investors who were willing to take on fraud risk. Those investors would not be able to earn a fraud risk premium.

The costs of obtaining relevant data and estimating fraud risk may be lower than the benefits for investors as a whole. However, the costs may be too high for a single investor to bear, and even if the single investor were to bear it, the other investors who benefit would not likely contribute to the costs (i.e., free-rider problem).

Finally, the benefits of identifying investments with fraud risk are limited because of short sale restrictions. In those instances, the investor only accrues an indirect benefit because the investor is limited to only avoiding such investments.

LO 93.c: Discuss ways to improve investors' ability to use disclosed data to predict fraud.

Although the public could always access a firm's current Form ADV, historical filings were not always made available. Previously, the SEC disclosed only the most recent Form ADV so investors were unaware that fraudulent activity (one-time or repeated) had occurred at the firm in the past. To improve transparency, the SEC began providing historical filings for potential investors.

However, the format in which the SEC provided the historical information was not in a user-friendly format, in that each Form ADV had to be downloaded manually in an encoded format and then the data had to be extracted. As a result, it was impractical for many investors to do so. The SEC has since started to provide the information in a standardized format that would allow most investors to access the relevant information in the Form ADV for due diligence purposes.



MODULE QUIZ 93.1

1. In the U.S., investment advisors are required to file Form ADV annually or whenever there is a material change when they manage at least how much in assets?
 - A. \$10 million.
 - B. \$25 million.
 - C. \$50 million.
 - D. \$100 million.
2. Which of the following factors would lead to an increased likelihood of fraud perpetrated by an investment firm?
 - A. Clients who are agents.
 - B. The firm serving as a custodian.
 - C. Existence of soft dollar arrangements.
 - D. Firms that are majority owned by employees.
3. Which of the following factors would lead to a decreased likelihood of fraud perpetrated by an investment firm?
 - A. Firms with larger clients.
 - B. Firms that manage hedge funds.
 - C. Firms with an appointed chief compliance officer (CCO).
 - D. Firms that are registered under the Investment Company Act of 1940.
4. The free-rider problem suggests that the costs of obtaining relevant data and estimating fraud risk may be:
 - A. higher than the benefits for investors as a whole and for single investors.

- B. higher than the benefits for investors as a whole and lower than the benefits for single investors.
 - C. lower than the benefits for investors as a whole and for single investors.
 - D. lower than the benefits for investors as a whole and higher than the benefits for single investors.
5. Which Form ADV and format is now made available for investors by the SEC?
- A. Current Form ADV in encoded format.
 - B. Current Form ADV in standardized format.
 - C. Current and historical Form ADV in encoded format.
 - D. Current and historical Form ADV in standardized format.

KEY CONCEPTS

LO 93.a

In addition to past fraudulent activity, items on Form ADV of which potential investors should be aware include: conflicts of interest the firm may have with investors, soft dollar arrangements, associations with broker/dealers, firms serving as custodians, registration under the ICA of 1940, the existence of a CCO, firms that are majority employee owned, investor size, firms with clients who are agents, and firms that manage hedge funds.

LO 93.b

In the past, cost considerations pertaining to fraud likely kept the emphasis on disclosure, but in the post-Bernie Madoff environment, more attention has been given to securities law enforcement. Strong fraud prediction methods would prevent some investors from earning a fraud risk premium. Fraud detection methods also suffer from a free-rider problem. The benefits of identifying investments with fraud risk are limited because of short sale restrictions.

LO 93.c

Providing historical and current Form ADVs in a standardized format has improved investors' ability to use disclosed data to predict fraud.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 93.1

- 1. **B** In the U.S., investment advisors who manage more than \$25 million are required to file Form ADV annually or whenever there is a material change. (LO 93.a)
- 2. **A** Firms that have more clients who are agents (e.g., pension fund manager who is the client but not a direct beneficiary of the funds) are more likely to engage in fraud. (LO 93.a)
- 3. **A** The smaller (larger) the firm's investors, the more (less) likely fraud will occur. (LO 93.a)

4. **D** The costs of obtaining relevant data and estimating fraud risk may be lower than the benefits for investors as a whole. However, the costs may be too high for a single investor to bear, and even if the single investor were to bear it, the other investors who benefit would not likely contribute to the costs (i.e., free-rider problem). (LO 93.b)
5. **D** To improve transparency, the SEC has now made the current and historical Form ADVs available to investors in a user-friendly (standardized) format. (LO 93.c)

The following is a review of the Current Issues in Financial Markets principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Board of Governors of the Federal Reserve System.

READING 94

REVIEW OF THE FEDERAL RESERVE'S SUPERVISION AND REGULATION OF SILICON VALLEY BANK

Study Session 15

EXAM FOCUS

This topic focuses on the events leading up to the failure of Silicon Valley Bank (SVB) and the regulatory shortcomings observed. For the exam, understand the history behind how SVB rose from its founding in 1983 to disaster in 2023. From the perspective of supervision, focus on what happened between 2018 and 2021 that created a poor transition in regulatory burdens for SVB. There were many events that led up to the ultimate collapse of SVB. Be familiar with the bank's deposit concentration, the type of deposits, the realities with their held-to-maturity (HTM) and available-for-sale (AFS) portfolios, their contingent funding efforts, and attempts to raise capital. With respect to governance and risk management, understand the three listed areas of deficiency, what happened with the firm's chief risk officer (CRO), and why supervisors were too lax in their dealings with SVB. For their liquidity position, focus on the bank's noted deficiencies, their 30-day liquidity buffer, the ways in which management began to address liquidity concerns, and the lack of timely regulatory response. Finally, be aware of management's use of the net interest income (NII) model and the economic value of equity (EVE) model.

MODULE 94.1: SVB BACKGROUND AND RISK EXPOSURES

Evolution of Silicon Valley Bank

LO 94.a: Describe the events leading up to the failure of Silicon Valley Bank.

Headquartered in Santa Clara, CA, Silicon Valley Bank Financial Group (SVBFG) was founded in 1983. They were a financial services firm with primary operations in the United States. Their largest subsidiary was **Silicon Valley Bank (SVB)**, which was a California state-chartered bank under the supervision of the Federal Reserve. Their clients were primarily U.S.-based emerging growth or mature firms in the technology and life sciences sectors. SVB's strategy involved attracting early-stage (or start-up) firms as clients and then retaining them as they grew through the industry lifecycle. Nearly half of all venture capitalists in the technology and life sciences sectors were clients of SVB.

From 1983 to 2019, SVB grew deposits gradually from \$18 million to \$58 billion. These deposits tripled to \$209 billion by 2021. This rapid growth was fueled by clients who enjoyed liquidity events like IPOs, secondary offerings, SPACs, acquisitions, and other fundraising activities. From year-end 2018 to year-end 2021, SVB grew assets by 271% as compared to the banking sector's average of 29%. This outsized deposit growth was due to them targeting start-up firms and venture capitalists as clients. These depositors put so much of their money at SVB that, as of 2022, roughly 94% of all SVB deposits were not insured by the Federal Deposit Insurance Corporation (FDIC).

SVB's positive net deposit growth lulled them into a mindset that this trend would continue. However, in the second half of 2022, venture capital activity fell sharply due to declining investor risk appetites resulting from interest rate increases. This decline in VC activity caused SVB's once stellar deposit growth rate to become negative (i.e., a net outflow of deposits). SVB customers were making withdrawal requests to fund their own business operations.

This sparked a significant issue because SVB had been investing deposits in long-term, HTM government securities or agency-issued mortgage-backed securities (MBSs). They were counting on their deposit growth from 2019 to 2021 to continue, and they were trying to boost short-term returns. It is important to understand that these long-dated MBSs were classified as HTM because SVB fully intended on not selling them until the fixed income securities matured. As of December 31, 2022, most of their HTM portfolio had a maturity of 10 years or longer and a weighted average duration of 6.2 years. HTM assets can be carried on a bank's books at their historical cost, not at the current market value (i.e., mark-to-market). According to accounting conventions, if a portion of the HTM portfolio were sold to cover customer withdrawal requests, then the entire HTM portfolio would be reclassified as available-for-sale (AFS) and marked-to-market at a loss. Management did not want to do this. It should also be noted that SVB's ratio of securities owned (i.e., AFS and HTM portfolios) to total assets was more than double that of the average large banking organization.

In 2023, SVB's deposit outflows accelerated as their customer's need for cash increased. Depositor patience was further reduced when the *Financial Times* ran an article discussing SVB's exceptionally high securities-to-assets ratio. On March 8, SVB announced a sale of \$21 billion from its AFS portfolio, which realized an after-tax loss of \$1.8 billion. On the same day, they disclosed plans for a \$2.25 billion equity offering to buy them some time for their HTM portfolio to recover, and both Moody's and S&P revealed a potential rating downgrade for SVB. Between social media and the reality that most of SVB's client base were personally connected because they were in the same industry, depositor withdrawal requests jumped to \$40 billion on March 9 alone. Management knew that they could not keep up with this pace of deposit outflows. In the evening of March 9, they communicated to banking supervisors that they expected outflow requests of \$100 billion on March 10 and they did not have the capital to meet that level of demand. On the morning of March 10, the California Department of Financial Protection and Innovation (CDFPI) closed SVB and delegated the FDIC as the receiver.

Supervisory Oversight

LO 94.b: Describe shortfalls and deficiencies in the Federal Reserve's supervisory oversight of Silicon Valley Bank during the period that the bank transitioned from the Fed's Regional Banking Organization (RBO) portfolio to its Large and Foreign Banking Organization (LFBO) portfolio.

Banks are classified under one of two categories based on the size of their assets. A **regional banking organization (RBO)** has assets between \$10 billion and \$100 billion, while a **large and foreign banking organization (LFBO)** either has assets exceeding \$100 billion or it is a foreign bank operating in the United States. Both RBOs and LFBOs are supervised by regional Reserve Banks with some level of oversight by the Board of Governors of the Federal Reserve System.

Supervision of RBOs involves a plan to address any identified concerns. The plan should specifically include financial and managerial strengths along with emerging risks. Supervisors then use the *Bank Exams Tailored to Risk (BETR) program* to shift examination hours from low-risk banks to high-risk banks. Risk reviews are periodically scheduled to consider unique risks associated with a given bank. Continuous monitoring involves regular meetings with senior bank leadership, review of key internal management reports, leveraging control functions (e.g., internal loan reviews, internal audits), and coordination with other regulators.

Supervision of LFBOs involves more depth and breadth. Horizontal exams are used to compare one bank against its peers relative to capital, liquidity, and cybersecurity. On an annual basis, the Federal Reserve rates these banks according to the *large financial institution (LFI) rating system*, which considers capital planning, liquidity and risk management, and governance controls. Continuous monitoring for LFBOs requires a current understanding of the bank's risk management practices. It is more in depth than the continuous monitoring used for RBOs because it is intended to function as an

early warning system before risk factors escalate. Through this more stringent process, larger banks are subject to *enhanced prudential standards (EPS)*.

The Division of Supervision and Regulation (DSR) reports to the Board of Governors and performs a pivotal role in banking sector risk detection. The DSR is required to provide an annual assessment letter to Reserve Bank presidents detailing the status of all banks, in aggregate, within the regional Reserve Bank's purview. The available ratings associated with this letter are "strong," "effective," "marginally effective," and "requires improvement." SVB's relevant regulator was the Federal Reserve Bank of San Francisco (FRBSF). Figure 94.1 shows the ratings for the aggregate portfolio of banks under the watch of FRBSF between 2019 and 2022.

Figure 94.1: Ratings Issued to FRBSF by the DSR

Year	RBO Supervisory Program	LFBO Supervisory Program
2019	Effective	Effective
2020	Strong	Effective
2021	Strong	Effective
2022	Effective	Strong

In July 2018, the Board of Governors raised the threshold for classification as LFBO from \$50 billion in assets to \$100 billion. Had the threshold been left at \$50 billion, then SVB would have shifted into LFBO status in 2018. This shift delayed the supervision change for SVB to February 2021. The transition was not managed well. Complicating matters was an examination pause during the COVID-19 pandemic. SVB staff described the shift to LFBO as a cliff event. One day they were treated as RBO and the next day it was LFBO without a gradual upshift. It would have been far better to have begun the transition as the \$100 billion threshold was approaching. Regulators were hesitant to pull forward more stringent EPS requirements until they were certain that SVB would achieve LFBO status. In February 2021, the RBO supervision cycle was well underway. It did not conclude until July 2021. Regulators agreed to allow the RBO supervision cycle to conclude and then the LFBO team would begin their work. SVB had to very quickly acquire the skills, processes, and teams to meet the EPS demands of the LFBO supervisors.

Some changes were also occurring with the regulatory system. From 2016 to 2022, banking sector assets grew by 37% while the headcount of regulatory supervisors declined by 3%. In 2021, a culture change began at the DSR. The vice chair for supervision looked to reduce regulatory burdens on RBO banks, meet a higher burden of proof on supervisory conclusions, and demonstrate due process related to recommended actions. The result was a delay in actions from supervisory staff and a reluctance to escalate potential concerns. When SVB was labeled as RBO, regulators allocated few resources to the bank, but this changed rapidly when they shifted to LFBO status.

When SVB was moved into the LFBO portfolio, regulatory examinations covered key EPS areas like risk management, liquidity, and governance. Before 2021, risk management was rated as "Satisfactory-2" but after the shift to the LFBO standards,

they were downgrade to “Less than Satisfactory-3.” Liquidity followed a similar story. Before 2021, it was rated as “Strong-1” but after the switch it was downgraded to “Satisfactory-2.” Governance was first checked in the August 2022 review (which began in late 2021), and it was rated as “Deficient-1.” When a deficiency is noted, a *memorandum of understanding (MOU)* is issued to denote an informal enforcement action. This process began in August 2022, but there are too many stakeholders that need to review the MOU before it becomes official. It needed to go through the FRBSF, the DSR, The Board of Governor’s legal department, and the California Department of Financial Protection and Innovation. By the time the MOU made it through this process, SVB had already failed.

SVB Risk Exposures

LO 94.c: Identify Silicon Valley Bank’s specific risk issues which led to and accelerated its failure including deposit concentration, type of deposits, held-to-maturity securities, available-for-sale securities, the bank’s contingent funding plan and capacity, and its capital raising efforts.

There were three critical weaknesses noted in SVB’s failure. They involved governance and risk management, interest rate risk as it related to their portfolio, and liquidity risk management. The multiplier was the banks’ rapid growth. SVB’s board of directors did not have the expertise to lead a large bank with over \$200 billion in assets. They were focused on short-term measures of profitability without fully understanding their longer-term risk exposures. They were also slow to address the rapid changes in risk management. Supervision from the Federal Reserve failed to keep pace with the rate of change SVB was experiencing. Specific risks for SVB could be grouped into the following categories:

- *Deposit concentration.* SVB’s depositor base was highly concentrated in start-ups, venture capitalists, and private equity firms in the technology and life science sectors. Due to industry concentration, they all experienced business issues at the same time and demand for withdrawals (i.e., deposit outflows at SVB) accelerated all at once.
- *Type of deposits.* Deposits were large and 94% of deposits were uninsured (i.e., above the FDIC coverage limit). Initially, the deposits were noninterest bearing, but in 2022 they switched and became interest bearing. This changed the duration of the bank’s liabilities and caused liquidity stress testing results to worsen.
- *Held-to-maturity (HTM) securities.* The HTM portfolio was largely invested in long-dated MBSs. Most of these securities had a maturity of 10 years or longer and a weighted average duration of 6.2 years. Management removed hedges against rising rates in order to focus on short-term profitability. When rates continued to rise, the losses in the HTM portfolio became substantial. HTM assets are carried on the books at historical cost, which kept the true losses out of the public eye. At one point, accounting rules dictated that selling a portion of HTM assets to cover capital needs would require the entire HTM portfolio to be reclassified as AFS. This would mean an instant shift from historical cost accounting to mark-to-market, which would

recognize substantial losses and spook investors and depositors. By the end of 2022, this rule was relaxed, and management was considering a one-time transfer of funds out of the HTM portfolio, which now would not have required a complete reclassification as AFS.

- *Available-for-sale (AFS) securities.* Assets classified as AFS are marked-to-market, which means that the balance sheet will reflect a timelier perspective of the value of the bank's liabilities. In late 2022, management considered selling some AFS assets to cover deposit demands. The losses would have been sizable because the bank had invested in long-dated fixed income securities before the interest rate increase cycle.
- *Contingent funding and capacity plans.* SVB management did not have sufficient plans in place to access capital from contingent sources. They needed to expand repo funding and bilateral relationships. They began to address this in the fourth quarter of 2022, but by that time it was too late.
- *Capital raising.* SVB management attempted to offer promotions to retain deposits, but this did not work. They considered selling HTM assets, but the losses were substantial and could not fully support the pace of withdrawal requests.



MODULE QUIZ 94.1

1. Silicon Valley Bank (SVB) was shut down in 2023 because:
 - A. the Federal Reserve raised interest rates too high.
 - B. foreign depositors withdrew all of their money at the same time.
 - C. they did not align their assets with potential customer withdrawal needs.
 - D. accounting conventions did not allow SVB to properly adjust to meet depositor withdrawal requests.
2. From the perspective of regulation, what was the issue that caused a delay in SVB receiving elevated regulatory scrutiny?
 - A. SVB hid assets from regulators.
 - B. The regulatory culture became more intense.
 - C. The asset threshold change for being considered LFBO.
 - D. Held-to-maturity assets were no longer involved in regulatory reviews.
3. Which of the following items was a factor that contributed to the failure of SVB?
 - A. Short duration on HTM securities.
 - B. Liquidity downgrades from regulators.
 - C. Concentration of depositors in two industries.
 - D. Realized losses from transferring assets from HTM to AFS.

MODULE 94.2: SVB MANAGEMENT DEFICIENCIES AND FAILURES

SVB Governance and Risk Management

LO 94.d: Identify and describe the failures and shortfalls of Silicon Valley Bank in the areas of governance and risk management including those related to the CRO

position and the bank's internal audit function.

Corporate governance is the system of rules, processes, and procedures that enables a firm to navigate economic and operational uncertainty. The end goal of effective governance is strong risk management. SVB's governance and risk management functions were lacking when they reached \$50 billion in assets, and they were deficient at the end of an unexpected growth cycle resulting in over \$200 billion in assets.

When SVB was below the \$100 billion asset mark (i.e., still in RBO status), their CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity to market risk) rating was listed as "Satisfactory-2." This happened in 2018, 2019, and 2020. In the review for the 2020 supervisory cycle (dated May 3, 2021), they were listed as "Satisfactory-2" despite several noted issues. Supervisors saw issues with credit risk management, internal loan reviews, and technology weaknesses. They also noted that management was reactive to issues identified by supervisors rather than proactively looking for risk factors independently. Yet, the supervisors still maintained a rating of effective (i.e., "Satisfactory-2") because they were focused on the bank's strong financial performance up to this point.

In late 2021, the LFBO supervisory team had their first look at SVB's governance and risk management structure. Meetings with management confirmed that they were not equipped to handle the rapid growth that SVB was enjoying. In conjunction with the DSR office, the LFBO supervisors waived a governance rating for 2021 to allow them time to gather more information (i.e., due diligence) to support their rating.

In a report issued on May 31, 2022, the supervisors itemized the following three *matters requiring immediate attention (MRIAs)*:

1. *Board effectiveness.* The board, management, and the CRO did not adequately shift from RBO standards to the more rigorous EPS requirements associated with larger banks (i.e., LFBOs). They hired consultants to help in 2020, but that still did not accomplish the goal. For these reasons, the supervisors found that the board (and management) did not have the necessary experience to run a large financial institution.
2. *Risk management programs.* The risk management system was found to be ineffective. Some specific deficiencies included a lack of a formal risk appetite standard, no mechanism for tracking or reporting root causes of risks, a lack of enterprise-wide policies and procedures, no enterprise-wide internal control framework, an incentive structure whose rewards were not linked to risk management outcomes, and an underdeveloped talent acquisition and performance review process.
3. *Internal audit effectiveness.* The internal audit (IA) function did not have a methodology that was sufficient to challenge management. They did not provide the audit committee with timely and sufficient reporting, and they did not ensure timely analysis of critical risk management functions.

The CRO recognized that she did not have the skillset needed to navigate SVB through the growing pains of becoming a large bank. This individual left the firm in April 2022.

Supervisors did not raise any concerns that the bank did not have a CRO because they were actively searching for one. This follows the pattern in which the supervisors were not citing issues either because the bank was highly profitable or because they were working toward a solution. While the CRO search was ongoing, risk management was supervised by a team of internal senior risk managers who were all very new to their roles. A new CRO was finally hired in December 2022.

SVB Liquidity Risk Management Deficiencies

LO 94.e: Identify the scope of Silicon Valley Bank's liquidity risk management deficiencies and shortfalls, including its modeling and stress testing of its 30-day liquidity buffer, as well as the actions that management and regulators considered to address these specific liquidity issues.

Liquidity is a financial institution's ability to meet its cash and collateral obligations at a reasonable cost. Liquidity risk is a natural part of the banking relationship because they use depositor funds, which are short-term in nature, to lend over longer periods of time. This so-called *maturity transformation* is an expected part of the business.

Regulation focuses on strong liquidity management practices including board oversight, liquidity risk tolerance parameters, internal liquidity stress testing, and contingency funding plans. SVB's liquidity risk management practices were fundamentally flawed and not suited for a firm of their size. This was partly due to their rapid growth from 2019 to 2021. Regulatory supervisors did not adequately identify the changes in SVB's risk profile.

When SVB was still in the RBO portfolio (i.e., assets less than \$100 billion), supervisors issued consistently positive reviews of their liquidity risk management practices. Their balance sheet was growing and heavily skewed toward large and uninsured deposits from VC-backed and private equity clients. Liquidity risk appeared to be minimal given the high growth in depositor assets and a large portfolio of low-credit risk securities in the HTM and AFS portfolios. The low perception of risk led the supervisors to use the RBO "risk-focused guidelines" to conduct very light reviews of SVB's liquidity risk, which was clearly underestimated. In low-risk scenarios, a single supervisory staff member would often cover multiple risk centers (e.g., liquidity risk, interest rate risk, etc.).

A much more thorough review occurred when SVB was transitioned to the LFBO portfolio. Supervisors found weaknesses across all areas reviewed. They cited the following six specific issues:

1. SVB needed to develop a plan to address shortfalls in liquidity stress testing and contingency funding.
2. The internal audit function needed a redesign to provide sufficient risk management oversight.
3. Liquidity stress testing needed to use more realistic assumptions and reflect immediate stress testing rather than the expected evolution over time.

4. Stress testing with respect to deposits did not incorporate realistic risks.
5. Liquidity risk limits and the associated processes were not keeping pace with the increase in size that SVB was enjoying.
6. SVB did not adequately assess potential funding sources.

The result was a liquidity rating of “Conditionally Meets Expectations.” Supervisors were again relying on the cash reserves and securities held on SVB’s balance sheet to create a buffer for weak risk controls. The cited six weaknesses were likely more in line with a deficient rating rather than the one provided.

A timeline of the awareness of the deterioration in SVB’s liquidity was as follows:

- **2022 Q2:** SVB’s concentrated client base began withdrawing deposits to meet their business needs. The bank began offering promotional deals to retain deposits with limited effectiveness.
- **2022 Q3:** In response to the LFBO review’s six identified issues, SVB began to update their liquidity stress testing process. As of August 21, 2022, they reviewed their 30-day liquidity buffer and modeled it to be a shortfall of \$18 billion. Additionally, they modeled a 90-day liquidity shortfall of \$23 billion. They incorrectly treated these as an “operational shortfall” when they should have been taken more seriously. Combining their internal findings of deficiency in their 30-day liquidity buffer and their 90-day liquidity buffer, these modeled shortfalls represented a material safety-and-soundness concern. Despite the negative modeled 30-day liquidity buffer, supervisors did not materially change their risk ratings.
- **2022 Q4:** Management began to recognize the problems. The board of directors was informed of the details in their November 2022 strategy meeting. They still continued to consider the modeled 30-day liquidity shortfall as “operational” rather than as a violation of EPS standards, which it was. Management made several attempts to address the liquidity issues including increasing Federal Home Loan Bank advances, increasing their repurchase agreement capacity, and changing the assumptions used in the 30-day liquidity shortfall modeling. This last option decreased the shortfall by \$5 billion, but assumptions should never be relaxed to produce a more favorable outcome. This does not change reality. Supervisors were aware of the change in liquidity stress testing assumptions, but they planned to delay their comments until the 2023 review cycle.
- **2023:** Supervisors conducted a horizontal review of SVB relative to peer banks. Their preliminary finding was that SVB was not meeting expectations. One serious concern was SVB’s insufficiently supported deposit outflow speed assumption. Management shifted their assumption to outflows not occurring until Days 31–90. SVB’s management made key assumptions (e.g., lending commitments and intraday assumptions) less conservative. It was also noted that management needed to make serious improvements to expand capacity through repo funding and bilateral relationships.

Supervisors did not adequately rate SVB’s risks or keep pace with the rate of deterioration in their processes and models. They primarily focused comments on SVB’s liquidity risk management practices rather than on the liquidity positions

themselves. Supervisors had limited interaction with SVB during the beginning of 2023. This changed on March 8 and March 9 when the music stopped for the bank.

SVB Interest Rate Risk Management Deficiencies

LO 94.f: Describe the deficiencies in Silicon Valley Bank's interest rate risk management process, including its modelling process, and explain how proper use of metrics such as net interest income (NII) at risk and economic value of equity (EVE) could have improved its management of interest rate risk.

Interest rate risk is inherent in the banking industry, but banks need to have adequate risk management practices to control for associated risks. There are two key metrics used to model interest rate risk and they both use deposit assumptions to model expected cash flow needs of the bank. These metrics are as follows:

- *Net interest income at risk.* **Net interest income (NII)** measures potential volatility over short periods of time. It tests the effect of large basis point moves in the yield curve and potentially incorporates yield curve twists. The goal is to quantify potential short-term cash flow impacts. This involves *deposit betas*, which are the assumed pass-through of market movement to deposit accounts.
- *Economic value of equity.* The **economic value of equity (EVE)** is the present value of assets minus the present value of liabilities. It measures the potential structural mismatch on the bank's balance sheet with the goal of estimating the longer-term impact of interest rate risk on the bank's securities.

SVB's risk appetite statement (RAS) only included NII and ignored EVE. They estimated that NII would increase during a rising rate environment because they failed to consider the impact of rising rates on their customer base (i.e., they assumed low deposit betas). Only a limited number of interest rate changes were tested. They did not conduct backtesting and only had limited sensitivity testing.

Management focused almost exclusive on the NII measurement. When they saw the yield curve invert, they anticipated a recession and falling, not rising, rates. They began to position their balance sheet for a falling rate cycle, which never arrived. Beginning in March 2022 and finalizing in July 2022, management removed their hedges against rising rates because they would have hurt profits if rates fell. This action lengthened the duration of the bank's securities, which harmed the EVE metric. At the same time, the bank's deposits were shifting from noninterest bearing to interest bearing. This altered the duration of the bank's liabilities (i.e., the deposits) and made the EVE even worse.

While EVE was excluded from the RAS, it was included in the review of the Risk Committee on the board of directors. Still, the full board was not aware of the deterioration the EVE would have shown. The Asset/Liability Committee (ALCO) only considered parallel shifts in the yield curve while ignoring nonparallel shifts that would have indicated trouble brewing. Additionally, management altered the assumed withdrawal rate for deposits and implemented other unsubstantiated assumption changes, which made the EVE outcomes look more favorable. This did not alter economic reality. It only made things appear better than they really were.

From 2018 to 2022, the supervisor's CAMELS rating for market risk was "Satisfactory-2." The 2022 CAMELS review cycle was vetted in November 2022. The planned new market risk rating was downgraded to "Less than Satisfactory-3." This review was not officially finalized until after SVB had already failed.



MODULE QUIZ 94.2

1. The governance and risk management function at SVB was:
 - A. supervised by a very qualified board of directors.
 - B. not regulated properly due to political connections.
 - C. supervised by a very qualified chief risk officer (CRO).
 - D. not able to keep up with the rate of growth the bank was enjoying.
2. SVB management attempted to solve their modeled 30-day liquidity buffer shortfall by:
 - A. selling AFS holdings at a loss.
 - B. selling HTM holdings at a loss.
 - C. borrowing money through the discount window.
 - D. changing assumptions to make them more favorable.
3. What action did SVB's management take that made their economic value of equity (EVE) model results much worse?
 - A. They shortened the duration of deposits.
 - B. They removed hedges against falling rates.
 - C. They removed hedges against rising rates.
 - D. They brought EVE deterioration to the board's attention.

KEY CONCEPTS

LO 94.a

Silicon Valley Bank (SVB) was founded in 1983. They targeted start-up firms in the tech and life sciences sectors. The bank grew gradually until 2019. However, deposits tripled from 2019 to 2021. The bank expected this rate of depositor growth to continue (or at least stay positive). They chased short-term returns and invested a substantial portion of investor deposits in long-dated mortgage-backed securities. When the Fed began its interest rate increase cycle in 2022, these assets lost considerable value. This would not have been a huge problem had SVB's customers not also begun to demand withdrawals to fund cash needs in their businesses. The withdrawal demands increased to a point where SVB could not meet the requests and they were shut down by regulators and turned over to the FDIC.

LO 94.b

Banks with assets between \$10 billion and \$100 billion are classified as regional banking organizations (RBOs). Banks with assets over \$100 billion (or a foreign bank operating in the United States) are classified as large and foreign banking organizations (LFBOs). Due to SVB's rapid deposit growth, they crossed from RBO to LFBO. However, a regulatory change altered the threshold to \$100 billion from the previous \$50 billion in 2021. This change delayed the update in status by three years. RBO banks are still supervised, but there is a much lower burden of analysis. LFBOs are monitored much

more heavily. The regulatory transition from RBO to LFBO was not managed well by the Federal Reserve. In 2021, there was also a culture change in the Division of Supervision and Regulation that lessened the regulatory burden for RBO banks. This widened the gap between what was being reviewed when SVB had less than \$100 billion in assets to when it was reclassified as LFBO.

LO 94.c

Several factors contributed to the failure of SVB. They include the high concentration of depositors (i.e., primarily two industries and the same overall subset of clients), the type of deposits (i.e., large and mostly uninsured deposits), the heavy losses in their HTM and AFS portfolios due to poor interest rate risk management, their lack of contingency funding plans, and their difficulty raising capital.

LO 94.d

SVB struggled with their governance and risk management policies. Many of the policies were lacking or nonexistent. The board, management, and the chief risk officer (CRO) were not equipped to navigate the bank through the rapid growth they were experiencing. LFBO status had much more stringent standards, and the board was simply not prepared. The CRO recognized her experience shortfall and stepped down in April 2022. The bank did not find a replacement until December 2022. While they were searching, risk management was led by a group of senior risk managers who were also underqualified for the role. Supervisors cut the bank some slack because they were very profitable, and they were working on some of the needed changes. In hindsight, this lax supervision was an incorrect decision.

LO 94.e

SVB had several notable liquidity concerns. When they were still in the RBO portfolio, regulators did not address these issues using the “risk-focused guidelines” to allow their deposit growth and the size of their security portfolio to outweigh the observed risks in processes and positions. Shifting to the LFBO portfolio brought a more in-depth review, with six noted issues. One of which was a 30-day liquidity buffer with an \$18 billion shortfall. Once management accepted the reality of the issues, they began to expand capacity and contingency planning. They also changed the assumptions used in their 30-day liquidity buffer models to make things appear more favorable. Supervisors chose to wait for the 2023 review cycle to comment on these new developments.

LO 94.f

It is commonly understood that as interest rates rise, bond prices fall, and longer-dated bonds fall more than shorter-term ones. SVB had a few issues with respect to interest rate risk management. First, management chose to focus on the short-term-focused net interest income (NII) model rather than the longer-term economic value of equity (EVE) model. In other words, they prioritized short-term profitability over long-term solvency. Second, they assumed, as many did, that the inverted yield curve was a harbinger of a recession. This led them to anticipate falling not rising rates. They removed hedges to protect against rising rates to boost expected short-term profits. They also changed some assumptions related to deposits in order to make the interest

rate risk stress testing look more favorable. Supervisors delayed their rating downgrades until it was too late.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 94.1

1. **C** SVB was shut down in 2023 because they invested depositor money into long-dated mortgage-backed securities and failed to adequately anticipate customer withdrawal requests. The Federal Reserve's interest rate increase cycle compounded the problem, but it did not cause it. There are accounting conventions that cause management to not want to sell portions of their held-to-maturity portfolio, but that was management's choice in conjunction with their previous bad decisions to focus on high duration assets. Withdrawal requests came from a consortium of depositors and most of SVB's clients were U.S. based. (LO 94.a)
2. **C** The core issue that caused a delay in SVB receiving a higher level of regulatory scrutiny was the change in 2018 that altered the threshold from \$50 billion to \$100 billion in assets for shifting from RBO to LFBO. This shift happened in 2021 for SVB. At the same time, the vice chair for supervision changed the regulatory culture to lessen the burden on RBO banks. This meant that SVB was shifting from a more lax RBO process to the more stringent LFBO process all at once rather than gradually over a multiyear transition period. (LO 94.b)
3. **C** Silicon Valley Bank failed for many reasons. These reasons included deposit contraction, long durations on HTM securities, massive losses on HTM assets, which were not sold or transferred to AFS, and delays in regulatory rating downgrades. (LO 94.c)

Module Quiz 94.2

1. **D** The governance and risk management function at SVB was not able to keep up with the rate of growth that the bank was enjoying. Broadly speaking, neither the board nor the CRO had the requisite skillset to adequately manage risk for a large financial institution. Regulators cut them slack because the bank was profitable (i.e., no risks were materializing yet) and they were working toward some solutions. (LO 94.d)
2. **D** SVB's modeled 30-day liquidity buffer issues were very real. They amounted to a modeled \$18 billion shortfall in the fourth quarter of 2022. Rather than raise capital or sell securities at a substantial loss, management chose to relax the assumptions used. This reduced the modeled shortfall by \$5 billion but did nothing to alter reality. (LO 94.e)
3. **C** Management made the decision to remove hedges against rising rates. This protected short-term profitability measures, but it made the losses in the HTM and AFS portfolios more pronounced. Their EVE metric was harmed by removing

the hedges against rising rates. At the same time, they lengthened the assumption on the duration of deposits to make EVE look better than it really was. (LO 94.f)

The following is a review of the Current Issues in Financial Markets principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Bolton, Jiang, Kartasheva.

READING 95

THE CREDIT SUISSE COCO WIPEOUT: FACTS, MISPERCEPTIONS, AND LESSONS FOR FINANCIAL REGULATION

Study Session 15

EXAM FOCUS

In this topic, we examine contingent convertible bonds (CoCos), which have been popular instruments to help recapitalize and deleverage banks in distress. CoCos have grown in popularity and are favored by large institutional players. For the exam, understand CoCo terminology and key design features, including the different triggers and the loss absorption mechanism, and why CoCos provide important protections to struggling financial institutions. Also, be able to contrast the rescue of Credit Suisse by Swiss regulators in 2023 with the rescue of Bear Stearns by U.S. regulators in 2008, and be able to identify the rationales for the write-down of the Credit Suisse CoCos. Finally, be able to explain why the write-down of the Credit Suisse CoCos has created significant market challenges and negative investor reaction, and be familiar with the characteristics proposed to reform the CoCo structure.

MODULE 95.1: CONTINGENT CONVERTIBLE BONDS

LO 95.a: Describe the features and mechanics of contingent convertible bonds (CoCos) and explain the rationale for banks to issue them.

Contingent convertible bonds (CoCos) are a type of fixed income security that helps stabilize the distress of an issuer by offering immediate recapitalization. CoCos were borne out of the distresses of the global financial crisis of 2007–2009 in order to limit systemic risk. They are a form of convertible bonds where the conversion can happen automatically if a trigger event occurs. For example, if a bank's core Tier 1 regulatory capital ratio falls below the minimum regulatory requirement. The conversion infuses the issuer with equity and reduces the need for capital injection by a government during a crisis. CoCos are often used to recapitalize *going concern* banks, that is, banks

that expect to remain viable following the CoCo write-down (by contrast, a *gone concern* bank is one in liquidation or bankruptcy).

CoCos are similar in some respect to *total loss absorbing capacity (TLAC) bonds* because they both protect equity and minimize losses. However, TLAC bonds are designed to ensure that the holding companies of global systemically important banks (GSIBs) have sufficient equity to absorb losses and allow their operating affiliates to continue operations during a resolution process and not allow their liabilities to be at risk.

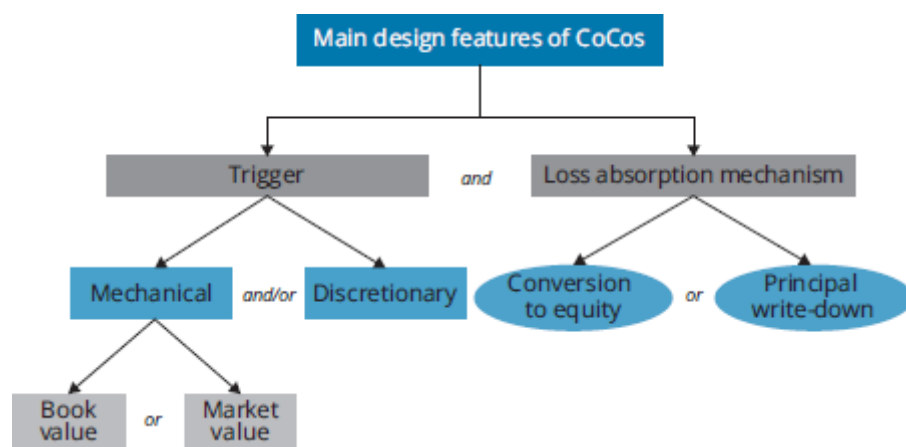
CoCo Mechanisms

CoCos have two key features: (1) the trigger that activates them, and (2) the loss absorption mechanism. These design features are illustrated in Figure 95.1. There can be multiple triggers that can be either *mechanical*, which are predefined based on the capital ratio, or *discretionary* (also called point of nonviability, or PONV), which is based on the judgment of a supervisory body or regulator if the bank's capital has weakened sufficiently to the point that it requires recapitalization.

The mechanical trigger is activated when the bank's capital (either based on book or market value) drops below a certain percentage of risk-weighted assets (RWAs). The discretionary trigger is activated based on the supervisor's view of the bank's insolvency. Basel III rules require that all regulatory capital CoCos have a discretionary (PONV) trigger.

When any of the triggers is breached, the loss absorption mechanism is activated. CoCos absorb losses either through a *conversion to equity* at a predefined conversion rate, or through a *write-down of principal*. The equity conversion can be specified based on the market price of the stock, or based on a predetermined price, or a combination of the two where the predetermined price acts as the floor. If the mechanism is a conversion to equity, the existing shareholders' position is diluted, who therefore have an incentive to avoid conversion. The principal write-down can be partial or full.

Figure 95.1: CoCo Features



Source: Avdjiev, Stefan, Bilyana Bogdanova, and Anastasia Kartasheva. 2013. "CoCos: A Primer." *BIS Quarterly Review*, September 2013, 43–56.

For CoCos to qualify as regulatory capital under Basel III, they must meet two criteria: (1) include a discretionary (PONV) trigger, and (2) have a minimum regulatory capital-to-risk-weighted assets ratio (i.e., the going concern rule). This is meant to protect capital levels and therefore improve a bank's resilience to external shocks. Overall, CoCos are designed to allow banks to easily recapitalize during financial stress without requiring external intervention (e.g., a bailout).

The CoCo Market

CoCos offer investors higher yields than traditional bonds. They have a lower priority of claim and will typically be attractive to investors with a long time horizon who are looking for a yield premium. For example, a 2013 study assessed CoCos to have average yield-to-maturities that were 2.8% above subordinated bonds and 4.7% above senior unsecured bonds of the same bank.

The CoCo market has grown significantly over the last two decades. Based on a 2020 analysis, there have been over 700 CoCo issuances raising around \$500 billion between 2009 and 2015. Over half of CoCos include a mechanical trigger with a principal write-down. The majority of CoCos up until 2015 were Additional Tier 1 (AT1) capital instruments issued by developed market banks.

CoCos are typically favored by well-capitalized, larger banks, for three reasons. First, because these banks are healthy, the risk of conversion is low, providing a yield benefit to the issuing bank. Second, the senior bondholders of the bank benefit from CoCo issuances given the risk-reduction benefits they offer. Third, the issuance of CoCos has historically not impacted the issuing bank's share price negatively; in fact, share prices often rose. However, this gives rise to a moral hazard problem because it can lead shareholders to take excessive risks favoring write-down CoCos because losses would be absorbed by the investors in the write-down CoCos, and not by shareholders.



MODULE QUIZ 95.1

1. A bank's contingent convertible bond (CoCo) has a point of nonviability (PONV) trigger. The PONV trigger would be activated:
 - A. at the discretion of the issuing bank if its capital weakened significantly.
 - B. automatically if the bank's liquidity balances fell below a specified value.
 - C. automatically if the bank's Tier 1 capital ratio fell below a specified value.
 - D. at the discretion of a regulator if the conversion would avoid the bank's nonviability.
2. Which of the following contingent convertible instruments would most likely qualify as regulatory capital under Basel III rules?
 - A. Bond 1 has a mechanical trigger based on book value, and a maximum regulatory capital-to-risk-weighted assets ratio.
 - B. Bond 2 has a mechanical trigger based on market value, and a minimum regulatory capital-to-risk-weighted assets ratio.
 - C. Bond 3 has a discretionary (point of nonviability, or PONV) trigger, and a minimum regulatory capital-to-risk-weighted assets ratio.

D. Bond 4 has a discretionary (PONV) trigger, and a maximum regulatory capital-to-risk-weighted assets ratio.

MODULE 95.2: CREDIT SUISSE CoCos

LO 95.b: Explain the rescue of Credit Suisse by Swiss regulators in 2023 and compare it to the rescue of Bear Stearns by U.S. regulators during the financial crisis in 2008.

2008 Bear Stearns Rescue

The March 2008 collapse of U.S. investment bank Bear Stearns was one of the key events of the global financial crisis of 2007–2009. The bank suffered heavy losses mainly due to its exposure to securitized investments that were backed by subprime mortgages. The bank's troubles forced the U.S. Federal Reserve and the U.S. Treasury to rescue the bank by facilitating a merger with JPMorgan. Given that both the Federal Reserve and the Treasury had limited powers to force a merger, the Federal Reserve needed to invoke a rarely used section of the Federal Reserve Act (section 13(3)), which allowed it to provide liquidity support to an institution but only under unusual and urgent circumstances.

As part of its bailout arrangement, the Federal Reserve persuaded JPMorgan to buy all Bear Stearns shares at an initially proposed price of \$2/share, subject to shareholder approval, and assume all Bear Stearns liabilities. The Federal Reserve then set up a collateralized special purpose vehicle (SPV) jointly with JPMorgan, financed with a \$1 billion junior tranche by JPMorgan and a \$29 billion senior tranche by the Federal Reserve, and used the SPV to buy up to \$30 billion of troubled Bear Stearns assets. The Federal Reserve subsequently had to make concessions to Bear Stearns shareholders by raising the share price to \$10/share. The deal was finalized in May 2008.

2023 Credit Suisse Rescue

Fifteen years following the events of Bear Stearns, U.S.-based Silicon Valley Bank (SVB) collapsed in March 2023, which created systemic risk and contagion in the global banking system. The collapse destabilized banks with spillover into Europe, creating significant liquidity problems for the Swiss banking giant Credit Suisse. Credit Suisse's liquidity problem was so large that it was forced to request up to CHF50 billion in liquidity support from the Swiss National Bank, Switzerland's central bank. The Swiss central bank and the Swiss Financial Market Supervisory Authority (FINMA) then helped organize a takeover of Credit Suisse by the Swiss investment bank UBS. As part of the arrangement, nearly CHF16 billion in Credit Suisse CoCos had been fully written down, helping to recapitalize the bank's balance sheet.

LO 95.c: Explain the rationale for the write-down of Credit Suisse CoCos that was engineered by regulators during the rescue of Credit Suisse and its takeover by

In contrast to the more limited authority of the U.S. government, the Swiss federal government had significantly more authority under Swiss emergency law. Specifically, the Swiss federal government had the authority to trigger the write-down of the CoCos, thereby avoiding the more complex route of setting up an SPV. The CoCo bond documents gave FINMA the authority to trigger a conversion to equity if Credit Suisse reached the PONV. As a result, when FINMA triggered the conversion and helped recapitalize Credit Suisse, the CoCos functioned exactly as they were intended.

The write-down of the CoCos (AT1 instruments) in March 2023 helped to quickly deleverage the Credit Suisse balance sheet. Further, the intervention of the Swiss government did not involve a complete bailout of creditors. This helped alleviate market concern that credit bailouts create a moral hazard problem by encouraging more risk taking by large financial institutions that reasonably expect they would be bailed out when facing significant distress.

LO 95.d: Describe the reactions by market participants to the write-down of the CoCos, and explain and evaluate different arguments and lessons learned related to the decision to write down the CoCos.

Market Reaction

The complete write-down of the Credit Suisse CoCo bonds, without any material adverse impact on shareholders, caught many investors by surprise and caused significant negative market reaction. Based on an analysis of Credit Suisse's credit default swap (CDS) bond spreads, the market did not anticipate a write-down of the AT1 until the day before the takeover. Bondholders and many market participants viewed the events as a subversion of the traditional creditor hierarchy, under which bondholders have a more senior claim than shareholders in a distress scenario. While investors understood that CoCos were riskier than traditional bonds—as evidenced by their higher yields—they likely did not fully understand the risks, including that write-down CoCos are, in essence, junior in claims to equity.

The adverse reaction was significant enough that it caused several non-Swiss European regulatory bodies to publicly distance themselves from the action of the Swiss government, in part to avoid contagion to the \$250 billion European AT1 market. For example, shortly following the FINMA decision, the EU's Single Resolution Board, the European Banking Authority (EBA), and the European Central Bank (ECB) issued a statement explicitly stating that in the event of a banking crisis, the regulatory intervention would lead to common shareholders absorbing losses first, and only then followed by a write-down of AT1 capital.

Further, both the adverse market reaction and the statements by the non-Swiss regulatory bodies incentivized several lawsuits against Credit Suisse, in large part arguing that the write-down of CoCos contradicted generally accepted creditor hierarchy, and that FINMA did not act in good faith in its intervention.

Lessons Learned

The key lesson from the Credit Suisse CoCo write-down was that despite a strong argument that the CoCo write-down worked exactly as it was intended, it created significant uncertainty and confusion, even among experienced investors. Many investors were unaware of the difference between going concern and gone concern CoCos, and were also unaware that these CoCos included multiple triggers: an automatic trigger when the common equity Tier 1 (CET1) ratio was breached, and a discretionary trigger based on the financial regulator's assessment that the PONV had been reached. This confusion was further exacerbated by the lack of clear criteria to determine what is the PONV.

This ambiguity also created pricing uncertainty, which increased costs to investors. The statement put out by the EBA and the ECB stating that they would not deviate from the traditional creditor priority further fueled confusion and created uncertainty in the CoCo market.

Reforming and simplifying the CoCo structure would help reduce these concerns. A proposed model for CoCos calls for its design to be simple, have off-the-shelf pricing, and allow for an easy recapitalization process. These CoCos would not include an automatic or discretionary trigger, but instead would give the issuer an option to convert the CoCo's fixed income component into equity. Because CoCos provide equity capital exactly at a time when a struggling bank requires capital infusion, CoCos also act as a countercyclical instrument. Their countercyclical nature is a key advantage to the issuing banks.

Under the redesigned structure, the conversion would be based on a specified strike price, which would provide the issuing bank with the ability to raise equity capital at predefined terms. Because the right to convert is an option for the issuer, pricing the CoCos would be based on a predictable option pricing model. This simpler model would also remove regulatory involvement in determining whether a trigger is met.



MODULE QUIZ 95.2

1. Which of the following features did not help trigger the Credit Suisse contingent convertible bonds (CoCos) write-down in March 2023?
 - A. Requiring the Swiss federal government to act under Swiss emergency law.
 - B. Involving the Swiss federal government to trigger the write-down of the CoCos.
 - C. Setting up a collateralized special purpose vehicle to facilitate the recapitalization.
 - D. Giving the Swiss Financial Market Supervisory Authority (FINMA) the authority to trigger a conversion to equity.
2. Following the Credit Suisse CoCo write-down in March 2023, there has been a push to reform the CoCo structure. Which of the following features is not one of the elements proposed to reform the CoCo structure?
 - A. Simplicity.
 - B. Off-the-shelf pricing.
 - C. Easy recapitalization process.

D. Automatic, rather than discretionary, triggers.

KEY CONCEPTS

LO 95.a

Contingent convertible bonds (CoCos) were introduced following the global financial crisis of 2007–2009 to limit systemic risk. CoCos help recapitalize a distressed bank and reduce the need for a bank bailout by a government during a crisis.

CoCo features include a trigger that activates them, and a loss absorption mechanism. The trigger can be mechanical (based on a capital ratio trigger), or discretionary (also called point of nonviability, or PONV) based on the judgment of a supervisory body. The loss absorption mechanism converts the bonds to equity (based on a predetermined price, market price, or a combination of both) at a predefined conversion rate or improves capital through a write-down of principal.

For CoCos to qualify as regulatory capital under Basel III, they must meet two criteria: (1) include a discretionary (PONV) trigger, and (2) have a minimum regulatory capital-to-risk-weighted assets ratio (going concern rule).

CoCos have higher yields than traditional bonds, because they have a lower priority of claim. They are generally attractive to well-capitalized, larger banks.

LO 95.b

Bear Stearns suffered heavy losses due to its exposure to subprime mortgages. The bank was bailed out by the U.S. Federal Reserve and the U.S. Treasury in 2008 through a merger with JPMorgan. The bailout was facilitated by setting up a collateralized special purpose vehicle (SPV), with a \$1 billion infusion by JPMorgan and a \$29 billion infusion by the Federal Reserve. The Federal Reserve then used the SPV to buy up to \$30 billion of troubled Bear Stearns assets.

Following systemic risk and contagion from the global banking system in 2023, the Swiss bank Credit Suisse faced significant liquidity problems that required it to request liquidity support from the Swiss National Bank. Credit Suisse was then taken over by the Swiss investment bank UBS. As part of the takeover arrangement, the Swiss financial regulator FINMA triggered a nearly CHF16 billion write-down of Credit Suisse CoCos, which helped to recapitalize the bank's balance sheet.

LO 95.c

The Swiss federal government had significant authority under Swiss emergency law to trigger the write-down of the Credit Suisse CoCos. Specifically, the regulator FINMA had the authority to trigger a conversion to equity when it determined that Credit Suisse reached the PONV. A key argument of this arrangement was that the CoCos functioned exactly as they were designed, by infusing equity and helping to deleverage Credit Suisse's balance sheet.

LO 95.d

The write-down of the Credit Suisse CoCos created several problems as follows:

1. It caught many investors by surprise who did not anticipate a write-down of the CoCos.
2. It caused significant negative market reaction because market participants viewed the events as disregarding the accepted creditor hierarchy that bondholders have a more senior claim than shareholders in a distress scenario.
3. Several non-Swiss European regulatory bodies distanced themselves from FINMA's action, and publicly stated that in the event of a banking crisis, common shareholders would absorb losses first, rather than bondholders.
4. Several lawsuits were brought against Credit Suisse.
5. It increased the costs to investors by creating pricing uncertainty.

To remedy these problems in the future, there is a push to reform the CoCo structure. A proposed structure would require CoCos to be: (1) simple, (2) have off-the-shelf pricing, and (3) allow for an easy recapitalization process. Under this structure, the issuer would be given the option to convert the CoCos based on a specified strike price, rather than based on automatic or discretionary triggers.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 95.1

1. **D** CoCos typically have multiple triggers, including a mechanical trigger based on a predefined capital ratio, and a discretionary trigger called PONV trigger. A PONV trigger is based on the judgment of a supervisory body or regulator that the bank's capital has weakened and that without the CoCo conversion the bank could become insolvent or nonviable. (LO 95.a)
2. **C** For CoCos to qualify as regulatory capital under Basel III rules, they must include a discretionary (PONV) trigger, and have a minimum regulatory capital-to-risk-weighted assets ratio (going concern rule) to protect capital levels. (LO 95.a)

Module Quiz 95.2

1. **C** FINMA, the Swiss federal financial regulator, under Swiss emergency law, had the authority to trigger the write-down of the Credit Suisse CoCos and trigger a conversion to equity. This helped avoid a more complex scenario of setting up a special purpose vehicle to bail out the bank. (LO 95.c)
2. **D** Reforming the CoCo structure recommends their design to be (1) simple, (2) have off-the-shelf pricing, and (3) allow for an easy recapitalization process. CoCos would not have triggers (either automatic or discretionary), but instead give the issuer an option to convert the CoCo's fixed income component into equity. (LO 95.d)

The following is a review of the Current Issues in Financial Markets principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Federal Reserve Bank of Richmond.

READING 96

ARTIFICIAL INTELLIGENCE AND BANK SUPERVISION

Study Session 15

EXAM FOCUS

This topic focuses on the adoption, advantages, disadvantages, and regulation of artificial intelligence (AI) in the financial sector. For the exam, understand the five advantages and disadvantages offered by AI for banks and their customers. Also, pay attention to how these factors might inform potential regulatory responses, including the potential for bias and its connection with alternative data.

MODULE 96.1: ARTIFICIAL INTELLIGENCE AND BANK SUPERVISION

AI Applications in the Financial Sector

LO 96.a: Describe historical evolution and common types of AI-based applications used in the financial sector.

Artificial intelligence (AI) is a form of intelligence that is generated directly by computers rather than humans. The concepts underlying AI were first developed by Alan Turing in a paper from 1950. In 1997, almost 50 years later, IBM's Deep Blue supercomputer used these ideas to win a chess match against world champion Garry Kasparov. These advances have harnessed the power of **machine learning**, where computers write their own code, and **neural networks**, which classify text, images, and other data. Some familiar applications of these innovations include Google searches, the movie recommendation engine for Netflix, and the advances in human-computer interaction via ChatGPT.

AI innovations have not escaped the notice of financial institutions. A survey conducted by McKinsey and Co. in 2019 found that 36% of firms in the financial sector are automating back-office processes using AI, 32% deploy AI chatbots for customer service, and 25% use AI for fraud detection and the evaluation of creditworthiness.

Cornerstone Advisors produced another survey in 2022 showing that 25% of financial sector firms were using AI for process automation and 18% were using AI chatbots. These two surveys present somewhat different numbers, but they both point to two realities. First, the financial sector is adopting AI technologies. Second, the adoption rate may be slow right now, but it is expected to accelerate considerably in the near future.

Regulators are anticipating this potential growth in AI adoption by financial institutions. In March 2021, the Federal Reserve Board of Governors, the Office of the Comptroller of the Currency (OCC), the Consumer Financial Protection Bureau, the Federal Deposit Insurance Corporation (FDIC), and the National Credit Union Administration jointly issued a request for information (RFI) to gather perspectives on the potential implications of AI on the financial sector.

AI Advantages

LO 96.b: Explain the advantages of implementing AI-based applications to the banking services companies and their customers.

Banks are increasingly considering how to use AI technologies as a value-add in their customer experience. The key advantages that AI introduces to the banking sector are as follows:

1. *Cost cutting.* Deloitte estimates that the top 2,000 U.S.-based banks spend \$250 billion annually on customer support inquiries. Using AI chatbots could cut this cost considerably.
2. *On-demand customer service.* Banking customers can have on-demand access to answer questions using a chatbot. They can also use text recognition to deposit checks and digital services to transfer money or pay bills. No longer is a trip to a brick-and-mortar location, during their operating hours, a necessity.
3. *Fraud detection and prevention.* Customers can be notified when an algorithm flags a transaction or a request for new service may be fraudulent. One example is when a credit card issuer alerts their customers regarding out-of-state or online transactions that could be the result of fraud. There is also an identity theft component to this advantage. Fraudsters are constantly evolving their path of attack, which makes machine learning and AI a potentially solid defense in that battle.
4. *Anti-money laundering (AML).* The Patriot Act, from 2001, places significant AML responsibilities on financial institutions. Since complexity is very high, machine learning and AI are very useful tools for banks to deploy. Companies like Symphony AyasdiAI have developed software that banks can use to help flag potential AML alerts that require human review.
5. *Know your customer (KYC) rules.* Banks must also verify the identity of their customers. This data would include address information. New York-based Socure has developed AI tools to help banks streamline these verification requirements.

AI Disadvantages

LO 96.c: Discuss the disadvantages and difficulties for financial companies using AI.

The use of AI technologies will also present challenges when applied to financial institutions. The key disadvantages and difficulties are as follows:

1. *Loss of personal touch.* Some customers prefer a high-touch and in-person experience. They like talking with a human on the phone to answer very specific questions and connecting with a bank teller to handle transactional needs. Shifting to an AI-based interaction has the potential to commoditize the customer experience and reduce differentiation between banks.
2. *Lack of control over potential biases.* Artificial intelligence gathers data points and evaluates them. All data is essentially reduced to an algorithm. Some of these algorithms may contain unintended biases based on the data inputs (e.g., when using credit scores).
3. *Opaque models.* Many AI-managed processes operate in a manner that is opaque to users. This makes it difficult to determine how results are achieved, which may pose barriers to acceptance and use. This also presents a challenge when individual credit applicants look for an explanation for credit decisions. An opaque model may not provide enough feedback to help consumers evaluate their next steps.
4. *Evolving sophistication of fraudsters.* People and entities who pursue fraud are finding some traditional paths blocked. This means that they will invest even more energy in finding alternative ways to conduct their fraudulent schemes. AI will need to constantly evolve to match the sophisticated rate of change in fraudulent creativity.
5. *Uncertain regulatory landscape.* Regulators are still struggling with the potential impacts of AI in the financial sector. Some factors present risks due to their complexity. Regulation will be an evolving and uncertain backdrop until AI innovations are better understood.

Regulatory Issues

LO 96.d: Clarify the specific issues faced by banks and regulators arising from utilizing AI in modeling and valuation.

Regulation covering the use of AI in modeling and valuation within the financial sector is a developing process. Broadly speaking, regulators view risk management as the desired outcome. They are treating traditional modeling and AI-infused models as existing along a spectrum rather than as two binary options. The end goal of regulation is for risk to be appropriately monitored, controlled, and documented.

Federal regulators are considering several potential issues that are marginally connected with the disadvantages of using AI in the financial sector. Specific issues faced by banks and regulators are as follows:

- *Potential for bias.* Several regulations already exist to protect consumers. One such law is the Equal Credit Opportunity Act. There is the potential for **algorithmic bias** (a.k.a. algorithmic discrimination), which occurs when the computer code running the AI process is not explicitly designed with fairness in mind. In other words, the AI process could unintentionally perpetuate a human bias like racism. The algorithms are designed to crunch numbers, not to ensure equal access to credit.
- *Evolution in credit scores.* Traditional models consider a relatively small number of inputs when deriving a credit score. With AI as a copilot, credit scores can adapt to consider a much wider dataset. This could potentially expand access to credit, which is a good thing, but it all hinges on what data is considered.
- *Use of alternative datasets.* AI models need data in order to analyze it and make better decisions. Some nontraditional data sources include the timeliness of rent and utility payments as well as cash flow patterns in a credit applicant's bank accounts. The use of these alternative data sources has the potential to expand access to credit. This sounds harmless, but there is also the potential for privacy concerns. The need for alternative data could push into social media and web browsing activity. This would raise concerns of consumer privacy rules and potentially stifling free expression. Certain mortgage regulators (e.g., the Federal Housing Agency, the Department of Housing and Urban Development, Fannie Mae, and Freddie Mac) expressly exclude alternative data sources from mortgage decisions. There are many other credit applications available that do not currently have such limitations.
- *Model opacity.* The algorithms that propel AI processes operate in a black box. The outcome may be known but the way in which the outcome was derived is not. This presents a natural challenge for regulators who have been focused on the process as well as the outcome.
- *The urge to overregulate.* The potential exists for regulators to impose excessive constraints on a system they do not fully understand yet. Overregulation could stifle innovation. To their credit, many regulators are currently seeking understanding, but there is the potential to impose a higher level of scrutiny on AI-driven processes than on traditional ones.



MODULE QUIZ 96.1

1. Artificial intelligence (AI) is:
 - A. a new idea.
 - B. not yet a focus of regulators.
 - C. expanding its reach into the financial sector.
 - D. highly adopted for use in customer service at financial institutions.
2. Which of the following statements is not an advantage for banks to deploy AI technologies as a part of their customer experience?
 - A. Use algorithms to stop all fraudulent transactions.
 - B. Save money while speeding up the customer service experience.
 - C. Use algorithms to verify customer information and identification.
 - D. Use algorithms to flag suspicious transactions that could have anti-money laundering implications.
3. Customers might be hesitant to embrace AI in the banking sector because:

- A. fraud prevention methods are too restrictive.
 - B. regulation creates too many checkpoints for customers.
 - C. they know exactly how the algorithm makes credit decisions.
 - D. they prefer human interaction to solve their unique challenges.
4. Which of the following items is an underlying concern related to the potential for biases when using AI in the banking sector?
- A. The use of transparent models.
 - B. Algorithmic bias through dataset selection.
 - C. Algorithmic bias through intentional model design.
 - D. Traditional credit scores being more favorable for those with low access to credit.

KEY CONCEPTS

LO 96.a

The genesis of artificial intelligence (AI) was a paper by Alan Turing in 1950. In 1997, IBM used a supercomputer (and AI) to beat the world chess champion. Machine learning, neural networks, and generative AI are more modern applications. AI adoption by the financial sector has started slowly but it has been accelerating. Regulators are beginning to take notice and consider the potential implications of AI on the financial sector.

LO 96.b

There are five key advantages for a bank to use AI as a part of their customer experience. They include cost cutting, providing on-demand customer service, fraud detection and prevention, anti-money laundering compliance, and know your customer compliance.

LO 96.c

There are five key disadvantages for a bank to use AI. They include a loss of personal touch, lack of control over potential biases, model opacity, the evolving sophistication of fraudulent actors, and an uncertain regulatory landscape.

LO 96.d

The end goal of financial regulation is to protect consumers and for risk to be appropriately monitored, controlled, and documented. With a rapidly evolving new technology, the risks are sometimes less certain. Some of the known regulatory concerns include the potential for bias, evolution in credit score calculations, the use of alternative datasets, model opacity, and the potential to overregulate.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 96.1

1. **C** Artificial intelligence adoption is building within the financial sector but less than one-third of firms are currently using this technology for customer service. Since

this area is growing so rapidly, regulators are beginning to take notice and trying to understand the potential pitfalls. (LO 96.a)

2. **A** Banks are using AI to cut costs and enhance customer service. They are also using it to detect fraud, but this is a constantly evolving area. They cannot catch all fraud, because fraudsters are perpetually probing new attacks. Banks also use AI for identify verification and to flag AML concerns that need human review. (LO 96.b)
3. **D** One reason that customers may be hesitant to embrace AI in their banking experience is a preference for personal touch. Fraud prevention is a key enhancement that AI brings to banking customers. AI models are opaque, so customers do not have a good understanding of why credit decisions are being made. That is a potential regulatory concern, but regulation over the use of AI in the financial sector is in its infancy. (LO 96.c)
4. **B** The potential for bias when using AI in the financial sector is a serious concern. The algorithms are designed to crunch numbers, not promote fairness. There is the potential that those algorithms will unintentionally promote bias through the data inputs. AI algorithms draw upon alternative datasets that may disadvantage certain credit applicants and have the potential to include inputs that could violate freedom of expression laws. While AI algorithms do not intentionally try to promote discrimination, regulators are not able to make a determination on this important element because AI models are opaque. (LO 96.d)

The following is a review of the Current Issues in Financial Markets principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Morgenthal, Hein, and Papenbrock.

READING 97

FINANCIAL RISK MANAGEMENT AND EXPLAINABLE, TRUSTWORTHY, RESPONSIBLE AI

Study Session 15

EXAM FOCUS

This topic focuses on the responsible use of artificial intelligence (AI) in the context of financial risk management. For the exam, be familiar with the areas of interest surrounding self-learning models and the discussed ethical considerations. Also, understand how the concept of fairness could be influenced through AI model design. In addition, be aware of how model explainability and archiving model elements are both needed for understanding the fairness outcomes of a model. From the perspective of trustworthy AI, be aware of the elements related to privacy concerns, explainable AI (XAI), and the process of optimization. Finally, understand how clustering and documentation can enhance trust in the explainability of AI model decisions.

MODULE 97.1: AI IMPLEMENTATION IN FINANCIAL RISK MANAGEMENT

Model Bias and Ethical Considerations

LO 97.a: Describe the challenge posed by potential model bias and the ethical and responsible considerations surrounding the implementation of AI-driven solutions in financial risk management.

The European Union (EU) has proposed the first major regulation targeting artificial intelligence, known as the **Artificial Intelligence Act (AIA)**. The goal of this proposal is to encourage ethical and unbiased behavior in the application of AI models, which are often highly complex. This is connected to Pillar 1 (capital requirements) of the Basel regulatory framework. The proposed oversight applies traditional regulatory model approval techniques because the areas of interest are model development, periodic validation processes, and model change methods. This review includes both static and

self-learning models, and it separates model changes into material and immaterial actions.

Three areas related to self-learning models that have drawn interest are as follows:

1. *Fraud detection.* Anti-money laundering (AML) is a subset of fraud detection. It is known that fraudulent actors are constantly evolving their methods. Using AI, and particularly self-learning AI, can help keep pace with the rate of change in AML detection and risk mitigation. Another benefit is an expected reduction in false alert flags.
2. *Trading models.* In a trading context, AI is involved with algorithmic trading and robo advising. These models are making capital allocation decisions, with low levels of explainability to associated stakeholders. Without the ability to review the process, the outcome is less trustworthy.
3. *Credit risk.* This area has major ethical components. AI is used to extend or deny credit to applicants. The opaqueness of this process has explainability issues that raise ethical and fairness concerns. It is simply not known to what extent biases may be involved in this process.

Ethical considerations are also important with the recent innovations of AI in financial risk management. One concern is called **wanted bias**, which relates to bias in most risk prediction models. There is also the possibility for unwanted **algorithmic bias** to sneak into the methodology underlying the AI models. In this context, discriminatory or other unfair limitations may exist based on inputs to the models or the models themselves. A root cause for bias could be errors in the training data. For example, there could be outliers or noise affecting the learned outcomes. Also, if populations are underrepresented or misrepresented in the training data, then that could introduce unethical biases to AI models.

Maintaining Fairness

LO 97.b: Analyze the potential benefits and challenges of utilizing AI while maintaining fairness and preventing biases in risk assessment and decision-making.

Most people agree that “fairness” in AI models is an admirable goal, but quantifying this abstract concept in the context of AI model development can be challenging. Considering this aspect of model design can minimize reputational risk for firms deploying AI in financial risk management. One way to define **fairness** is ensuring an equitable and just allocation of benefits (and costs) while also minimizing the impact of bias, discrimination, and stigmas. An effective fairness process will also provide the opportunity to contest or seek restitution for any unfair decisions made. In other words, AI models must have a layer of accountability built in.

At a high level, two considerations for fairness outcomes are as follows:

1. *Fairness outcomes can be influenced.* One common example is when there are different health insurance premiums for men versus women. This may be based on

actual cost data, but there is the potential for bias based on the data used for analysis. This fairness outcome could be influenced through AI model design.

2. *Fairness outcomes cannot be influenced.* Some fairness issues originate from a problem outside the control of the AI model. For example, there could be spillover gender bias that exists in the credit approval process. There is the possibility for women to receive higher credit denial rates than men. This may not be a fairness outcome that can be influenced through model design if it originates from gender-based differences. This is a bigger issue to address that is outside of AI model design.

Validation of Decision-Making Algorithms

LO 97.c: Explain the proposed considerations for the technical validation of decision-making algorithms to check for potential unfairness.

The technical validation of models needs to expand beyond conceptual soundness and model plausibility to include checks for fairness. Factors in this process include the following:

- Training data needs to be reviewed for potential outliers, noise, and bias.
- Model testing should focus on the problems that need to be solved rather than just the process. For example, testing of anti-money laundering alert models should be less focused on calibration and more focused on false positive ratios.
- Areas that are difficult to test (e.g., historically unprecedented financial market scenarios) should be identified and explicitly considered.
- The financial institution's risk appetite needs to be considered relative to allowable residual risk.
- Explainability becomes increasingly important for decisions with a critical impact on end users. The main drivers of model results should be understood. The level of explainability relates to fairness where low levels of explainability (i.e., transparency) may point to unfairness issues that need to be addressed.
- To test both validity and fairness, models must have the ability to be audited.

The ability to audit a model can be very challenging when it is self-learning. By its very nature, self-learning models are constantly evolving, which makes explaining how a specific outcome was historically derived problematic. This makes it vitally important to retain the full data history, all parameters, and all metadata for self-learning models. Jumps in model parameters need to be examined across versions of the model. This is the only way to rewind the model to understand why a certain outcome was achieved. Understanding what the model did is a critical step in considering any issues of fairness in model design.

The following six steps will likely need to be used when self-learning models are deployed:

1. Create an incident management system to determine what happens when the model is not working properly.
2. Develop an infrastructure to monitor model or data drift.

3. Rigidly stress test the model.
4. Develop an infrastructure to archive all relevant historical information (i.e., data, parameters, and versions) to promote explainability.
5. Evaluate the model using the Pareto optimum (i.e., the efficiency line showing tradeoffs between competing goals).
6. Incorporate identified risk and side effects into existing processes and requirements.

Trustworthy AI

LO 97.d: Describe the approaches and technologies that should be considered in the implementation and assessment of Trustworthy AI.

Adopting ethical AI practices involves harnessing the innovative potential of AI in a way that promotes trust, fairness, and explainability. There are various high-level concerns, which include privacy, transparency, and optimization.

Compliance with domestic privacy laws and regulations is a very important consideration for **trustworthy AI**. Three areas of note are as follows:

1. *Differential privacy*. When looking at the dataset used by an AI model, users should not be able to identify the inclusion or exclusion of any individual data point. This is both an antidiscrimination checkpoint and a roadblock to ensuring that fairness is being implemented appropriately.
2. *Federated learning*. This feature enables multiple entities to share and combine training data used in the testing of machine learning models without physically sharing the source information.
3. *Synthetic data*. This data is generated by AI and used to support collaboration between unaffiliated parties. Using a *generative adversarial network (GAN)*, synthetic datasets can be created that replicate the statistical properties of the real data they are representing. This enables privacy to be preserved, but it does not explicitly address fairness concerns.

Explainable AI (XAI) attempts to resolve fairness concerns related to the low transparency inherent in AI models. One XAI approach is to make models easy to interpret by design. A flaw in this approach is that it requires tremendous simplification, which is not sufficient for addressing complex relationships in the underlying data. Another option is to use post-hoc approaches (e.g., local interpretable model-agnostic explanations [LIME] or SHapley Additive exPlanations [SHAPs]) that apply game theory to interpolate what the model may have been doing. These methods are more about making the model less opaque than making it transparent. Using either a LIME or SHAP approach to address explainability requires a significant amount of computing power and it introduces a second model into the workflow. An innovative new approach in an investment context is to generate forecasted market data that is highly correlated to actual market data. Using this approach allows users to explore simulated outcomes to search for explainability with synthetic data that closely resembles real-world data.

The pursuit of explainable and trustworthy AI can be extended through **optimization**. One approach is to optimize four variables: explainability, fairness, model accuracy, and model speed (i.e., efficiency). Despite the usual conflict between explainability and accuracy, this method can mathematically produce an efficient Pareto frontier of choices. This frontier will be nonsmooth and nonconvex. Each point on the frontier can be displayed as a single line in a parallel-coordinate plot. Users can then select the line with the most desirable set of tradeoffs. In other words, this optimization method can be used to find a compromise solution that achieves the most desirable combination of the four variables for the user's unique situation.

Many companies are now incorporating formal ethics reviews in their use of AI. This includes internal and external algorithmic audits. The purpose is to ensure that models are producing verified results that are sufficiently trustworthy and explainable. The Frankfurt Institute for Risk Management and Regulation is working on solutions in trustworthy AI through a project called Financial AI Cluster (FAIC).

AI Case Study in Credit Risk Management

LO 97.e: Examine the application of Explainable AI (XAI) in the field of credit risk management as presented in the use case of a European insurance group.

The EU Horizon 2020 project presented a use case scenario to begin addressing concerns related to explainable AI in the context of a European insurance group. It drew upon resources from the FIN-TECH (FINancial supervision and TECHnology compliance training) program, and it applied SHAP-based XAI to the credit risk management process. The goal was to consider ways to make black-box AI models more explainable to reduce any inhibitors to innovation.

An interesting approach is to utilize SHAP clustering in which portfolio constituents with similar outcomes or explanatory variables are grouped (i.e., clustered) together. The post-hoc explanatory information is clustered together based on observed similarities. Data points at the intersection of two clusters deserve further investigation because they represent “fuzzy” decision-making. Any clusters with nearly equal “yes” and “no” outcomes could indicate an error in the self-learning AI model. The data can also be correlated with input features and predictions. This process is associated with Shapley value decomposition to breakdown the contribution of each input variable to the ultimate outcome of a self-learning AI model. It is important to note that this approach is model agnostic.

This entire process can be performed using only GPU memory, which means that it is very efficient to generate explanations quickly. The FIN-TECH program used RAPIDS, which is a suite of open-source Python libraries to speed up calculation times. Visual dashboards can be used by customer service representatives to answer customer inquiries about financial decisions (e.g., a decision to underwrite or not underwrite a policy).

The goal is to reduce opacity and increase accountability such that AI model users can consider a specific data point and understand the corresponding inputs that

contributed to it. Part of this accountability process is traceability through documentation of datasets, data labeling, and decisions derived from the AI model. The explanation of the outcome must be understandable to all users (i.e., model developers, model validators, risk managers, senior managers, supervisors, and customers). In this context, *explainability* means that further information would not increase model understanding, the user can sufficiently explain AI decisions to a third party, documentation exists for traceability, the process fosters trust, and supervisors have what they need to ensure compliance with laws and regulations.



MODULE QUIZ 97.1

1. An ethical concern related to the use of artificial intelligence (AI) in the context of financial risk management is the:
 - A. enacted Artificial Intelligence Act (AIA).
 - B. potential bias in credit decision-making.
 - C. increase in false anti-money laundering flags.
 - D. inability to keep pace with anti-money laundering violations.
2. Fairness is a key consideration in AI model design. Which of the following actions is not something that model developers can control?
 - A. Ensuring that models limit unfair biases.
 - B. Creating an option for model accountability.
 - C. Ensuring that no bias is in the final outcome of a model.
 - D. Providing an equitable distribution of costs through the model.
3. From the perspective of technical validation, it is important to:
 - A. discard model metadata periodically.
 - B. ignore potential outliers in the training data.
 - C. measure only the model outcomes for fairness.
 - D. archive historical data, parameters, and versions of self-learning models.
4. Regarding the ethical implications of artificial intelligence, which of the following statements is most accurate?
 - A. Synthetic data can be used to mask actual user information.
 - B. Synthetic data is not advisable when two unaffiliated entities are collaborating on an AI model.
 - C. Federated learning ensures that no individual user can be identified in the dataset used in a model.
 - D. Optimization methods are not helpful because the variables of interest are often at opposite ends of the risk spectrum.
5. From the EU Horizon 2020's use case example, it can be seen that:
 - A. clustering by influential input variables can increase model explainability.
 - B. there are no adequate options available to increase trust in model explainability.
 - C. methods to increase trust for risk managers do not address trust issues for supervisors.
 - D. clusters with an equal assortment of "yes" and "no" decisions inspire trust in model accuracy.

KEY CONCEPTS

LO 97.a

Regulation is being considered to address model bias and ethical issues related to the use of artificial intelligence in financial risk management. Many of the concerns relate to self-learning models and the approval and vetting of models. AI models are not transparent (i.e., they have low explainability). This concern leaves the potential for unwanted biases. There are also some specific ethical concerns related to model training data.

LO 97.b

Broadly defined, fairness means ensuring an equitable and just distribution of benefits and costs while minimizing biases and discrimination. There is the potential for AI model design to address some issues of fairness.

LO 97.c

Technical validation is important for any model. In this context, fairness also needs to be considered. There can be issues in the training data or in the lack of explainability in the model. It is important to archive historical data, parameters, and model versions so that results can be reverse engineered to better understand if there are any fairness issues in the outcome of the model.

LO 97.d

It is very important for users to trust that AI models are producing outcomes that consider ethical issues. Matters related to privacy concerns include differential privacy, federated learning, and synthetic data. The explainability of an AI model can be monitored using AI auditing, but there is a tradeoff between some variables of interest (e.g., explainability and model accuracy). Optimization using a Pareto frontier is one way to address this concern.

LO 97.e

The EU Horizon 2020 project presented a use case in explainable AI using an EU insurance group. They attempted to enhance explainability by clustering outcomes and the associated input variables that led to those outcomes. The process allowed them to use AI to deal with complex situations while better understanding the elements that derived the AI decisions. Documentation is essential to increase trust and to provide developers, risk managers, supervisors, and customers with a level of trust that they understand how AI decisions were made.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 97.1

1. **B** The potential for bias is a material ethical concern relative to AI models. These models are opaque, and stakeholders are often not adequately informed about the reasons for credit decisions. Regulation (e.g., the AIA) has been proposed, but not yet enacted. Two benefits of using AI in financial risk management are a reduction in false AML flags and the ability to keep pace with the evolution in

criminal activity. Still, the lack of explainability poses unknown ethical concerns. (LO 97.a)

2. **C** Model developers can create an accountability system, promote equitable and just distribution of benefits and costs, minimize bias and discrimination, and focus on model outcomes that can be controlled. However, some model outcomes cannot be controlled because biases exist in such a way that model design cannot address. (LO 97.b)
3. **D** A vital element in technical validation is to check for fairness as well as model accuracy. To check for fairness, explainability needs to be a factor and historical archives of all data, parameters, and model versions are necessary. (LO 97.c)
4. **A** There are three areas of note connecting privacy laws to trustworthy AI. Differential privacy relates to not identifying any single user in a larger dataset. Federated learning relates to sharing data between unaffiliated parties, and synthetic data creates an artificial dataset with the same statistical parameters as the original dataset. There is a tradeoff between too much model simplification and the accuracy of the output. Optimization using a Pareto frontier is one way to choose a combination of competing priorities. (LO 97.d)
5. **A** The EU Horizon 2020's use case identified clusters of influential input variables and outcomes to enhance explainability. This method uses low computing power and can produce fast results. Enhancing trust through the use of clusters and documentation can provide model developers, model validators, risk managers, supervisors, and customers with detailed information that increases their trust in the decisions made by an AI model. (LO 97.e)

The following is a review of the Current Issues in Financial Markets principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—National Institute of Standards and Technology.

READING 98

ARTIFICIAL INTELLIGENCE RISK MANAGEMENT FRAMEWORK

Study Session 15

EXAM FOCUS

A key objective for artificial intelligence (AI) risk management is to reduce the negative aspects of AI systems while benefiting from the positive ones. For the exam, understand the challenges for AI risk management, especially risk measurement. Also, an understanding of the AI lifecycle and the key dimensions of an AI system is beneficial for understanding the specific tasks of AI actors. In addition, be able to identify and describe the characteristics of trustworthy AI systems and understand how there are inevitable tradeoffs between those characteristics. Finally, be familiar with the four key functions for AI risk management: govern, map, measure, and manage.

MODULE 98.1: AI RISKS AND TRUSTWORTHINESS

LO 98.a: Describe how organizations can frame the risks related to AI and explain the challenges that should be considered in AI risk management.

Risk Management

The **AI risk management framework (AI RMF)** views AI systems as having both positive and negative impacts. In the context of risk management, the goal is to minimize negative impacts and maximize positive impacts. This contrasts with traditional risk management approaches, which focus almost exclusively on minimizing the negative impacts. Proper AI risk management in the form of understanding the weaknesses in the current AI systems and wanting to correct them should result in the development of more reliable AI systems that are beneficial to society.

The AI RMF seeks to manage and mitigate the risks of potential harms from implementing AI technology. These potential harms include compromising individuals' rights and safety, discrimination of specific groups, reputational damage, financial losses, and environmental damage.

Risk Measurement

Risk measurement challenges to consider for AI in the context of risk management include the following:

- *Using third-party data and computer systems* is often necessary, but it adds complexity to risk measurement because additional risks are introduced for the organization using the AI system. For example, the organization developing the AI system may not be forthcoming about the risk metrics it employs, or it employs different risk metrics than the organization that will be using the AI system. Additionally, customers (of the organization using the AI system) may not always have the necessary controls in place when interacting with third-party data and systems, which adds further complexity to risk measurement.
- *Identifying and measuring new/incremental risks* is a challenge for the obvious reason that little or nothing is known about such emergent risks.
- Currently, there are *no universally accepted processes or metrics* specifically tailored for AI risk measurement. Using generic metrics or forcing the use of metrics (that are ultimately unsuitable) may provide misleading conclusions or results that are useful only in specific contexts. Some measurement processes lack sophistication or are potentially biased.
- Risk measurement may yield *completely different results depending on which stage of the AI lifecycle* is being measured, because some risks develop (or even decline) over time. An AI system could be thought of as having three stages: design, development, and deployment, and the approach to risk measurement would likely differ in each of those stages.
- The actual level of AI risk in a *real-world setting may differ substantially from what is measured in a theoretical (e.g., pre-deployment) setting*.
- Some AI systems are complicated and difficult to understand; the *nontransparent nature* of such systems makes risk measurement a greater challenge.
- The purpose of AI systems is to *act as a substitute for human decision-making*, so proper measurement and comparison requires benchmark (i.e., baseline) metrics. However, what AI systems do and the way they do it is not the same as it is for humans; therefore, such metrics would be difficult or impossible to determine.

Risk Tolerance

The level of risk tolerance for an organization or AI actor will depend on the context and the specific use case, which limits comparisons. As policies and conventions evolve in the AI world, risk tolerances will continually change. Among organizations, risk tolerances will also differ due to differing business objectives. For these reasons, attempting to specify risk tolerances will complicate the AI risk management process.

Risk Prioritization

It is not optimal to reduce all negative risks to zero simply because some negative events are too costly or impossible to eliminate. The challenge for AI risk management is to determine which risks are the greatest for the AI systems. These risks will initially

be given the highest priority in terms of risk mitigation tactics. Excessive risk that comes with potentially significant losses or harm may require temporary cessation of development and deployment activities. If a risk is determined to be low, then it will be given less priority, but the pitfall here is that the risk may actually be greater than expected.

Another facet of prioritization comes in the form of AI systems dealing with sensitive and confidential information, or those that clearly impact people—both types of AI systems are given higher risk prioritization. In contrast, if the AI system does not deal with private information or it impacts only nonhumans, then it might initially be given a lower risk prioritization. Regardless, the challenge remains to engage in ongoing assessment and risk prioritization because such lower-risk AI systems may still possess peripheral impacts that negatively affect humans and the environment.

Additionally, there will likely be *residual risk* remaining after attempting to mitigate risks. The system provider will need to fully document these residual risks. This documentation will help advise end users of the risks from using the AI system.

Organizational Integration

An additional challenge with AI risks is that they need to be thought of in an integrated manner within the organization. For example, AI risks could be considered with other key risks, such as cybersecurity, to result in a more unified organizational risk management strategy. In that regard, efficiencies may occur because there are similar risks (e.g., privacy, data and systems security, environmental effects) that impact AI systems and those of non-AI systems.

To achieve this integration, organizations face the initial challenge of setting up and monitoring the necessary controls, duties, and incentives. However, the greater challenge comes in the form of potential changes required in the organizational culture, which comes from the senior levels of management (i.e., “tone from the top”).

As a general note, small- and medium-sized organizations are likely to face different challenges than large organizations when it comes to mitigating AI risks and implementing the AI RMF.

AI Actors

LO 98.b: Identify AI actors across the AI lifecycle dimensions and describe how these actors work together to manage risks and achieve the goals of trustworthy and responsible AI.

AI lifecycle stages have been segregated into five key categories by the Organization for Economic Co-operation and Development (OECD). These categories are as follows: (1) People and Planet, (2) Application Context, (3) Data and Input, (4) AI Model, and (5) Task and Output. Figure 98.1 displays these categories in the two inner circles as well as the corresponding AI lifecycle stages in the outer circle. In the best-case scenario, the risk management process begins at the *Plan and Design* lifecycle stage and proceeds clockwise around the outer circle.

Figure 98.1: AI System Categories and Lifecycle Stages¹



Notably, the *People and Planet* category is focused on human rights and overall societal and environmental well-being. AI actors involved in this category include end users, advocacy and environmental groups, and researchers. These actors could help analyze AI impacts, advise suggested limits for employing AI, provide guidance on AI risk management, and discuss the cost-benefit analysis of the AI system in terms of the economy, the environment, and society.

Key **AI actors** within the AI lifecycle are shown in Figure 98.2. The big picture here is that these AI actors cooperate with each other in a risk management context, with the ultimate objective to create a trustworthy and responsible AI. Note that *TEVV* refers to the test, evaluation, verification, and validation processes that occur on an ongoing basis during the corresponding AI lifecycle stage. TEVV could be useful for determining and monitoring potential or new risks. The ongoing use of TEVV provides opportunities to fix problems early and to prevent or reduce them in the future.

Figure 98.2: AI Actors Across Lifecycle Stages

Lifecycle Stage	TEVV Tasks	AI Actors
Plan and Design	Audit and impact assessment	System operators; end users; AI designers; impact assessors
Collect and Process Data	Internal and external validation	Data scientists; data engineers; data providers
Build and Use Model	Model testing	Modelers; model engineers; data scientists; developers
Verify and Validate	Model testing	Modelers; model engineers; data scientists; developers
Deploy and Use	Integration, compliance testing, and validation	System integrators; developers; system and software engineers
Operate and Monitor	Audit and impact assessment	System operators, end users, and practitioners; AI designers
Use or Impacted By (People and Planet)	Audit and impact assessment	End users; general public; policymakers; advocacy and environmental groups; researchers

Proper risk management will be achieved with the cooperation of all AI actors. The greater the level of diversity among the actors (e.g., professions, experiences), the greater the potential for more productive brainstorming sessions—including the ability to find creative solutions for existing and new risks.

Trustworthy AI

LO 98.c: Describe the characteristics of trustworthy AI and analyze the proposed guidance to address them.

There are seven specific characteristics of **trustworthy AI** that will be described in detail in this section. These characteristics are as follows:

1. Valid and reliable
2. Safe
3. Secure and resilient
4. Accountable and transparent
5. Explainable and interpretable
6. Privacy enhanced
7. Fair (by managing harmful bias)

A trustworthy AI system has an appropriate mix of these characteristics depending on the purpose of the AI system. *Accountable and transparent* is a fundamental characteristic, as it is pervasive to all other characteristics. *Valid and reliable* is also fundamental as it relates to all characteristics (except *accountable and transparent*).

The characteristics of trustworthy AI systems need to be measured using appropriate metrics and there needs to be appropriate thresholds for success and failure. This

measurement has some subjective elements—and, therefore, requires human judgment. It is also important to recognize that the approach to the characteristics must be holistic in nature and not done merely on an individual basis. In practice, some characteristics will have higher priority than others in certain scenarios, and the overall concept of trustworthiness follows the principle of “it is only as good as its weakest link.”

Furthermore, the concept of tradeoffs between trustworthy characteristics is always a challenge when managing AI risks. There are many tradeoff scenarios to consider, including: higher interpretability will often come at the cost of lower privacy, and higher accuracy may come at the cost of lower interpretability. The ultimate consideration with tradeoffs is highly dependent on what decision needs to be made, and because managing the tradeoff is largely subjective, it should be adequately justified.

Valid and Reliable

A *valid AI system* has gone through an unbiased testing process to confirm that the system works properly for its planned use.

A *reliable AI system* takes the elements of validation as a starting point. It then goes further by considering the time frame (e.g., life of the AI system) and performance without failure.

An *accurate AI system* will compute amounts that are close to the true values (subject to a stated acceptable margin of error). Additionally, accuracy must consider other factors such as Type I and Type II errors and the ability to generalize to the “real world” as opposed to merely the testing scenarios. Accuracy may be enhanced when disaggregating and disclosing the results for numerous datasets, but this may come at the price of privacy because there is greater transparency with disaggregation.

A *robust (or generalizable) AI system* can function at the acceptable performance level in different scenarios. Not only must the system perform properly for its planned use (i.e., validity), but it may also need to perform well for other unplanned (but reasonable) uses. It should have low potential harm in the unplanned cases. There is a natural tradeoff between accuracy and robustness—so in simplified terms, a more accurate system may only be usable in a narrow range of circumstances, but a less (but acceptably) accurate system may be usable in a wider range of circumstances.

Safe

A *safe AI system* will not harm and will not place individuals, property, or the environment in danger. AI systems that are safe will have been designed, developed, and deployed with safety in mind during the earliest stages as well as provide guidelines on how to use the systems responsibly to avoid misuse or dangerous situations. Any incidents involving safety need to be clearly documented, along with the resulting risks to attempt to prioritize and rigorously mitigate the most critical ones (e.g., major threats to health or risk of death).

AI safety can be further enhanced through extensive testing and simulation that is done on an ongoing basis as well as more intense oversight. A “kill switch” to automatically shut down or modify an AI system may be desirable if there is suspected misuse of the system.

Secure and Resilient

A *resilient AI system* can survive or continue to operate normally in the face of unplanned negative events or changes.

A *secure AI system* is one that is sufficiently guarded to only allow authorized use and access. It can, therefore, provide service continuity, confidentiality, and with integrity despite threats (internal or external) to access.

Security addresses resilience, but security also encompasses controls to prevent, deal with, or recover from cyberattacks. Resilience addresses robustness (described previously), but resilience also encompasses misuse or abuse of the system or data.

Accountable and Transparent

For AI to be accountable, it must first be transparent. *Transparent AI* provides all system participants with relevant information regarding the system and the output. The information may be targeted and personalized to specific AI actors or users. The idea is to increase understanding by individuals, which will enhance their confidence in the system. Furthermore, transparency is very useful when investigating inaccurate output or negative impacts in an effort to determine cause(s) and to prevent future occurrences.

Accountability refers to the outcomes of the AI systems, and in that regard, the focus is on the AI actors. There is a positive correlation between risks taken and the associated level of accountability. Thus, when there are potentially serious and damaging outcomes from AI use (i.e., increased risk), then accountability (and by implication, transparency) must be increased. By having risk management processes in place, accountability within the organization can be strengthened.

Attempts to improve accountability and transparency need to consider key issues such as the availability of resources and the need to protect sensitive and confidential data. In that regard, there is an obvious tradeoff between transparency and privacy. Documentation that clearly ties the training data to the decisions made by the AI system enhances accountability and transparency when such documentation is needed when rectifying any AI system failures.

Explainable and Interpretable

AI is *explainable* if the operational details behind the AI system can be easily conveyed. It is *interpretable* if the output can be linked to the planned purpose and the output may be assessed and conveyed to interested parties.

By combining explainability and interpretability, developers and users are able to clearly understand the steps involved with how the AI system works—and, therefore, increase their trust in the system and its output. Different levels of explanations should

be provided, as the various users will likely have a wide range of roles and levels of understanding. For interpretability, providing clear justifications to support the AI system's conclusions will help users interpret results.

In short, in the context of AI systems and their output, transparency can be thought of as *what happened*, explainability as *how it happened*, and interpretability as *why it happened*.

Privacy Enhanced

Privacy covers concepts including anonymity and confidentiality, which should be accounted for during all stages of the AI lifecycle. A more sophisticated AI system may allow for inferencing by using multiple data sources, which could ultimately unmask the identity of individuals or remove the privacy from what was formerly private information.

Privacy and security go together—an increase in one will result in an increase in the other. The use of privacy-enhancing technologies (PETs) comes with tradeoffs, however, as increased aggregation of data is privacy enhancing, but it reduces transparency. In addition, when data is scarce, increasing privacy could reduce accuracy and fairness.

Fair

In an AI context, *fairness* deals with avoiding harmful bias and discrimination. The concept of fairness is problematic, as what is considered fair is not the same over a large population, and it may change depending on the situation or context. The removal of harmful bias does not always make for a fair system. For example, a geographic bias may be removed, but that action could leave an element of unfairness if there is still a lack of access to AI systems by those who are economically disadvantaged.

Biases can be subdivided into the following types: (1) systemic bias, (2) computational and statistical bias, and (3) human-cognitive bias. *Systemic bias* could be embedded in data and in the AI lifecycle in general. *Computational and statistical bias* likely occurs from poor sampling processes that contain too many unrepresentative items. *Human-cognitive bias* is the most pervasive of the biases and refers to how humans may (incorrectly) think that AI uses information to make conclusions or (incorrectly) think what the objective is of the AI system.

Because of the prevalence of bias, it may be embedded in an automated AI system. The power of AI systems could inadvertently spread such biases, potentially increasing harm to certain individuals or groups of society.



MODULE QUIZ 98.1

1. Regarding risk management challenges, the AI risk management framework (AI RMF) deals directly with which of the following risk features?
 - A. Risk prioritization and risk tolerance.
 - B. Risk measurement and risk tolerance.
 - C. Risk measurement and risk prioritization.

- D. Risk measurement, risk prioritization, and risk tolerance.
2. Which of the following AI lifecycle stages is most likely covered within the key dimension of application context?
- A. Deploy and Use.
 - B. Plan and Design.
 - C. Verify and Validate.
 - D. Build and Use Model.
3. Within the context of AI systems and their output, which of the following characteristics of trustworthy AI systems is best thought of as “what happened in an AI system?”
- A. Fairness.
 - B. Explainability.
 - C. Transparency.
 - D. Interpretability.

MODULE 98.2: FUNCTIONS OF AI RISK MANAGEMENT

AI RMF Effectiveness

LO 98.d: Explain the potential benefits of periodically evaluating AI risk management effectiveness.

A periodic evaluation process of AI risk management effectiveness considers an organization's policies, processes, actions, implementation plans, metrics, expected outcomes, and other factors. Benefits of this periodic evaluation include the following:

- More robust approach regarding overseeing, mapping, measuring, and mitigating AI risks, together with more thorough outcome documentation practices.
- Greater understanding of how trustworthiness characteristics, sociotechnical approaches, and AI risks are integrated, together with the inevitable tradeoffs that need to be made.
- Clearer decision-making framework regarding designing and deploying AI systems.
- Developed set of rules and best practices regarding accountability within the organization for AI risks.
- Stronger AI risk culture within the organization that places importance on determining and mitigating AI risks to reduce the negative impacts to individuals and society.
- Enhanced system of communicating relevant information (internally and externally) regarding AI risks, risk management roles, and best practices.
- Better understanding of downstream risks.
- Enhanced TEVV processes for AI systems.

AI RMF Core Functions

LO 98.e: Describe specific functions applied to help organizations address the risks of AI systems in practice.

There are four key functions for AI risk management: (1) govern, (2) map, (3) measure, and (4) manage. The *govern* function is designed to be pervasive across the other three functions. AI risk management should be ongoing and carried out during the entire AI system lifecycle. With the establishment of proper governance, it is often the case that *mapping* is the first step, followed by *measuring* and then *managing*; however, the order of the functions is discretionary and based on what is most beneficial to the user.

Govern

The **govern function** involves the establishment of an internal risk management culture as well as the creation of a process for determining risks, analyzing the implications of those risks, and mitigating the risks from the AI system. Effective governance results in the AI risk management process as well as the technical aspects of the AI system and design being consistent with the organization's operational and business objectives. In addition, governance deals with the entire product lifecycle and related issues, such as the use of third-party information and computer systems.

As mentioned, the govern function is comprehensive and ongoing. In that regard, certain areas within governance, such as compliance and evaluation, might be seen in the other three functions as well. For example, a robust system of governance must come from the top (e.g., board of directors) in terms of setting the mission, values, goals, and risk tolerance. From there, senior management implements an appropriate risk management culture throughout the organization. Management staff at lower levels should clearly show the connection between the technical details of risk management and organizational policies.

Categories pertaining to the govern function² include the following:

- Policies, processes, procedures, and practices across the organization related to the mapping, measuring, and managing of AI risks are in place, transparent, and implemented effectively.
- Accountability structures are in place so that the appropriate teams and individuals are empowered, responsible, and trained for mapping, measuring, and managing AI risks.
- Workforce diversity, equity, inclusion, and accessibility processes are prioritized in the mapping, measuring, and managing of AI risks throughout the lifecycle.
- Organizational teams are committed to a culture that considers and communicates AI risk.
- Processes are in place for robust engagement with relevant AI actors.
- Policies and procedures are in place to address AI risks and benefits arising from third-party software and data and other supply chain issues.

Map

The **map function** sets up the risk framing within the AI system in terms of determining context and then the risks in relation to the context. In the absence of mapping, the various AI actors involved in the different lifecycle stages are not likely to know of or be able to influence what is being done in the other stages. Given that there is significant integration between the AI actors and their tasks, it creates the problem that the work done in an earlier stage of the lifecycle by one AI actor is inadvertently compromised in a later stage by another AI actor. The end result is a disorganized risk management process with which users are unable to place any confidence.

By being able to determine (negative) risks and their causes during the map function, the objective is to use that information to reduce risk and determine whether an AI system is required or not.

The map function requires the input of a wide range of opinions from both sources internal and external to the organization. Gathering as much information as possible results in benefits to the organization, such as (1) knowing the strengths and weaknesses of their AI system, (2) knowing what real-life constraints on AI can lead to undesirable results, (3) predicting potential undesirable results when AI is used as planned, and (4) knowing the risks that may occur when AI is used for purposes that are unintended. Ultimately, these benefits will hopefully allow for better risk prevention and the creation of trustworthy AI systems.

From the conclusions obtained during the map function, the process moves to the measure and manage functions, assuming that the relevant decision-makers provide approval to continue with using the AI system. At the same time, the map function is not a one-time process, and it may continue to be used as AI systems and their features change over time.

Categories pertaining to the map function³ include the following:

- Context is established and understood.
- Categorization of the AI system is performed.
- AI capabilities, targeted usage, goals, and expected benefits and costs compared with appropriate benchmarks are understood.
- Risks and benefits are mapped for all components of the AI system, including third-party software and data.
- Impacts to individuals, groups, communities, organizations, and society are characterized.

Measure

The **measure function** is a mixture of both numerical and nonnumerical methods. It is a bridging step that uses the information on AI risks from the map function and develops conclusions that will subsequently be used during the manage function.

Measuring risk includes documentation of system functionality and trustworthiness. In addition, metrics for trustworthy features and social impact should be used. Within the measure function, tasks performed include thoroughly testing software and developing

performance assessment techniques. Regarding assessment, actual-to-budget comparisons and standardized reporting documentation are necessary.

As noted previously, trustworthy characteristics may require tradeoffs, so the process of measurement provides an objective basis to make such tradeoff decisions. Based on those decisions, the AI system may require a partial or complete overhaul, or it may be taken out of use completely. Alternatively, a range of controls (e.g., detective, recovery) may be put in place.

A full set of metrics (both quantitative and qualitative) should be developed and available for use by the end of the measure function. Consideration must always be given to developing new metrics as the AI environment evolves. The results from the measurement function will be carried over to the manage function for risk oversight purposes. Similar to the earlier discussion on the map function, the measure function is not a one-time process and may continue to be used as AI systems and their features change over time.

Categories pertaining to the measure function⁴ include the following:

- Appropriate methods and metrics are identified and applied.
- AI systems are evaluated for trustworthy characteristics.
- Mechanisms for tracking identified AI risks over time are in place.
- Feedback about efficacy of measurement is gathered and assessed.

Manage

Within the **manage function**, things come full circle as risk resources are distributed to the mapped and measured risks within the context of the govern function. At this step, risks are prioritized, and risk mitigation takes place. In addition, action plans are developed for addressing any incidents and errors.

Data from the govern and map functions are used in the manage function to help reduce the risk of system failures. The documentation processes from the govern, map, and measure functions assist with the risk management process and promote greater transparency and accountability. Analyzing new risks would also occur during the manage function. As usual, the manage function is not a one-time process and may continue to be used as AI systems and their features change over time.

Categories pertaining to the manage function⁵ include the following:

- AI risks based on assessments and other analytical output from the map and measure functions are prioritized, responded to, and managed.
- Strategies to maximize AI benefits and minimize negative impacts are planned, prepared, implemented, documented, and informed by input from relevant AI actors.
- AI risks and benefits from third-party entities are managed.
- Risk treatments, including response and recovery, and communication plans for the identified and measured AI risks are documented and monitored regularly.



MODULE QUIZ 98.2

1. Which of the following reasons is least likely to be considered a direct benefit of frequent periodic evaluations of AI risk management effectiveness?
 - A. Stronger risk culture.
 - B. Reduction in operating costs.
 - C. More thorough documentation processes.
 - D. Better understanding of downstream risks.
2. Which of the following functions within AI risk management specifically focuses on risk prioritization?
 - A. Map.
 - B. Govern.
 - C. Manage.
 - D. Measure.

KEY CONCEPTS

LO 98.a

The AI risk management framework (AI RMF) views AI systems as having positive and negative impacts. Within the context of risk management, the goal is to both minimize the negative impacts and maximize the positive impacts.

Risk measurement challenges to consider for AI include the use of third-party data and computer systems, determining and measuring new or incremental risks, the lack of universally accepted processes or metrics specifically tailored for AI risk measurement, different risk measurement results depending on the stage of the AI lifecycle, differing measurement levels in real-world versus theoretical settings, complicated AI systems that are difficult to understand, and the lack of appropriate benchmark metrics.

Additional challenges exist regarding risk tolerance, risk prioritization, and organizational integration.

LO 98.b

AI lifecycle activities have been segregated into the following five categories:

1. People and Planet (Use or Impacted By)
2. Application Context (Plan and Design; Operate and Monitor)
3. Data and Input (Collect and Process Data)
4. AI Model (Build and Use Model; Verify and Validate)
5. Task and Output (Deploy and Use)

AI actors cooperate with each other in a risk management context with the ultimate objective to create a trustworthy and responsible AI system.

LO 98.c

The seven specific characteristics of a trustworthy AI are as follows:

1. Valid and reliable

2. Safe
3. Secure and resilient
4. Accountable and transparent
5. Explainable and interpretable
6. Privacy enhanced
7. Fair

LO 98.d

Benefits of a periodic evaluation of risk management effectiveness include the following:

- More robust approach regarding overseeing, mapping, measuring, and mitigating AI risks.
- Greater understanding of how trustworthiness characteristics, sociotechnical approaches, and AI risks are integrated.
- Clearer decision-making.
- Development of a set of rules and best practices regarding accountability within the organization for AI risks.
- Stronger AI risk culture within the organization.
- Enhanced system of communicating relevant information.
- Better understanding of downstream risks.
- Enhanced TEVV processes.

LO 98.e

There are four key functions for AI risk management:

1. Govern
2. Map
3. Measure
4. Manage

The govern function involves the establishment of an internal risk management culture as well as the creation of a process for determining risks, analyzing the implications of those risks, and mitigating the risks from the AI system. Effective governance results in the AI risk management process as well as the technical aspects of AI system and design being consistent with the organization's objectives.

The map function sets up the risk framing within the AI system. By being able to determine (negative) risks and their causes during this function, the objective is to reduce risk and determine whether or not an AI system is required. The map function requires the input of a wide range of opinions from both sources internal and external to the organization. Gathering as much information as possible results in benefits to the organization, such as (1) knowing the strengths and weaknesses of their AI systems, (2) knowing what real-life constraints on AI can lead to undesirable results, (3) predicting potential undesirable results when AI is used as planned, and (4) knowing the risks that

may occur when AI is used for purposes that are unintended. Ultimately, these benefits will hopefully allow for better risk prevention and the creation of trustworthy AI systems.

The measure function is a mixture of both numerical and nonnumerical methods. It is a bridging step that uses the information on AI risks from the map function and develops conclusions that will be used during the manage function. Measuring risk includes documentation of system functionality and trustworthiness. In addition, metrics for trustworthy characteristics and social impact should be used. Within the measure function, tasks performed include thoroughly testing software and developing performance assessment techniques.

Within the manage function, things come full circle as risk resources are distributed to the mapped and measured risks within the context of the govern function. At this step, risks are prioritized, and risk mitigation takes place. In addition, action plans are developed for addressing any incidents and errors.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 98.1

1. **C** Although the AI RMF is used for risk prioritization, it does not prescribe risk tolerance. Risk tolerance is determined independently of the AI RMF. Risk measurement is a crucial part of the AI RMF. (LO 98.a)
2. **B** Within the key dimension of application context, the two applicable lifecycle stages are *Plan and Design* and *Operate and Monitor*. (LO 98.b)
3. **C** Within the context of AI systems and their output, transparency can be thought of as *what happened*, explainability as *how it happened*, and interpretability as *why it happened*.

Transparent AI provides all system participants with relevant information regarding the system and the output. The idea is to increase understanding by individuals, which will enhance their confidence in the AI system. Furthermore, transparency is very useful when investigating inaccurate output or negative impacts in an effort to determine cause(s) and to prevent future occurrences. (LO 98.c)

Module Quiz 98.2

1. **B** Although a reduction in operating costs may be an indirect benefit in future periods, it is least likely to be a direct benefit in the current period. In fact, there are incremental costs incurred in performing more frequent evaluations. (LO 98.d)
2. **C** Within the manage function, risks are prioritized, and risk mitigation takes place. In addition, action plans are developed for addressing any incidents and errors. (LO 98.e)

¹ OECD (2022), OECD Framework for the Classification of AI Systems—OECD Digital Economy Papers.

² NIST AI RMF Playbook, https://airc.nist.gov/AI_RMF_Knowledge_Base/Playbook

³ NIST AI RMF Playbook, https://airc.nist.gov/AI_RMF_Knowledge_Base/Playbook.

⁴ NIST AI RMF Playbook, https://airc.nist.gov/AI_RMF_Knowledge_Base/Playbook.

⁵ NIST AI RMF Playbook, https://airc.nist.gov/AI_RMF_Knowledge_Base/Playbook.

READING 99

CLIMATE-RELATED RISK DRIVERS AND THEIR TRANSMISSION CHANNELS

Study Session 15

EXAM FOCUS

This reading examines the potential impacts of climate risk drivers on banks through both physical and transition risk drivers. For the exam, understand the distinction between physical and transition risk drivers. Also, be familiar with the analysis of various transmission channels, including both microeconomic channels (which link a bank's financial risks to counterparty risk) and macroeconomic channels (which link a bank's financial risks to broad market factors). The two main risks facing banks from these exposures are credit risk and market risk, however, other risks like liquidity risk and operational/reputational risks can also be prominent. Finally, understand the impact of geographic location, and the effect of amplifiers (which increase risks) and mitigants (which decrease risks) in the context of a bank's financial risk exposures.

MODULE 99.1: CLIMATE-RELATED RISK DRIVERS

This reading discusses how financial risks of banks and the global financial system are impacted by climate-related risks, which is one of many risks that affect the operations of financial institutions. The findings are based on the work of the Basel Committee on Banking Supervision (Basel Committee), including its Task Force on Climate-Related Risks (TFCR). The relationship between climate-related changes and a bank's financial risks is based on the following components:

1. *Climate-related risk drivers* are climate risks that could create financial risks for banks and are classified as either physical or transition risks. Physical risks include weather and climate changes, including both acute and chronic physical hazards. Transition risks relate to changes in risks from moving to a low carbon economy.
2. *Transmission channels* relate to the channels that directly or indirectly impact a bank's financial risks, including microeconomic and macroeconomic drivers of climate risk.

3. *Causes of variability* refer to factors that influence the probability and magnitude of the impact of climate risks on banks, including the location of banks or their assets, and the impact of other climate risk drivers.
4. *Financial risks* include credit, market, liquidity, and operational risks, which are the four key risk categories that can be impacted by climate risks.
5. *The Basel framework* provides the regulatory backdrop and refers to the core principles of the four financial risks.

LO 99.a: Describe climate-related risk drivers and explain how those drivers give rise to different types of risks for banks.

Climate-related risk drivers refer to climate changes that impact economies but could also create financial risks for banks. They are usually segmented into physical risks and transition risks. Physical risks arise from weather and climate-related changes and include both acute and chronic physical hazards. Transition risks relate to societal changes from a transition from a carbon-reliant to a low carbon economy.

Global temperatures have risen notably over the last century, and will continue to rise given the increase in greenhouse gas (GHG) emissions globally caused by human activity. Rising GHG emissions cause increased environmental risks, including rising sea levels, increasing rainfalls, and resulting mudslides, but also increases in droughts and wildfires. As a result, curbing the increase in GHGs is critical and will require regulatory and technological changes as well as changes in human behavior.

Physical and Transition Risk Drivers

LO 99.b: Compare physical and transition risk drivers related to climate change.

Physical risk drivers include weather and climate-related changes and can be further classified into acute physical risk drivers related to extreme weather events (e.g., floods, wildfires) and chronic physical risk drivers related to climate shifts (e.g., rising temperatures, rising sea levels). Predicting these risks is challenging due to differences in the frequency and severity of risks and because they may appear with a lag. About 60% of global natural disasters are meteorological or climatological in nature.

Acute physical risks are related to extreme weather events. These include heatwaves, floods, wildfires, and a range of storms (e.g., hurricanes and cyclones). There is significant scientific evidence that rising global temperatures create acute risks generating both higher incidence of heatwaves and wildfires, while also generating more severe rainfalls and flooding since warmer air can hold more moisture. Acute risks often compound, with some economies experiencing wildfires followed by flooding or droughts. These events can, and have, caused significant damages to local economies creating considerable financial losses.

Chronic physical risks are related to climate shifts, and include rising sea levels and temperatures and ocean acidification. Prolonged periods of rising temperatures can cause changes in ecosystems and can contribute to lands becoming deserts. They may

also contribute to migration of large amounts of population within a country from dry to less dry areas. Melting ice and glaciers can also contribute to rising sea levels and the erosion of coastal land and cities.

The impact of physical risks will be different globally and may impact some regions more severely than others.

Transition risk drivers relate to societal changes from transitioning from a carbon-reliant to a low carbon economy. Several factors contribute to these changes, including: (1) changes in public climate policies, (2) innovation and changes in technology, (3) changes in investor behavior, and (4) changes in consumer attitudes toward green initiatives.

Climate policies relate to public measures and regulations to control pollution and facilitate energy transition toward a low carbon economy. Approaches differ globally, and may include explicit policies to reduce GHG emissions or to provide subsidies for electric vehicle production.

Technology relates to changes toward greener, less polluting and more energy efficient technologies. This could create additional costs for corporations to adopt more expensive technologies or to operate with higher costs (e.g., due to carbon levies). Technology also relates to adopting technologies that result in greater environmental efficiency, for example, improved energy storage or the adoption of new climate-resilient infrastructure.

Investor sentiment relates to a greater awareness and changing attitudes by investors to climate risk, and directly incorporating climate considerations into their investment decisions. Large investors like fund managers can also act as activists by pushing for greater change in climate considerations.

Consumer sentiment relates to changes in human behavior toward climate risks and changes in consumer preferences. This could relate to more efficient energy use or more climate-friendly consumption. A bank's clients or investors may request the bank to adopt more climate-friendly policies or to direct its investments toward greener products.

Assessing climate risks needs to account for a significant amount of uncertainty in the evolution of these risks and should incorporate the four factors just described. Relying on historical experience may be incorrect given the speed at which changes occur. The nonlinearity of risks (frequent changes) and geographical diversification also increases uncertainty.

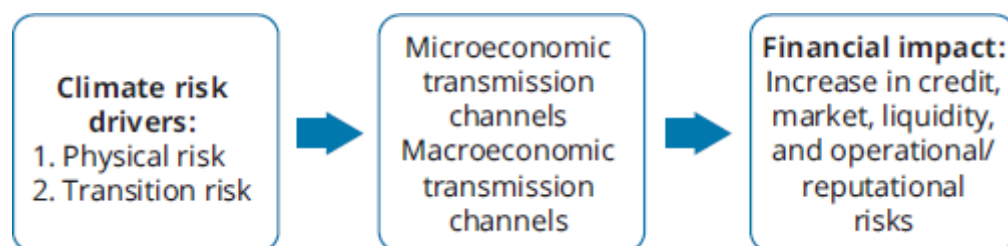
Microeconomic and Macroeconomic Drivers

LO 99.c: Assess the potential impact of different microeconomic and macroeconomic drivers of climate risk.

Transmission channels link climate event impacts to a bank's financial risk. Two classifications are microeconomic channels, which link climate risk to a bank's financial risk and exposure to individual counterparties, and macroeconomic channels,

which link climate risk to a bank's exposure to broad economic factors. Climate risks impact credit, market, liquidity, and operational/reputational risks as shown in Figure 99.1.

Figure 99.1: Financial Impact of Climate Risk Drivers and the Role of Transmission Channels



Microeconomic Transmission Channels

Microeconomic transmission channels look at how banks can be impacted both directly and indirectly from climate risk factors through their exposures to counterparties and financial assets.

Credit Risk

Both physical and transition risk drivers from climate events can create credit risk for banks. They can affect a borrower's capacity to service or repay debt obligations, or they can result in a bank's inability to fully recover losses from borrower default if the pledged collateral values are insufficient.

Physical risk drivers primarily impact banks indirectly through counterparty risk. Physical hazards can reduce counterparty capital and asset values, resulting in lower wealth, lower income, and lower cash flows to service debt. Transmission channels of physical risk drivers include households, corporates, and sovereigns and subnational institutions.

- **Households:** Severe physical climate risk events like floods and hurricanes often lead to property value declines from damaged real estate and land. This is especially true for properties located in coastal areas or in floodplains. Lower property prices lower bank equity ratios and increase nonperforming asset ratios.
- **Corporates:** Weather events primarily impact corporates through reduced profitability. Natural disasters can lead to lower corporate sales, reduced equity values, and contribute to supply chain issues. Risk varies greatly by reliance on energy use, geography, and sector. For example, the rise in global temperatures will have a greater impact on the agricultural sector given declining crop yields and lower corporate profitability. In turn, this increases credit risk to lenders including banks.
- **Sovereigns and subnational institutions:** Sovereigns and subnational governments are impacted by severe climate events through lower tax income from weaker corporates and households, and through higher spending to boost the economy. This could lead to higher borrowing costs and weaker access to debt capital markets, resulting in elevated credit risk to banks from their sovereign and regional and municipal exposures.

Transition risk drivers have a more limited impact on a bank's credit risk since banks historically have not experienced significant losses from a transition to a low-carbon economy. Credit risk is therefore mainly evaluated through scenario analysis of government policy, technological change, and consumer and market sentiment.

- *Government policy:* Changes in government policy can directly impact corporate profitability and credit quality. For example, new or increased taxes on GHG emissions reduce corporate earnings and profitability, thereby reducing their creditworthiness which could lead to limited funding access or higher funding costs, and reduced ability to repay debt obligations. In turn, this increases the credit risk for banks that lend to these corporates. Corporates may also be faced with stranded assets from reserves that have been rendered uneconomical given the transition to a low carbon economy. If these reserves are used as collateral for bank lending, banks face increased credit risk of collateral value declines.
- *Technological change:* Adopting new technologies is expensive which reduces profitability; however, the new technologies can give adopters a competitive advantage over other entities. Banks lending to entities that are unable to adopt to new technologies will face higher potential credit losses.
- *Consumer and market sentiment:* Both consumer and market sentiment (expectations) may change as a result of a transition to low-carbon economies. An increased expectation of future negative climate events can lead to greater awareness of climate change and more action to prevent adverse outcomes. Those sectors that continue to produce in less environmentally friendly ways may see a negative effect on brand recognition, and less favorable funding conditions. These changing expectations can influence investment decisions with fund managers and investors favoring "green" investment alternatives and divesting investments in entities with high carbon emissions. Increased litigation against fossil fuel producers has also resulted in increased costs and reduced wealth for these entities, elevating a bank's credit risk through higher borrower default risk.

Market Risk

Market risk captures the change in value of financial assets to changes in economic factors like interest rates, exchange rates, price shocks, and changes in volatility. Climate transition risks can reduce asset market values, lead to a breakdown in correlations and diminish the effectiveness of hedges. All of this can raise the risk to banks.

- Physical risk drivers can manifest in higher volatility given climate events. As a result, the stock options of entities located in regions impacted by climate risks can experience higher implied volatilities, increasing market risk.
- Transition risk drivers relating to transitioning to a low-carbon economy impact the creditworthiness of bank borrowers resulting in higher risk premiums for carbon-intensive borrowers and may result in lower valuations of financial assets that are used as collateral.
- Pricing of climate risks can help corporates and banks minimize their sensitivity to future significant weather events. There is some evidence that asset prices, including

municipal and corporate debt and some equities, already reflect transition risk, although pricing is challenged by a lack of consistent methodologies and disclosures. The evidence for pricing-in transition risk for corporations is mixed, although equity prices of corporates in carbon-intensive industries is seen as partly reflecting climate risks. Similarly, evidence is mixed whether residential real estate prices reflect flood risk and sea level rise risk. A bank's use of historical data to assess market risk and price assets may not be appropriate given the significant increase in recent risks in the market.

Liquidity Risk

Climate risks can raise the liquidity risk of banks given their capacity to raise funds, their ability to liquidate assets in order to meet obligations, and given reduced liquidity levels. Following severe climate events—including the 2011 earthquake in Japan—banks often offer significant liquidity support to the market to ensure stability, but this can negatively impact liquidity buffers. The liquidity needs of a bank's counterparties can also reduce bank liquidity. For example, following a natural disaster, households and corporations may require liquidity from banks to finance their obligations and pay for the recovery, however, this puts a strain on bank liquidity.

Operational and Reputational Risk

Operational risk arises mainly from inadequate controls within a bank, employee mistakes, and breakdowns in internal processes and systems. These factors can impact a bank's reputation. There is mixed evidence if climate events result in operational and reputational risks, although there are some indications that corporations and banks could be subject to legal and regulatory compliance risk, especially from climate-related lawsuits.

Macroeconomic Transmission Channels

Macroeconomic transmission channels look at how banks can be impacted from climate risk factors through their ties to the economy. Macroeconomic factors primarily impact banks through credit risk and market risk (further research is needed for liquidity and operational impacts).

Credit Risk

A bank's credit risk can be impacted through multiple ways, including impaired asset values or stranded assets, higher debt costs, challenges in accessing debt markets, or weaker counterparty creditworthiness. Here we look at the impact of both physical and transition risk drivers on banks.

Physical risk drivers:

- Climate changes can lead to lower economic output given higher mortality rates, lower labor productivity, lower supply, and lower agricultural output. In fact, it is estimated that global economic output will decline by \$2 trillion by 2030. Research indicates that natural disasters disproportionately affect poorer municipalities since

they lead to higher costs from physical damage, higher borrowing costs, and even higher underwriting fees to reflect the greater credit risk of these municipalities.

- Similarly, major climate risk events impact developing countries to a greater degree than developed countries. Climate risks have already increased borrowing costs in developing countries, which can lead to higher taxes, lower government spending, and lower economic activity. In turn, these weaknesses impact a bank's country-level credit assessment.
- Climate risks can also impact socioeconomic changes that in turn have an effect on economic growth. Although it is difficult to quantify the impact of extreme weather factors like droughts and flooding, they can lead to significant damages that could create physical conflicts. These impacts could affect the creditworthiness of a bank's borrowers, raising the bank's credit risk.

Transition risk drivers:

- Transition risk drivers impact banks primarily through weakened counterparty creditworthiness following climate risk events. The transition away from fossil fuels toward a low-carbon economy could result in impaired sovereign incomes and stranded assets. For example, an estimated 80% of fossil fuel reserves globally will become stranded. In turn, this could impair their ability to service debt and result in weaker credit ratings, impacting bank ratings within that country. Lower country GDP could also negatively impact household income, resulting in lower borrower credit quality.

Market Risk

While the implications of physical climate events for market risk are less evident, two of the key impacts are a sovereign's ability to access debt capital markets and rising debt costs. Higher sovereign risk can result in credit downgrades, which could impact the creditworthiness of banks within that country. Banks could also be exposed to higher losses on government debt, reduced collateral values, and increased funding costs.

Factors That Impact Climate-Related Risks

LO 99.d: Describe and assess factors that can amplify the impact of climate-related risks on banks as well as potential mitigants for these risks.

In addition to the risk drivers we have discussed so far, the probability and magnitude of the impact of climate-related risks on banks is also determined by three other factors: (1) geographical heterogeneity that determine the extent of the impact, (2) amplifiers of the impacts, and (3) mitigants that moderate the impacts.

Geographical Heterogeneity

A bank's exposure to climate risks varies by geographical location, by sector, by jurisdiction, and by the nature of the bank's business. Broad geographical factors like

weather patterns and even political systems could influence risks as well as differences in markets, economies, and financial systems.

Both top-down and bottom-up analysis could be used to assess a bank's climate-related financial risks, which could vary based on the geographical location of the bank and of their assets.

Climate Risk Drivers

While average global temperatures are rising, there is considerable variance in temperatures across jurisdictions. Banks that operate in higher acute or chronic physical risk locations, including those impacted by frequent droughts or flooding, will experience more adverse impacts on their operations. Research suggests that banks in many jurisdictions including in Asia, Europe, and Latin America have been impacted by extreme weather events over the last few decades.

Differences in transition risk also exist due to differences in government policies, technological innovation, and changes in attitudes. The speed and type of transition also influence exposure. For example, jurisdictions that were early adopters of low-carbon policies are likely less exposed to sudden changes in risk factors later. However, there needs to be more detailed research on these risk drivers since they are not widely discussed in existing research.

Economic and Market Structure

Differences in economic and social policies, differences in financial systems, and the availability of insurance can have a significant impact on a country's—and therefore its financial system's—exposure to climate risks. Emerging market economies have been shown to be significantly more exposed to physical and transition risks than advanced market economies. In addition, characteristics like low debt-to-GDP factors could provide resilience to risks in the short term, while a flexible exchange rate, strong infrastructure, financial sector stability, and low levels of inequality can provide long-term resilience to climate risks.

Societal and sector differences also matter in the level of risk exposure. Because rising temperatures adversely impact variables like mortality and labor supply, countries that depend more on labor and outside activities (e.g., agriculture-dependent economies) will be more adversely impacted by weather changes. Countries and regions that are more dependent on fossil fuels or on tourism would also be negatively impacted by adverse climate changes. However, insurance can offer an important mitigating role in transferring risks.

Financial Systems

The structure of financial markets and a country's banking system and the state of development of its capital markets can impact a bank's exposure to climate risks. For example, countries which have a large share of small local banks relative to the share of large banks are likely to see increased lending to consumers and businesses following adverse weather events. While this improves local economic development, it could increase concentration risk if lending is to a small number of borrowers. Some banks

may securitize these climate-risk related assets, especially in the U.S., which could help with redistributing credit risk. Additional research would be needed in this area to gain a better picture of recent exposures.

At the same time, developing economies typically have less liquid financial and capital markets, which could increase exposures and risks to climate risks. Capital markets also differ by the level of development of their derivatives markets. Derivatives are critical in hedging risks, including credit exposures to climate events. For example, it has been shown that the U.S. options market is sensitive to transition risks, such as the uncertainty from climate regulations, as well as physical risks, such as forecasts for hurricane activity.

Amplifiers

Amplifiers are risk factors that enhance climate risks' impacts on banks. They include: (1) interactions between risk drivers, (2) financial amplifiers, and (3) the combined impact of multiple risk drivers.

Interactions Between Risk Drivers

While it is important to understand the individual physical and transition risk drivers, their combined impact on a bank's risks is also important. For example, this could include the concurrent adoption of climate-related policies and the introduction of technological developments. When multiple climate events occur at the same time, they can also increase a bank's climate risk.

Financial Amplifiers

Financial amplifiers are those factors whose combined interaction amplifies a bank's financial risk and could create or enhance losses. The behavior of market participants and interaction with the economy are two examples of amplifiers, often through feedback loops. For example, during heightened climate risk periods, banks may reduce lending or reduce providing insurance coverage to participants. Reduced insurance coverage may make insurance more scarce and more expensive in the future, further exposing individuals and businesses to climate-related risks. Government intervention may be necessary to control the rise in risk premiums for climate insurance (e.g., flood, fire). The lack of insurance itself can amplify financial risks. For example, banks may require borrowers to have insurance policies in place, but they would not monitor if these borrowers renew their insurance policies annually, magnifying the banks' financial risks.

Combined Impact of Risk Drivers

Amplification of risks could also occur when a risk driver impacts a bank through multiple channels, including through both microeconomic and macroeconomic channels. For example, a climate risk event could impair the bank's economic environment (macro factor) while at the same time reduce the creditworthiness of the bank's clients (micro factor). Another example is through feedback loops, for example, when a physical risk event like a hurricane damages homes and therefore reduces household wealth, this will in turn increase a bank's credit risk to those homeowners.

Mitigants

Mitigants reduce a bank's exposure to financial risks and can be both *proactive*, which are done preemptively (e.g., diversification) and *reactive*, which are put in place after a risk event (e.g., hedging and insurance). Banks cannot take the availability, reliability, and costs of mitigants for granted. Many mitigants, especially derivatives and insurance used in hedging, must be rolled over or renewed in the future, but the availability of these instruments and their future costs (insurance premiums may rise) are not known upfront and therefore create uncertainty.

The availability and level of mitigants vary by country and by the state of development of a country. Climate-related risks are often pushed to less developed segments of the financial markets which increases financial stability risk. The state of a country's physical and financial infrastructure can significantly help in mitigating a country's exposure to climate risks. Countries with well-built and interconnected infrastructure can reduce the financial losses from damages from climate events, while strong access to financial markets can reduce borrowing costs and provide greater access to funding when needed.

Financial mitigants can be categorized as: (1) bank behavior and business models, (2) insurance, (3) capital markets and securitization, and (4) hedging.

Bank Behavior and Business Models

Historically, a bank's investment strategies looked at climate risk as an idiosyncratic risk affecting individual assets, and therefore banks hedged this risk through diversifying their portfolios. However, diversification proved much less effective in the face of rising temperatures and extreme weather events. The magnitude of a bank's exposure to both physical and transition risks depends on the bank's business models and on the type of risk event. For example, borrowers in high-risk exposure sectors including transportation, fuel-based industries, and manufacturing that is energy intensive are much more sensitive to a sudden shift to a low-carbon economy, which could adversely impact their balance sheets (creating stranded assets) and financial positions. Reducing these exposures is therefore critical. This can be done through a steady reduction in climate-sensitive assets or an increase in financial resources.

Banks can engage in risk reduction in several ways:

- Following climate risk events, especially natural disasters, banks have been observed to reduce their exposure to climate-sensitive assets. For example, following Hurricane Sandy in 2012, banks in affected areas tightened their lending requirements to corporations by lending at higher interest rates and requiring greater collateral from borrowers.
- Banks can strengthen their balance sheet by improving their capital positions. For example, many banks in hurricane-impacted areas strengthened their capital buffers by improving their risk-based capital ratios following the hurricanes.
- Banks have also been observed to reduce their exposure to transition risk drivers by lending at shorter loan maturities, reducing access to permanent forms of financing,

and corporations are subject to increasing shadow bank participation in syndicated deals.

Not all banks focus on risk reduction. Following natural disasters, many banks focus on economic support by increasing lending in the most severely impacted areas, often by reallocating credit from less impacted areas.

Insurance

Insurance is important because it mitigates losses against physical risks and improves a bank's financial resilience since the insurer would pay for covering financial losses.

Insurance can be used directly by a bank against losses, or by a bank's counterparties.

Insurance can significantly reduce the losses resulting from natural disasters. Evidence suggests that financial and economic costs from natural catastrophes are primarily from uninsured losses. Insured losses in areas impacted by flooding or earthquakes and other natural disasters are typically small, and the economic impact can even be positive if payouts help support reconstitution efforts in an area.

Market participants' perception of risk and the mandatory nature of risks in many instances can increase insurance coverage. For example, insurance demand in coastal areas in the U.S. and in flood-risk areas is typically price inelastic and more open to bank lending requirements. However, how much insurance individual borrowers will buy depends on their perception of risk, their age, and their level of education.

A significant shortcoming of insurance is that historically it has only covered a portion of losses, including 51% in 2018 but only 38% in 2019 of global economic losses from natural disasters. This protection gap is even more acute in developing nations, covering only 10% of losses on average. To protect against capital losses, several insurance-linked products have been developed, including catastrophe bonds, which act as a risk transference mechanism for climate risk-related losses. Types of insurance-linked securities include parametric catastrophe bonds, which are based on a trigger event like wind speed or rainfall, rather than on loss measures. The advantage of these bonds is that their triggers can be determined quickly and precisely. This market has gained prominence in recent years, growing to \$40 billion in 2020.

Capital Markets and Securitization

Securitization is a form of risk transference, and securitizing riskier loan portfolios can significantly help banks reduce their on-balance sheet risk. This practice is particularly popular in the U.S. where banks often hold geographically concentrated portfolios. These banks often increase lending following natural disasters to help with economic growth, and then securitize these loans. In fact, mortgage and loan lending in disaster-impacted areas may increase specifically as a result of the climate events. Because these loans and mortgages carry significantly higher risk of default and payment delinquencies, these products are often securitized below the conforming loan limit (this is the dollar threshold for mortgage buying by government-sponsored enterprises).

Hedging

A special derivatives type used for hedging weather-related risks are **weather derivatives**. These are derivatives whose payout depends on weather events, and are popular among the insurance, energy, tourism, and entertainment sectors. While they provide useful protection against localized risks, they have proven to be less effective against providing protection of global climate risk events. However, recent innovation in climate derivatives like environmental, social, and governance (ESG) futures and carbon derivatives help bridge some of this gap.



MODULE QUIZ 99.1

1. A bank's risk officer is evaluating climate-related risk drivers that could create financial risk for the bank. The risk officer has classified rising global temperatures and wildfires as acute physical risks. The risk officer is correct with respect to:
 - A. wildfires only.
 - B. rising global temperatures only.
 - C. both wildfires and rising global temperatures.
 - D. neither wildfires nor rising global temperatures.
2. A Canadian bank is worried that a corporate borrower will be unable to successfully adopt to new technologies to address rising climate risks, resulting in higher losses on its loans to the corporation. This risk is best characterized as:
 - A. credit risk through microeconomic channels.
 - B. operational risk through microeconomic channels.
 - C. credit risk through macroeconomic channels.
 - D. operational risk through macroeconomic channels.
3. A risk manager notes that macroeconomic transmission channels primarily relate to the impact of severe climate events on a bank's credit and market risks, while microeconomic transmission channels primarily relate to liquidity and operational risks. The manager is correct with respect to:
 - A. microeconomic channels only.
 - B. macroeconomic channels only.
 - C. both microeconomic and macroeconomic channels.
 - D. neither microeconomic nor macroeconomic channels.
4. Following a recent hurricane, a local U.S. bank reduced the overall amount of lending to corporate borrowers impacted by the hurricane, and began charging higher loan interest rates. The bank's actions are examples of:
 - A. financial amplifiers.
 - B. financial mitigants.
 - C. stranded assets.
 - D. risk transference.
5. Which of the following factors least likely provides long-term resilience to climate risks?
 - A. Low debt-to-GDP.
 - B. Flexible exchange rate.
 - C. Low levels of inequality.
 - D. Well-built infrastructure.

KEY CONCEPTS

LO 99.a

Climate-related risk drivers in the context of the banking sector refer to climate changes that impact economies but could also create financial risks for financial institutions. Two types of climate-related risks include: (1) physical risks and (2) transition risks.

LO 99.b

Physical risks include weather and climate-related changes, and can be classified into acute physical risks related to extreme weather events and chronic physical risks related to climate shifts.

Acute physical risks relate to extreme weather events like heatwaves, floods, and wildfires. These risks often compound and can cause significant economic damages and financial losses.

Chronic physical risks are related to climate shifts like rising temperatures causing droughts, but also rising sea levels from changes like melting ice and glaciers.

Transition risks relate to societal changes from the transition to a low carbon economy. These changes are influenced by: (1) changes in public climate policies, (2) innovation and changes in technology, (3) changes in investor behavior, and (4) changes in consumer attitudes toward green initiatives.

Evaluating climate risks should factor in the evolution of risks which create uncertainty in outcomes, and should also account for other factors like the nonlinearity of risks which increases uncertainty.

LO 99.c

Transmission channels link climate event impacts to a bank's financial risk and can be classified as microeconomic channels (a bank's financial risk and exposure to individual counterparties) and macroeconomic channels (a bank's exposure to broad economic factors).

Microeconomic transmission channels look at how banks can be impacted both directly and indirectly from climate risk factors through their exposures to counterparties and financial assets. These mainly result from physical climate risk and can impact a bank's credit, market, liquidity, and operational/reputational risks.

- Credit risk can arise from a bank's exposure to weaker counterparties following severe climate events given counterparties' lower asset (including real estate) values, lower profitability, collateral value declines, and higher borrower default risk.
- Market risk is the risk of a change in value of financial assets given changes in broad economic factors.
- Liquidity risk relates to a bank's capacity to raise funds, its ability to liquidate assets on time, and the risk of reduced liquidity levels. Following a natural disaster, banks also often supply liquidity to impacted households and corporations which puts a strain on bank liquidity.
- Operational risk can arise from inadequate controls within a bank, employee errors, and breakdowns in internal processes and systems, all of which could adversely

impact a bank's reputation.

Macroeconomic transmission channels mainly impact banks indirectly through their ties to the economy and the sovereign. The two key risks facing banks are credit risk and market risk.

- Credit risk arises from physical and transition risk drivers that affect national and local economies, leading to impaired asset values, higher borrowing costs, and reduced economic output, especially in developing countries. These in turn could impact the creditworthiness of a bank's borrowers and raise the bank's credit risk.
- Market risk arises from climate risks that impact a sovereign's ability to access debt capital markets and could lead to higher debt costs. Banks in turn could be exposed to losses on government debt, reduced collateral values, and increased funding costs.

LO 99.d

The impact of climate-related risks on banks is determined by: (1) geographical heterogeneity, (2) amplifiers of the impacts, and (3) mitigants that moderate the impacts.

Geographical heterogeneity refers to the variability to a bank's exposure to climate risks by geographical location, by sector, by jurisdiction, and by the nature of the bank's business. Geographical heterogeneity depends on:

- Climate risk drivers which include factors like rising global temperatures, droughts, and flooding and their impacts on a bank's operations. These drivers are also impacted by differences in government policies, technological innovation, and changes in attitudes.
- Economic and market structure which looks at how differences in economic and social policies and financial systems and the availability of insurance impact a financial system's exposure to climate risks.
- Financial systems which look at how the structure of financial markets and a country's banking system and the state of development of its capital markets can impact a bank's exposure to climate risks.

Amplifiers are risk factors that enhance climate risk impacts on banks, and include:

- Interactions between risk drivers which look at the combined impact of risks on a bank's exposure to climate risks.
- Financial amplifiers which look at the combined interaction of risk drivers specifically on a bank's financial risk.
- Combined impact of risk drivers which look at the impact on banks from multiple microeconomic and macroeconomic channels.

Financial mitigants are risk offsets that reduce a bank's exposure to financial risks.

Categories of mitigants include:

- Bank behavior and business models, which suggest that exposure to both physical and transition risks depends on a bank's business models and on the type of risk event, and a bank's efforts to reduce these exposures (tighter lending conditions, lending with shorter maturities, improved capital positions).

- Insurance which mitigates losses against physical risks. Insurance improves financial resilience by covering financial losses. Market participants' perception of risk and lender requirements can increase insurance purchases. However, insurance only covers a fraction of the costs in most countries (worse in developing countries).
- Capital markets and securitization, which looks at the role of securitizing as a form of risk transference, especially for banks which hold geographically concentrated portfolios.
- Hedging which primarily relates to the widening use of weather derivatives which protect against localized risks.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 99.1

1. **A** Wildfires are correctly classified as acute physical risks, which relate to extreme weather events. Rising global temperatures should be classified as chronic physical risks, which relate to climate shifts like rising global temperatures and rising sea levels. (LO 99.b)

2. **A** The risk of climate events impacting counterparties is captured under microeconomic transmission channels. If the corporation is unable to adopt to new technologies to address climate risks, this will result in higher credit risk (the risk of nonpayment/default on an obligation by one of its borrowers) and potential losses for the bank.

Macroeconomic transmission channels involve a bank's exposures to broad economic factors like interest rates, inflation, commodity prices, and exchange rates. Market risk is the risk of asset value changes in response to changing economic conditions. (LO 99.c)

3. **B** The two main risk impacts from climate events via both microeconomic and macroeconomic channels are credit risk and market risk. Microeconomic channels also include the impact on liquidity risk and operational/reputational risks. (LO 99.c)

4. **B** Financial mitigants are risk offsets that reduce a bank's exposure to financial risks. Categories of mitigants include bank behavior and business models, insurance, capital markets and securitization, and hedging. A bank's tightened credit lending is an example of bank behavior within financial mitigants.

Financial amplifiers are amplifiers that look at the combined interaction of risk drivers specifically on a bank's financial risk. Stranded assets are assets that no longer earn their expected economic return usually as a result of moving to a low-carbon economy. Risk transference refers to offloading risk from one party to another, either in the form of insurance or securitization. (LO 99.d)

5. **A** Low debt-to-GDP could provide short term resilience to climate risks. A flexible exchange rate, strong infrastructure, financial sector stability, and low levels of

inequality can provide long-term resilience to climate risks. (LO 99.d)

The following is a review of the Current Issues in Financial Markets principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Basel Committee on Banking Supervision.

READING 100

CLIMATE-RELATED FINANCIAL RISKS—MEASUREMENT METHODOLOGIES

Study Session 15

EXAM FOCUS

The focus of this reading is on climate-related risks and how banks and supervisors measure and model these risks to understand financial and operational impacts. For the exam, be familiar with the primary issues to date in identifying and measuring financial risks tied to climate change. Understanding these risks requires unique data and the various methodologies used to analyze the data. Also, know the methodologies already in place and those in development, as well as strengths and weaknesses of those approaches. Finally, modeling frameworks present several gaps and challenges that must be addressed going forward.

MODULE 100.1: INTRODUCTION TO CLIMATE-RELATED FINANCIAL RISKS

In April 2021, the Basel Committee on Banking Supervision issued the “Climate-Related Financial Risks—Measurement Methodologies” report. This report reflects on five critical findings:

1. Forward-looking and granular measurement methodologies are needed due to the unique features of climate-related financial risks.
2. Banks and supervisors have focused on mapping near-term transition risk drivers into portfolio and counterparty exposures.
3. Banks and supervisors have primarily focused on credit risk assessments as they try to translate climate-related exposures into financial risk categories.
4. The pace of translating climate-related risks into quantifiable financial risks has increased.
5. Future analytical exploration is needed due to measurement gaps in data and risk classification methods. More exploration is also needed for developing

methodologies suitable for assessing nonstandard, long-term climate issues.

There continues to be uncertainty in the following three primary areas:

1. Intrinsic uncertainty related to physical and transition risk driver projections
2. Data gaps, which create measurement uncertainty
3. Model-based uncertainty

Climate change could have profound financial and economic impacts on banking institutions. To effectively manage the risk associated with climate change, these institutions should be identifying significant climate transmission channels and risk drivers, mapping and measuring climate-related exposures, and translating climate-related risks into financial risk metrics that can be quantified.

A risk assessment framework for banks and supervisors should incorporate the following:

- Risk identification (risk drivers and transmission channels)
- Exposure mapping and measurement (metrics and indicators, as well as methodological considerations)
- Risk quantification (applying various measurement approaches to economic models and forward-looking assessments)
- Risk management (determining how to manage and mitigate risk exposures)

Considerations and Issues With Climate-Related Financial Risks

LO 100.a: Describe main issues in identifying and measuring climate-related financial risks.

Banks and supervisors should consider the following concepts when mapping exposures and measuring financial risks.

Transition and Physical Risk Measurement

Transition risk results from the shifting of a high-carbon economy to a low-carbon economy. **Physical risk drivers** or hazards are applied to financial exposures through damage functions. These functions define specific hazard impacts on activities and real assets which generate financial cash flows. The impacts can be incorporated into risk models that apply financial risk parameters. A damage function is driven by time horizon, hazard severity, location uniqueness, and relevant sectors.

While transition and physical risks may be assessed separately, their dependencies may merit assessment in tandem. As physical risks increase, government policymakers may feel pressure to reduce future physical risks—and, therefore, increase the level of transition risk.

Exposure Granularity

Clients and counterparties for banks include individual households, corporations, government entities, and other financial institutions. Banks must determine the level of exposure granularity needed for each of these parties. The factors that influence these decisions include relevant data availability, specific risk drivers (transition and physical), the higher computational complexity associated with increased granularity, and the risk management decisions that the assessment is supporting.

While higher granularity methodologies are more valuable for valuations and underwriting, they can be highly complex and suffer from input data quality. Lower granular methodologies may be better suited to portfolio allocations and strategic planning, as they are less complex and more tolerant of sporadic data.

Top-Down and Bottom-Up Approaches

Top-down approaches begin by measuring risk at an aggregated (general) level and then breaking it down into its component parts. Industry or sector averages are determined and then applied to individual exposures. The logic is that any differences between companies within an industry and sector are minimal.

Bottom-up approaches measure risks at a component level and aggregate them up to a consolidated view. Risk exposure assessments at individual loan or counterparty levels may be summed to provide banking sector-views or portfolio level-views of risk exposure. A challenge with this approach is understanding any potential correlations among risk exposures.

Incorporating Risk Mitigation and Risk Reduction

Potential risk mitigation is a key element of climate-related financial risk measurement. While financial risks may be mitigated through government regulations and measures, banks and supervisors can take proactive measures as well. Gross exposure is a measure of climate risk impacts without the incorporation of mitigation strategies. Net exposure is a measure of climate risk impacts after accounting for proactive offsetting strategies, such as derivatives hedging and insurance. Gross exposure is important to capture not only because seeing the magnitude can help inform decisions, but also because mitigation strategies may change, lapse, or fail to function as designed. Modeling gross exposures and mitigation strategies separately may be helpful.

Heterogeneities

Because of the uniqueness of each individual bank's portfolio and the exposures that each client and counterparty faces (which differ due to factors such as geography, technology, political environment, sector, location, etc.), heterogeneities impact the selection of measurement approaches. Geography is significant because climates are different in various locations. Climate risks tend to be similar across a given sector, even if individual clients within that sector are impacted differently. Policies and regulations may also differ by individual client or counterparty.

Sources of Uncertainty

Though modeling for climate risk has improved considerably, there is still the risk of misestimating risk. In particular, there is the concern of underestimating climate risk. Inevitably, assumptions must be made regarding interactions between the climate, human activity, economic activity, and government activity. Forecasting future emissions, technology, policy actions, and economic consequences is a challenge to the development of a model. In addition, climate change has a long time horizon that requires significant assumptions regarding timing and discounting.



MODULE QUIZ 100.1

1. A CFO estimates a climate-related financial risk of \$35 million. She incorporates insurance with a benefit of \$15 million and derivatives contracts that would provide a benefit of \$3 million. The gross exposure associated with the financial risk is closest to:
 - A. \$17 million.
 - B. \$20 million.
 - C. \$32 million.
 - D. \$35 million.
2. If a bank wants to employ a risk management methodology for strategic planning purposes, it will most likely use a model that has:
 - A. low granularity and low complexity.
 - B. high granularity and low complexity.
 - C. low granularity and high complexity.
 - D. high granularity and high complexity.

MODULE 100.2: DATA NEEDS AND MEASUREMENT METHODOLOGIES

Unique Data Needs

LO 100.b: Identify unique data needs inherent in the climate-related risks and describe candidate methodologies that could be used to analyze these types of data.

Unlike the data that is used for traditional financial risk analysis, the data needed for climate-related financial analysis is unique and may not be fully available. In addition, historical relationships between climate impact and financial or economic consequences may not be particularly useful in developing forward projections, given the dynamics of climate change.

The following three data categories are useful for assessing climate-related financial risk:

1. *Data related to physical and transition risk drivers translated into economic risk factors.*

This category includes information on climate and current/projected hazard events. Driver data can come from commercial third parties, academic organizations, and

government agencies. Survey agencies also provide “off-the-shelf” data useful in identifying geographic areas exposed to physical hazards and areas that are particularly vulnerable to physical hazards. Data can be used to project macroeconomic and microeconomic variables for different climate-related scenarios.

2. Data linking climate-adjusted economic risk factors to exposures.

Vulnerability to risk factors is critical for banks and supervisors to understand. Exposure vulnerability data can then be used to translate economic risk factors to financial exposures. Geospatial counterparty locations are used to determine bank exposure vulnerabilities to physical risks, while counterparties’ economic activity within specific areas can drive bank exposure vulnerabilities to transition risk.

With physical risks, the concern is damage to inventory and property, value chain disruptions, and declines in region or sector activity. Aligning physical hazards to vulnerable locations will help in estimating economic impact and cash flow disruptions. Understanding how corporate, retail, and municipal borrowers are interconnected can help with determining how weather events can negatively impact a local economy. For transition risk exposures, assessing vulnerability requires data on counterparty sectors and their associated carbon sensitivities.

3. Financial exposure data used to translate economic risks into financial risks.

Data used to project valuations, prices, and cash flows may be useful in translating exposure vulnerabilities into estimates of financial loss. Exposure at the bank system level requires data on portfolio composition and relevant counterparty information. The impact of climate risk on bank liquidity requires data on funding provider behaviors in response to climate risk drivers.

Climate risk classifications are helpful in ranking exposures based on their vulnerability to and associated financial risk from climate-related economic risk factors. Traditional financial risk measures such as leverage ratios, debt service coverage ratios, and loan-to-value ratios may have unique exposures to climate-related risks. The ability to incorporate dynamic features into a classification scheme that accounts for technological advances and ever-changing climate impacts is critical.

In creating a climate risk classification scheme, geographic location is important, as it is a primary driver of physical risk differentiation. Proximity and vulnerability to physical hazards are obviously significant factors. For transition risk, jurisdiction plays the most important role, as laws, regulations, energy grids, and markets drive an individual or entity’s ability to manage this risk. Another useful differentiating factor is economic sector. While greenhouse gas emissions may be quantified as an average number by sector, individual entities within the sector are likely to have unique exposures and vulnerabilities.

Current and Developing Methodologies

LO 100.c: Describe current and developing methodologies for measuring

climate-related financial risks employed by banks and supervisors.

LO 100.e: Identify strengths and weaknesses of the main types of measurement approaches.

Because climate-related financial risk estimations are still in the early stages of development, there are various modeling approaches used in practice. Examples include the following:

- **Integrated assessment models (IAMs).** These models combine economic growth, energy, and climate modeling. They have been very popular in linking projections of greenhouse gas emissions and transition risk drivers to impacts on economic growth. Unfortunately, IAMs have not captured extreme weather economic impacts as they are more useful in capturing recurring physical risks. IAMs also tend to understate severity associated with future climate events, which is exacerbated by the lack of historical data and precedent.
- **Sensitivity analysis.** This analysis has been useful in illustrating the volatility of projected losses due to probabilities assigned to tail events, although underestimating tail events remains a big concern because of the potential financial impact it can have on banks.
- **Input-output models.** These models are useful in estimating the economic impact of policy shifts from climate change events on a region or industry.
- **Computable general equilibrium (CGE) models.** These models are useful in testing policy experiments with complex interactions among agents and sectors in the economy, although the results may be somewhat limited by overall complexity.
- **Dynamic stochastic general equilibrium (DSGE) models.** These models are also used to manage complexity, although the computational burden is significant.
- **Overlapping generation (OLG) models.** These models are a more transparent way to model long-term macroeconomic evolutions.
- **Agent-based models (ABMs).** These models are useful in measuring climate-related impacts because they can better manage complexity and uncertainty. The modeler employs decision-making rules and creates simulations where individuals and institutions within the economy interact with one another. Technological change and damage functions are incorporated into the model. Although ABMs are highly flexible, they require a significant amount of computation and data demand, and the underlying mechanisms driving model outputs are somewhat unclear.
- **Stress testing.** This activity has been useful in linking bank portfolio risk to the modeling approaches just listed.

As mentioned, traditional measures of financial risk do not always apply well to climate-related financial risk assessments due to limited historical data, inconsistent time horizons, and the range of impact uncertainties. Banks and supervisors are currently using the following broad risk measurement approaches:

- **Climate risk ratings/scores.** Risk scores are used to rate climate risk exposures of countries, companies, portfolios, and even assets. Grading criteria (qualitative and quantitative) are used to assign a quality score to the various exposures.

- **Scenario analysis.** This analysis is used to identify projected impacts on individual exposures and portfolios through the identification of physical and transition risk scenarios, the linking of the scenarios to financial risks, the assessment of sensitivities to risks, and the extrapolation of sensitivities to determine an aggregate measure of potential losses and exposure. Forms of scenario analysis include the following:
 - *Stress testing* is used to evaluate an entity's resiliency to economic shocks. Macroprudential stress tests measure the impact of financial shocks to financial systems, while microprudential stress tests are used to evaluate an entity's solvency. Climate-related stress testing assesses the resiliency of an entity and/or its systems to the impacts of severe climate scenarios.
 - *Sensitivity analysis* is used to evaluate the impact of a specific variable on economic outcomes. This approach is particularly useful in evaluating transition risk based on individual policy changes.
- **Natural capital analysis** (third-party approach). This analysis is useful in assessing how the degradation of nature and natural resources (which are finite) negatively impacts an institution. This portfolio-level assessment is carried out by identifying relevant sectors, assets, and geographies; relevant natural capital assets like forests and water; potential disruptions to nature; and sectors, assets, and geographies that are the most vulnerable.
- **Climate value at risk (VaR)** (third-party approach). This risk assessment applies the VaR framework to climate change impacts to an entity's balance sheet.

Risk scenario analyses use forward-looking information in conjunction with historical data to assess the impact of financial and macroeconomic variables from climate-economy models. Baseline scenarios are used as a comparison to scenarios reflecting different risks associated with climate change. Steps in scenario analysis include choosing the scenario, modeling economic impacts, and translating those impacts into financial risk drivers.

While macroeconomic stress testing focuses more on financial statements across a two-to-five-year time frame and is useful for estimating capital needs, climate-related scenario analysis focuses more on specific exposures over various time frames and can help modelers identify the impact to an entity's risk profile and strategy.

The two scenario dimensions for climate-related financial risks are (1) the climate outcome (temperature increase) and (2) the type of transition (gradual or abrupt reductions in emissions and the availability of technology). Banks and supervisors can rely either on in-house developed models or externally generated scenarios coming from the Network for Greening the Financial System (NGFS) or the International Energy Agency (IEA). Drawbacks to these scenarios include the lack of detailed economic damages from physical risk drivers and the lack of extreme event impacts in the model.

Time horizon poses a challenge to models, as physical climate risks extend well beyond the typical capital planning and strategic planning horizons (of two to three years and

three to five years, respectively). Modeling uncertainty is a natural result of this longer time horizon and the necessity of making assumptions about a distant future.

Banks and supervisors must make assumptions about their balance sheets over a given time horizon. The balance sheet can either be static or dynamic. While static balance sheets help illustrate climate risk in current exposures, a static balance sheet becomes less reliable as the time horizon extends beyond current asset durations. Dynamic balance sheets are useful in assessing the impacts from possible strategy adjustments needed to reduce climate risk exposures.



MODULE QUIZ 100.2

1. In gathering the data needed for taking physical and transition risk drivers and translating them into economic risk factors, a bank will likely gather the information from each of the following sources except:
 - A. government agencies.
 - B. retail brokerage firms.
 - C. academic organizations.
 - D. commercial third parties.
2. Which of the following statements regarding broad risk management approaches is most accurate?
 - A. Stress testing is a form of scenario analysis.
 - B. Assigning climate risk ratings and scores is an external third-party approach.
 - C. The VaR framework cannot be applied to a climate-related financial risk model.
 - D. Sensitivity analysis is very useful when assessing the impact of policy changes on physical risk.

MODULE 100.3: CLIMATE-MEASUREMENT STRENGTHS AND WEAKNESSES

LO 100.d: Compare and contrast climate-measuring methodologies utilized by banks, regulators, and third-party providers.

At this point, banks are still in the early stages of developing models and methodologies to quantify and translate exposures into climate-related financial risks. Supervisors have focused more on transition risks and third parties (e.g., academics, rating agencies, think tanks) have also worked on methodologies to assess climate-related financial risks.

Exposure Mapping and Measurement

Bank-level methodologies can be categorized as follows:

- **Portfolio and sector exposures.** Identifying material risk transmission channels within a bank's exposure profile is critical. Key indicators include physical vulnerability of collateral based on location, measures of carbon and emission intensity, and energy label or efficiency real estate distributions. Both qualitative and quantitative data is relevant.

- **Transition risks.** Banks must understand the impact to certain sectors of transitioning to a low-carbon economy. Sectors most at risk include utilities, transportation, oil and gas, construction, and metals and mining. Determining the carbon footprint of bank assets is also valuable in estimating transition risks, and this can be done at a full balance sheet level or just for sectors sensitive to transition risk. *Financed emissions* related to lending and investment activities and *greenness* of financial assets are also valuable measures of transition risk. *Alignment* approaches measure the gap between current portfolios and portfolios consistent with a specific climate target. *Green activities* (like renewable energy) and *brown activities* (like coal and gas-fired power) are also areas for banks to analyze, with a goal of increasing the former and reducing the latter.
- **Physical risks.** Management of physical risk at the portfolio level will focus on geographic concentration of risk, hazard types, probability, and severity. Physical risk scores and geospatial mapping help banks assess where they are exposed to flood risks.
- **Client or project ratings and scores.** Banks are increasingly looking at transition and physical exposures to climate-related financial risk at the individual counterparty level. A bank will evaluate climate-related risks and opportunities for companies that are currently financed or may be financed in the future by banks. This analysis, which includes both qualitative and quantitative components, helps to inform the bank's lending decision. Ratings and scoring mechanisms may be done in-house or by external entities. Qualitative measures include assessing the client's climate strategy, management capabilities in dealing with risks, and vulnerability areas. An example of a risk assessment is a traffic light approach, where *red* indicates high exposure to climate risk and *green* indicates low exposure. Traditional credit scores may be complemented with climate risk scores.

Supervisor-level methodologies use similar methodologies to the bank methodologies just described. High-level categories of transition and physical risk assessments are as follows:

- **Transition risk.** This is assessed and matched to bank exposures using ad hoc surveys or regulatory information. Corporate (non-real estate) portfolios are often assessed by supervisors using indicators related to carbon footprint, emission intensity, and sensitivity to climate policies. A sector-level view is taken to assess which sectors in the economy are most vulnerable to policy change. Real estate tends to have fewer analyses related to transition risk.
- **Physical risk.** Supervisors assess most relevant hazards in their jurisdictions and the specific regions within their jurisdiction that are most vulnerable. Publicly available information and climate risk scores from commercial vendors are helpful in this regard. Key indicators include flood maps, water stress levels from industrial facility locations, populations below specific elevations, agriculture percentage as a component of gross domestic product, and areas most vulnerable to climate change. Exposures to physical risk can be done at a counterparty, activity, sector, or country level depending on the granularity of available data. A supervisor will use the outputs

of these models to assess the percentage of the portfolio in higher risk areas and exposure to specific climate events.

Risk Quantification

Bank-level methodologies employ stress testing and scenario analysis to both inform strategic planning and to quantify climate-related financial risks. Scenario analysis for climate-related financial risks is used to understand the potential impact on specific portfolios, to assess limitations and benefits, and to refine methodologies. Transition risk analysis looks at impacts to credit parameters for counterparties in specific sectors. Physical risk analysis looks at household (mortgage) and corporate exposures, where a key exposure is for sectors that are highly sensitive to long-term weather pattern changes. Chronic physical risks are converted into productivity changes, while acute physical risks are viewed through the lens of extreme weather impacts on real estate collateral.

A series of outreach workshops in 2020 were conducted to gather data on how banks are managing climate-related financial risks. The following are the key takeaways from the workshops:

- Climate-related financial risks are still not fully understood, and there is only initial evidence that these risks are priced into financial markets.
- For transition risks, banks have focused on specific economic sectors. The most relevant sectors were the ones with high carbon footprints (utilities, energy/oil, transportation, gas, construction, etc.).
- For physical risks, banks have focused on economic sectors and geographic regions. The most relevant sectors here were utilities, agriculture, real estate, and retail mortgages.
- Key data sources include public data sources (company annual reports), external rating agency and commercial third-party data, internal client data, regular data collections, and ad hoc requests.
- Heatmaps have been helpful in monitoring and managing portfolio and sector climate-related financial risks.
- Banks are still in the piloting stages in terms of methodologies to quantify climate-related risks, and there is still uncertainty regarding incorporating this information into existing risk management processes.
- Scenario analyses used to assess transition risks have focused on emission-intensive sectors, while analyses used to assess physical risks have focused on the impact of hazards like floods and water-related risks on individual portfolios or on groups of clients.
- Insurance coverage is critical to risk mitigation, although these coverages may change over time.

Supervisors can use stress tests and scenario analysis to inform macroprudential policies and for microprudential supervision. For macroprudential policies, the size and distribution of climate risks to the financial system must be assessed. For microprudential, business model vulnerability and the quantification of financial

exposures vulnerable to climate risk drivers are the focal points. Because of the relatively long time horizon and the nature of these risks, stress tests and climate scenario analysis tend to be more exploratory in nature. The two main objectives served by these analytical approaches are to help supervisors understand the impact of climate change on business and risk management strategies and to raise awareness and incentivize the development of appropriate risk models and the filling of data gaps.

Climate scenarios are translated into financial market and macroeconomic variables. Sector-level calibrations can help supervisors enhance granularity and differentiate risk profiles across sectors. Multicountry, multisector frameworks can also be deployed. Borrower-level analyses may be used to complement macro- and sector-level approaches.

The following are key takeaways from supervisors responding to a survey in 2020:

- Transition risk is a higher priority than physical risk.
- Supervisors are focused on banking system (macro) and individual bank (micro) dimensions.
- Mapping and measuring climate risk driver exposures are the primary methodologies deployed, with stress tests and scenario analyses also heavily used.
- Third-party scenarios and methodologies are the most used, with in-house methodologies planned for the future.
- The primary sources for data are public or borrower reports.
- A sectoral approach is the primary tool used to assess transition risk, with emissions or emission intensity as the primary indicator.
- An aggregate climate risk indicator is the most commonly used metric to assess physical risk, with primary drivers being storms, floods, and droughts. Most physical risks are assessed at the country and regional levels.
- Short time horizons are used for transition risk only, while long time horizons are used for both transition and physical risk. Time ranges are from 5 to 30 years.

Third-party approaches continue to be heavily used by banks and supervisors. Features of these methodologies include scenario selection, exposure mapping, transition and physical risk shocks, and impact assessments on financial performance. Sensitivity and vulnerability to climate events is included in these approaches via the identification of resources impacted by climate change and facility exposures. Models may also incorporate adaptive capacities, alternative production sources, and damage functions.



MODULE QUIZ 100.3

1. Based on the bank outreach workshops in 2020, the most relevant sectors for assessing physical risks included all of the following sectors except:
 - A. utilities.
 - B. agriculture.
 - C. construction.
 - D. retail mortgages.
2. Given the challenges of time horizons with climate-related financial risk modeling, supervisors are least likely to apply which of the following time horizons to their risk

modeling?

- A. Long time horizons for physical risks.
- B. Long time horizons for transition risks.
- C. Short time horizons for physical risks only.
- D. Short time horizons for transition risks only.

MODULE 100.4: GAPS AND CHALLENGES IN MODEL FRAMEWORK DESIGN

LO 100.f: Assess gaps and challenges in designing a modeling framework to capture climate-related financial risk.

As noted earlier, the methodologies used to measure climate-related financial risks are still in their infancy in terms of development. Future analytical development will focus on three areas: (1) data availability, (2) modeling complexity, and (3) conceptual challenges.

Risk Differentiation

Transition and physical risk classifications are established to group exposures among identification criteria that approximate and reflect these risks. Aggregate risk classifications, which can be made using publicly available information on vulnerable sectors and countries, are simple to determine and useful to banks and supervisors. However, the identification criteria used may not truly be specific or granular enough for counterparty differentiation. Adding ratings/scores as complements can help, although they may be hard to integrate into day-to-day counterparty risk management. Quality, data availability, and a lack of an established link to financial risk parameters continue to present modeling challenges.

A risk classification system should have “comparability” across banks’ exposures. However, the desire to compare and aggregate bank exposures must be balanced with the desired level of risk differentiation and complexity or granularity needed. The obstacles toward developing comparability include the following:

- **Jurisdiction comparability.** A baseline set of common standards is needed for the methods used to assess risk drivers like physical and transition risks, as well as the metrics used to evaluate the risks. Developing useful, cross-jurisdictional standards is a challenge.
- **Aggregation perspectives.** Determining the risk profile of large banks is challenging, as financial services and bank lending provided to corporations that are active across multiple jurisdictions is difficult to model given the differences unique to each jurisdiction. Consistent bank decisions in line with group-wide policy may conflict with climate risks unique to location.

The biggest obstacle to climate risk measurement process development is data and information availability. Specific limitations include the following:

- Data for transition and physical risk drivers can lie outside the ranges for traditional financial data. The data may be incomplete, lack sufficient granularity, or not be

updated with appropriate frequency.

- While third-party (external) providers try to fill data gaps and assign ratings/scores to transition and physical risks, end users have limited insight into the data scrubbing process or the accuracy of the data itself. Comparability and standardization among providers are also challenges for input reconciliations.
- Some of the data gaps may be filled by counterparties providing proprietary, nonpublic information to banks as part of the lending process. Smaller counterparties may provide more qualitative data than quantitative, which presents challenges. Also, banks may not be able to easily update data after underwriting and onboarding processes have concluded. Public information (both quality and quantity) is often tied to company size, making it difficult to compare counterparties. Common accounting standards and principles related to climate data are nonexistent.
- Supervisors can augment the data gathered by banks via supervisory reports, which provide standardized and recurring data. However, supervisory reporting may not be as granular as needed to truly assess transition and physical risk.

Climate-Related Financial Risk Complexity

Ideally, a modeling framework will allow banks and supervisors to translate climate risk drivers into financial risk parameters in a prospective manner. Three big challenge areas for developing this framework are (1) fully capturing economic effects from climate scenarios, (2) managing long-term horizon assessments, and (3) operational assessment challenges.

- **Capturing economic effects.** Because climate-related financial risks are complex, banks and supervisors have a difficult time incorporating scenarios into existing risk processes and structures. There is significant uncertainty regarding climate risk drivers and the nonlinearity of the climate system. A robust analysis must consider many alternative scenarios and different modeling approaches using the same scenario. Existing models typically fall short of estimating the full impact of climate change, as chronic physical risks like temperature increases can be modeled but extreme weather events or future disruptive changes are not usually captured. In addition, damage functions from physical risks are typically not granular enough to be fully captured. Historical observations tend to be either missing or not good predictors of the future, which means empirical risk parameters cannot be derived by financial models. There are work-arounds to this challenge, but they require gathering complete data and necessary assumptions regarding the relationship of scenario-related variables and the financial impact to counterparties.
- **Long-term horizon assessments.** Climate-related financial risks require extending time horizons out far longer than typical modeling—in some cases, it is up to 30 years. Reliably assessing risk outcomes is difficult because economic and financial projections become more uncertain the further out the horizon extends. Observed (actual) results are more likely to differ from estimations as time horizons extend. Future efforts should be geared toward addressing increased uncertainty about future portfolio composition.

- **Operational assessment challenges.** Both banks and supervisors must continue to assess climate-related risks by building operational capabilities. The quality of information technology systems and the ability to manage significant quantities of data will drive the bank's ability to assess its overall exposure to climate risks. Climate-related financial risk measurement is interdisciplinary and will require significant investments in both in-house functional expertise and outsourced climate-specific expertise.

Key Takeaways

The following is a list of the key overall takeaways from this report on climate-related financial risks:

- Much of the work done by banks and supervisors to map and measure climate-related risk exposures has focused on transition risk over physical risk.
- Remaining gaps in creating models include climate-relevant risk data from counterparties, disclosure quality, and data consistency across jurisdictions and portfolios.
- As many banks and supervisors have limited in-house expertise, there is a market for third-party provider services and data.
- Areas of focus have been more on credit risk and market risk than on liquidity risk and operational risk.
- Climate-related metrics are used by banks to communicate their strategies on climate change to their stakeholders.
- Translating climate change scenarios into financial risks remains a challenge. Climate and financial risk are often difficult to model together, exacerbated by uncertainty regarding transition and physical risk projections across very lengthy time horizons.
- Insurance mechanisms may be in place to mitigate climate risks, although protection gaps and potential scope changes may limit future coverage.
- Supervisors for internationally active banks face an even greater challenge to be consistent across jurisdictions.
- For measurement, risk classification is a key issue. Classification approaches, to date, have been geared toward disclosure and stakeholder communication. Classifications embedded within risk metrics have also presented comparison challenges. Globally consistent climate risk classifications and sufficient risk mapping are goals.
- Data gaps are also a key measurement concern. Although data collection is expensive, banks and supervisors need accurate and comprehensive data to understand and model counterparty vulnerability, sector and region vulnerability, and real economic and financial system impacts of climate change to translate climate risk into financial risk.
- For methodologies, investments are needed to develop models that can measure forward-looking climate-related risks. Models that can accommodate longer time horizons are important.



MODULE QUIZ 100.4

1. Which of the following statements is most accurate regarding climate-related data gaps?
 - A. Established accounting standards help bridge any data gaps.
 - B. Supervisory reports often lack the granularity needed to assess climate risk.
 - C. Banks can get updated data long after the underwriting process is complete.
 - D. Relatively small counterparties tend to have more quantitative than qualitative data.
2. The obstacles toward achieving comparability across risk classification systems include all of the following except:
 - A. jurisdictional comparability.
 - B. complexity and granularity needs.
 - C. undefined damage functions related to physical risks.
 - D. establishing consistent bank decisions in line with group-wide policies.

KEY CONCEPTS

LO 100.a

Banks and supervisors should consider the following concepts when mapping exposures and measuring financial risks:

- **Transition and physical risk measurement:** Transition risks result from the shifting of a high-carbon to a low-carbon economy. Physical risk drivers or hazards are the specific hazards that impact future financial cash flows.
- **Exposure granularity:** Banks and supervisors must find a balance between the granularity needed and the associated complexity and cost.
- **Top-down and bottom-up approaches:** The former measures risk at an aggregated (general) level and then breaks it down into its component parts. The latter measures risks at a component level and aggregates them up to a consolidated view.
- **Incorporating risk mitigation and risk reduction:** While financial risks may be mitigated through government regulations and measures, banks and supervisors can take proactive measures as well. Gross exposure is a measure of climate risk impacts without the incorporation of mitigation strategies. Net exposure is climate risk impacts after accounting for proactive offsetting strategies, such as derivatives hedging and insurance.
- **Heterogeneities:** Because of the uniqueness of each individual bank's portfolio and the exposures each client and counterparty faces (which differ due to factors such as geography, technology, political environment, sector, location, etc.), heterogeneities impact the selection of measurement approaches.
- **Sources of uncertainty:** Assumptions must be made regarding interactions between the climate, human activity, economic activity, and government activity, which can lead to underestimating risk.

LO 100.b

Three data categories useful for assessing climate-related financial risk are as follows:

- Data related to physical and transition risk drivers translated into economic risk factors
- Data linking climate-adjusted economic risk factors to exposures
- Financial exposure data used to translate economic risks into financial risks

Climate risk classifications are helpful in ranking exposures based on their vulnerability to and associated financial risk from climate-related economic risk factors. In creating a climate risk classification scheme, geographic location is important as it is a primary driver of physical risk differentiation. For transition risk, jurisdiction plays the most important role, as laws, regulations, energy grids, and markets drive an individual or entity's ability to manage this risk.

LO 100.c

Examples of the types of climate-related financial risk estimation models include the following:

- Integrated assessment models (IAMs)
- Sensitivity analysis
- Input-output models
- Computable general equilibrium (CGE) models
- Dynamic stochastic general equilibrium (DSGE) models
- Overlapping generation (OLG) models
- Agent-based models (ABMs)
- Stress testing

Banks and supervisors are currently using the following broad risk measurement approaches:

- Climate risk ratings/scores
- Scenario analysis
- Stress testing
- Sensitivity analysis
- Natural capital analysis (third-party approach)
- Climate value at risk (third-party approach)

Risk scenario analyses use forward-looking information in conjunction with historical data to assess the impact of financial and macroeconomic variables from climate-economy models.

The two scenario dimensions for climate-related financial risks are (1) the climate outcome (temperature increase) and (2) the type of transition (gradual or abrupt reductions in emissions and the availability of technology).

Time horizon poses a challenge to models, as physical climate risks extend well beyond the typical capital planning and strategic planning horizons.

Banks and supervisors must make assumptions about their balance sheets (which can either be static or dynamic) over a given time horizon.

LO 100.d

Bank-level methodologies can be categorized as follows:

- Portfolio and sector exposures
- Transition risks
- Physical risks
- Client or project ratings and scores

Bank-level methodologies employ stress testing and scenario analysis to both inform strategic planning and to quantify climate-related financial risks. Supervisor-level methodologies use similar methodologies to the bank methodologies and are categorized based on managing transition risk versus physical risk.

Key takeaways from the 2020 workshops involving banks include the following:

- Climate-related financial risks are still not fully understood.
- For transition risks, banks have focused on specific economic sectors with high carbon footprints (utilities, energy/oil, transportation, gas, construction, etc.).
- For physical risks, banks have focused on economic sectors and geographic regions. The most relevant sectors here were utilities, agriculture, real estate, and retail mortgages.
- Key data sources include public data sources (company annual reports), external rating agency and commercial third-party data, internal client data, regular data collections, and ad hoc requests.
- Heatmaps have been helpful in monitoring and managing portfolio and sector climate-related financial risks.
- Banks are still in the piloting stages in terms of methodologies to quantify climate-related risks, and there is still uncertainty regarding incorporating this information into existing risk management processes.
- Scenario analyses used to assess transition risks have focused on emission-intensive sectors, while analyses used to assess physical risks have focused on the impact of hazards like floods and water-related risks on individual portfolios or on groups of clients.
- Insurance coverage is critical to risk mitigation, although these coverages may change over time.

Key takeaways from supervisors responding to a survey in 2020 include the following:

- Transition risk is a higher priority than physical risk.
- Most supervisors are focused on banking system (macro) and individual bank (micro) dimensions.
- Mapping and measuring climate risk driver exposures are the primary methodologies deployed, with stress tests and scenario analyses also heavily used.

- Third-party scenarios and methodologies are the most used, with in-house methodologies planned for the future.
- The primary sources for data are public or borrower reports.
- A sectoral approach is the primary tool used to assess transition risk, with emissions or emission intensity as the primary indicator.
- An aggregate climate risk indicator is the most commonly used metric to assess physical risk, with primary drivers being storms, floods, and droughts. Most physical risks are assessed at the country and regional levels.
- Short time horizons are used for transition risk only, while long time horizons are used for both transition and physical risk. Time ranges are from 5 to 30 years.

Third-party approaches continue to be heavily used by banks and supervisors.

LO 100.e

Much of the climate-related financial risk modeling is in its infancy in terms of development, and the longer time horizons make estimations more difficult. However, banks and supervisors are learning how to translate these model outputs into financial risks and to incorporate the outputs into overall risk assessment and strategic planning.

LO 100.f

Future analytical development will focus on three areas: (1) data availability, (2) modeling complexity, and (3) conceptual challenges.

A risk classification system should have “comparability” across banks’ exposures. The desire to compare and aggregate bank exposures must be balanced with the desired level of risk differentiation and complexity or granularity needed. The obstacles toward developing comparability include jurisdiction comparability and aggregation perspectives.

The biggest obstacle to climate risk measurement process development is data and information availability. Specific limitations include incomplete and inconsistent data, the opaqueness of externally provided data, differences in data stemming from counterparty size, and a lack of granularity in supervisory reports.

Three big challenge areas for developing a comprehensive modeling framework are (1) fully capturing economic effects from climate scenarios, (2) managing long-term horizon assessments, and (3) operational assessment challenges.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 100.1

1. **D** The gross exposure represents the amount of risk independent of any risk mitigation strategies employed. The net exposure is the amount of risk remaining after taking mitigation actions. Here, the gross exposure is \$35 million, as the insurance and derivatives contracts only impact the net exposure. (LO 100.a)

2. **A** The more granular a model, the more complexity it will have. So, the combination must be high/high or low/low. For strategic planning purposes, the model does not have to be highly complex. So, the most likely model will have low granularity and low complexity. (LO 100.a)

Module Quiz 100.2

1. **B** While data related to physical and transition risk factors often comes from government agencies, academic organizations, and commercial third parties, retail brokerage firms typically don't provide this information. (LO 100.b)
2. **A** Stress testing and sensitivity analysis are both forms of scenario analysis. Assigning climate risk ratings and scores is done in-house by banks and supervisors. The VaR framework can be applied to climate-related financial risk models. Sensitivity analysis is useful in assessing the impact of policy changes on transition risk. (LO 100.c)

Module Quiz 100.3

1. **C** Construction is a sector with a high carbon footprint, and that is going to be more relevant with transition risk. Utilities, agriculture, and retail mortgages are more relevant sectors for physical risk assessment. (LO 100.d)
2. **C** No matter how good a methodology or approach is, the time horizon will always be a challenge. Long time horizons may be applied for physical and transition risks, while shorter time horizons are typically only for transition risks (and not physical risks). (LO 100.d)

Module Quiz 100.4

1. **B** Supervisory reports are helpful in augmenting the data gathered by banks, but they often lack the granularity needed to truly assess transition and physical risks. Common accounting standards do not yet exist on climate. Banks often cannot get updated data after the underwriting process is completed. Relatively small counterparties tend to have more qualitative data than quantitative data. (LO 100.f)
2. **C** Specific obstacles toward achieving comparability include jurisdictional comparability (risks are different in different places), complexity and granularity needs, and bank decision consistency in line with group-wide policies. Damage functions may not be as robust as needed for physical risk assessments, but that is not a specific obstacle to comparability. (LO 100.f)

The following is a review of the Current Issues in Financial Markets principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Basel Committee on Banking Supervision.

READING 101

PRINCIPLES FOR THE EFFECTIVE MANAGEMENT AND SUPERVISION OF CLIMATE-RELATED FINANCIAL RISKS

Study Session 15

EXAM FOCUS

The effects of climate change could include physical and transition risks that negatively impact the stability and continuity of all banks (regardless of size) as well as the overall financial system. This reading outlines 12 broad principles on managing climate risks followed by 6 broad principles pertaining to supervisors.¹ This material is general in nature—so, for the exam, focus on understanding the principles described.

MODULE 101.1: GOVERNANCE, CONTROL, CAPITAL, LIQUIDITY, AND RISK MANAGEMENT PRINCIPLES

LO 101.a: Describe the principles for managing climate-related financial risks related to corporate governance and internal control framework.

Corporate Governance

There are three distinct principles to consider pertaining to corporate governance.

Principle 1: Banks should develop and implement a sound process for understanding and assessing the potential impacts of climate-related risk drivers on their businesses and on the environments in which they operate. Banks should consider material climate-related financial risks that could materialize over various time horizons and incorporate these risks into their overall business strategies and risk management frameworks.

Key physical and transition risks must be included in formulating business plans, including an assessment of their potential effects on the robustness of the bank's business model over various time periods as well as effects on meeting the bank's goals.

This requires a determination of how climate risks will affect a bank's response to changes in the economy, the financial system, and the competitive environment.

The entire assessment should include the board and senior management in the stages where they are needed; the assessment methodology adopted by the board must be thoroughly articulated to the nonsenior bank management and staff. The impact of climate risks could be extended to account for possible amendments to employee remuneration within the bank.

Internal risk management policies must be congruent with statements made externally by the bank regarding initiatives and objectives pertaining to climate risk.

Principle 2: The board and senior management should clearly assign climate-related responsibilities to members and/or committees and exercise effective oversight of climate-related financial risks. Further, the board and senior management should identify responsibilities for climate-related risk management throughout the organizational structure.

Clearly assigning management responsibilities allows key climate risks to be accounted for within the bank. Furthermore, it is necessary to ascertain that the various groups within the bank have the needed knowledge and tools to properly perform their duties in managing climate risks.

The board and senior management must possess sufficient knowledge of climate risks, and senior management must be able to handle such risks. On an as-needed basis, consultation with external specialists and technical training for board members and senior management on climate risks may be desirable.

Principle 3: Banks should adopt appropriate policies, procedures, and controls that are implemented across the entire organization to ensure effective management of climate-related financial risks.

Policies and procedures throughout all areas of the bank (e.g., new client acceptance, audits of material transactions) should account for climate risk management.

Internal Control Framework

Principle 4: Banks should incorporate climate-related financial risks into their internal control frameworks across the three lines of defense to ensure sound, comprehensive, and effective identification, measurement, and mitigation of material climate-related financial risks.

The first line of defense involves risk assessments whereby employees must have the technical knowledge and experience to properly spot climate-related risks when accepting new clients, reviewing credit applications and renewals, and setting up new products.

The second line of defense involves the bank's risk department, and it should operate its risk analysis and review processes separate from the first line. For example, this may involve questioning the analysis performed by the first line.

The third line of defense involves the bank's internal audit department, and it takes a high-level view of the internal control process, together with a review of the first and second lines. Additionally, the third line considers data quality.

LO 101.b: Describe the principles for managing climate-related financial risks related to capital and liquidity adequacy and the risk management process.

Capital and Liquidity Adequacy

Principle 5: Banks should identify and quantify climate-related financial risks and incorporate those assessed as material over relevant time horizons into their internal capital and liquidity adequacy assessment processes, including their stress testing programs where appropriate.

There should be analysis of solvency (e.g., debt-to-equity ratio, interest coverage ratio) in the context of potential and significant negative climate events during the longer-term capital planning period. Specifically, the concern here is the potential reduction in equity.

A parallel analysis should occur for impacts on liquidity in the short term arising from climate risks in both normal and stressed situations. Liquidity risk, here, refers to significant negative net cash outflows and/or significant deterioration in the current or quick ratio, for example.

To clarify the term *incorporate*, it would mean including such risks in the firm's business model, business strategy, and stress testing plans. Stress testing accounts for circumstances that are extreme but still possible for the firm to encounter.

Climate risks are likely to be incorporated gradually and, in that regard, firms must be able to determine the key climate risks that could significantly impair their balance sheets. In addition, firms should identify the relevant risk indicators and metrics to help with quantification of exposures. Finally, it is important to be able to determine any important connections between climate risks and regular financial risks.

Risk Management Process

Principle 6: Banks should identify, monitor, and manage all climate-related financial risks that could materially impact their financial condition, including their capital resources and liquidity positions. Banks should ensure that their risk appetite and risk management frameworks consider all material climate-related financial risks to which they are exposed and establish a reliable approach to identifying, measuring, monitoring, and managing those risks.

Significant climate risks must be clearly described and accounted for in the risk appetite framework.

A thorough analysis of climate risks should be performed on a periodic basis. Defining materiality in terms of overall risk management within the firm is important, especially to ensure that all material risks are addressed. Concentration risk is the most crucial to consider regarding sectors, industry, geography, and counterparties. Climate

risks, like all key risks, need to be identifiable through risk factors to ensure that they can be subsequently mitigated.

Risk management tools should be employed such as setting exposure limits to market, credit, liquidity, and operational risk.

Climate risks are constantly changing, and there may be new ways for such risks to enter the mainstream financial system. This means that banks must attempt to comprehend such risks and their evolution so as to minimize any negative impacts on other key financial risks.



MODULE QUIZ 101.1

1. Within the internal control framework, the third line of defense most likely covers which of the following areas?
 - A. Internal audit and data quality only.
 - B. Internal audit and risk assessments only.
 - C. Risk assessments and data quality only.
 - D. Risk assessments, internal audit, and data quality.
2. Which of the following types of material risk is most directly associated with climate-related financial risk?
 - A. Concentration risk.
 - B. Counterparty risk.
 - C. Default risk.
 - D. Liquidity risk.

MODULE 101.2: COMPREHENSIVE MANAGEMENT AND SUPERVISION PRINCIPLES

LO 101.c: Describe the principles for the management of climate-related financial risks related to management, monitoring, and reporting, comprehensive management of credit risk and other risks, and scenario analysis.

Management Monitoring and Reporting

Principle 7: Risk data aggregation capabilities and internal risk reporting practices should account for climate-related financial risks. Banks should seek to ensure that their internal reporting systems are capable of monitoring material climate-related financial risks and producing timely information to ensure effective board and senior management decision-making.

When accumulating data across the firm, the process should have controls in place to account for climate risks to spot the exact risks, to spot any concentration of such risks, and to spot any new risks. Underlying all of this is to have controls in place to produce data that is correct and usable. The bank may need to improve its existing IT systems to allow for the proper accumulation, processing, and centralization of the data to allow for the proper evaluation of key climate risks.

Taking a further step, banks might involve some of their clients or counterparties by gathering data from them regarding their transition strategies. Sometimes, it is not possible to obtain relevant data on climate risk—and, in those instances, it may be necessary to use approximations and best assumptions to derive *temporary* data.

Reporting of climate risks must occur periodically and be done without delay; therefore, the choice of what constitutes *periodic* must be carefully considered.

Metrics should be used that are both numerical and nonnumerical in nature to allow climate risks to be properly managed. The bank should fully and publicly disclose any restrictions it may encounter in being able to sufficiently analyze climate risk data.

Comprehensive Management of Credit Risk

Principle 8: Banks should understand the impact of climate-related risk drivers on their credit risk profiles and ensure that credit risk management systems and processes consider material climate-related financial risks.

Credit policies and procedures should be thorough and meticulously defined to allow proper assessment and management of significant climate risks and their effects on the bank's credit portfolio. In the context of lending activities, banks should ensure that there is sufficient assessment of potential climate risks for all borrowers before granting credit as well as continuous review of such risks during the credit term.

When identifying climate concentration risks, tools such as heatmaps could be used to measure and illustrate any geographic pockets of risk.

When managing climate risks, tools such as tightening underwriting standards, shorter-term loans, lower lending amounts, and greater collateral requirements could be used. In addition, banks could explicitly limit their exposures regarding industries, sectors, and geography if exposures in specific areas are not deemed desirable.

Comprehensive Management of Market, Liquidity, Operational, and Other Risks

There are three principles related to the comprehensive management of market, liquidity, operational, and other risks.

Principle 9: Banks should understand the impact of climate-related risk drivers on their market risk positions and ensure that market risk management systems and processes consider material climate-related financial risks.

When looking at the securities in their own asset portfolios, banks must determine the impact of climate risks on portfolio value. Also, they must assess the impact of losses and higher standard deviation of returns on portfolio value and devise methods to minimize the impacts.

Shock scenario analysis might be an effective approach to model the effect of climate risks on the bank's portfolio value. One way to do so could be to shock the liquidity of assets that have climate risk exposure and make differing assumptions on the timeliness and ability to offset those exposures. In addition, hedges may become very

costly or impossible to obtain depending on the nature and severity of any climate events.

Principle 10: Banks should understand the impact of climate-related risk drivers on their liquidity risk profiles and ensure that liquidity risk management systems and processes consider material climate-related financial risks.

Banks must consider how climate risks will affect their cash outflows in the form of borrowers drawing more from their credit lines or depositors withdrawing their funds suddenly, for example. In addition, banks should consider the effects on the value of the underlying assets in their liquidity buffers.

Principle 11: Banks should understand the impact of climate-related risk drivers on their operational risk and ensure that risk management systems and processes consider material climate-related risks. Banks should also understand the impact of climate-related risk drivers on other risks and put in place adequate measures to account for these risks where material. This includes climate-related risk drivers that might lead to increasing strategic, reputational, and regulatory compliance risk, as well as liability costs associated with climate-sensitive investments and businesses.

Climate risks need to be considered in terms of their effect on overall business operations in normal times and on business continuity during potential material climate events. The effects of any material climate events should also be considered in the context of other business risks such as strategic, reputational, compliance, and general liability.

Scenario Analysis

Principle 12: Where appropriate, banks should make use of scenario analysis to assess the resilience of their business models and strategies to a range of plausible climate-related pathways and determine the impact of climate-related risk drivers on their overall risk profile. These analyses should consider physical and transition risks as drivers of credit, market, operational, and liquidity risks over a range of relevant time horizons.

Scenario analysis could consider any or all of the following goals:

- Determining the effects of climate change and a move to low-carbon usage on the bank's business operations and operational resiliency
- Determining climate risk factors that apply most to the bank
- Quantifying exposures and expected losses from climate risks
- Determining any restrictions regarding data and analysis of climate risks
- Formulating methods of managing climate risk

Scenario analysis should focus only on the pertinent risks that apply to the bank's operations. It should include a wide enough breadth of potential and realistic situations and account for pros and cons as well as document the underlying assumptions in each of the situations. The scope of scenario analyses of climate risks should be commensurate with the bank's size and operational complexity.

Scenario analysis should cover short- and long-term periods. Shorter terms could cover specific climate risks that are more likely to materialize in the future and could be quantified with greater certainty. Longer terms could cover more general climate risks that are much less certain to quantify, but such analysis could still assess the bank's ability to adapt to long-term economic changes or adapt to changing climate risks.

It is important to ensure that any scenario analysis does not remain static given the constant and frequent changes expected in climate science. Therefore, all scenarios must be revised for changes and updated to reflect current circumstances on an ongoing basis.

LO 101.d: Describe the principles for the supervision of climate-related financial risks related to prudential regulatory and supervisory requirements for banks and responsibilities, powers, and functions of supervisors.

Prudential Regulatory and Supervisory Requirements for Banks

There are three principles related to prudential regulatory and supervisory requirements for banks.

Principle 13: Supervisors should determine that banks' incorporation of material climate-related financial risks into their business strategies, corporate governance, and internal control frameworks is sound and comprehensive.

All roles and duties pertaining to climate risks should be transparent, sufficient, thoroughly documented, and delegated to appropriate individuals.

In being able to properly oversee the bank's climate risks, the board and senior management must be able to receive relevant and reliable internal information on significant climate risks.

Supervisors must maintain regular and adequate lines of communication with the board and senior management to ensure management has a plan to deal with climate risks in a proactive manner with an eye to the future.

Supervisors should ensure that banks look at the effects of climate risks in their business models and strategies, including resiliency in both the short term and long term. In the context of climate risk, supervisors need to be in a position to question the underlying assumptions used in establishing business strategies.

Additionally, climate risks must be included in corporate governance and internal controls. This would include having relevant regulations in place in all three lines of defense as well as having periodic audits of the bank's general internal control process.

Principle 14: Supervisors should determine that banks can adequately identify, monitor, and manage all material climate-related financial risks as part of their assessments of banks' risk appetite and risk management frameworks.

Supervisors should check the scope and frequency of the bank's determination of the materiality of climate risks. In addition, they should check for methods proposed to

minimize such risks.

Supervisors should examine the bank's risk management process to ensure that they include the significant climate risk exposures. Also, banks need to determine whether their data processing and reporting abilities are sufficient to allow for the effective determination of current and future risks and prompt execution of various risk mitigation measures.

Principle 15: Supervisors should determine the extent to which banks regularly identify and assess the impact of climate-related risk drivers on their risk profile and ensure that material climate-related financial risks are adequately considered in their management of credit, market, liquidity, operational, and other types of risk. Supervisors should determine that, where appropriate, banks apply climate scenario analysis.

Banks should consider multiple methods of climate risk mitigation. Any such risks deemed to be significant should be included when computing internal capital and liquidity sufficiency for a given time period.

As stated earlier, a process of scenario analysis should exist that is commensurate to the size and operational complexity of the bank. Supervisors should be in a position to question any underlying assumptions and processes behind the scenario analyses performed.

Responsibilities, Power, and Functions of Supervisors

There are three principles related to responsibilities, powers, and functions of supervisors.

Principle 16: In conducting supervisory assessments of banks' management of climate-related financial risks, supervisors should utilize an appropriate range of techniques and tools and adopt adequate follow-up measures in case of material misalignment with supervisory expectations.

Requirements should be established, accounting for the scope and operational complexity of the bank. International cooperation between banking groups is strongly encouraged in the form of exchanging information on climate risk and making use of existing information from other groups to create efficiencies and to avoid duplication of work.

Principle 17: Supervisors should ensure that they have adequate resources and capacity to effectively assess banks' management of climate-related financial risks.

Supervisors must always be cognizant of their current abilities and what will be required in the future based on industry trends. This will involve setting aside time regularly to enhance their knowledge and skills. For tasks delegated to external parties, supervisors must still possess sufficient knowledge to interpret the work performed by the external parties.

A wide and varied group of stakeholders possessing complementary knowledge and experience should be assembled to measure and analyze climate risks.

Supervisors could use publicly available information from past years to determine the significance of climate risks. Should there be any deficiencies in the data, supervisors could then turn to banks and request their internal information in an attempt to obtain complete information.

Principle 18: Supervisors should consider using climate-related risk scenario analysis to identify relevant risk factors, size portfolio exposures, identify data gaps, and inform the adequacy of risk management approaches. Supervisors may also consider the use of climate-related stress testing to evaluate a firm's financial position under severe but plausible scenarios. Where appropriate, supervisors should consider disclosing the findings of these exercises.

Note that scenario analysis and stress testing, specifically, are targeted toward larger and international banks, for example—and these tools will not necessarily be used for smaller local banks.

Supervisors should state their goals for climate scenario analysis, some of which may include the following:

- Determining the effects of climate change and a move to low-carbon usage on the bank's business operations and operational resiliency
- Determining climate risk factors that apply most to the bank or overall financial system
- Promoting information exchange between banking entities; determining any existing deficiencies regarding data and analysis of climate risks
- Determining the sufficiency of risk management systems in banks as well as their abilities to manage climate risk

Stress testing might be employed to examine the sufficiency of the balance sheet in an extreme, but possible, situation.

As stated earlier, scenario analysis should cover short- and long-term periods. Shorter terms could cover specific climate risks that are more likely to materialize in the future and could be quantified with greater certainty. Longer terms could cover more general climate risks that are much less certain to quantify, but such analysis could still assess the bank's ability to adapt to long-term economic changes or adapt to changing climate risks.

Supervisors possess enough knowledge and skill to supervise the scenario analyses. As mentioned before, a wide and varied group of stakeholders possessing complementary knowledge and experience should be assembled to create scenarios that allow for a complete range of analysis of climate risks. In addition, supervisors must be up to date in their knowledge of current issues in scenario analysis.

Because scenario analysis is constantly changing, supervisors must always be aware that such analysis may not always be relevant in all circumstances or may have become obsolete with time. This will have a significant impact on interpretation and reporting of the analysis. Bank supervisors are encouraged to interact with other supervisors and

other banks to stay abreast of and acquire more in-depth knowledge of climate risks affecting banks—and how to develop methods to manage such risks.

In reporting the results of scenario analysis, care must be taken in view of the lack of certainty that is the nature of scenario analysis. For example, the reporting might occur at a higher level with less specific detail on underlying assumptions and methods of analysis given the lack of certainty.

As stated earlier, international cooperation between banking groups is strongly encouraged in the form of exchanging information on climate risk and making use of existing information from other groups to create efficiencies and avoid duplication of work. Exchanging best practices as well as the results of scenario analyses and using consistent scenarios are highly encouraged.



MODULE QUIZ 101.2

1. Which of the following actions by a bank is least likely to mitigate material climate-related credit risk?
 - A. Targeting specific clients.
 - B. Engaging in long-term lending.
 - C. Using discounted asset valuations.
 - D. Lowering the debt-equity ratio covenant.
2. Which of the following statements regarding the role of a supervisor in climate-related risk management is most accurate?
 - A. Supervisors should ensure that climate-related risk assessments are done on an internal basis.
 - B. Supervisors should focus on engaging quantitative specialists to design robust and accurate scenario analyses.
 - C. Supervisors should focus more on stress testing and scenario analysis if they are responsible for a larger bank versus a smaller bank.
 - D. Supervisors should use sensitivity analysis to specifically assess the sufficiency of a bank's financial position under a severe but possible situation.

KEY CONCEPTS

LO 101.a

Banks should develop and implement a sound process for understanding and assessing the potential impacts of climate-related risk drivers on their businesses and on the environments in which they operate. Banks should consider material climate-related financial risks that could materialize over various time horizons and incorporate these risks into their overall business strategies and risk management frameworks.

The board and senior management should clearly assign climate-related responsibilities to members and/or committees and exercise effective oversight of climate-related financial risks. Further, the board and senior management should identify responsibilities for climate-related risk management throughout the organizational structure.

Banks should adopt appropriate policies, procedures, and controls that are implemented across the entire organization to ensure effective management of climate-

related financial risks.

Banks should incorporate climate-related financial risks into their internal control frameworks across the three lines of defense to ensure sound, comprehensive, and effective identification, measurement, and mitigation of material climate-related financial risks.

LO 101.b

Banks should identify and quantify climate-related financial risks and incorporate those assessed as material over relevant time horizons into their internal capital and liquidity adequacy assessment processes, including their stress testing programs where appropriate.

Banks should identify, monitor, and manage all climate-related financial risks that could materially impact their financial condition, including their capital resources and liquidity positions. Banks should ensure that their risk appetite and risk management frameworks consider all material climate-related financial risks to which they are exposed and establish a reliable approach to identifying, measuring, monitoring, and managing those risks.

LO 101.c

Risk data aggregation capabilities and internal risk reporting practices should account for climate-related financial risks. Banks should seek to ensure that their internal reporting systems are capable of monitoring material climate-related financial risks and producing timely information to ensure effective board and senior management decision-making.

Banks should understand the impact of climate-related risk drivers on their credit risk profiles and ensure that credit risk management systems and processes consider material climate-related financial risks.

Banks should understand the impact of climate-related risk drivers on their market risk positions and ensure that market risk management systems and processes consider material climate-related financial risks.

Banks should understand the impact of climate-related risk drivers on their liquidity risk profiles and ensure that liquidity risk management systems and processes consider material climate-related financial risks.

Banks should understand the impact of climate-related risk drivers on their operational risk and ensure that risk management systems and processes consider material climate-related risks. Banks should also understand the impact of climate-related risk drivers on other risks and put in place adequate measures to account for these risks where material. This includes climate-related risk drivers that might lead to increasing strategic, reputational, and regulatory compliance risk, as well as liability costs associated with climate-sensitive investments and businesses.

Where appropriate, banks should make use of scenario analysis to assess the resilience of their business models and strategies to a range of plausible climate-related pathways and determine the impact of climate-related risk drivers on their overall risk profile.

These analyses should consider physical and transition risks as drivers of credit, market, operational, and liquidity risks over a range of relevant time horizons.

LO 101.d

Supervisors should determine that banks' incorporation of material climate-related financial risks into their business strategies, corporate governance, and internal control frameworks is sound and comprehensive.

Supervisors should determine that banks can adequately identify, monitor, and manage all material climate-related financial risks as part of their assessments of banks' risk appetite and risk management frameworks.

Supervisors should determine the extent to which banks regularly identify and assess the impact of climate-related risk drivers on their risk profile and ensure that material climate-related financial risks are adequately considered in their management of credit, market, liquidity, operational, and other types of risk. Supervisors should determine that, where appropriate, banks apply climate scenario analysis.

In conducting supervisory assessments of banks' management of climate-related financial risks, supervisors should utilize an appropriate range of techniques and tools and adopt adequate follow-up measures in case of material misalignment with supervisory expectations.

Supervisors should ensure that they have adequate resources and capacity to effectively assess banks' management of climate-related financial risks.

Supervisors should consider using climate-related risk scenario analysis to identify relevant risk factors, size portfolio exposures, identify data gaps, and inform the adequacy of risk management approaches. Supervisors may also consider the use of climate-related stress testing to evaluate a firm's financial position under severe but plausible scenarios. Where appropriate, supervisors should consider disclosing the findings of these exercises.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 101.1

1. **D** The third line of defense involves the bank's internal audit department, and it takes a high-level view of the internal control process, together with a review of the first and second lines (risk assessments). Additionally, the third line considers data quality. (LO 101.a)
2. **A** Concentration risk is particularly specific to climate-related financial risk. With concentrations, the risk could be related to industry, economic sectors, or geographic regions. The other risks, although relevant to climate effects, are not as strongly associated with climate-related financial risk. (LO 101.b)

Module Quiz 101.2

1. **B** Banks often resort to shorter-term loans to reduce their credit risk. Long-term loans would increase credit risk, all things being equal.

Targeting clients with higher credit ratings, for example, is a means of reducing credit risk. In valuing collateral using a steeper discount rate, the lower asset valuations (likely comfortably below market or net realizable value) for a given loan principal amount provide some additional protection to the bank as the market or realizable value will likely exceed the outstanding loan balance by a comfortable margin. This would decrease the bank's credit risk. A debt-equity covenant represents a maximum amount of debt permitted by the bank. Lowering the ratio is, therefore, a tightening of credit standards and would decrease the bank's credit risk. (LO 101.c)

2. **C** Stress testing and scenario analysis are more targeted toward larger banks but some use of those tools may be useful even for smaller banks.

Some aspects of climate-related risk assessments may be specialist in nature; therefore, they would need to be outsourced externally. Supervisors should collaborate with a broad and diverse set of stakeholders, including climate scientists. Supervisors should use stress testing (not sensitivity analysis) to assess the sufficiency of a bank's financial position under a severe, but possible, situation. (LO 101.d)

¹ <https://www.bis.org/bcbs/publ/d532.pdf>.

The following is a review of the Current Issues in Financial Markets principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Basel Committee on Banking Supervision.

READING 102

THE CRYPTO ECOSYSTEM: KEY ELEMENTS AND RISKS

Study Session 15

EXAM FOCUS

The crypto ecosystem is a complex and ever-evolving network of digital technologies and applications. Key elements include cryptocurrencies, stablecoins, smart contracts, and decentralized finance (DeFi). For the exam, be familiar with these main crypto market characteristics. Also, know the structural flaws inherent in the crypto ecosystem, including congestion and fragmentation, the need for a stablecoin anchor, and the false claims of decentralization. Finally, be able to identify the risks that crypto poses to different parties, such as investors, governments, regulators, and financial institutions.

MODULE 102.1: KEY ELEMENTS OF THE CRYPTO ECOSYSTEM

LO 102.a: Describe the key elements of the crypto ecosystem, including unbacked crypto, stablecoins, smart contracts, and DeFi services.

The crypto ecosystem is built on a foundation of blockchain technology, which is a methodology that provides a secure and decentralized way to record transactions and store data. A **blockchain** is essentially a digital ledger that keeps track of transactions in an open yet unalterable manner. An interconnected network of computers stores the information, and (unlike traditional databases, which are centralized) every participant has a copy of the same ledger. When any new transaction is transmitted to the network, numerous validators (called “miners”) compete using complex algorithms to verify the transaction. A successful miner is compensated with a small amount of cryptocurrency.

By broadcasting all information publicly, the crypto ecosystem ensures that every transaction is in agreement with the history of transactions on the blockchain; for example, that a piece of digital currency actually belongs to the seller, and that the cryptocurrency has not been spent previously. Once a transaction has been confirmed,

it is placed in the ledger as a “block,” and cannot be changed without the network’s approval. The actual names of the transacting parties (i.e., the crypto wallet owners) are kept private. However, the history of every transaction is publicly viewable; thus, blockchain transactions are referred to as pseudo-anonymous.

Key elements of the crypto ecosystem include (1) unbacked crypto, (2) stablecoins, (3) smart contracts, and (4) decentralized finance (DeFi).

Unbacked crypto refers to the original variety of cryptocurrency, first created in 2009 with Bitcoin. Unbacked crypto is not supported by any other assets, such as precious metals or fiat currency. Rather, the value of unbacked crypto is based on supply and demand. Well-known examples of unbacked cryptocurrency include Bitcoin and Ethereum.

Unbacked cryptocurrencies offer numerous advantages, such as decentralization, security, and transparency. However, unbacked crypto is sometimes criticized for its lack of intrinsic value as well as its volatility. These cryptocurrencies are subject to extreme price volatility (i.e., market risk), which makes them inappropriate as a medium of exchange. As a result, stablecoins have been developed as a means of allowing cryptocurrencies to potentially serve as payment instruments.

Stablecoins are a relatively new kind of cryptocurrency that are pegged to an independent asset, such as a fiat currency, (commonly the U.S. dollar [USD]), or occasionally to an asset such as gold, or even to other cryptoassets. This pegging feature is intended to make stablecoins more liquid and less volatile than unbacked cryptocurrencies. The underlying assumption is that a user can exchange a stablecoin relatively quickly for USD (or other asset) at par value. For this reason, stablecoins have become an essential medium of exchange in the crypto ecosystem. *Asset-backed stablecoins* are supported by bonds or bank deposits and are managed by a centralized intermediary that invests the underlying collateral and manages coin redemptions. *Algorithmic stablecoins*, on the other hand, are not backed by real-world assets, but use arbitrage with a volatile token to maintain their value. (The failed TerraUSD is an example of an algorithmic stablecoin.)

Smart contracts are pieces of self-executing code that are stored on a blockchain (used by Ethereum and other newer blockchains). Smart contracts trigger some predetermined task if specific conditions are met. These computer programs can be used to automate a range of financial agreements and will perform transactions directly with another party without the use of an intermediary.

In traditional finance, a collateralized loan would require the parties to agree on terms, transfer the collateral to a third party, and process the loan through multiple steps involving intermediaries. Smart contracts eliminate the need for intermediaries, by embedding the loan agreement into a program stored on the blockchain. The blockchain automatically executes the contract between the two parties. *Composability* in smart contracts refers to the possibility of combining several smart contracts together to produce a more-sophisticated transaction.

Decentralized finance (DeFi) refers to financial services that are built on blockchain technology. DeFi services are intended to replicate conventional methods of managing

money, allowing users to borrow, lend, trade, and invest assets without the need for a central intermediary. DeFi services have the potential to be more open, accessible, transparent, and efficient than traditional financial services.

However, DeFi is still in its early stages of development, and faces some potential downsides. DeFi protocols are complex and can be vulnerable to hacks and other security vulnerabilities. Additionally, the DeFi market is very volatile given that the value of crypto assets can fluctuate wildly. Furthermore, there is no central authority to offer users recourse in the event of fraud or theft.

Though the crypto ecosystem is still in its initial stages of development, it is rapidly evolving and growing. Crypto is seen as having the potential to revolutionize the way we exchange value and interact with the financial system.



MODULE QUIZ 102.1

1. Which of the following statements about the key elements of the crypto ecosystem is least accurate?
 - A. Algorithmic stablecoins are backed by real-world assets.
 - B. The code underlying smart contracts is publicly available.
 - C. Smart contracts cannot directly access “off-chain” information.
 - D. Most DeFi transactions do not finance activity in the real economy.
2. Regarding the key elements of the crypto ecosystem, which of the following statements is most accurate?
 - A. Smart contracts can fetch real-world data for themselves.
 - B. Unbacked crypto is managed by a centralized intermediary.
 - C. Stablecoin values are susceptible to volatility and speculation.
 - D. DeFi services operate independently from any centralized authority.

MODULE 102.2: STRUCTURAL FLAWS IN THE CRYPTO ECOSYSTEM

LO 102.b: Describe the structural flaws inherent in various elements of the crypto ecosystem.

Because the crypto ecosystem is in its early stages of development, there are numerous inadequacies that still need to be addressed. These flaws include (1) congestion and fragmentation, (2) the need for a nominal anchor for stablecoins, and (3) the false claims of decentralization.

Congestion

Crypto **congestion** occurs when a cryptocurrency network is not processing transactions quickly enough to meet demand. During periods of congestion, fees for validators rise as a glut of users seek to have their transactions processed. This results in higher transaction fees, as users are forced to pay more to use congested blockchains.

The blockchain capacity problem is one result of the **scalability trilemma**, whereby blockchains can only achieve two of the following three properties:

1. *Security*. The ability to protect the network from attack.
2. *Decentralization*. The distribution of power not only in the hands of a few, but across the network.
3. *Scalability*. The ability to process many transactions per second.

Fragmentation

One result of a lack of scalability is that additional blockchains with greater capacity (such as the Terra blockchain) have been introduced. Essentially, with **fragmentation**, the cryptocurrency market divides into many different and often incompatible blockchain networks. These newer blockchains often feature a higher ability to process transactions, but weaker security and increased centralization. This has led to a proliferation of different blockchains and applications, each with its own unique set of rules and features.

Crypto fragmentation can lead to several problems, including the following:

- *Lack of interoperability*. Assets or data cannot be easily exchanged between blockchains.
- *Risk of hacks and theft*. Hackers may be able to exploit vulnerabilities in smaller and less secure blockchains.
- *Decreased network effects*. Users are less likely to use a blockchain if it is incompatible with other popular blockchains.

Stablecoins Anchors

As mentioned previously, stablecoins are cryptocurrencies that are pegged in value to some other asset, sometimes called a **nominal anchor**. This property is intended to make stablecoins less volatile than unbacked cryptocurrencies; and for this reason, stablecoins are considered in the crypto ecosystem to be a superior medium for transactions. However, the search for an appropriate nominal anchor can be challenging. Stablecoin issuers are incentivized to invest in risky assets in order to generate higher returns, which would increase demand for the stablecoins, and diversify their reserves. However, in order for the stabilization effects of a nominal anchor to be effective, the reserve assets must be transparent and consist of high-quality assets.

False Decentralization Claims

Another structural flaw inherent to the crypto ecosystem is that decision-making power tends to be more concentrated than proponents of cryptocurrencies would have you believe. Advocates of cryptocurrencies and DeFi applications promote these items as decentralized. However, in practice, the nature of operational costs results in power being shared by just a small number of validators.

In a **proof-of-work (PoW)** consensus mechanism, potential validators compete with one another to solve cryptographic puzzles in order to be allowed to add transactions to the distributed ledger. The congestion resulting from PoW systems, which are decentralized platforms, leads to a concentration of power.

In newer **proof-of-stake (PoS)** blockchains, validators are required to post their coins as collateral in order to increase their chances of being chosen to validate the next block. This consensus mechanism reduces congestion; however, it too can lead to a concentration of power and coins among a small number of validators because operational costs are largely fixed.

Another factor that contributes to centralization is the issue of “contract incompleteness,” which refers to the impossibility of creating agreements that anticipate all possible future scenarios. Centralization can help to overcome the issue of contract incompleteness. An equivalent idea in the DeFi space is “algorithm incompleteness,” referring to the inability to write code that will account for all potential situations and outcomes. For this reason, DeFi platforms require some degree of centralized governance to address any gaps in their code. This aspect of centralization is typically represented by **governance tokens**, which are tokens issued as an incentive to platform developers and other insiders, who are then part-owners and can vote on proposals. A downside of validators becoming more centralized is that the risk of attacks and hacks increases.

Centralization is also exhibited in cryptocurrency trading, where investors primarily use **centralized exchanges (CEXs)** over **decentralized exchanges (DEXs)**. DEXs employ automated market-maker protocols to match transaction counterparties. Meanwhile, CEXs maintain off-chain records of outstanding orders placed by traders (analogous to limit order books in the world of traditional finance). One reason that CEXs attract more trading activity than DEXs is because of the lower costs for CEXs.

Crypto CEXs lack the regulatory oversight of traditional financial exchanges, despite replicating some of their functions. Not surprisingly, these exchanges have repeatedly demonstrated inadequate risk management controls and governance standards, sometimes resulting in the loss of users’ investments.



PROFESSOR’S NOTE

Notable examples of failures of centralized exchanges include the November 2022 FTX bankruptcy, which led to billions in customer losses, as well as the 2014 Mt. Gox bitcoin exchange failure, which resulted in over \$400 million in bitcoin losses.

False decentralization claims can also have a negative impact on the security of cryptocurrencies and DeFi applications. DeFi platforms tend to require some amount of central governance in order to deal with unforeseen eventualities. However, if a cryptocurrency or DeFi application is not truly decentralized, then it is more vulnerable to attacks and hacks.

The Oracle Problem

Smart contracts rely on *oracles* to provide them with real-world data (such as the outcome of an event, or the price of an asset), because smart contracts cannot otherwise access information that resides “off-chain.” The **oracle problem** in decentralized finance refers to the challenge of providing accurate data to smart contracts (that may be used for lending, borrowing, and trading).

When an oracle is compromised or malicious, it can provide false data to smart contracts. If the self-executing code is deceived into making an incorrect decision based on inaccurate data, it can result in market manipulation, and losses for market participants. Because oracles are typically decentralized, there is little recourse should an oracle provide wrong information. The cost of errors and illegal actions related to oracles can be very high.

Technological efforts are being made to address the oracle problem and make DeFi systems more secure by making it more difficult for users to manipulate the oracle network.

Addressing Crypto Ecosystem Flaws

There are numerous initiatives to address the structural flaws mentioned in this section. For example, there are cross-chain bridges in development that should allow users to trade DeFi assets and cryptocurrencies across different blockchains. Also in the works are new kinds of stablecoins that are intended to be less volatile and more secure.



MODULE QUIZ 102.2

1. The scalability trilemma refers to the problem that permissionless blockchains can achieve only two of which of the following three properties?
 - A. Security, decentralization, and scalability.
 - B. Localization, robustness, and immutability.
 - C. Openness, representation, and auditability.
 - D. Monetization, centralization, and reliability.
2. Which of the following statements about stablecoins is most accurate?
 - A. Stablecoins require a central bank currency as a nominal anchor.
 - B. True to their name, stablecoins typically exhibit low price volatility and high liquidity.
 - C. Asset-backed stablecoins automatically rebalance the coin's supply to maintain the peg value.
 - D. Stablecoins that peg their value to the U.S. dollar (USD) must be directly convertible into the USD.

MODULE 102.3: RISKS POSED BY THE CRYPTO ECOSYSTEM

LO 102.c: Describe the risks crypto poses to parties including crypto investors, governments, regulators, and traditional financial institutions; and identify potential policy actions that can be taken to mitigate these risks.

Risks to Crypto Investors

Media coverage and rising cryptocurrency prices in recent years have drawn many investors (particularly small retail investors) into the crypto market. However, the majority of crypto investors have not profited. In fact, large investors have generally benefited at the expense of small investors.

The highly volatile nature of cryptocurrencies makes these assets a risky bet for investors. Furthermore, because the crypto industry is fairly new and mostly unregulated, it has become a target for scammers and fraudsters (e.g., investors can lose their cryptocurrency if their wallet is hacked). Furthermore, market participants face the risk of loss if the exchange they are using collapses or is compromised. The bankruptcy of a crypto firm often means that investors will not be able to recover their funds, or that it may take a long time to do so.

Risks to Governments

The advent of crypto can pose new risks to governments, such as the risks arising from money laundering and terrorist financing. Cryptocurrencies are a natural fit for such nefarious undertakings, because these assets can be transferred anonymously. Governments are exploring measures to address these risks, including anti-money laundering efforts and schemes to combat the financing of terrorism. The use of crypto presents particular challenges for law enforcement agencies, because of the difficulty of tracking and tracing crypto transactions.

Cryptocurrencies may also lead to a loss of tax revenue for governments if they are used to avoid paying taxes. Cryptocurrencies may additionally pose a risk to governments in terms financial stability if crypto becomes widely adopted as a means of payment. This is because cryptocurrencies are decentralized and not subject to government control.

Crypto's growth may furthermore pose a danger to monetary sovereignty.

Cryptoization occurs when crypto currencies are used in place of local currencies. This is especially likely to happen in emerging markets; in a period of economic instability, holders of a country's currency may decide to move into crypto, putting even more downward pressure on the domestic currency. Thus, cryptoization could lead to a loss of monetary sovereignty and have a negative impact on the country's financial stability.

Risks to Regulators

Regulators have struggled to keep up with the rapidly evolving cryptocurrency industry, which has made it difficult to develop and implement effective regulations. While cryptocurrencies are global in nature, laws vary from country to country. These jurisdictional challenges will make it difficult for regulators to enforce rules and coordinate with one another.

As noted earlier, the anonymity of crypto lends itself well to being used for illicit activities such as tax evasion, money laundering, and the financing of terrorism. Such misuse of crypto presents vital challenges for law enforcement, who typically find it difficult to trace crypto transactions.

Risks to Traditional Financial Institutions

Up to this point in time, there has been relatively limited interaction between the real economy and the crypto market, and between traditional finance and the crypto

market. In fact, crypto has so far mostly been used as a tool for speculation. However, it is expected that linkages between crypto and traditional finance will become stronger going forward.

Traditional banks increase their connections to crypto when they provide credit to customers with crypto exposures, or when they provide crypto-related services to their clients. This effect could be observed in the latter half of 2022, when a plunge in crypto prices spilled over into the traditional banking sector, leading to the demise of at least two banks with significant customers in the crypto world. That outcome demonstrated how any bank with significant exposure to crypto can be impacted by crypto failures.

Another way that the interconnections between traditional finance and crypto could grow is through the **tokenization** of claims on real-world assets, such as real estate or stocks. Tokenization is the process of adding a digital representation of an asset (i.e., a “token”) to a programmable ledger. These tokens can include information such as ownership details. Tokenization could result in the development of a complicated web of interconnections between crypto and traditional finance, leading to a substantial increase in the systemic importance of the crypto ecosystem.

The continuing evolution of the regulatory landscape in many jurisdictions means that it can be difficult for institutions to operate in the crypto space, which can also create uncertainty for investors. In the long run, cryptocurrencies may pose a competitive threat to traditional financial institutions, because cryptocurrencies can offer a faster and cheaper way to transfer money.

Potential Policy Actions to Mitigate Crypto Risks

Governments around the world are still struggling with how best to deal with the risks introduced by cryptocurrencies. Some countries have pursued containment strategies. Others have opted for regulation, while a few countries have even banned crypto altogether.

Containment refers to a policy intended to limit the connection between crypto and the traditional financial system. The idea of containment is to protect consumers from harm, while reducing the risk of crypto shocks to the financial system. Implementation could take numerous forms, including prohibiting banks from offering crypto services. This kind of containment can be effective, but can also stifle innovation, and make operating difficult for legitimate crypto businesses.

Regulation means establishing rules and oversight for the crypto industry. Regulation could be accomplished through measures such as licensing crypto exchanges or requiring crypto businesses to adopt procedures that hinder criminal activity. To effectively manage the risks associated with cryptocurrencies, comprehensive governance, anti-money laundering (AML), and combating the financing of terrorism (CFT) measures are essential. Robust governance frameworks can ensure that market participants operate in a transparent and accountable manner, which should help to preserve the integrity of the financial system. The Financial Action Task Force (FATF) has provided guidance on registration and licensing requirements for service providers

in the cryptocurrency ecosystem. A downside of regulation is that it can be expensive and complex to implement.

Bans involve prohibiting crypto entirely. This can be done through a variety of mechanisms, such as making it illegal to buy, sell, or trade crypto assets. The ultimate goal of a cryptocurrency ban is to eliminate the risks posed by crypto entirely. An unintended consequence of bans is that such prohibitions are likely to drive crypto activity into the underground economy.

If properly implemented, policy actions related to the crypto ecosystem could support private sector innovation, reduce the cost of transactions, increase financial inclusion, and improve the overall integrity of the financial system. Policy actions that could be taken to mitigate the risks that crypto poses include the following:

- *Develop and implement clear and comprehensive regulations.* Regulators and governments should develop detailed rules and laws for the cryptocurrency industry. Initiatives in this area are being coordinated by the Financial Stability Board (FSB), and steps are being taken to develop a consensus on appropriate regulatory actions.
- *Increase cooperation and coordination.* Governments and regulatory authorities should collaborate to develop a comprehensive framework for mitigating the jurisdictional issues associated with cryptocurrencies. There is already a proposed framework for the international regulation of crypto activities that is intended to promote consistency and comprehensiveness of regulation, supervision, and oversight of crypto activities and markets.
- *Encourage crypto innovation.* By encouraging positive innovation within the traditional financial system, the cost of payments should decrease, even as the quality of transactions increases. The FedNow instant payment service in the United States, which allows money to be sent and received instantly, is an example of such financial services innovation.
- *Develop central bank digital currencies (CBDCs).* CBDCs are digital currencies issued by central banks. They could potentially offer a safer and more efficient way to transfer money.



MODULE QUIZ 102.3

1. Which of the following actions is least likely to lead to interconnectedness between the crypto market and traditional financial institutions?
 - A. Crypto exchanges partnering with banks.
 - B. Banks offering cryptocurrency-related services.
 - C. Asset managers launching cryptocurrency funds.
 - D. Retail investors speculating on cryptocurrency values.
2. The most appropriate policy action for managing the risks that crypto poses would be to:
 - A. tax cryptocurrency transactions at a high rate.
 - B. encourage the use of central bank digital currencies.
 - C. outlaw initial coin offerings and decentralized exchanges.

D. impose strict regulations on crypto exchanges and wallets.

KEY CONCEPTS

LO 102.a

Unbacked cryptocurrencies, such as Bitcoin and Ethereum, are digital assets that are not backed by physical assets nor by fiat currencies. Ownership of unbacked crypto assets is verified by decentralized validators, with transactions recorded on a public ledger.

Stablecoins are a kind of digital currency whose value is pegged to an independent asset, often USD. Stablecoins can be advantageous in that they provide a robust medium of exchange and can be traded relatively quickly for currency at par value.

Smart contracts are self-executing computer programs stored on a blockchain that help automate functions and reduce reliance on intermediaries. Multiple instructions can be combined within a single contract, thereby facilitating complex transactions. Smart contracts increase transparency and reduce the risk that transactions can be manipulated.

DeFi refers to decentralized financial services (such as trading, investing, borrowing, and lending) that are built on blockchain technology. DeFi has allowed for the introduction of innovative financial products and services and has helped to make these new products and services accessible to everyone.

LO 102.b

Because the crypto ecosystem is in its early stages of development, there are numerous structural flaws that still need to be addressed.

Congestion occurs when a cryptocurrency network is not processing transactions quickly enough to meet demand. This results in higher transaction fees for users and may lead to new competing blockchains being introduced—a process called fragmentation. Fragmentation increases crypto ecosystem complexity and decreases overall security. Also, the lack of interoperability can make it more difficult to trade across different blockchains.

A stablecoin's "nominal anchor" should make it less volatile than unbacked cryptocurrencies. However, choosing an appropriate stablecoin anchor can be a challenge. Some common choices include: a fiat currency, gold, or a basket of goods and services. Each kind of nominal anchor has its own advantages and disadvantages; for example, fiat currencies can be volatile, while precious metals are not particularly liquid.

Cryptocurrencies and DeFi applications are generally perceived to be decentralized. However, this can be the result of false decentralization claims—sometimes these entities are actually controlled by a small group of people. False decentralization can have a negative impact on the security of cryptocurrencies and DeFi applications, making them vulnerable to attacks, and exposing them to possible manipulation or fraud.

Smart contracts must rely on oracles to provide them with real-world data, because smart contracts cannot otherwise access information “off-chain.” “The oracle problem” in DeFi refers to the challenge of providing accurate data to smart contracts. A compromised or malicious oracle could provide false or inaccurate data to smart contracts, resulting in market manipulation, with high losses and little recourse for users.

LO 102.c

Risks crypto poses to investors include:

- Cryptocurrency prices are volatile and can fluctuate wildly, making them a risky investment.
- The crypto industry is relatively new and unregulated, which makes it a target for fraud and scams.
- Investors can lose their cryptocurrency to theft if their wallet is hacked or if the exchange is compromised.

Risks crypto poses to governments include:

- Cryptocurrencies can pose a risk to financial stability because they are decentralized and not subject to government control.
- Crypto assets are ideal for laundering money and financing terrorism because they are anonymous and difficult to track.
- Governments can lose revenues when cryptocurrencies are used to avoid paying taxes.

Risks crypto poses to regulators include:

- Regulators have struggled to develop and implement effective rules and laws that keep up with the rapidly evolving cryptocurrency industry.
- Cryptocurrencies are global in nature, so jurisdictional challenges make it difficult for countries to enforce regulations.

Risks crypto poses to traditional financial institutions include:

- Cryptocurrencies pose a competitive threat to traditional financial institutions because this technology offers a faster and cheaper way to move money.
- Financial institution instability is possible if banks take on significant exposure to volatile cryptocurrencies.

Potential policy actions to mitigate crypto risks include:

- Develop and implement clear and comprehensive regulations to protect investors and reduce the risks of fraud, scams, and hacking.
- Increase cooperation and coordination among governments and regulators to address cryptocurrency jurisdictional challenges.
- Educate investors about the risks of cryptocurrencies to help them to make informed decisions.

- Develop central bank digital currencies (CBDCs) to offer a safer and more efficient way to transfer money compared to current cryptocurrencies.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 102.1

1. **A** Algorithmic stablecoins (such as the failed TerraUSD stablecoin) maintain a peg using a paired volatile token; they are typically not backed by real-world assets. The other statements are accurate. Most DeFi activities involve only exchanging one crypto coin for another. (LO 102.a)
2. **C** Stablecoins are susceptible to the same market forces (such as volatility and speculation) that affect other cryptocurrencies. Asset-backed stablecoins (not unbacked crypto) are managed by a centralized intermediary that invests the underlying collateral and coordinates the coins' redemption. Stablecoins are typically issued by centralized entities that are responsible for maintaining the peg of the stablecoin to the underlying asset. In order to import data into a blockchain that can be accessed by a smart contract, it is necessary to involve a third party (the oracle). (LO 102.a)

Module Quiz 102.2

1. **A** The scalability trilemma of a blockchain implies that permissionless blockchains can achieve some combination of scalability, security, and decentralization, but not all three together. Security is achieved through incentives and decentralization, but congestion and high fees, which are needed to provide incentives, limit scalability. (LO 102.b)
2. **B** Stablecoins aim to minimize volatility, and typically have relatively high levels of liquidity. Asset-backed stablecoins use an intermediary to manage and invest the collateral to maintain the peg value, while algorithmic stablecoins use a feedback loop to automatically rebalance supply to maintain the peg value. Stablecoins that peg their value to a currency like the USD can be convertible to another cryptoasset instead of the currency directly (e.g., the TerraUSD stablecoin was convertible to another token named Luna to maintain the 1:1 peg value with the USD). Stablecoins often peg their value to a conventional currency such as the USD, but they can also peg their value to other assets, including commodities or precious metals. (LO 102.b)

Module Quiz 102.3

1. **D** The interplay between the crypto market and traditional finance (as well as with the real economy) has remained limited so far, because for the most part, crypto has mainly been used for speculation. However, there are signs of growing interconnectedness between crypto and traditional financial institutions. For example, Coinbase has partnered with PayPal to allow users to buy and sell cryptocurrencies using their PayPal accounts, BlackRock launched a private

Bitcoin trust in 2021, and JPMorgan Chase now offers its clients access to Bitcoin and other cryptocurrencies. (LO 102.c)

2. **B** Encouraging the use of central bank digital currencies should reduce costs of payment, enhance financial inclusion, and bolster the integrity of the financial system. Outlawing certain types of crypto activity would be difficult to enforce, and it would stifle innovation in the crypto space. Imposing heavy regulations on crypto exchanges and wallets would make it difficult and expensive for people to buy and sell crypto and would discourage the adoption of crypto. Taxing crypto transactions at a high rate would make it less attractive as an investment and would discourage people from using it. (LO 102.c)



The following is a review of the Current Issues in Financial Markets principles designed to address the learning objectives set forth by GARP®. Cross-reference to GARP assigned reading—Martinez.

READING 103

DIGITAL RESILIENCE AND FINANCIAL STABILITY: THE QUEST FOR POLICY TOOLS IN THE FINANCIAL SECTOR

Study Session 15

EXAM FOCUS

Digital resilience is crucial for maintaining financial stability as financial institutions move toward an almost completely digital system. For the exam, understand the complexities involved with the extensive and concentrated use of cloud computing by financial institutions. There has been some progress with increasing cyber resilience regarding the mitigation of risks posed by third-party service providers. However, challenges in fundamental measurement principles and standardization still exist. Finally, understand the role of macroprudential measures such as circuit breakers and cooperation arrangements for dealing with cyber threats across the banking industry.

MODULE 103.1: CYBER THREATS AND DIGITAL RESILIENCE

LO 103.a: Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions.

Cyber and ICT Risks

Cyber risk refers to the loss of confidentiality, integrity, and availability (CIA) of data and data systems that result from a cyberattack. **Information and communications technology (ICT) risk** refers to disruptions pertaining to information and communication systems that may have similar effects as cyber risk—but, strictly speaking, do not stem from cyberattacks.

The overwhelming level of digitalization at financial institutions, combined with the essential nature of their services, places financial institutions in a vulnerable position regarding cyber and ICT risks. The financial sector is responsible for managing an immense amount of funds together with the expectation of providing uninterrupted transaction services and maintaining confidentiality. This suggests a greater risk of cyberattacks and ICT breakdowns.

Based on a general increase in serious cybersecurity incidents, some financial institutions have ranked cyber risk as more significant than market risk. To date, there have been major examples of confidentiality breaches of client information, theft of funds, and denial-of-service (DoS) attacks experienced by financial institutions worldwide. Reliable estimates of losses arising from cyber risks are generally not possible to compute due to a lack of data. As a result, the development of cyber insurance has been significantly hindered.

Cloud Computing

Cloud computing can be thought of as the provision of computer resources (e.g., storage, servers, software) through the internet (i.e., “the cloud”). The cloud operates on a shared system with multiple users, and its benefits include lower costs and greater operating flexibility. Cloud service providers are expected to have extremely robust controls in place. Financial institutions have strongly embraced cloud computing in recent years, which has resulted in a corresponding decrease in cyber risks; however, it has also led to an increase in ICT risks. The main issue is that the use of cloud computing moves some, but not all, of an organization’s cyber risks to cloud service providers because the organization’s providers and customers might not always have adequate controls.

The significant adoption of cloud computing by financial institutions has resulted in the integration of cyber and ICT risks to an extent that there could be systemic implications (discussed in the next section). Furthermore, cloud computing has needed to evolve (e.g., multiclouds) to deal with concentration risks, for example. The use of multiclouds limits the amount of damage that could occur from a cyberattack because the computing assets are spread out over multiple clouds, rather than just one. All major banks have adopted the use of multiclouds (compared to less than half for asset management companies and insurance companies). However, such evolution comes with unintended additional risks, such as weaker stewardship over the distributed computing processes.

ICT risks in the form of physical outages have become a key concern, with about 80% of cloud customers suffering at least one outage within the prior three-year period. During that same period, roughly 20% of those outages were considered significant enough to noticeably disrupt operations. Furthermore, climate change and rising energy costs may be causing longer and more frequent outages. In short, cloud computing use is high—despite its numerous and significant drawbacks.

Interactions Between Cyber Risks and Financial Stability

LO 103.b: Assess the interactions between cyber and ICT risks and financial risks and explain how cyber and ICT risk events at financial institutions can lead to systemic financial risk.

Cyber vulnerabilities that are triggered can cause negative financial impacts such as illiquidity, excessive leverage, and loss of customer trust. The most important consideration is how technology risks can lead to the realization of financial risks.

The modern-day design of cyber and ICT connections is highly integrated, which opens the potential for significant contagion effects. Financial contagion can occur extremely rapidly through the financial industry with the onset of just one key triggering event. The effects of such systemic risks on the **CIA triad** (i.e., confidentiality, integrity, and availability) are wide-ranging. For example, severity seems to be lower for data confidentiality, but higher for data integrity and data/systems availability for operational use. Although the quantification of systemic risks is important, it is also necessary to look at the qualitative effects of systemic risks in terms of their impact on the CIA triad.

Because contagion effects are nonlinear in nature, it is challenging to predict the probability of key events with reasonable accuracy. The solution is to use **agent-based models**, which are computer simulations of actions and interactions among agents. These models quantify the impacts of a cyber incident in terms of the CIA effects and how they affect the liquidity and asset values of the financial institutions in question. However, such models are still in their infancy stage, but their basic principles could be applied to developing cyber and ICT stress tests.

Agent-based models are useful for tracing the source of systemic risks, as what appears to happen in real life between technology and finance is not always completely accurate. For example, a cyber incident that causes a run on bank deposits may not materialize because the bank's computer systems have been affected so as to not allow the actual withdrawals of funds to occur. In that regard, it would appear on the surface that technology was able to suppress the effects of the contagion—but this is clearly misleading because bank customers have other alternatives available, such as withdrawing funds they have on deposit with other banks.

Virtually all financial institutions are digitalized now, and they have become integrated and more reliant on each other due to cloud computing, networks, and shared software. Therefore, it is necessary for each participating institution to be reliable enough to quickly recover from a cyber incident to ensure the continuity of key functions in the general financial system. In that regard, given the importance of ICT functions in the financial system, each institution must be “tested” to ensure it is sufficiently reliable (which does not necessarily mean 100% reliability).

The ultimate result of cyber and ICT risks, combined with the vulnerabilities of financial institutions, is the overwhelming loss of confidence of customers. This

combination has been seen frequently in the past with runs on bank deposits stemming from customer panic or financial fundamentals. Building up liquidity buffers helps to mitigate the effects of bank runs caused by cyber shocks in theory, although history has shown that there has been a high level of liquidity in the financial system for a reasonable amount of time, thereby not requiring such buffers to be used. A more accurate assessment of the impact of a loss of confidence does require the use of agent-based models.

Financial Stability Policies and Tools

LO 103.c: Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one.

Digital resilience covers both technological and financial aspects and involves micro-oriented tools and framework policies. **Micro-oriented tools** deal with general operational guidance, while **framework policies** deal with fundamental rules pertaining to network and system operations. Due to the significant threats to digital resilience in recent years, framework policies have seen major revisions to strengthen the defenses against new cyber and ICT risks.

Policy Measures and Micro-Oriented Tools

The **International Telecommunications Union (ITU)** covers a wide range of legal, technical, and other measures to promote cyber resilience. However, due to the complexity involved, it is necessary to have certification and standardization of solutions as a starting point for building cybersecurity defenses. With the many entities involved with cyber resilience, there needs to be a well-coordinated effort to maximize efficiency and minimize redundancy; therefore, the result has been the consolidation of numerous cybersecurity entities when dealing with cybersecurity issues on a national level.

Compared to past rules, recent rules pertaining to operational resilience are more forward-looking and principles-based approaches. This is necessary to evolve with the growing complexity of the cyber resilience world. However, a principles-based approach is more useful when working with the risk for a given entity as opposed to systemic risk, which involves multiple entities.

Policies have long been in place for incident reporting and information sharing in the European Union (EU)—but, in taking it a step further, the **European Systemic Risk Board (ESRB)** has suggested the formation of a European-wide structure for coordinating incident reporting.

Also in the EU, the **Digital Operational Resilience Act (DORA)** has provisions to respond to ICT third-party risks associated with cloud computing. In addition, it includes provisions dealing with penetration stress testing (i.e., cyberattack simulation), which is a useful exercise to keep entities conscious of security issues on an ongoing basis.

Macprudential Tools

Circuit Breakers

Circuit breakers are essentially a temporary pause on normal rules and regulations, and they have been used more in the banking industry in recent years. In the context of major cyber incidents impacting the banking industry, it makes sense to have circuit breakers to minimize the negative impacts of customer loss of confidence and the resulting bank run on deposits. Therefore, circuit breakers supplement the controls already in place regarding bank deposits. Circuit breakers are analogous to a temporary stoppage to analyze the fallout from a major attack and to determine the upcoming steps for recovery.

The challenge, here, is for the bank to prove that a given cyberattack is sufficiently serious to be considered a *force majeure* event that will provide some recovery time to the bank after the attack. It could be argued that proper cyber controls and other preventative measures could have been put in place to avoid the negative effects of a cyberattack; therefore, it is not a *force majeure* event. At the same time, a compelling argument for the use of circuit breakers is their ability to control and minimize the negative effects of a potential bank run after a significant cyberattack on a bank.

Cooperative Arrangements

Cooperation among members may seem like an obvious solution for small- and medium-sized organizations, but it has taken time to occur in the financial sector where most organizations are large. With the development of information and knowledge-sharing resources within the financial sector, there is evidence of growing cooperation among organizations to develop more robust defenses against cyberattacks.

Within the United States, there has been progress on unifying the various cybersecurity standards and in transitioning from tick-the-box-style compliance requirements to a judgment-based system. The challenge remains to bring the level of standardization and thorough reporting to an international level, although cooperation within the realm of incident reporting has been on the rise in Europe as well as internationally.

ICT buffers to combat systemic risk are a good example of cooperation and allow for a much quicker recovery time for those banks that have suffered a major cyberattack. ICT buffers have been used in the past in the United States with its tri-party repo system. In the United States, there is a collectively owned entity (Sheltered Harbor, LLC) whereby any co-owner bank that suffers a serious cyberattack could allow its customers to transact on their accounts at another co-owner bank. With consistent data formats and operating procedures for storage and restoration of data, the risk to a co-owner bank from a cyberattack is substantially reduced.

Structural Measures

The reach of macroprudential regulation has inevitably spread to systemically important technological institutions (SITIs), given the numerous externalities arising from SITIs as well as the significant dependence of the financial sector on SITIs.

Alternatives to macroprudential regulation would include more thorough evaluations of critical providers, strongly encouraging multicloud services, and using private clouds.

Additionally, regulators could engage in “forced diversification” of firms to incentivize them to use multiclouds or multiple types of software. Penalties for concentrated software use could be used as negative motivators; however, penalties also introduce the issue of enforceability.



MODULE QUIZ 103.1

1. Which of the following statements regarding cloud storage and cloud computing is most accurate?
 - A. All major banks have adopted the use of multiclouds.
 - B. The increased use of cloud computing in recent years by banks has resulted in decreases in both cyber and ICT risks.
 - C. The advance of cloud storage and computing has allowed ICT risks and cybersecurity risks to be treated independently for analytical purposes.
 - D. A cloud services audit firm has recently reported that about 50% of cloud customers have experienced a physical outage in the past three years.
2. Which of the following statements regarding the interaction between cyber risks and financial stability is most accurate?
 - A. Agent-based models are useful to trace the source of systemic risks.
 - B. The severity of shocks tends to be higher for data integrity compared to data confidentiality.
 - C. The most important consideration is how financial risks can lead to the realization of technology risks.
 - D. Shocks to the CIA triad can trigger financial sector vulnerabilities that are focused on a few main areas, including profitability.
3. The concept of a bank being able to recover very quickly from a serious cyberattack by allowing its customers to access their accounts from another bank is best described as which of the following macroprudential tools?
 - A. Circuit breaker.
 - B. Structural measures.
 - C. Forced diversification.
 - D. Cooperative arrangement.
4. Which of the following statements regarding the use of circuit breakers is most accurate? Circuit breakers:
 - A. have been used extensively in the banking industry in the past.
 - B. are used to deal with the effects of a loss of confidence by bank customers.
 - C. are an improvement; therefore, they are a valid substitute to the controls in place regarding bank deposits.
 - D. are a long-term pause on normal rules and regulations to allow a bank sufficient time to recover from a serious cyberattack.

KEY CONCEPTS

LO 103.a

Cyber risk refers to the loss of confidentiality, integrity, and availability (CIA) of data and data systems that result from a cyberattack. ICT risk refers to disruptions pertaining to information and communication systems that may have similar effects as cyber risk, but do not stem from cyberattacks.

Financial institutions have strongly embraced cloud computing in recent years, which has resulted in a corresponding decrease in cyber risks but also an increase in ICT risks. The main issue is that the use of cloud computing moves some, but not all, of an organization's cyber risks to cloud service providers because the organization's providers and customers might not have adequate controls.

The significant adoption of cloud computing by financial institutions has resulted in the integration of cyber and ICT risks to an extent that there could be systemic implications.

LO 103.b

Cyber vulnerabilities that are triggered can cause negative financial impacts such as illiquidity, excessive leverage, and loss of customer trust. The most important consideration is how technology risks can lead to the realization of financial risks.

The modern-day design of cyber and ICT connections is highly integrated, which opens the potential for significant contagion effects. Financial contagion can occur extremely fast through the financial industry with the onset of just one key triggering event.

Because contagion effects are nonlinear in nature, it is challenging to predict the probability of key events with reasonable accuracy. The solution is to consider agent-based models to quantify the impacts of a cyber incident in terms of the CIA effects and how they affect the liquidity and asset values of the financial institutions in question.

LO 103.c

Regarding policy measures and micro-oriented tools, it is necessary to have certification and standardization of solutions as a starting point for building cybersecurity defenses. Given the many entities involved with cyber resilience, the result has been the consolidation of numerous cybersecurity entities. Compared to past rules, recent rules pertaining to operational resilience are more forward-looking and principles-based approaches. Other policy measures deal with coordinated incident reporting and information sharing.

Regarding macroprudential tools, circuit breakers are essentially a temporary pause on normal rules and regulations. In the context of major cyber incidents impacting the banking industry, it makes sense to have circuit breakers to minimize the negative impacts of customer loss of confidence and the resulting bank run on deposits. Another macroprudential tool is cooperation, and one aspect may involve unifying the various cybersecurity standards and transitioning from tick-the-box-style compliance requirements to a judgment-based system. ICT buffers to combat systemic risk are a good example of cooperation and allow for a much quicker recovery time for those banks that have suffered a major cyberattack. Finally, the reach of macroprudential regulation has inevitably spread to systemically important technological institutions

(SITIs) given the numerous externalities arising from SITIs as well as the significant dependence of the financial sector on SITIs.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 103.1

1. **A** Within the financial sector, all major banks have adopted multicloud computing. With more banks embracing cloud computing in recent years, it has resulted in a corresponding decrease in cyber risks but also an *increase* in ICT risks. The advance of cloud storage and computing has resulted in a *commingling* of ICT and cybersecurity risks with potential systemic impact. A cloud services audit firm has recently reported that about 80% (not 50%) of cloud customers have experienced a physical outage in the past three years. (LO 103.a)
2. **A** Agent-based models are useful for tracing the source of systemic risks, as what appears to happen in real life between technology and finance is not always completely accurate. However, such models are still in their infancy stage, but their basic principles could be applied to developing cyber and ICT stress tests. There tends to be a *lower* severity of shocks to data confidentiality and a higher severity to data integrity and availability of data and systems. The most important consideration is how *technology risks* can lead to the realization of *financial risks*. Shocks to the CIA triad can trigger financial sector vulnerabilities that are focused on a few main areas such as *liquidity, leverage, and trust*. (LO 103.b)
3. **D** In the United States, there is a collectively owned entity (Sheltered Harbor, LLC) whereby any co-owner bank who suffers a serious cyberattack could allow its customers to transact on their accounts at another co-owner bank. With consistent data formats and operating procedures for storage and restoration of data, the risk to a co-owner bank from a cyberattack is substantially reduced. This is a good example of a cooperative arrangement. (LO 103.c)
4. **B** In the context of major cyber incidents impacting the banking industry, it makes sense to have circuit breakers to minimize the negative impacts of customer loss of confidence and the resulting bank run on deposits. Circuit breakers have not been commonly used in the banking industry until recent years. Circuit breakers are a *complement* to the controls in place regarding bank deposits. Circuit breakers are a *temporary* (and, therefore, likely short-term) pause on normal rules and regulations to allow a bank sufficient time to recover from a serious cyberattack. (LO 103.c)

FORMULAS

Reading 83

investor risk premium: $E(R_M) - R_F = \bar{\gamma} \times \sigma_M^2$

security market line:

$$E(R_i) - R_F = \frac{\text{cov}(R_i, R_M)}{\text{var}(R_M)} \times [E(R_M) - R_F] = \beta_i \times [E(R_M) - R_F]$$

Reading 84

Fama-French three-factor model:

$$E(R_i) = R_F + \beta_{i,MKT} \times E(R_M - R_F) + \beta_{i,SMB} \times E(SMB) + \beta_{i,HML} \times E(HML)$$

Fama-French model with momentum effect:

$$E(R_i) = R_F + \beta_{i,MKT} \times E(R_M - R_F) + \beta_{i,SMB} \times E(SMB) + \beta_{i,HML} \times E(HML) + \beta_{i,WML} \times E(WML)$$

Reading 85

fundamental law of active management: $IR \approx IC \times \sqrt{BR}$

Reading 86

risk aversion = $\frac{\text{information ratio}}{2 \times \text{active risk}}$

marginal contribution to value added = (alpha of asset) - [2 × (risk aversion) × (active risk) × (marginal contribution to active risk of asset)]

Reading 87

diversified VaR: $\text{VaR}_p = Z_c \times \sigma_p \times P$

individual VaR: $\text{VaR}_i = Z_c \times \sigma_i \times |P_i| = Z_c \times \sigma_i \times |w_i| \times P$

standard deviation of a two-asset portfolio:

$$\sigma_P = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2}$$

VaR of a two-asset portfolio:

$$\text{VaR}_P = Z_c P \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2}$$

undiversified VaR: $\text{VaR}_P = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2 \text{VaR}_1 \text{VaR}_2} = \text{VaR}_1 + \text{VaR}_2$

standard deviation of equally weighted portfolio with equal standard deviations and correlations:

$$\sigma_P = \sigma \sqrt{\frac{1}{N} + \left(1 - \frac{1}{N}\right)\rho}$$

$$\text{marginal VaR: } \text{MVaR}_i = \frac{\text{VaR}}{P} \times \beta_i$$

$$\text{component VaR: } \text{CVaR}_i = (\text{MVaR}_i) \times (w_i \times P) = \text{VaR} \times \beta_i \times w_i$$

Reading 88

surplus = assets – liabilities

$$\Delta \text{surplus} = \Delta \text{assets} - \Delta \text{liabilities}$$

return on the surplus:

$$\begin{aligned} R_{\text{surplus}} &= \frac{\Delta \text{surplus}}{\text{assets}} = \frac{\Delta \text{assets}}{\text{assets}} - \left(\frac{\Delta \text{liabilities}}{\text{liabilities}} \right) \left(\frac{\text{liabilities}}{\text{assets}} \right) \\ &= R_{\text{asset}} - R_{\text{liabilities}} \left(\frac{\text{liabilities}}{\text{assets}} \right) \end{aligned}$$

Reading 89

$$\text{liquidity duration: } LD = \frac{Q}{(0.10 \times V)}$$

where:

LD = liquidity duration for the security on the assumption that the desired maximum daily volume of any security is 10%

Q = number of shares of the security

V = daily volume of the security

Reading 90

$$\text{Sharpe ratio: } S_A = \frac{\bar{R}_A - \bar{R}_F}{\sigma_A}$$

where:

$\bar{R}_A$ = average account return

$\bar{R}_F$ = average risk-free return

σ_A = standard deviation of account returns

$$\text{Treynor measure: } T_A = \frac{\bar{R}_A - \bar{R}_F}{\beta_A}$$

where:

$\bar{R}_A$ = average account return

$\bar{R}_F$ = average risk-free return

β_A = average beta

Jensen's alpha: $\alpha_A = R_A - E(R_A)$

where:

α_A = alpha

R_A = return on the account

$E(R_A) = R_F + \beta_A[E(R_M) - R_F]$

information ratio: $IR_A = \frac{\bar{R}_A - \bar{R}_B}{\sigma_{A-B}}$

where:

$\bar{R}_A$ = average account return

$\bar{R}_B$ = average benchmark return

σ_{A-B} = standard deviation of excess returns measured as the difference between account and benchmark returns

statistical significance of alpha returns: $t = \frac{\alpha - 0}{\sigma / \sqrt{N}}$

where:

α = alpha estimate

σ = alpha estimate volatility

N = sample number of observations

standard error of alpha estimate = $\sigma / \sqrt{N}$

asset allocation attribution: $\sum(\text{portfolio weight} - \text{benchmark weight}) \times R_B$

selection attribution: $\sum \text{portfolio weight} \times (R_P - R_B)$

aggregate contribution: $\sum(\text{portfolio weight} \times R_P) - \sum(\text{benchmark weight} \times R_B)$

APPENDIX

USING THE CUMULATIVE Z-TABLE

Probability Example

Assume that the annual earnings per share (EPS) for a large sample of firms is normally distributed with a mean of \$5.00 and a standard deviation of \$1.50. What is the approximate probability of an observed EPS value falling between \$3.00 and \$7.25?

If $\text{EPS} = x = \$7.25$, then $z = (x - \mu) / \sigma = (\$7.25 - \$5.00) / \$1.50 = +1.50$.

If $\text{EPS} = x = \$3.00$, then $z = (x - \mu) / \sigma = (\$3.00 - \$5.00) / \$1.50 = -1.33$.

For z -value of 1.50: Use the row headed 1.5 and the column headed 0 to find the value 0.9332. This represents the area under the curve to the left of the critical value 1.50.

For z -value of -1.33: Use the row headed 1.3 and the column headed 3 to find the value 0.9082. This represents the area under the curve to the left of the critical value +1.33. The area to the left of -1.33 is $1 - 0.9082 = 0.0918$.

The area between these critical values is $0.9332 - 0.0918 = 0.8414$, or 84.14%.

Hypothesis Testing—One-Tailed Test Example

A sample of a stock's returns on 36 nonconsecutive days results in a mean return of 2.0%. Assume the population standard deviation is 20.0%. Can we say with 95% confidence that the mean return is greater than 0%?

$H_0: \mu \leq 0.0\%$, $H_A: \mu > 0.0\%$. The test statistic = $z\text{-statistic} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$
 $= (2.0 - 0.0) / (20.0 / 6) = 0.60$.

The significance level = $1.0 - 0.95 = 0.05$, or 5%.

Since this is a one-tailed test with an alpha of 0.05, we need to find the value 0.95 in the cumulative z -table. The closest value is 0.9505, with a corresponding critical z -value of 1.65. Since the test statistic is less than the critical value, we fail to reject H_0 .

Hypothesis Testing—Two-Tailed Test Example

Using the previous assumptions, suppose that the analyst now wants to determine with 99% confidence that the stock's return is not equal to 0.0%.

$H_0: \mu = 0.0\%$, $H_A: \mu \neq 0.0\%$. The test statistic (z -value) = $(2.0 - 0.0) / (20.0 / 6)$
 $= 0.60$. The significance level = $1.0 - 0.99 = 0.01$, or 1%.

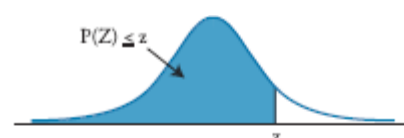
Since this is a two-tailed test with an alpha of 0.01, there is a 0.005 rejection region in both tails. Thus, we need to find the value 0.995 ($1.0 - 0.005$) in the table. The closest value is 0.9951, which corresponds to a critical z -value of 2.58. Since the test statistic is

less than the critical value, we fail to reject H_0 and conclude that the stock's return equals 0.0%.

Cumulative Z-Table

$P(Z \leq z) = N(z)$ for $z \geq 0$

$P(Z \leq -z) = 1 - N(z)$



z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9889
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

STUDENT'S *t*-DISTRIBUTION

Level of Significance for One-Tailed Test						
df	0.100	0.050	0.025	0.01	0.005	0.0005
Level of Significance for Two-Tailed Test						
df	0.20	0.10	0.05	0.02	0.01	0.001
1	3.078	6.314	12.706	31.821	63.657	636.619
2	1.886	2.920	4.303	6.965	9.925	31.599
3	1.638	2.353	3.182	4.541	5.841	12.294
4	1.533	2.132	2.776	3.747	4.604	8.610
5	1.476	2.015	2.571	3.365	4.032	6.869
6	1.440	1.943	2.447	3.143	3.707	5.959
7	1.415	1.895	2.365	2.998	3.499	5.408
8	1.397	1.860	2.306	2.896	3.355	5.041
9	1.383	1.833	2.262	2.821	3.250	4.781
10	1.372	1.812	2.228	2.764	3.169	4.587
11	1.363	1.796	2.201	2.718	3.106	4.437
12	1.356	1.782	2.179	2.681	3.055	4.318
13	1.350	1.771	2.160	2.650	3.012	4.221
14	1.345	1.761	2.145	2.624	2.977	4.140
15	1.341	1.753	2.131	2.602	2.947	4.073
16	1.337	1.746	2.120	2.583	2.921	4.015
17	1.333	1.740	2.110	2.567	2.898	3.965
18	1.330	1.734	2.101	2.552	2.878	3.922
19	1.328	1.729	2.093	2.539	2.861	3.883
20	1.325	1.725	2.086	2.528	2.845	3.850
21	1.323	1.721	2.080	2.518	2.831	3.819
22	1.321	1.717	2.074	2.508	2.819	3.792
23	1.319	1.714	2.069	2.500	2.807	3.768
24	1.318	1.711	2.064	2.492	2.797	3.745
25	1.316	1.708	2.060	2.485	2.787	3.725
26	1.315	1.706	2.056	2.479	2.779	3.707
27	1.314	1.703	2.052	2.473	2.771	3.690
28	1.313	1.701	2.048	2.467	2.763	3.674
29	1.311	1.699	2.045	2.462	2.756	3.659
30	1.310	1.697	2.042	2.457	2.750	3.646
40	1.303	1.684	2.021	2.423	2.704	3.551
60	1.296	1.671	2.000	2.390	2.660	3.460
120	1.289	1.658	1.980	2.358	2.617	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.291

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